

BOOK 2

KINEMATICS

2 Kinematics

2.1 Equations of motion

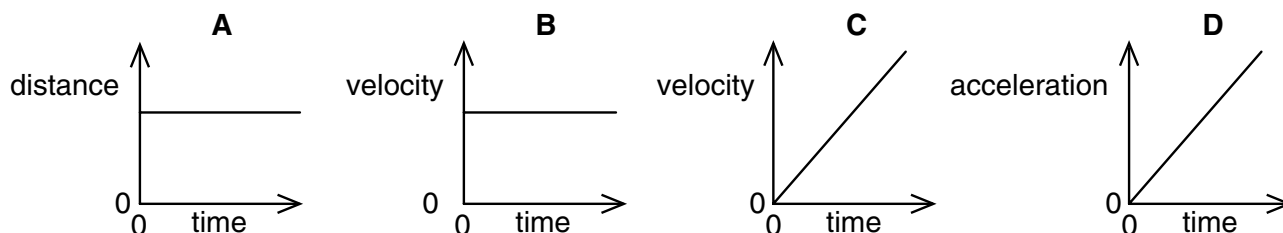
Candidates should be able to:

- 1 define and use distance, displacement, speed, velocity and acceleration
- 2 use graphical methods to represent distance, displacement, speed, velocity and acceleration
- 3 determine displacement from the area under a velocity–time graph
- 4 determine velocity using the gradient of a displacement–time graph
- 5 determine acceleration using the gradient of a velocity–time graph
- 6 derive, from the definitions of velocity and acceleration, equations that represent uniformly accelerated motion in a straight line
- 7 solve problems using equations that represent uniformly accelerated motion in a straight line, including the motion of bodies falling in a uniform gravitational field without air resistance
- 8 describe an experiment to determine the acceleration of free fall using a falling object
- 9 describe and explain motion due to a uniform velocity in one direction and a uniform acceleration in a perpendicular direction

1 9702/1/O/N/02/Q2*

A particle is moving in a straight line with uniform acceleration.

Which graph represents the motion of the particle?



2 9702/01/M/J/03/Q4

Which experimental technique reduces the systematic error of the quantity being investigated?

- A adjusting an ammeter to remove its zero error before measuring a current
- B measuring several internodal distances on a standing wave to find the mean internodal distance
- C measuring the diameter of a wire repeatedly and calculating the average
- D timing a large number of oscillations to find a period

3 9702/01/M/J/03/Q5

A student makes measurements from which she calculates the speed of sound as 327.66 m s^{-1} .

She estimates that her result is accurate to $\pm 3\%$.

Which of the following gives her result expressed to the appropriate number of significant figures?

- A 327.7 m s^{-1}
- B 328 m s^{-1}
- C 330 m s^{-1}
- D 300 m s^{-1}

4 9702/01/M/J/03/Q6

A steel rule can be read to the nearest millimetre. It is used to measure the length of a bar whose true length is 895 mm. Repeated measurements give the following readings.

length / mm	892, 891, 892, 891, 891, 892
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Are the readings accurate and precise to within 1 mm?

	results are accurate to within 1 mm	results are precise to within 1 mm
A	no	no
B	no	yes
C	yes	no
D	yes	yes

5 9702/13/O/N/12/Q3

A 1.5 V cell supplies 0.20 A to a lamp for seven hours before the lamp goes out.

What is a sensible estimate for the initial chemical energy content of the cell?

- A $1 \times 10^2 \text{ J}$
- B $1 \times 10^4 \text{ J}$
- C $1 \times 10^6 \text{ J}$
- D $1 \times 10^8 \text{ J}$

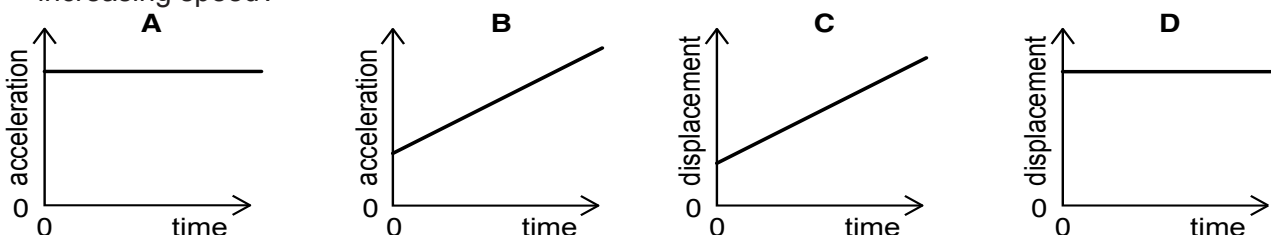
6 9702/1/M/J/02/Q3

Which formula could be correct for the speed v of ocean waves in terms of the density ρ of sea-water, the acceleration of free fall g , the depth h of the ocean and the wavelength λ ?

- A $v = \sqrt{g\lambda}$
- B $v = \sqrt{\frac{g}{h}}$
- C $v = \sqrt{\rho gh}$
- D $v = \sqrt{\frac{g}{\rho}}$

7 9702/1/M/J/02/Q7

Which graph represents the motion of a car that is travelling along a straight road with a uniformly increasing speed?



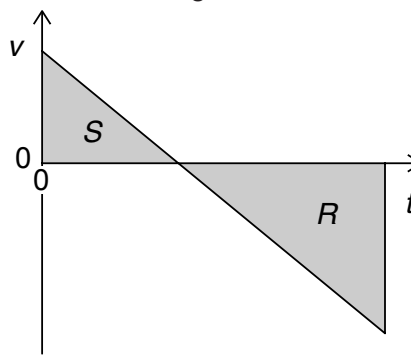
8 9702/1/M/J/02/Q8

A stone is thrown upwards from the top of a cliff. After reaching its maximum height, it falls past the cliff-top and into the sea.

The graph shows how the vertical velocity v of the stone varies with time t after being thrown upwards. R and S are the magnitudes of the areas of the two triangles.

What is the height of the cliff-top above the sea?

- A R B S
C $R + S$ D $R - S$



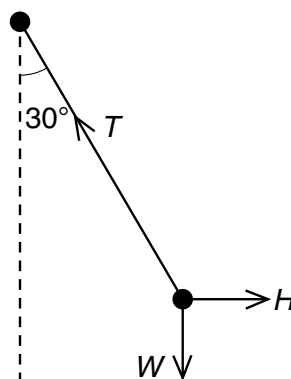
9 9702/1/O/N/02/Q3

A pendulum bob is held stationary by a horizontal force H . The three forces acting on the bob are shown in the diagram.

The tension in the string of the pendulum is T .
The weight of the pendulum bob is W .

Which statement is correct?

- A $H = T \cos 30^\circ$ B $T = H \sin 30^\circ$
C $W = T \cos 30^\circ$ D $W = T \sin 30^\circ$



10 9702/1/O/N/02/Q4

What is meant by the weight of an object?

- A the gravitational field acting on the object
B the gravitational force acting on the object
C the mass of the object multiplied by gravity
D the object's mass multiplied by its acceleration

11 9702/12/O/N/13/Q8

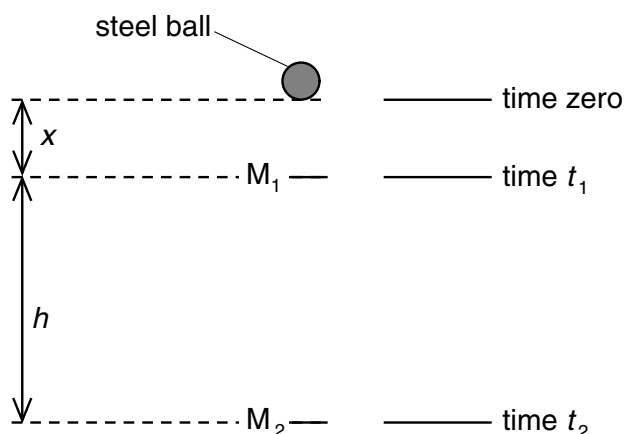
An aeroplane travels at an average speed of 600 km h^{-1} on an outward flight and at 400 km h^{-1} on the return flight over the same distance.

What is the average speed of the whole flight?

- A 111 ms^{-1} B 167 ms^{-1} C 480 km h^{-1} D 500 km h^{-1}

12 9702/12/M/J/12/Q8**

Two markers M_1 and M_2 are set up a vertical distance h apart.



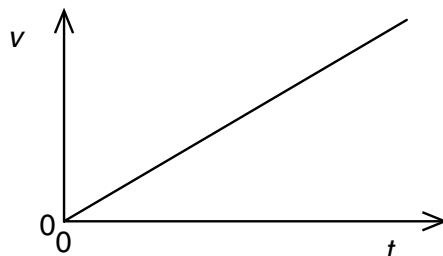
When a steel ball is released from rest from a point a distance x above M_1 , it is found that the ball takes time t_1 to reach M_1 and time t_2 to reach M_2 .

Which expression gives the acceleration of the ball?

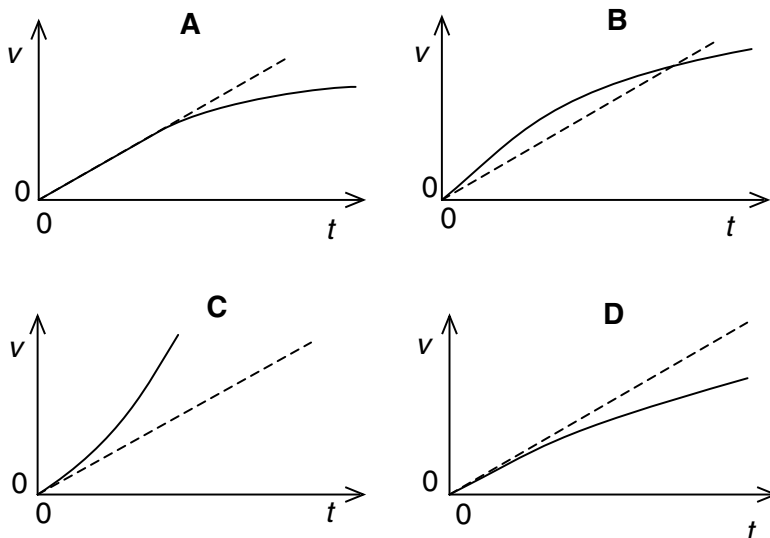
- A $\frac{2h}{t_2^2}$ B $\frac{2h}{(t_2 + t_1)}$ C $\frac{2h}{(t_2 - t_1)^2}$ D $\frac{2h}{(t_2^2 - t_1^2)}$

13 9702/1/O/N/02/Q10

A body falls from rest in a vacuum near the Earth's surface. The variation with time t of its speed v is shown below.



Which graph shows the variation with time t of the speed v of the same ball falling in air at the same place on Earth?



14 9702/01/M/J/03/Q7

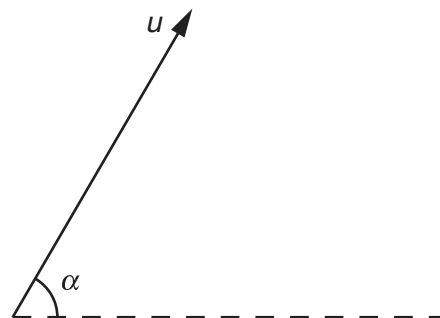
A projectile is fired at an angle α to the horizontal at a speed u , as shown.

What will be the vertical and horizontal components of its velocity after a time t ?

Assume that air resistance is negligible.

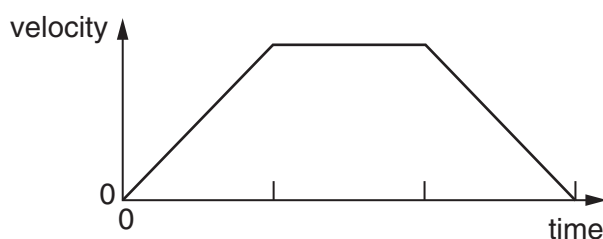
The acceleration of free fall is g .

	vertical component	horizontal component
A	$u \sin \alpha$	$u \cos \alpha$
B	$u \sin \alpha - gt$	$u \cos \alpha - gt$
C	$u \sin \alpha - gt$	$u \cos \alpha$
D	$u \cos \alpha$	$u \sin \alpha - gt$

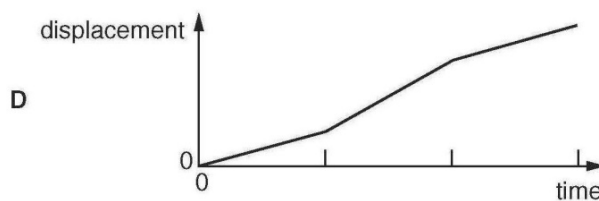
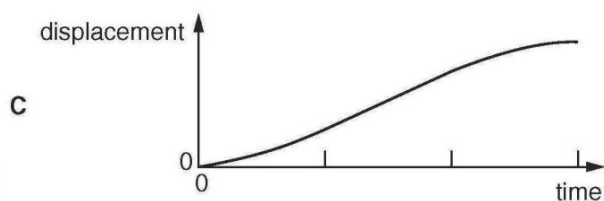
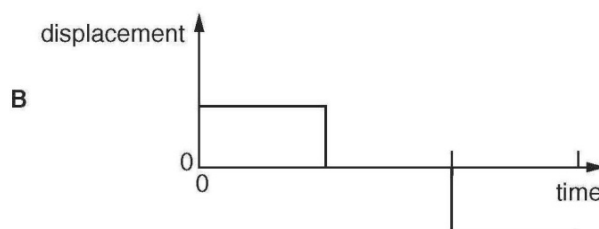
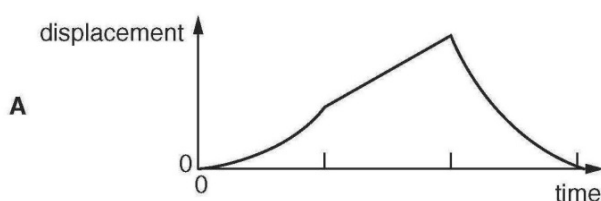


15 9702/12/O/N/12/Q9*

The graph of velocity against time for an object moving in a straight line is shown.



Which of the following is the corresponding graph of displacement against time?



16 9702/01/M/J/04/Q7

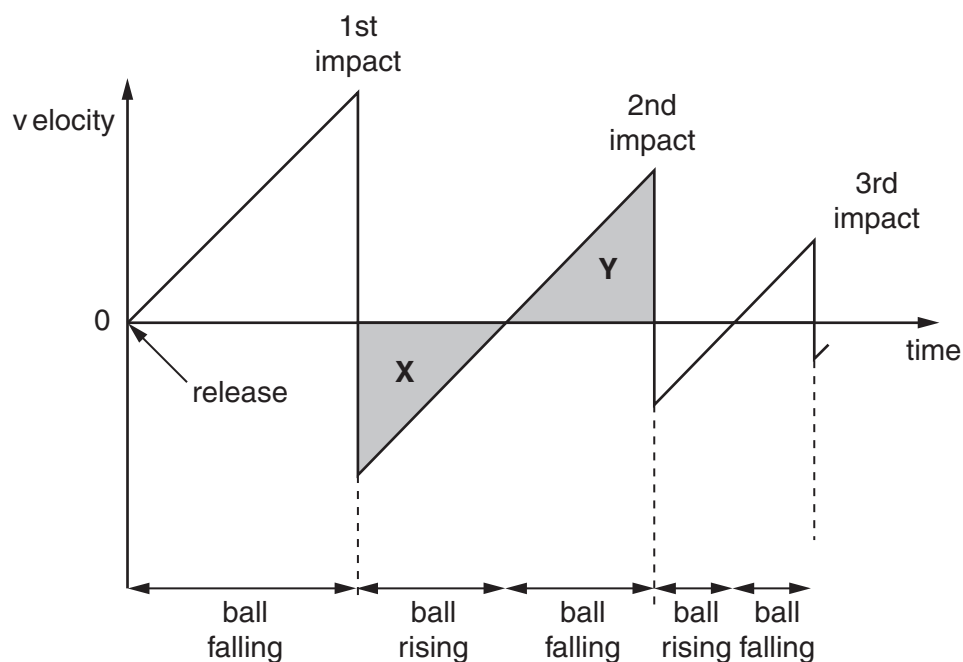
A car is travelling with uniform acceleration along a straight road. The road has marker posts every 100 m. When the car passes one post, it has a speed of 10 m s^{-1} and, when it passes the next one, its speed is 20 m s^{-1} .

What is the car's acceleration?

- A** 0.67 m s^{-2} **B** 1.5 m s^{-2} **C** 2.5 m s^{-2} **D** 6.0 m s^{-2}

17 9702/01/M/J/03/Q9

A ball is released from rest above a horizontal surface. The graph shows the variation with time of its velocity.



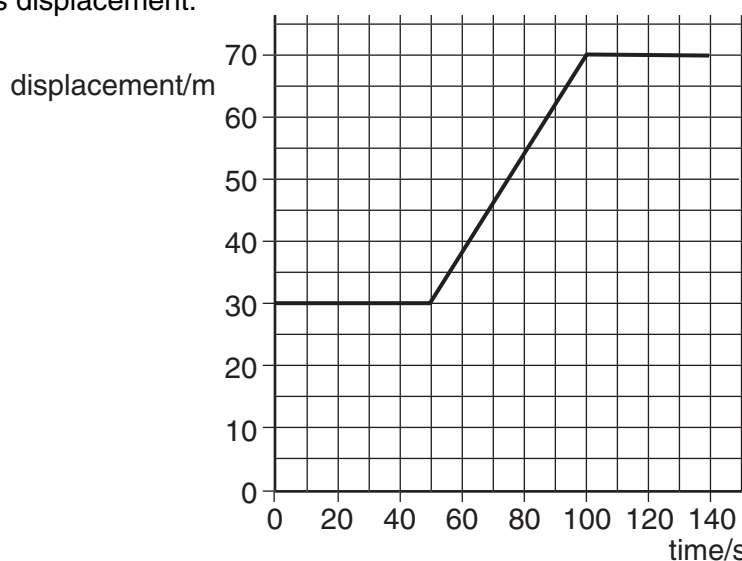
Areas **X** and **Y** are equal.

This is because

- A** the ball's acceleration is the same during its upward and downward motion.
- B** the speed at which the ball leaves the surface after an impact is equal to the speed at which it returns to the surface for the next impact.
- C** for one impact, the speed at which the ball hits the surface equals the speed at which it leaves the surface.
- D** the ball rises and falls through the same distance between impacts.

18 9702/01/O/N/03/Q7

A car at rest in a traffic queue moves forward in a straight line and then comes to rest again. The graph shows the variation with time of its displacement.



What is its speed while it is moving?

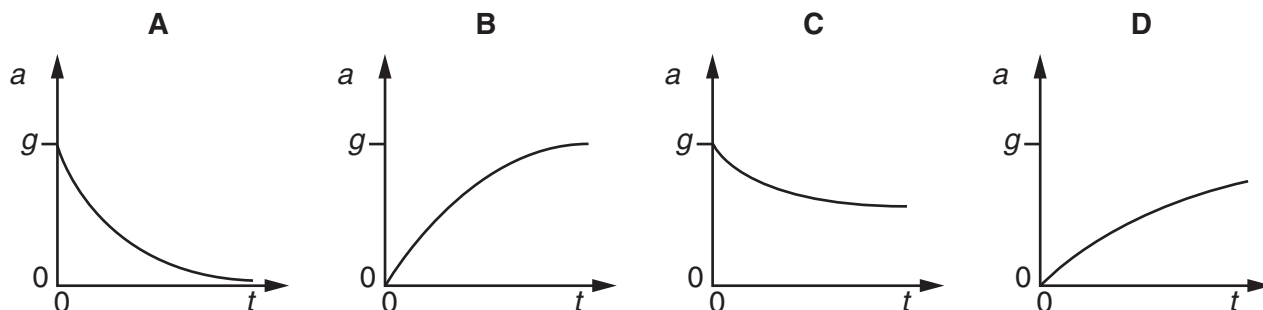
- A** 0.70 m s^{-1} **B** 0.80 m s^{-1}
- C** 1.25 m s^{-1} **D** 1.40 m s^{-1}

19 9702/01/O/N/03/Q8

An object is dropped from a great height and falls through air of uniform density.

The acceleration of free fall is g .

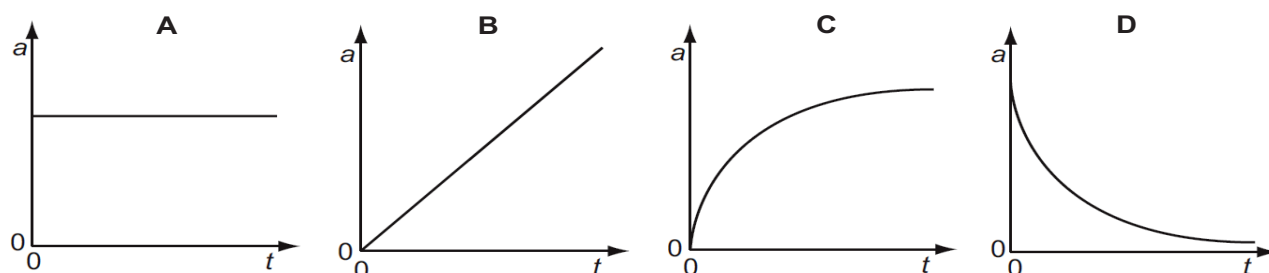
Which graph could show the variation with time t of the acceleration a of the object?



20 9702/01/M/J/04/Q8*

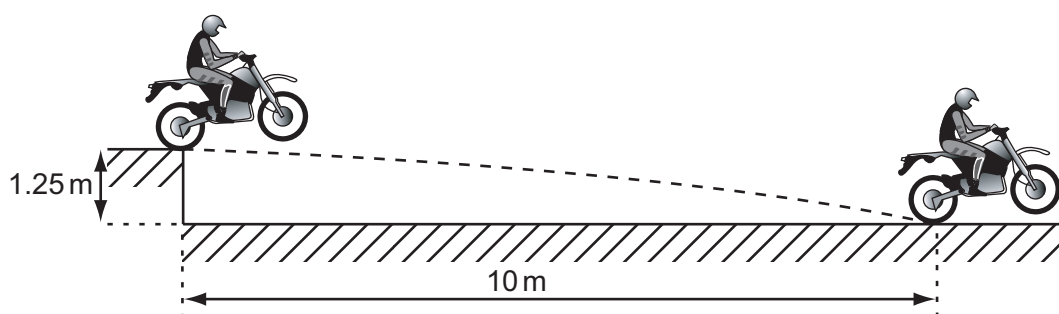
A tennis ball is released from rest at the top of a tall building.

Which graph best represents the variation with time t of the acceleration a of the ball as it falls, assuming that the effects of air resistance are appreciable?



21 9702/01/M/J/04/Q9

A motorcycle stunt-rider moving horizontally takes off from a point 1.25 m above the ground, landing 10 m away as shown.



What was the speed at take-off?

- A 5 ms^{-1} B 10 ms^{-1} C 15 ms^{-1} D 20 ms^{-1}

22 9702/01/O/N/07/Q7

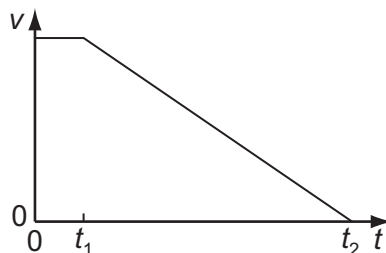
The symbol g represents the acceleration of free fall.

Which of these statements is correct?

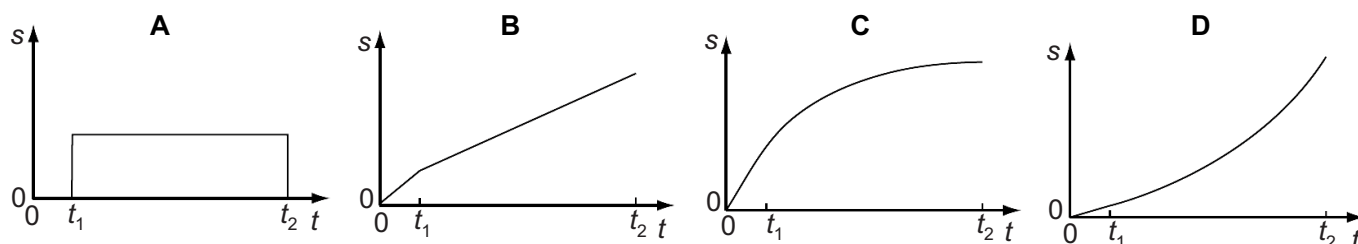
- A g is gravity. C g is the ratio weight / mass.
 B g is reduced by air resistance. D g is the weight of an object.

23 9702/01/O/N/04/Q8*

When a car driver sees a hazard ahead, she applies the brakes as soon as she can and brings the car to rest. The graph shows how the speed v of the car varies with time t after the hazard is seen.



Which graph represents the variation with time t of the distance s travelled by the car after the hazard has been seen?



24 9702/01/O/N/04/Q9

An object falls 10.0 m from rest before entering some water.

Assuming negligible air resistance, what is the time taken to reach the water and the speed with which the object reaches the water?

	time / ms	speed / m s^{-1}
A	1.02	10.0
B	1.02	14.0
C	1.43	10.0
D	1.43	14.0

25 9702/01/O/N/04/Q10

A constant mass undergoes uniform acceleration.

Which of the following is a correct statement about the resultant force acting on the mass?

- A** It increases uniformly with respect to time.
- B** It is constant but not zero.
- C** It is proportional to the displacement from a fixed point.
- D** It is proportional to the velocity.

26 9702/01/M/J/07/Q6

What gives the value of a body's acceleration?

- A** the area under its displacement-time graph
- B** the area under its velocity-time graph
- C** the gradient of its displacement-time graph
- D** the gradient of its velocity-time graph

27 9702/01/M/J/07/Q8

A stone is dropped from the top of a tower of height 40 m. The stone falls from rest and air resistance is negligible.

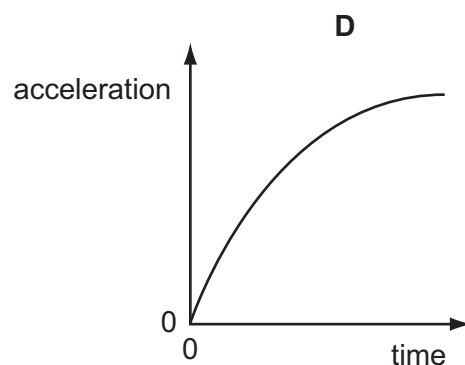
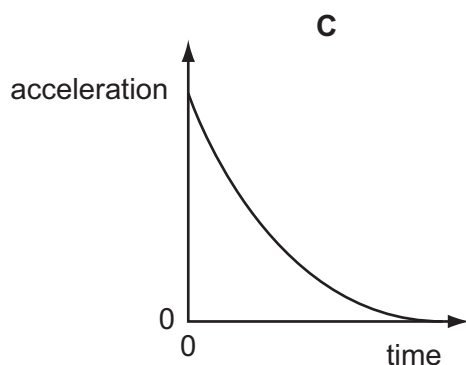
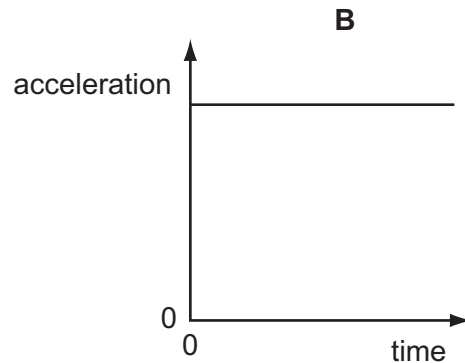
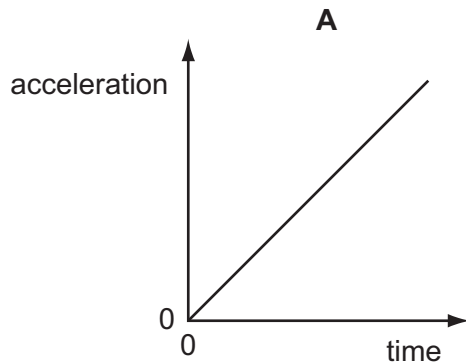
What time is taken for the stone to fall the last 10 m to the ground?

- A** 0.38 s
- B** 1.4 s
- C** 2.5 s
- D** 2.9 s

28 9702/01/M/J/08/Q8*

A football is dropped from the top of a tall building.

Which acceleration-time graph best represents the motion of the football through the air?



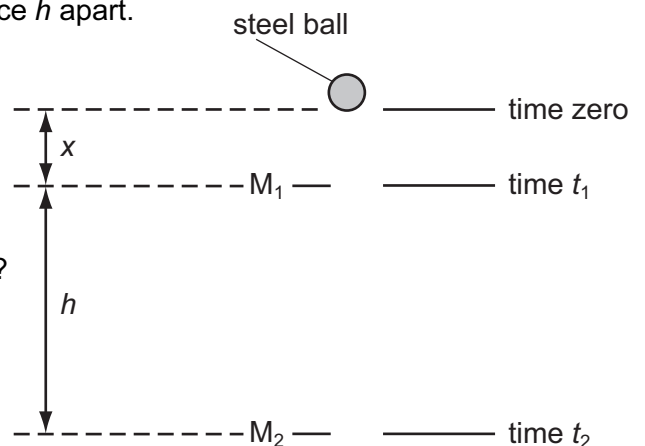
29 9702/12/M/J/12/Q8

Two markers M_1 and M_2 are set up a vertical distance h apart.

A steel ball is released at time zero from a point a distance x above M_1 . The ball reaches M_1 at time t_1 and reaches M_2 at time t_2 . The acceleration of the ball is constant.

Which expression gives the acceleration of the ball?

- A** $\frac{2h}{t_2^2}$ **B** $\frac{2h}{(t_2 + t_1)}$
- C** $\frac{2h}{(t_2 - t_1)^2}$ **D** $\frac{2h}{(t_2^2 - t_1^2)}$



30 9702/01/M/J/06/Q7

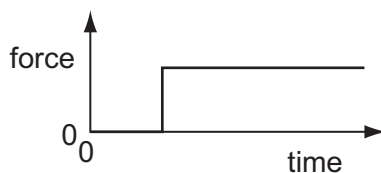
An experiment is done to measure the acceleration of free fall of a body from rest.

Which measurements are needed?

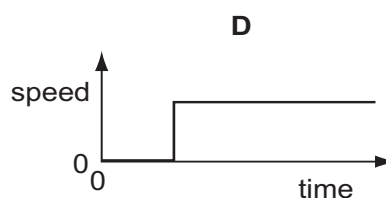
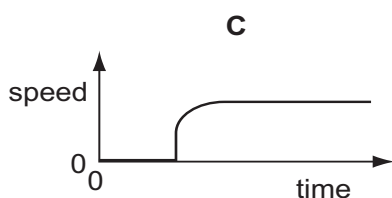
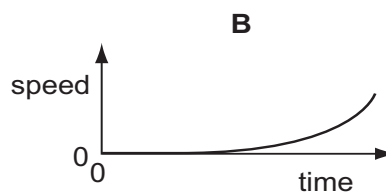
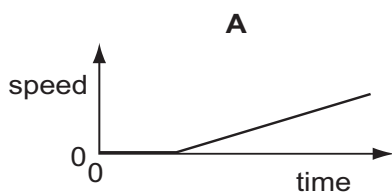
- A** the height of fall and the time of fall **C** the mass of the body and the height of fall
- B** the height of fall and the weight of the body **D** the mass of the body and the time of fall

31 9702/01/O/N/05/Q8

A car driver sharply presses down the accelerator when the traffic lights go green. The resultant horizontal force acting on the car varies with time as shown.



Which graph shows the variation with time of the speed of the car?

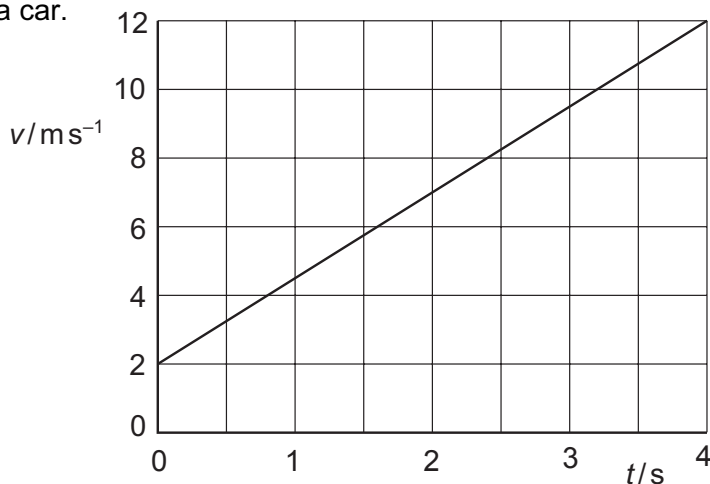


32 9702/01/M/J/05/Q8*

The diagram shows a velocity-time graph for a car.

What is the distance travelled between time $t = 0$ and $t = 4$ s?

- A** 2.5 m **B** 3.0 m
C 20 m **D** 28 m



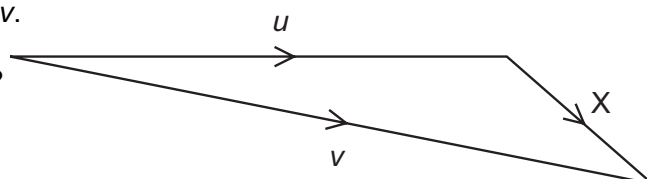
33 9702/01/M/J/07/Q7

An object has an initial velocity u . It is subjected to a constant force F for t seconds, causing a constant acceleration a . The force is **not** in the same direction as the initial velocity.

A vector diagram is drawn to find the final velocity v .

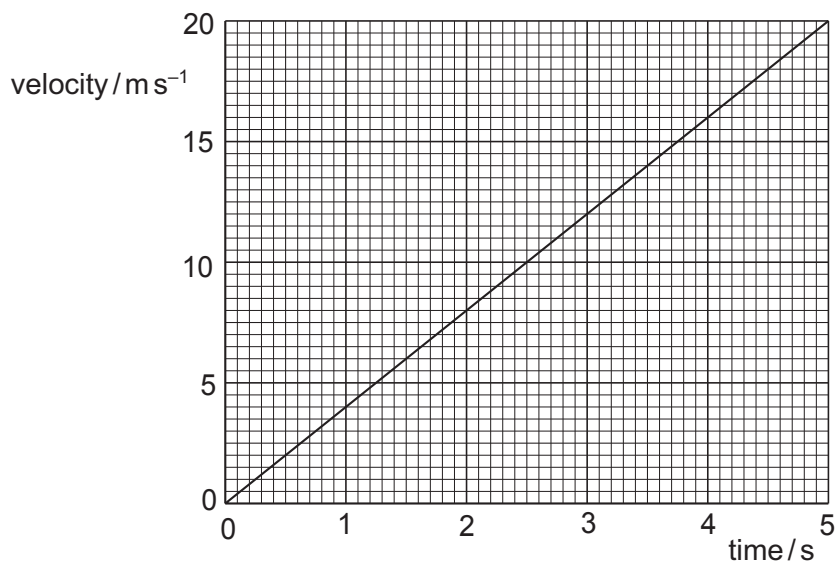
What is the length of side X of the vector diagram?

- A** F **B** Ft
C at **D** $u + at$



34 9702/12/O/N/11/Q6*

The velocity of an object during the first five seconds of its motion is shown on the graph.

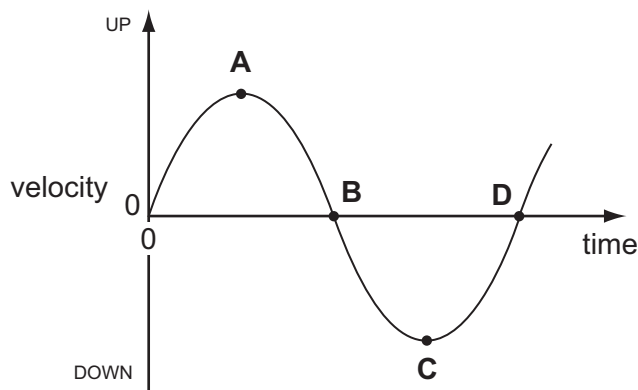


What is the distance travelled by the object in this time?

- A** 4 m **B** 20 m
C 50 m **D** 100 m

35 9702/13/O/N/14/Q7*

The diagram shows a velocity-time graph for a mass moving up and down on the end of a spring. Which point represents the velocity of the mass when at the lowest point of its motion?



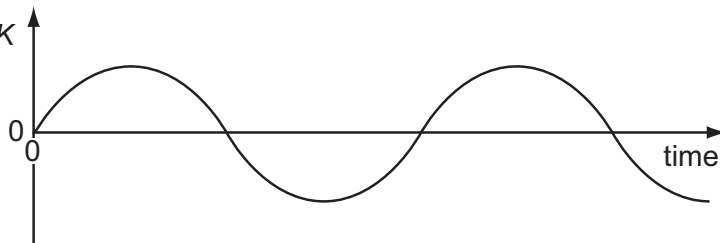
36 9702/01/O/N/07/Q8

A particle moves along a straight line. A particular property K of the particle's motion is plotted against time.

At any time, the slope of the graph is the K acceleration of the particle.

What is the property K ?

- A** the displacement of the particle
B the distance travelled by the particle
C the speed of the particle
D the velocity of the particle



37 9702/01/M/J/08/Q7

The acceleration of free fall on a planet P is $\frac{1}{6}$ of the acceleration of free fall on Earth.

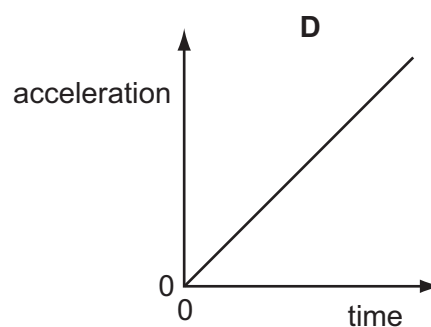
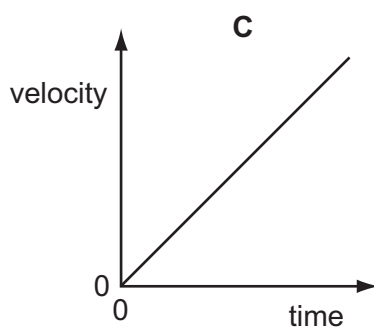
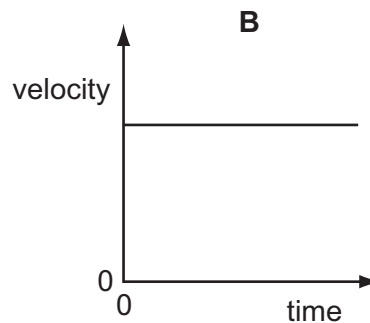
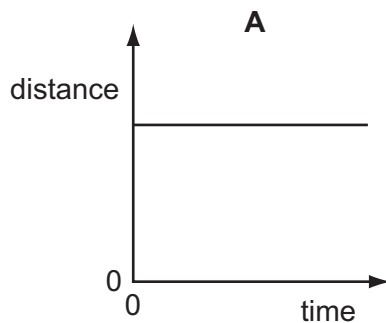
The mass of a body on planet P is 30 kg.

What is its weight on planet P?

- A** 4.9 N **B** 49 N **C** 180 N **D** 290 N

38 9702/1/O/N/02/Q2

A particle is moving in a straight line with uniform acceleration.
Which graph represents the motion of the particle?

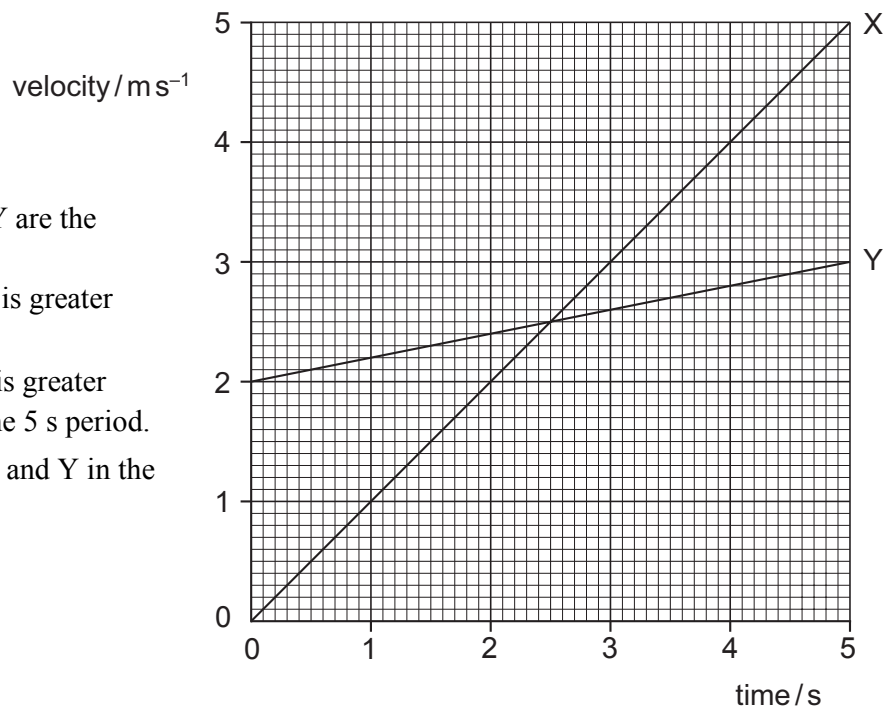


39 9702/01/O/N/06/Q8

The graph shows velocity-time plots for two vehicles X and Y. The accelerations and distances travelled by the two vehicles can be estimated from these plots.

Which statement is correct?

- A** The accelerations of X and Y are the same at 2.5 s.
- B** The initial acceleration of Y is greater than that of X.
- C** The distance travelled by X is greater than that travelled by Y in the 5 s period.
- D** The distances travelled by X and Y in the 5 s period are the same.

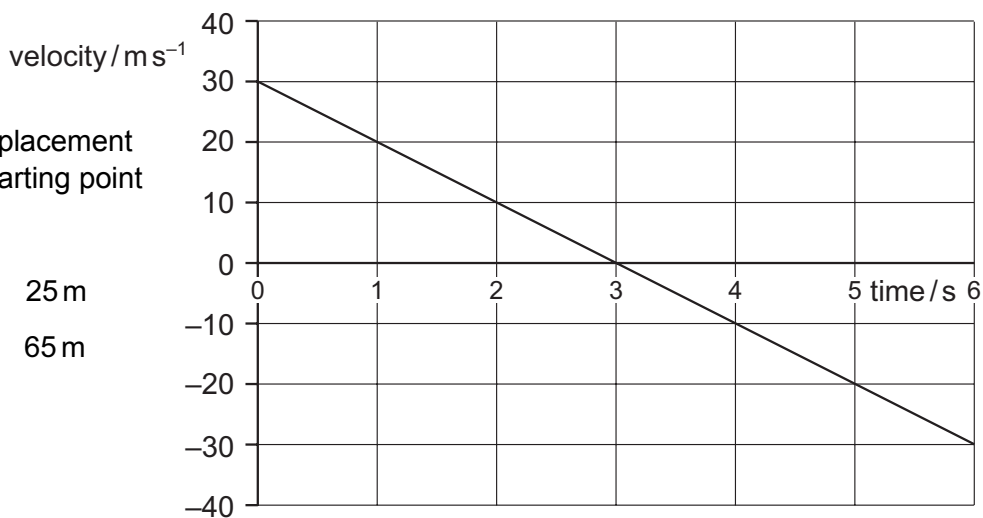


40 9702/01/O/N/07/Q9

A stone is thrown vertically upwards. A student plots the variation with time of its velocity.

What is the vertical displacement of the stone from its starting point after 5 seconds?

- A 20 m B 25 m
C 45 m D 65 m



41 9702/01/M/J/08 /Q6

An object accelerates in a direction that is always perpendicular to its motion.

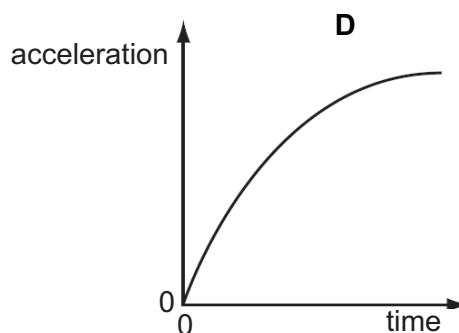
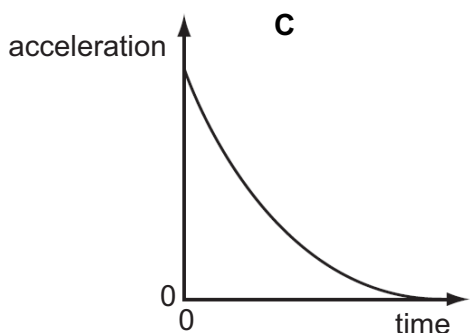
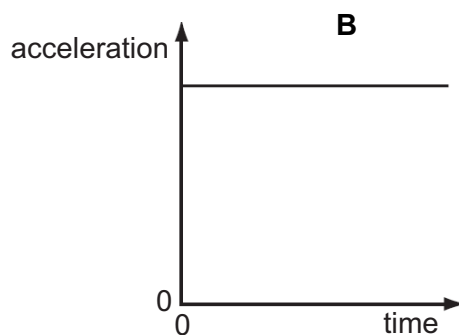
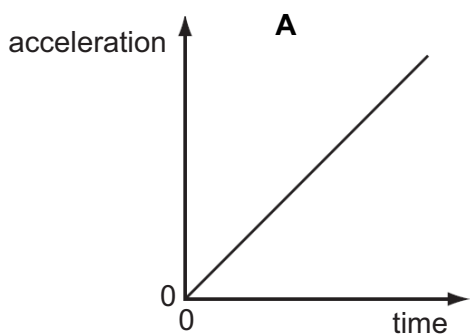
What is the effect, if any, of the acceleration on the object's speed and direction?

	speed	direction
A	changes	changes
B	changes	constant
C	constant	changes
D	constant	constant

42 9702/01/M/J/08/Q8

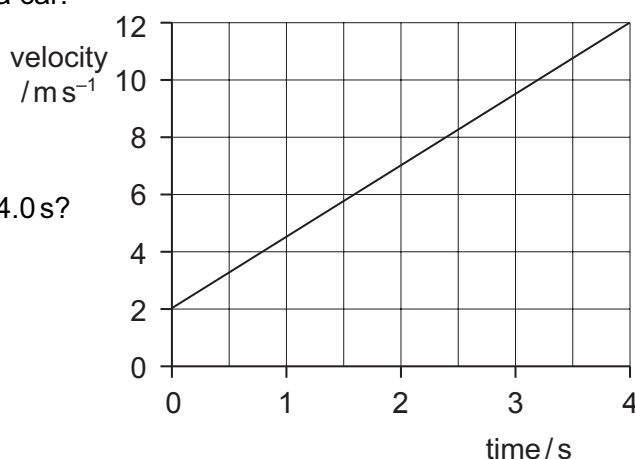
A football is dropped from the top of a tall building.

Which acceleration-time graph best represents the motion of the football through the air?



43 9702/01/O/N/08/Q6

The diagram shows a velocity-time graph for a car.



What is the distance travelled during the first 4.0 s?

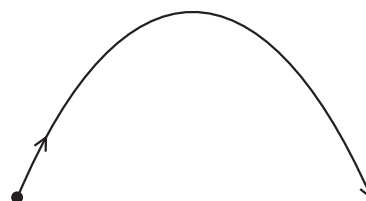
- A 2.5 m B 3.0 m
C 20 m D 28 m

44 9702/01/O/N/08/Q7

A stone is thrown upwards and follows a curved path.

Air resistance is negligible.

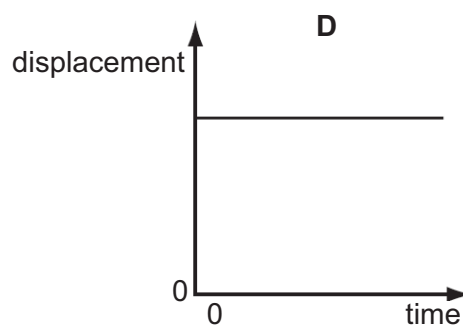
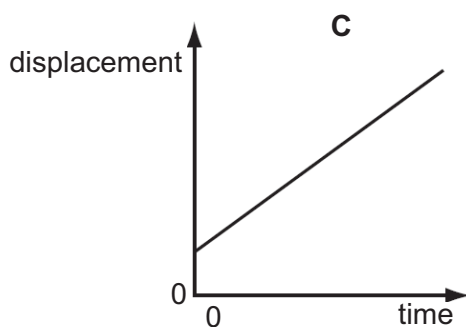
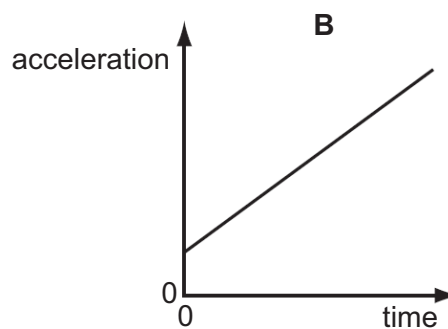
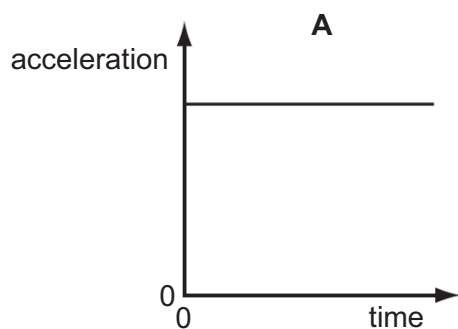
Why does the path have this shape?



- A The stone has a constant horizontal velocity and constant vertical acceleration.
B The stone has a constant horizontal acceleration and constant vertical velocity.
C The stone has a constant upward acceleration followed by a constant downward acceleration.
D The stone has a constant upward velocity followed by a constant downward velocity.

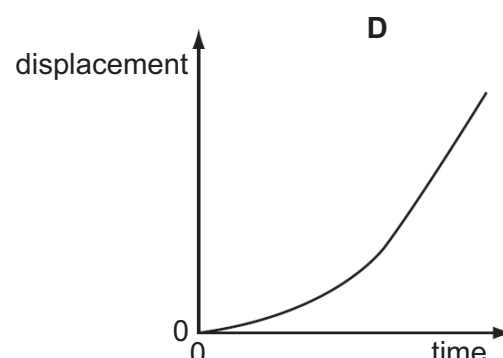
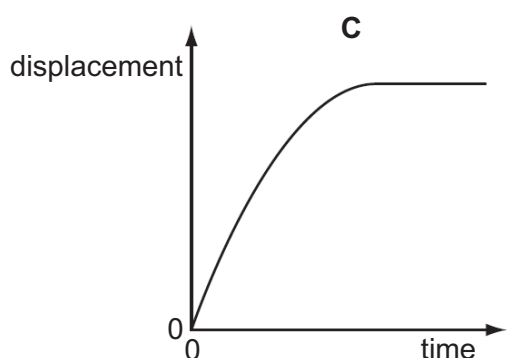
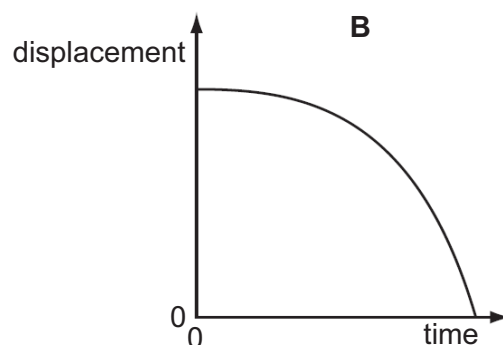
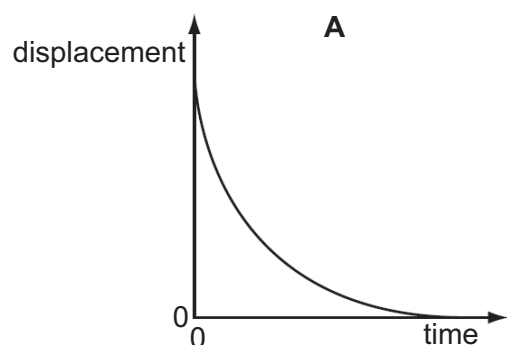
45 9702/11/O/N/14/Q6*

Which graph represents the motion of a car that is travelling along a straight road with a speed that increases uniformly with time?



46 9702/01/M/J/09/Q5

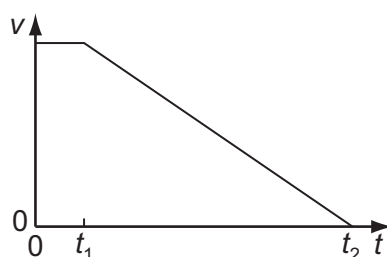
Which displacement-time graph best represents the motion of a falling sphere, the initial acceleration of which eventually reduces until it begins to travel at constant terminal velocity?



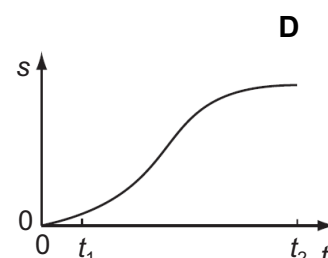
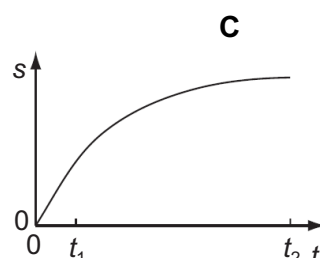
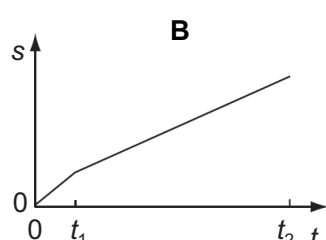
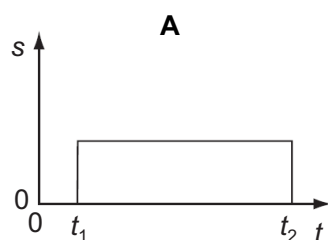
47 9702/01/M/J/09/Q6

When a car driver sees a hazard ahead, she applies the brakes as soon as she can and brings the car to rest.

The graph shows how the speed v of the car varies with time t after she sees the hazard.



Which graph represents the variation with time t of the distance s travelled by the car after she has seen the hazard?



48 9702/12/O/N/09/Q4*

On a particular railway, a train driver applies the brake of the train at a yellow signal, a distance of 1.0 km from a red signal, where it stops.

The maximum deceleration of the train is 0.2 m s^{-2} .

Assuming uniform deceleration, what is the maximum safe speed of the train at the yellow signal?

- A** 20 m s^{-1} **B** 40 m s^{-1} **C** 200 m s^{-1} **D** 400 m s^{-1}

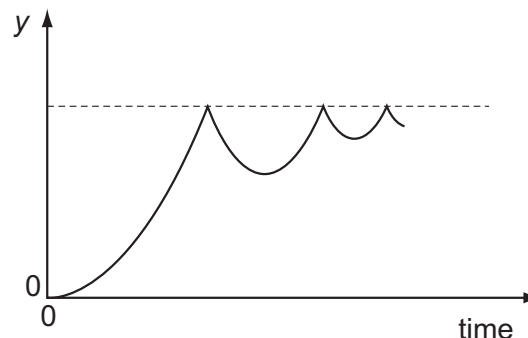
49 9702/12/O/N/09/Q5

A ball is released from rest above a horizontal surface and bounces several times.

The graph shows how, for this ball, a quantity y varies with time.

What is the quantity y ?

- A** acceleration **B** displacement
C kinetic energy **D** velocity

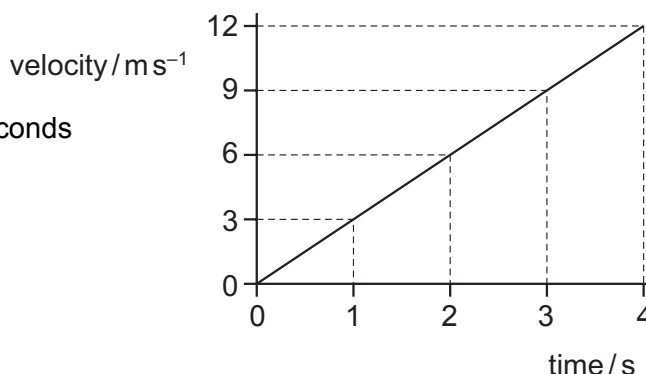


50 9702/12/O/N/09/Q6

The diagram shows a velocity-time graph.

What is the displacement during the last 2 seconds of the motion?

- A** 6 m **B** 12 m
C 18 m **D** 24 m



51 9702/12/M/J/10/Q4

An ion is accelerated by a series of electrodes in a vacuum. A graph of the power supplied to the ion is plotted against time.

What is represented by the area under the graph between two times?

- A** the change in kinetic energy of the ion
B the average force on the ion
C the change in momentum of the ion
D the change in velocity of the ion

52 9702/11/O/N/10/Q9

In order that a train can stop safely, it will always pass a signal showing a yellow light before it reaches a signal showing a red light. Drivers apply the brake at the yellow light and this results in a uniform deceleration to stop exactly at the red light.

The distance between the red and yellow lights is x .

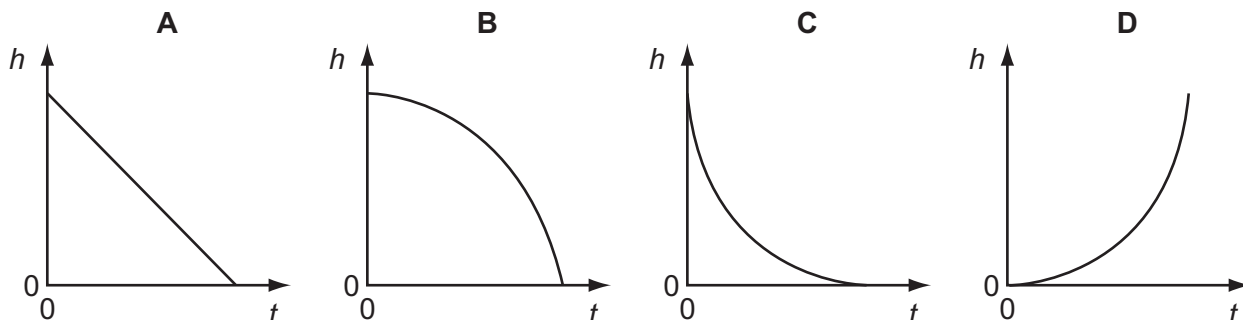
What must be the minimum distance between the lights if the train speed is increased by 20 %, without changing the deceleration of the trains?

- A** $1.20x$ **B** $1.25x$ **C** $1.44x$ **D** $1.56x$

53 9702/12/M/J/10/Q8

A small steel ball falls freely under gravity after being released from rest.

Which graph best represents the variation of the height h of the ball with time t ?



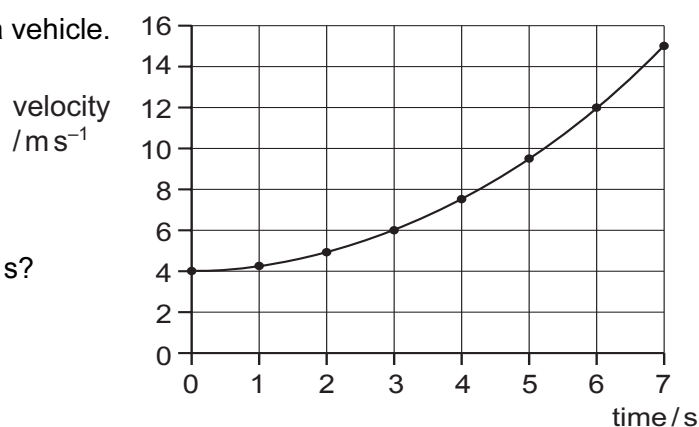
54 9702/12/M/J/10/Q9

The diagram shows a velocity-time graph for a vehicle.

The vehicle, moving at 4.0 m s^{-1} , begins to accelerate at time = 0.

What is the vehicle's acceleration at time = 3.0 s?

- A 0.67 m s^{-2} B 1.0 m s^{-2}
C 1.3 m s^{-2} D 2.0 m s^{-2}



55 9702/11/O/N/10/Q7

A ball is thrown horizontally in still air from the top of a very tall building. The ball is affected by air resistance.

What happens to the horizontal and to the vertical components of the ball's velocity?

	horizontal component of velocity	vertical component of velocity
A	decreases to zero	increases at a constant rate
B	decreases to zero	increases to a constant value
C	remains constant	increases at a constant rate
D	remains constant	increases to a constant value

56 9702/11/O/N/10/Q8

The velocity-time graph below is for a stone thrown vertically up into the air. Air resistance is negligible.

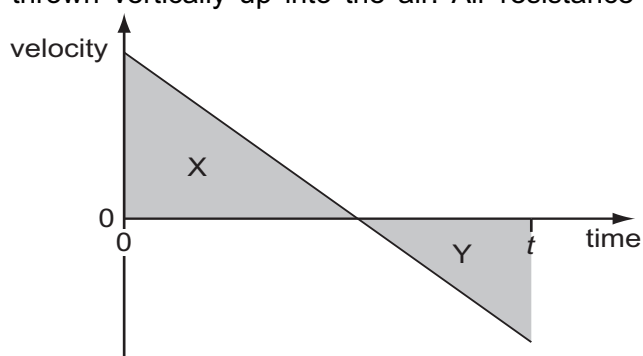
The stone is thrown up at time zero.

Area X represents a distance of 5 m.

Area Y represents a distance of 3 m.

What is the displacement of the stone from its initial position at time t ?

- A 2 m C 5 m B 3 m D 8 m



57 9702/12/O/N/10/Q6

A football is dropped from the top of a three-storey building. It falls through air until it reaches the ground.

What remains constant throughout the fall?

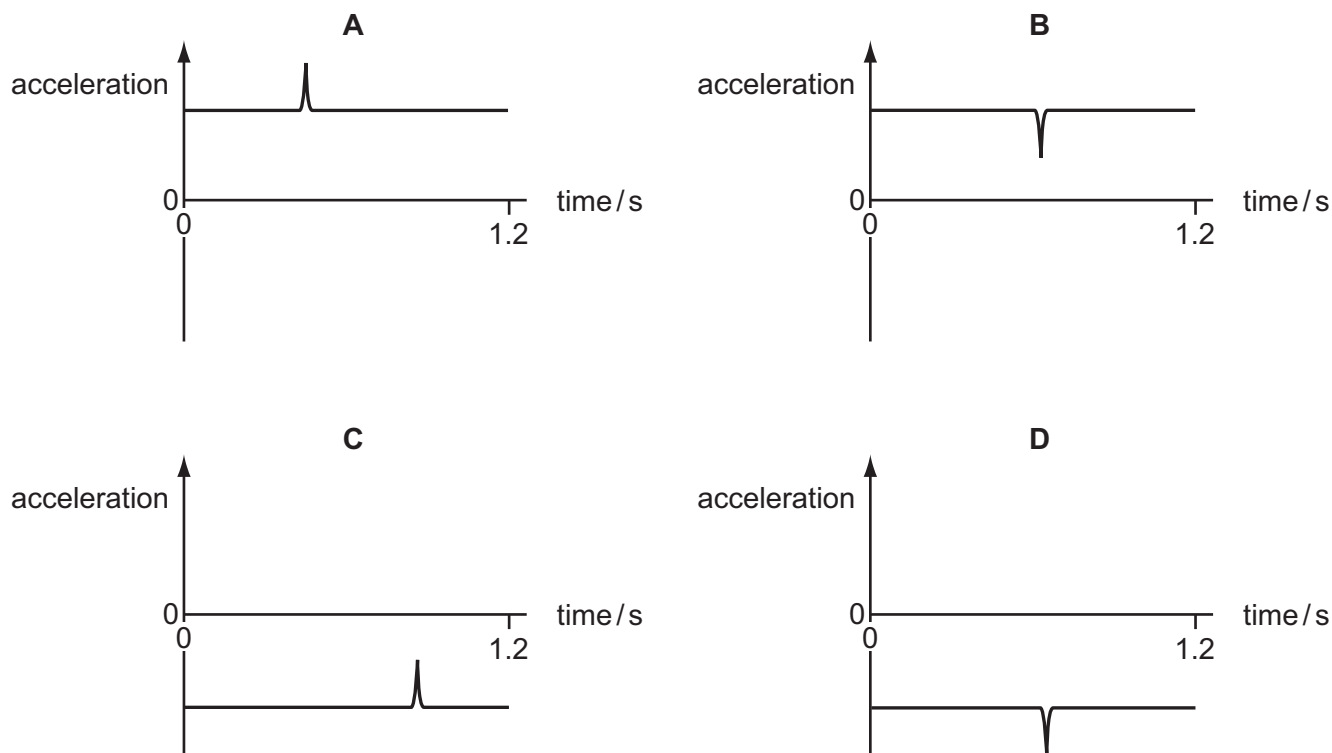
- A** acceleration of the football **C** velocity of the football
B air resistance on the football **D** weight of the football

58 9702/12/O/N/10/Q7

A student throws a ball in the positive direction vertically upwards.

The ball makes an elastic collision with the ceiling, rebounds and accelerates back to the student's hand in a time of 1.2 s.

Which graph best represents the acceleration of the ball from the moment it leaves the hand to the instant just before it returns to the hand?



59 9702/12/O/N/10/Q8

A moving body undergoes uniform acceleration while travelling in a straight line between points X, Y and Z. The distances XY and YZ are both 40 m. The time to travel from X to Y is 12 s and from Y to Z is 6.0 s.

What is the acceleration of the body?

- A** 0.37 ms^{-2} **B** 0.49 ms^{-2} **C** 0.56 ms^{-2} **D** 1.1 ms^{-2}

60 9702/13/O/N/13/Q9

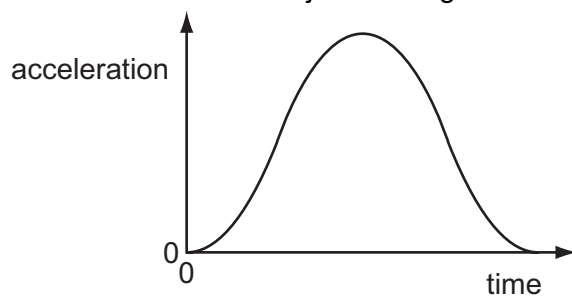
A person, travelling on a motorway a total distance of 200 km, travels the first 90 km at an average speed of 80 km h^{-1} .

Which average speed must be obtained for the rest of the journey if the person is to reach the destination in a total time of 2 hours 0 minutes?

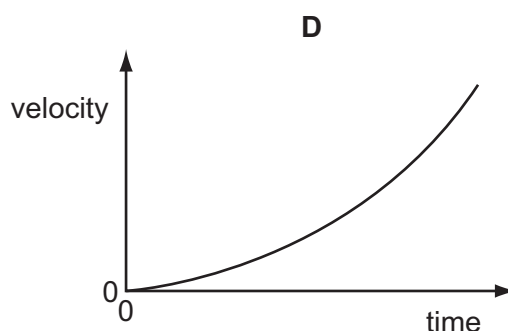
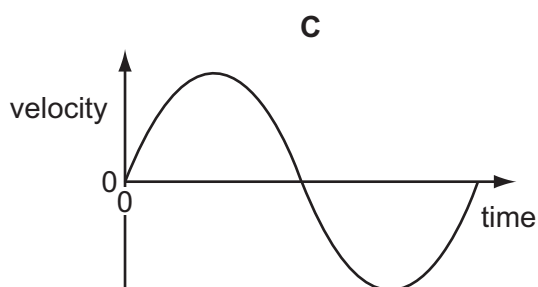
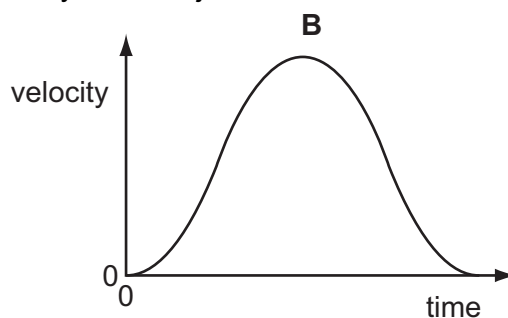
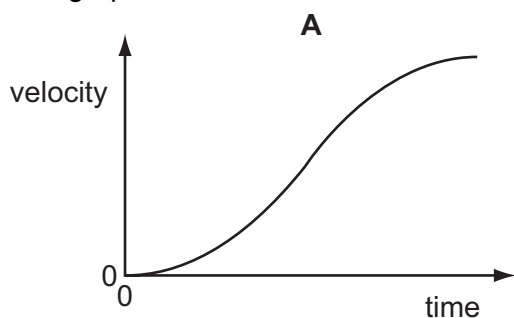
- A** 110 km h^{-1} **B** 120 km h^{-1} **C** 122 km h^{-1} **D** 126 km h^{-1}

61 9702/11/M/J/14/Q8*

The graph shows how the acceleration of an object moving in a straight line varies with time.



Which graph shows the variation with time of the velocity of the object?

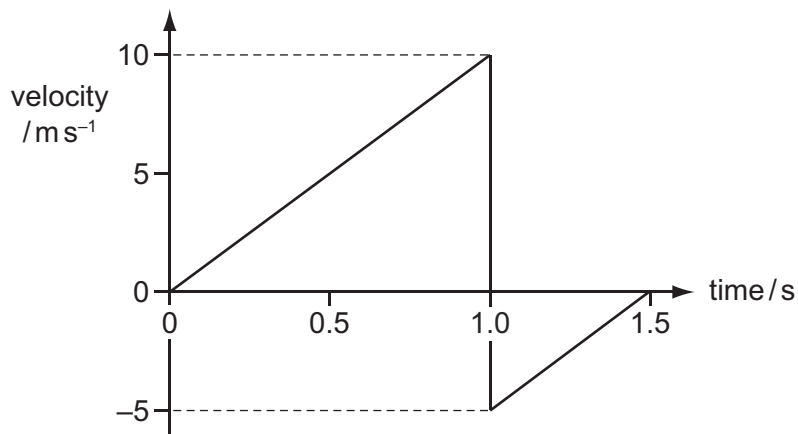


62 9702/11/M/J/11/Q7

A ball is released from rest at time zero. After 1.0 s it bounces inelastically from a horizontal surface and rebounds, reaching the top of its first bounce after 1.5 s.

What is the total displacement of the ball from its original position after 1.5 s?

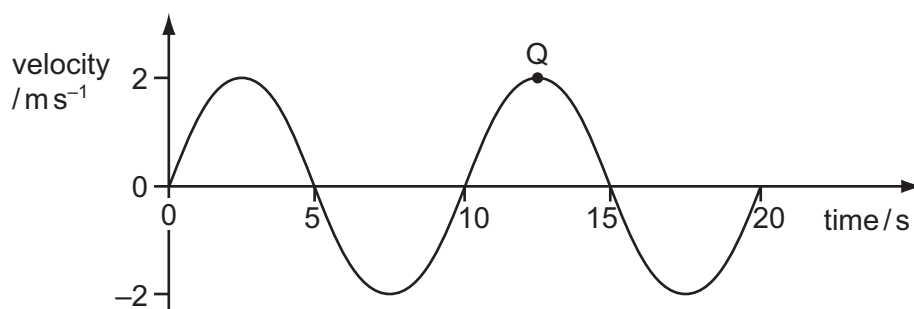
- A** 1.25 m **B** 3.75 m
C 5.00 m **D** 6.25 m



63 9702/12/M/J/11/Q7

A particle moves in the manner shown by the velocity-time graph.

The displacement of the particle has been measured so that it is zero at $t = 0$. Point Q refers to a point in its motion.



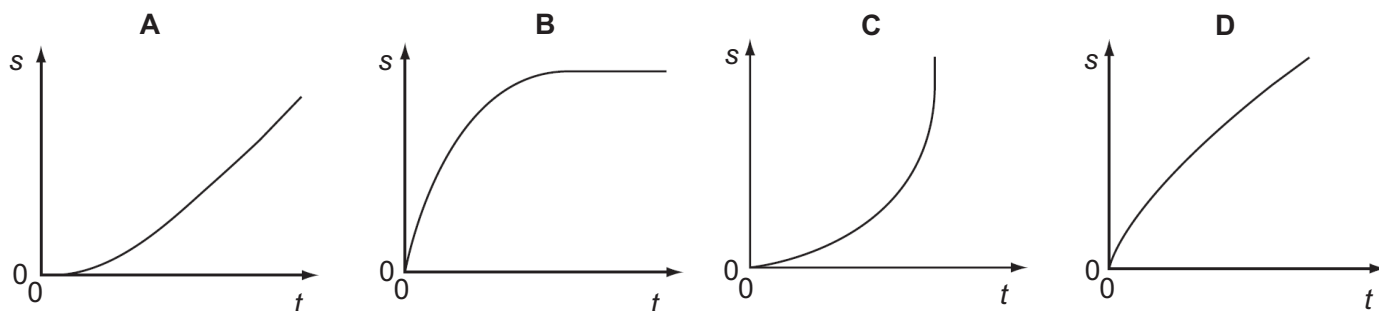
Which row of the table is correct?

	times for maximum displacement/s		acceleration at point Q / ms^{-2}
A	2.5	12.5	2
B	5	15	2
C	2.5	12.5	0
D	5	15	0

64 9702/12/M/J/11/Q8*

A tennis ball falls freely, in air, from the top of a tall building.

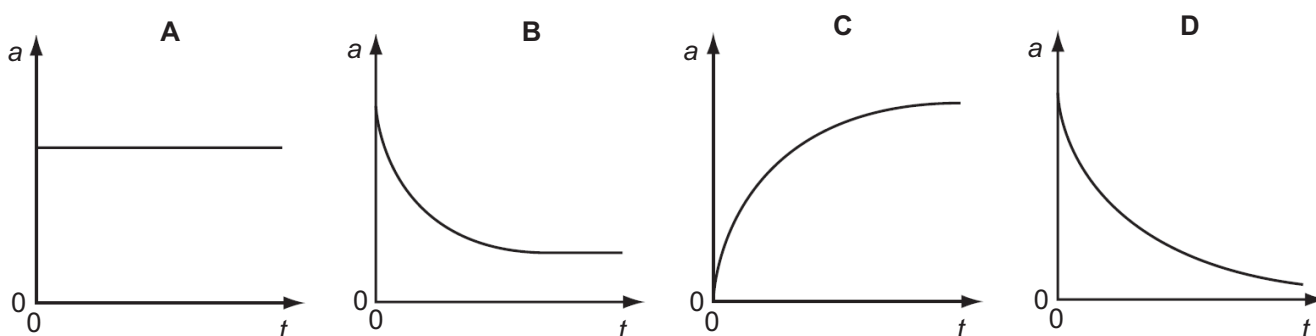
Which graph best represents the variation of distance s fallen with time t ?



65 9702/11/O/N/11/Q7

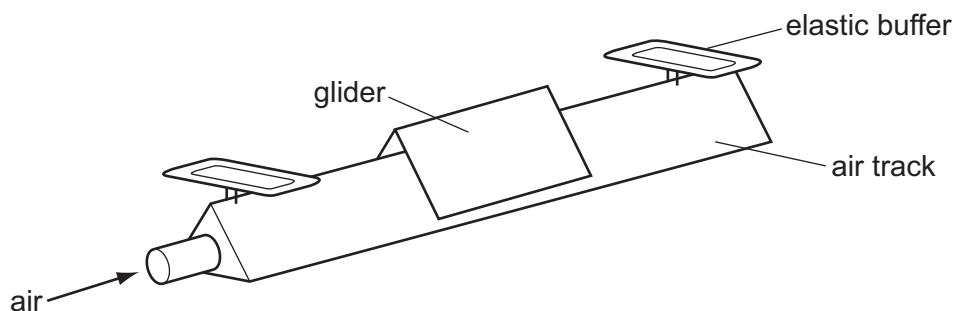
A tennis ball is released from rest at the top of a tall building.

Which graph best represents the variation with time t of the acceleration a of the ball as it falls, assuming that the effect of air resistance is **not** negligible?



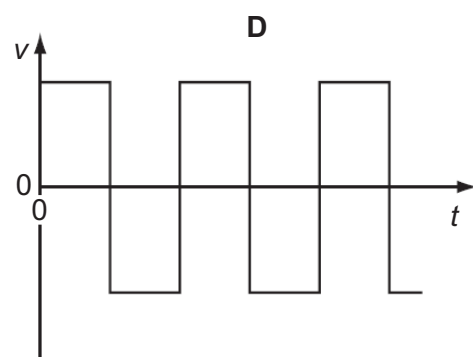
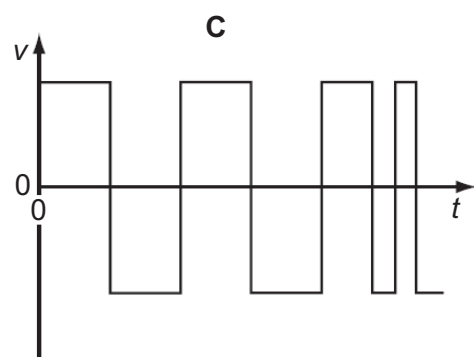
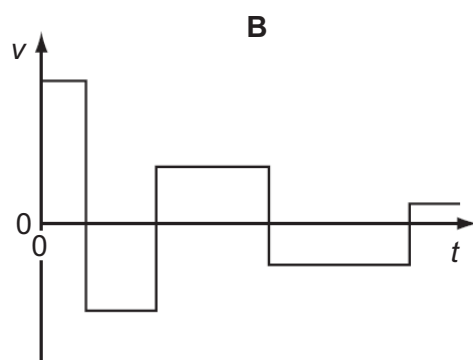
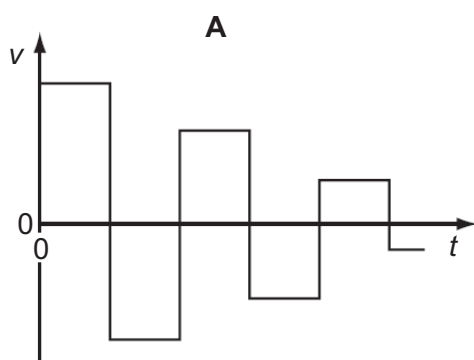
66 9702/12/M/J/11/Q9

A small glider moves along a friction-free horizontal air track as shown below.



At each end of the air track there is a perfectly elastic buffer.

Which graph represents the variation with time t of the velocity v of the glider as it moves between the two buffers?

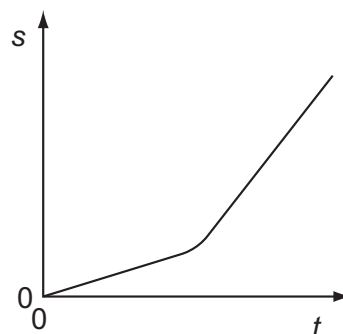


67 9702/12/O/N/11/Q8

The variation with time t of the distance s moved by a body is shown below.

What can be deduced from the graph about the motion of the body?

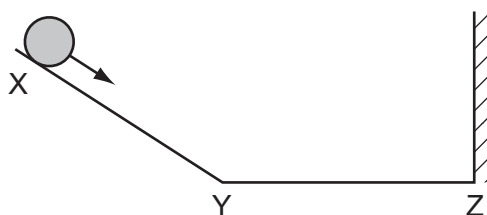
- A It accelerates continuously.
- B It starts from rest.
- C The distance is proportional to time.
- D The speed changes.



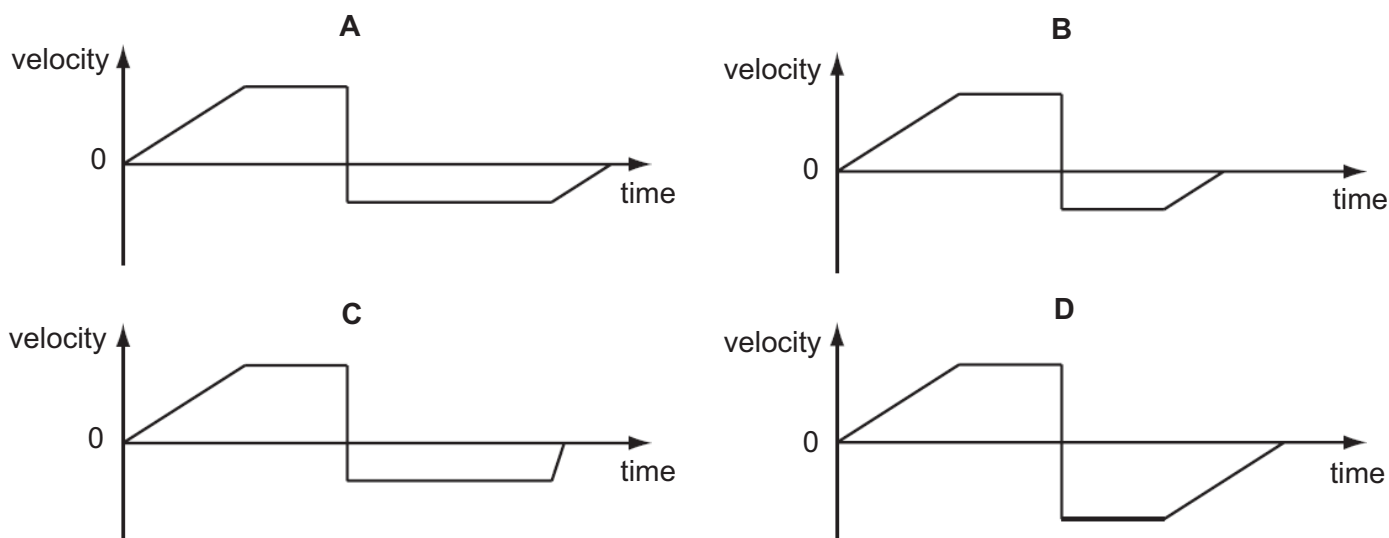
68 9702/13/O/N/14/Q8*

A ball is released from rest on a smooth slope XY.

It moves down the slope, along a smooth horizontal surface YZ and rebounds inelastically at Z. Then it moves back to Y and comes to rest momentarily somewhere on XY.



Which velocity-time graph represents the motion of the ball?



69 9702/12/O/N/15/Q7

A boy throws a ball vertically upwards. It rises to a maximum height, where it is momentarily at rest, and then falls back to his hands.

Which row gives the acceleration of the ball at various stages in its motion? (Take vertically upwards as positive. Ignore air resistance.)

	rising	at maximum height	falling
A	-9.81 ms^{-2}	0	$+9.81 \text{ ms}^{-2}$
B	-9.81 ms^{-2}	-9.81 ms^{-2}	-9.81 ms^{-2}
C	$+9.81 \text{ ms}^{-2}$	$+9.81 \text{ ms}^{-2}$	$+9.81 \text{ ms}^{-2}$
D	$+9.81 \text{ ms}^{-2}$	0	-9.81 ms^{-2}

70 9702/11/O/N/11/Q9*

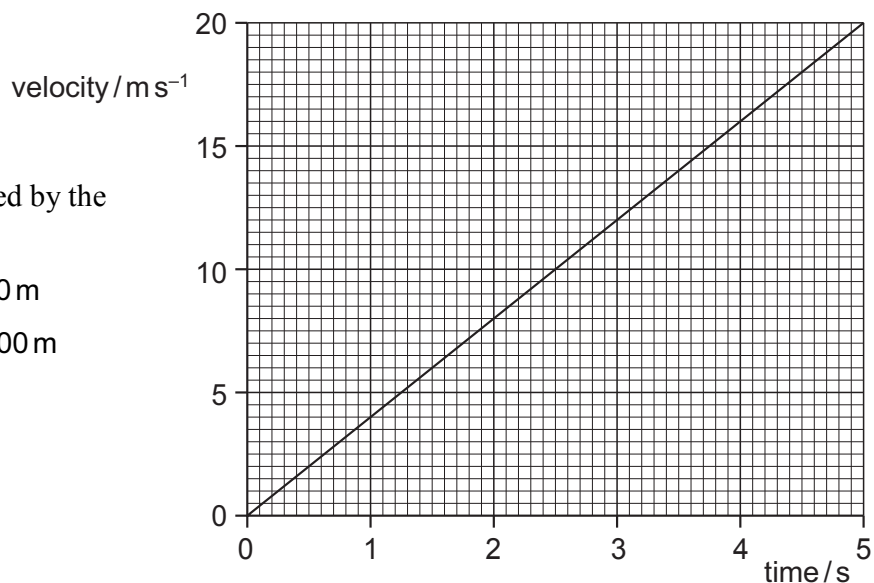
A body falling in a uniform gravitational field encounters air resistance. The air resistance increases until terminal velocity is reached.

Which factor does **not** affect its terminal velocity?

- A** the density of the air
- B** the height from which the body falls
- C** the mass of the body
- D** the shape of the body

71 9702/12/O/N/11/Q6

The velocity of an object during the first five seconds of its motion is shown on the graph.



What is the distance travelled by the object in this time?

- A 4 m B 20 m
C 50 m D 100 m

72 9702/12/O/N/11/Q7*

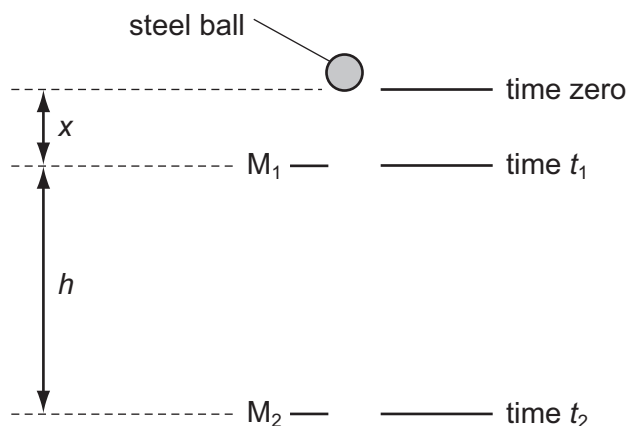
A stone of mass m is dropped from a tall building. There is significant air resistance. The acceleration of free fall is g .

When the stone reaches its terminal velocity, which information is correct?

	magnitude of the acceleration of the stone	magnitude of the force of gravity on the stone	magnitude of the force of air resistance on the stone
A	g	mg	mg
B	zero	mg	mg
C	zero	zero	mg
D	zero	zero	zero

73 9702/01/O/N/05/Q7

Two markers M_1 and M_2 are set up a vertical distance h apart.



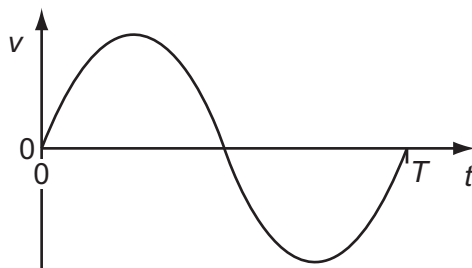
A steel ball is released at time zero from a point a distance x above M_1 . The ball reaches M_1 at time t_1 and reaches M_2 at time t_2 . The acceleration of the ball is constant.

Which expression gives the acceleration of the ball?

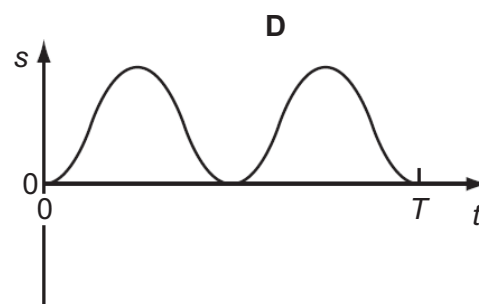
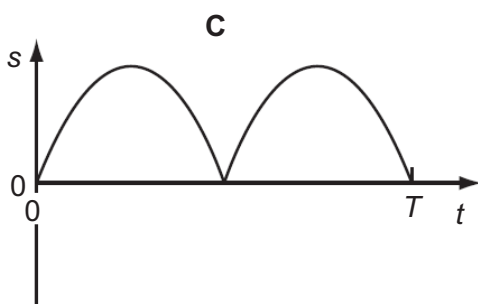
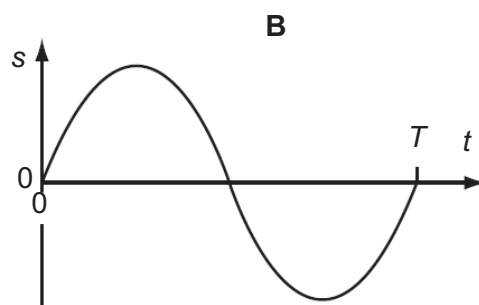
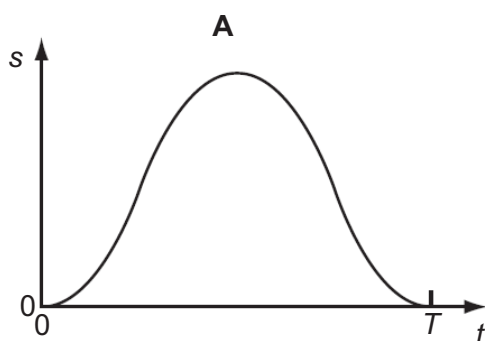
- A $\frac{2h}{t^2}$ B $\frac{2h}{(t_1 + t_2)}$ C $\frac{2h}{(t_1^2 - t_2^2)}$ D $\frac{2h}{(t_1^2 - t_2^2)}$

74 9702/12/M/J/12/Q7

The graph shows how the velocity v of an object moving in a straight line varies over time $t = 0$ to $t = T$.



Which graph represents the displacement s of the object in the time $t = 0$ to $t = T$?

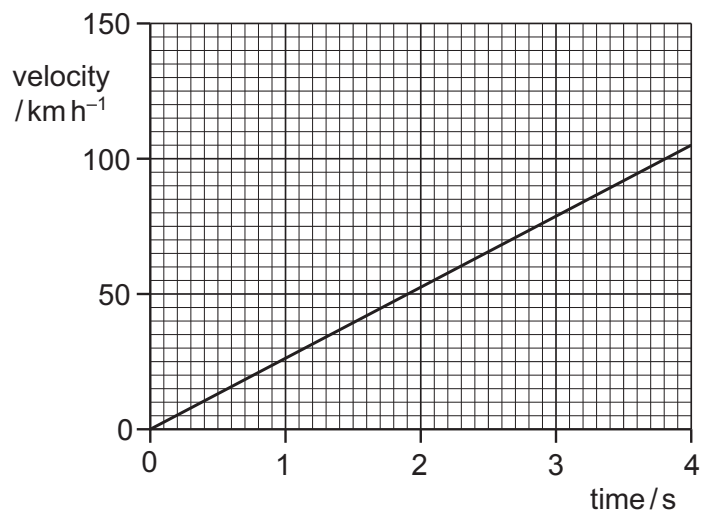


75 9702/13/M/J/15/Q8

The velocity of an electric car changes as shown.

What is the acceleration of the car?

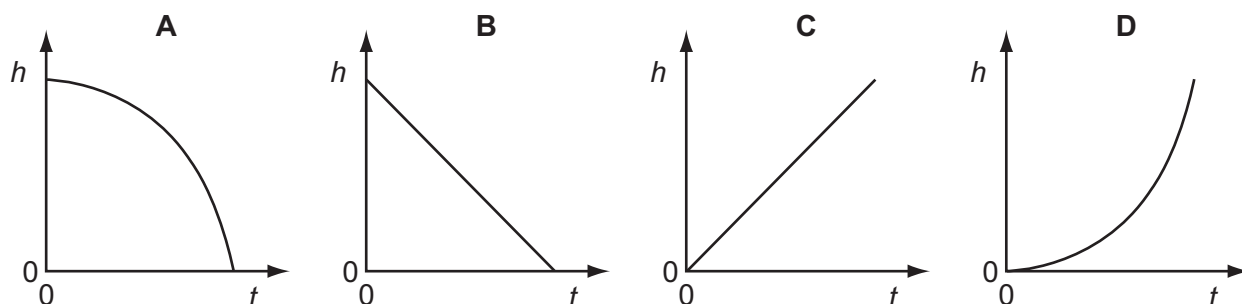
- A** 210 ms^{-2} **B** 58 ms^{-2}
C 26 ms^{-2} **D** 7.3 ms^{-2}



76 9702/12/M/J/12/Q9

A brick is dislodged from a building and falls vertically under gravity.

Which graph best represents the variation of its height h above the ground with time t if air resistance is negligible?



77 9702/13/M/J/12/Q6

Which feature of a graph allows acceleration to be determined?

- A the area under a displacement-time graph C the slope of a displacement-time graph
B the area under a velocity-time graph D the slope of a velocity-time graph

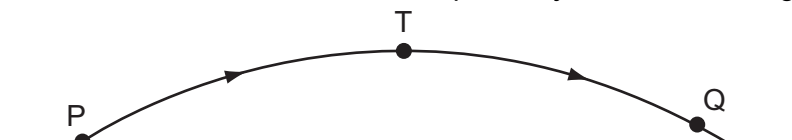
78 9702/13/M/J/12/Q7

In an experiment to determine the acceleration of free fall using a falling body, what would lead to a value that is too large?

- A air resistance C measured distance longer than true distance
B dimensions of the body are too large D measured time longer than true time

79 9702/13/M/J/12/Q8

In the absence of air resistance, a stone is thrown from P and follows a parabolic path in which the highest point reached is T. The stone reaches point Q just before landing.



The vertical component of acceleration of the stone is

- A zero at T. C larger at Q than at T.
B larger at T than at Q. D the same at Q as at T.

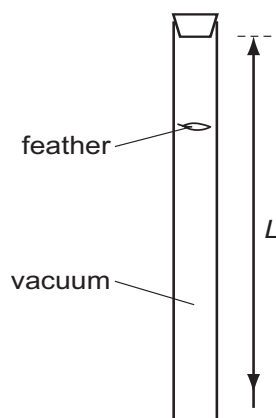
80 9702/13/M/J/12/Q9

The diagram shows a laboratory experiment in which a feather falls from rest in a long evacuated vertical tube of length L .

The feather takes time T to fall from the top to the bottom of the tube.

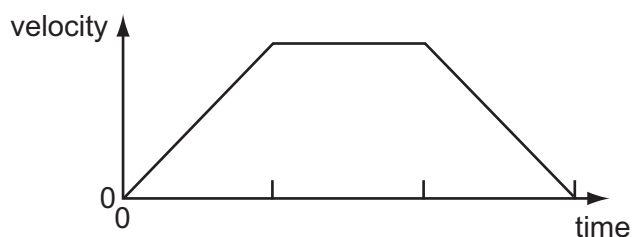
How far will the feather have fallen from the top of the tube in time $0.50 T$?

- A $0.13 L$ B $0.25 L$ C $0.38 L$ D $0.50 L$

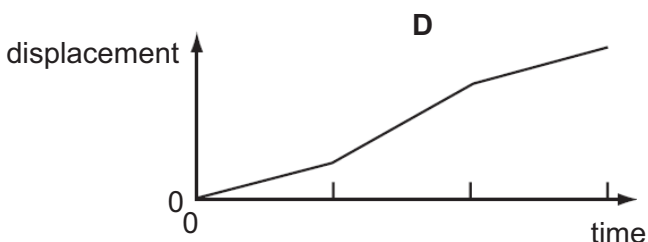
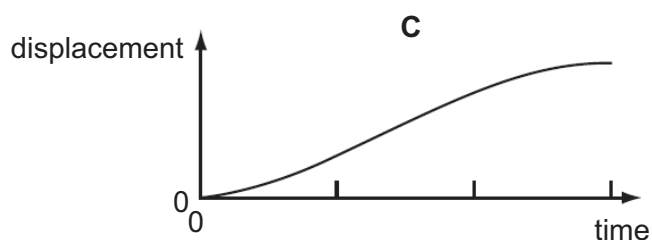
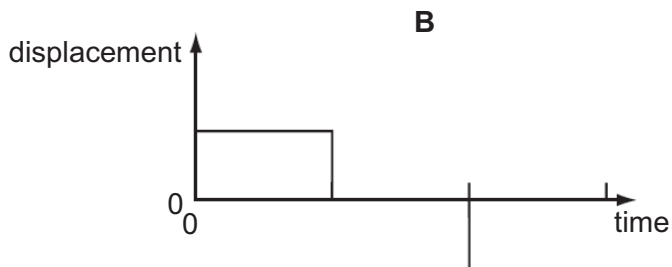
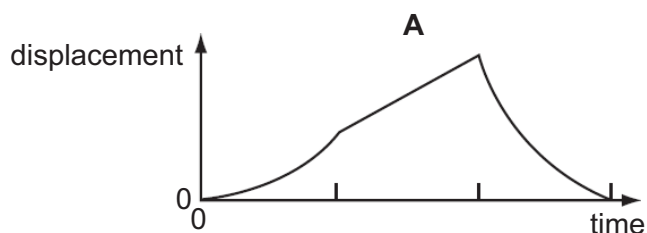


81 9702/12/O/N/12/Q9

The graph of velocity against time for an object moving in a straight line is shown.

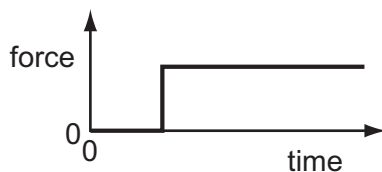


What is the corresponding graph of displacement against time?

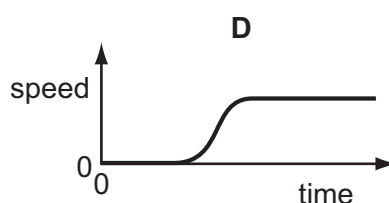
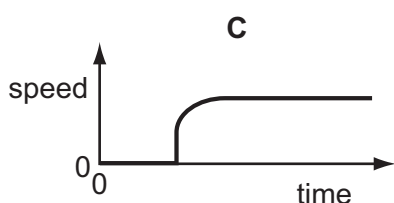
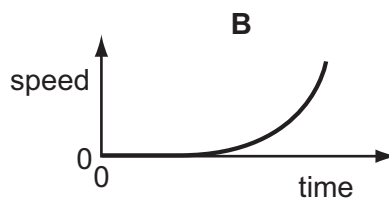
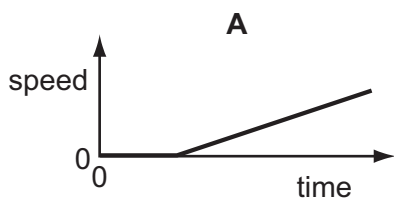


82 9702/11/M/J/13/Q9

A car is stationary at traffic lights. When the traffic lights go green, the driver presses down sharply on the accelerator. The resultant horizontal force acting on the car varies with time as shown.

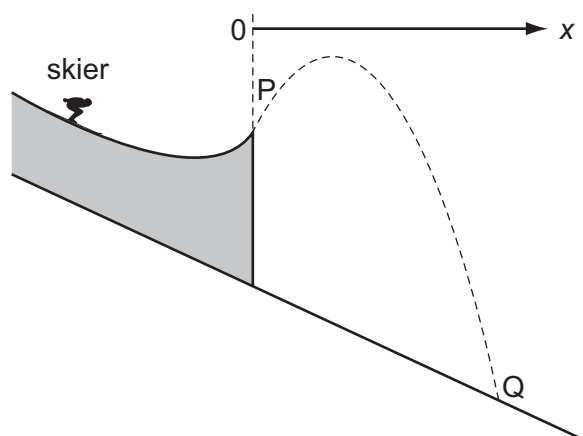


Which graph shows the variation with time of the speed of the car?

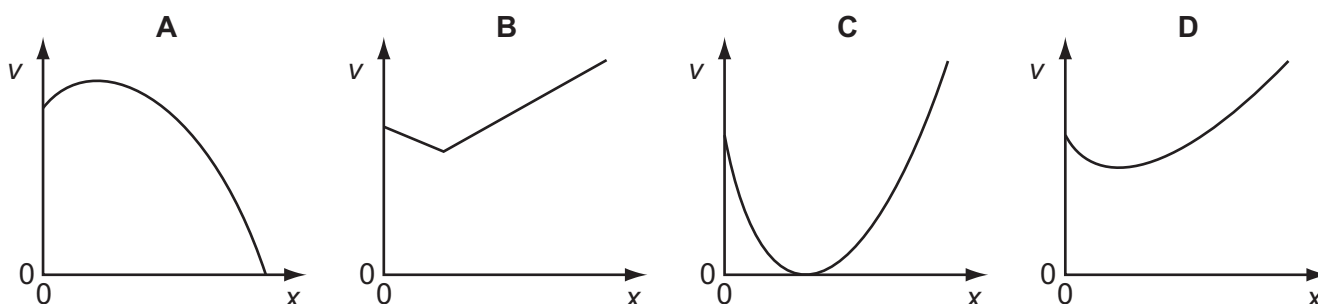


83 9702/12/O/N/12/Q10

The dotted line shows the path of a competitor in a ski-jumping competition.



Ignoring air resistance, which graph best represents the variation of his speed v with the horizontal distance x covered from the start of his jump at P before landing at Q?

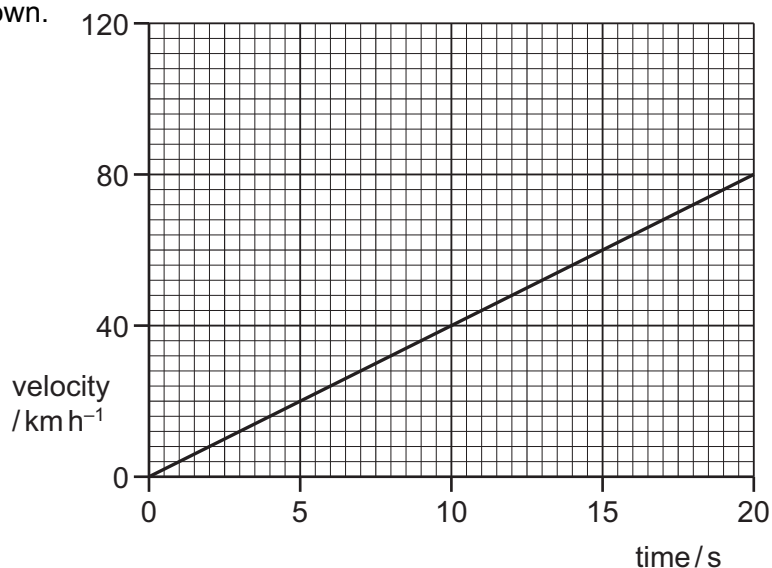


84 9702/12/O/N/12/Q11

The velocity of a car changes as shown.

What is the acceleration of the car?

- A 1.1 ms^{-2}
- B 4.0 ms^{-2}
- C 224 ms^{-2}
- D 800 ms^{-2}



85 9702/11/M/J/13/Q8

A goods train passes through a station at a steady speed of 10 ms^{-1} . An express train is at rest at the station. The express train leaves the station with a uniform acceleration of 0.5 ms^{-2} just as the goods train goes past. Both trains move in the same direction on straight, parallel tracks.

How much time passes before the express train overtakes the goods train?

- A 6 s
- B 10 s
- C 20 s
- D 40 s

86 9702/13/O/N/12/Q8

A bicycle brakes so that it undergoes uniform deceleration from a speed of 8 ms^{-1} to 6 ms^{-1} over a distance of 7 m.

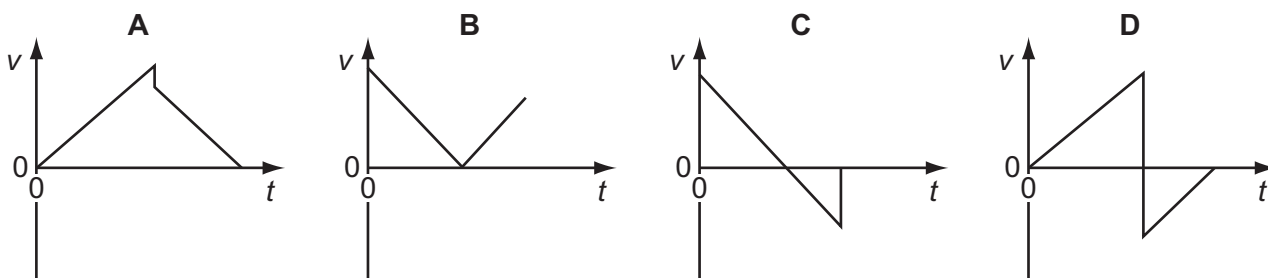
If the deceleration of the bicycle remains constant, what further distance will it travel before coming to rest?

- A 7 m B 9 m C 16 m D 21 m

87 9702/13/O/N/12/Q9

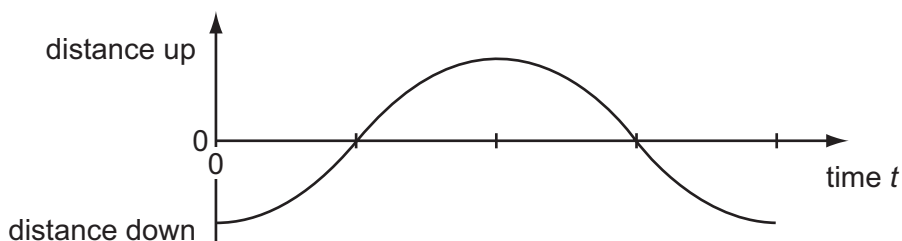
A ball is released from rest above a horizontal surface. It bounces once and is caught.

Which graph represents the variation with time t of the velocity v of the ball?

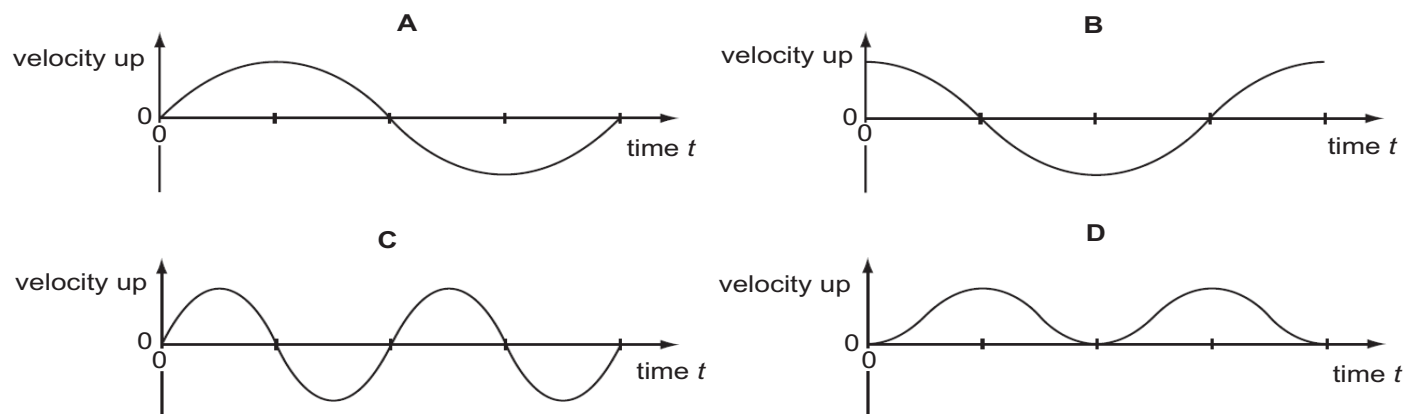


88 9702/13/O/N/12/Q10

A mass on the end of a spring bounces up and down as shown, after being released at time $t = 0$.



Which graph shows how the velocity varies with time?



89 9702/13/M/J/13/Q10

A stationary nucleus has nucleon number A .

The nucleus decays by emitting a proton with speed v to form a new nucleus with speed u . The new nucleus and the proton move away from one another in opposite directions.

Which equation gives v in terms of A and u ?

- A $v = \left(\frac{A}{4} - 1\right)u$ B $v = (A - 1)u$ C $v = Au$ D $v = (A + 1)u$

90 9702/11/M/J/13/Q7

A body is released from rest and falls vertically in air of constant density.

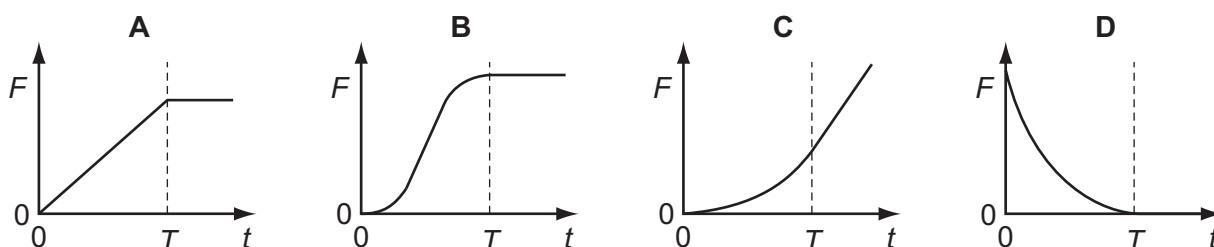
Which statement about the motion of the falling body is correct?

- A As it accelerates, its weight decreases so that its acceleration decreases until it travels with constant velocity.
- B It accelerates initially at 9.8 m s^{-2} but the drag force increases so its acceleration decreases.
- C Its velocity increases at a constant rate until its velocity becomes constant.
- D The drag force of the air increases continually and eventually the velocity decreases.

91 9702/12/M/J/13/Q6

A sky diver falls vertically from a stationary balloon. She leaves the balloon at time $t = 0$. At time $t = T$, she reaches terminal velocity. Beyond the time shown in the graphs, she opens her parachute.

Which graph shows the variation with time t of the force F due to air resistance?



92 9702/12/M/J/13/Q7

The diagram shows an arrangement to stop trains that are travelling too fast.



Trains coming from the left travel at a speed of 50 ms^{-1} . At marker 1, the driver must apply the brakes so that the train decelerates uniformly in order to pass marker 2 at no more than 10 ms^{-1} .

The train carries a detector that notes the times when the train passes each marker and will apply an emergency brake if the time between passing marker 1 and marker 2 is less than 20 s.

How far from marker 2 should marker 1 be placed?

- A 200 m
- B 400 m
- C 500 m
- D 600 m

93 9702/12/M/J/15/Q10

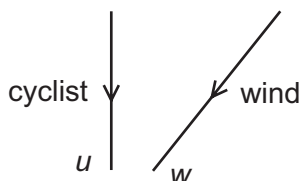
A firework rocket is fired vertically upwards. The fuel burns and produces a constant upwards force on the rocket. After 5 seconds there is no fuel left. Air resistance is negligible.

What is the acceleration before and after 5 seconds?

	before 5 seconds	after 5 seconds
A	constant	constant
B	constant	zero
C	increasing	constant
D	increasing	zero

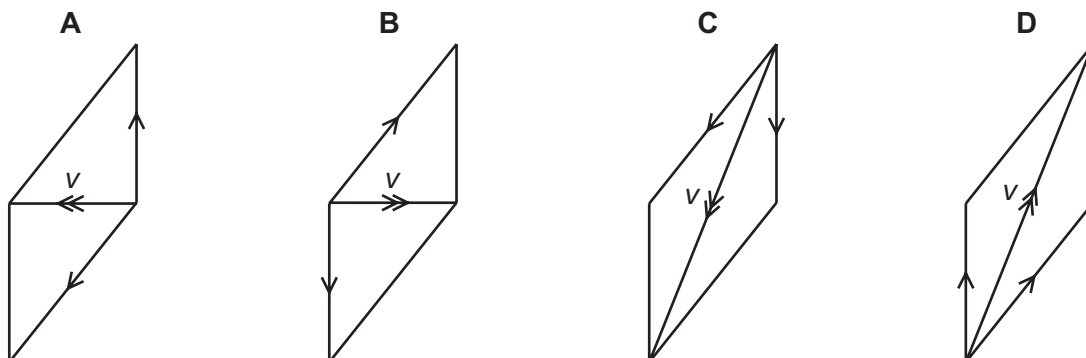
94 9702/13/M/J/13/Q4

A cyclist is travelling due south with velocity u . The wind is blowing from the north-east with velocity w .



The wind has a velocity v relative to the cyclist, where $v = w - u$.

Which vector diagram shows the magnitude and direction of velocity v ?

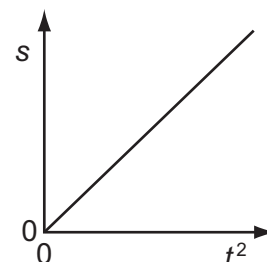


95 9702/13/M/J/13/Q8

At time $t = 0$, a body moves from rest with constant acceleration in a straight line. At time t , the body is distance s from its rest position. A graph is drawn of s against t^2 , as shown.

Which statement describes the acceleration of the body?

- A It is equal to half the value of the gradient of the graph.
- B It is equal to the value of the gradient of the graph.
- C It is equal to twice the value of the gradient of the graph.
- D It is equal to the reciprocal of the gradient of the graph.

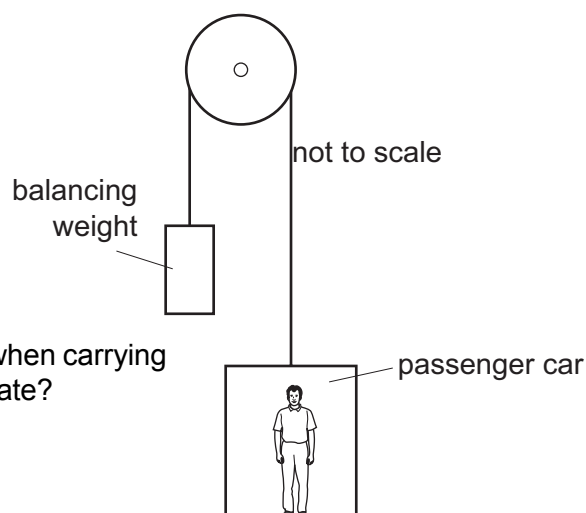


96 9702/13/M/J/13/Q9

A lift (elevator) consists of a passenger car supported by a cable which runs over a light, frictionless pulley to a balancing weight. The balancing weight falls as the passenger car rises.

Some masses are shown in the table.

	mass / kg
passenger car	520
balancing weight	640
passenger	80

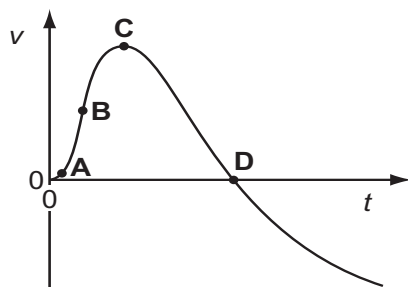


What is the magnitude of the acceleration of the car when carrying just one passenger and when the pulley is free to rotate?

- A 0.032 ms^{-2}
- B 0.32 ms^{-2}
- C 0.61 ms^{-2}
- D 0.65 ms^{-2}

97 9702/13/O/N/13/Q7

The graph shows how the velocity v of a firework rocket changes with time t .
At which point on the graph does the rocket have the greatest acceleration?



98 9702/13/O/N/13/Q8

On a particular railway, a train driver applies the brake of the train at a yellow signal, a distance of 1.0 km from a red signal, where the train stops.

The maximum deceleration of the train is 0.20 m s^{-2} .

Assuming uniform deceleration, what is the maximum safe speed of the train at the yellow signal?

- A 14 m s^{-1} B 20 m s^{-1} C 40 m s^{-1} D 400 m s^{-1}

99 9702/11/M/J/14/Q6

A tennis ball is thrown horizontally in air from the top of a tall building.

If the effect of air resistance is **not** negligible, what happens to the horizontal and vertical components of the ball's velocity?

	horizontal component of velocity	vertical component of velocity
A	constant	constant
B	constant	increases at a constant rate
C	decreases to zero	increases at a constant rate
D	decreases to zero	increases to a maximum value

100 9702/11/M/J/14/Q7

An object is thrown with velocity 5.2 m s^{-1} vertically upwards on the Moon. The acceleration due to gravity on the Moon is 1.62 m s^{-2} .

What is the time taken for the object to return to its starting point?

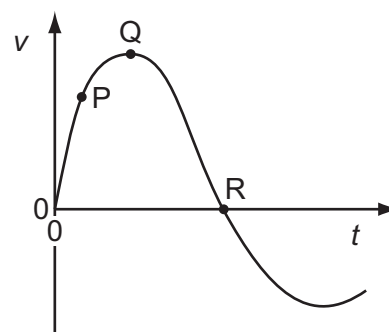
- A 2.5 s B 3.2 s C 4.5 s D 6.4 s

101 9702/12/O/N/13/Q7

The graph shows how velocity v varies with time t for a bungee jumper.

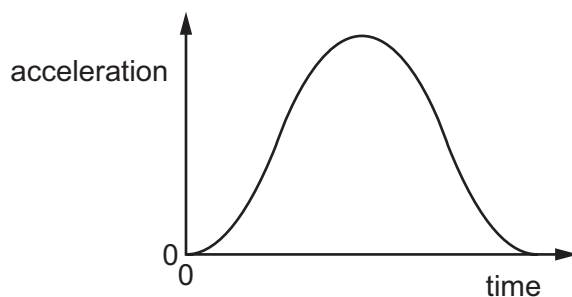
At which point is the bungee jumper momentarily at rest and at which point does she have zero acceleration?

	jumper at rest	jumper with zero acceleration
A	Q	P
B	Q	R
C	R	Q
D	R	R



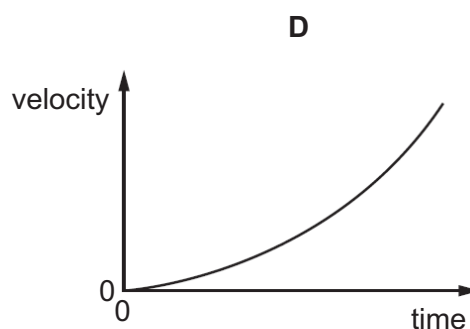
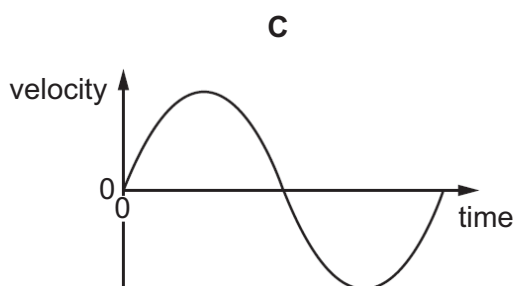
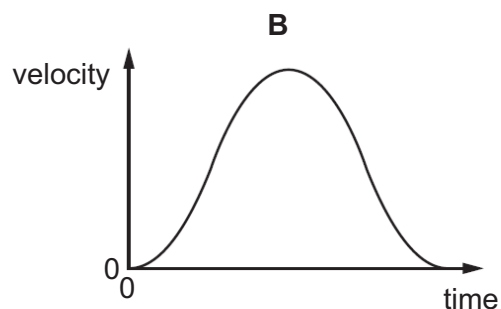
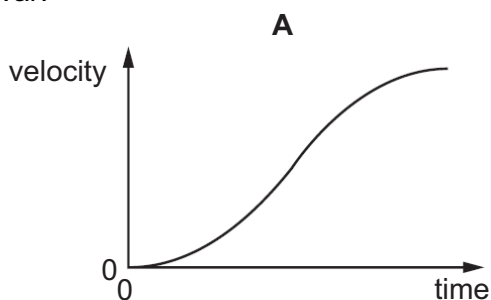
102 9702/11/M/J/14/Q8

The graph shows how the acceleration of an object moving in a straight line varies with time.



The object starts from rest.

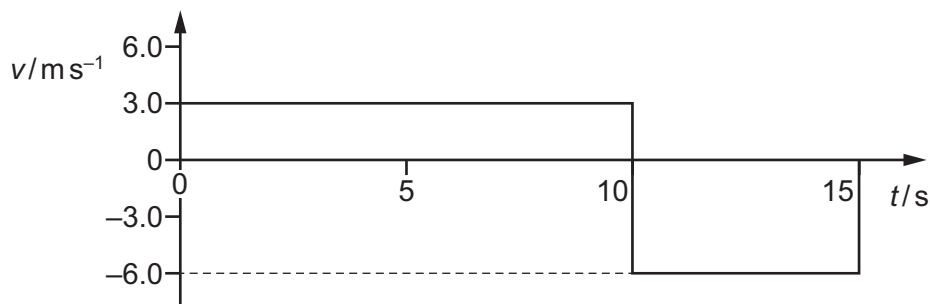
Which graph shows the variation with time of the velocity of the object over the same time interval?



103 9702/12/M/J/14/Q6

A radio-controlled toy car travels along a straight line for a time of 15 s.

The variation with time t of the velocity v of the car is shown below.

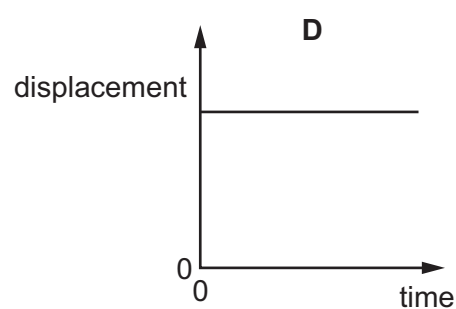
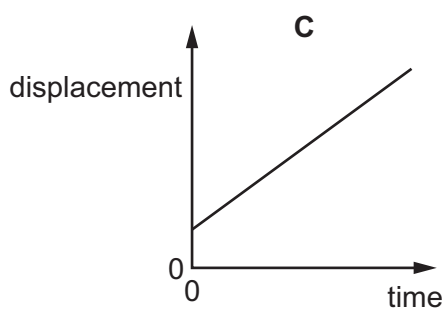
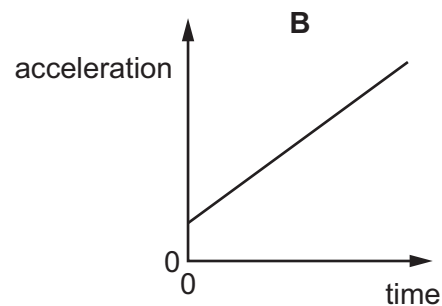
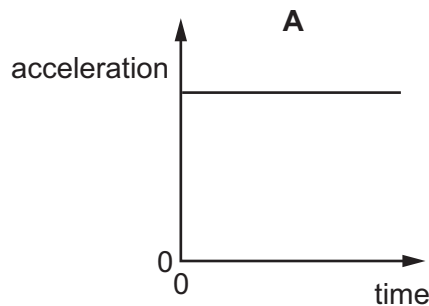


What is the average velocity of the toy car for the journey shown by the graph?

- A** -1.5 ms^{-1} **B** 0.0 ms^{-1} **C** 4.0 ms^{-1} **D** 4.5 ms^{-1}

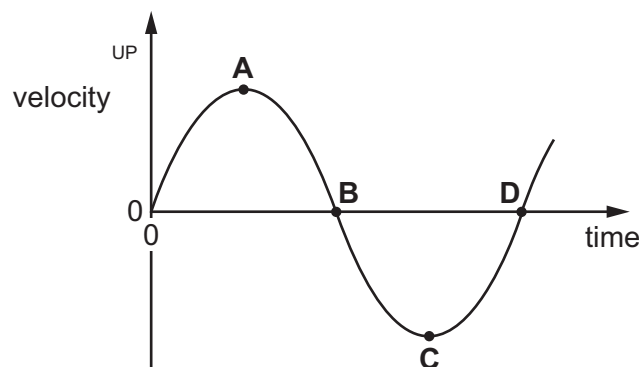
104 9702/12/O/N/14/Q6

Which graph represents the motion of a car that is travelling along a straight road with a speed that increases uniformly with time?



105 9702/13/O/N/14/Q7

The diagram shows a velocity-time graph for a mass moving up and down on the end of a spring. Which point represents the velocity of the mass when at the lowest point of its motion?



106 9702/13/M/J/15/Q9

A body falling in a uniform gravitational field encounters air resistance. The air resistance increases until terminal velocity is reached.

Which factor does **not** affect its terminal velocity?

- | | |
|---|--------------------------------|
| A the density of the air | C the mass of the body |
| B the height from which the body falls | D the shape of the body |

107 9702/12/M/J/15/Q9

A sprinter runs a 100 m race in a straight line. He accelerates from the starting block at a constant acceleration of 2.5 m s^{-2} to reach his maximum speed of 10 m s^{-1} . He maintains this speed until he crosses the finish line.

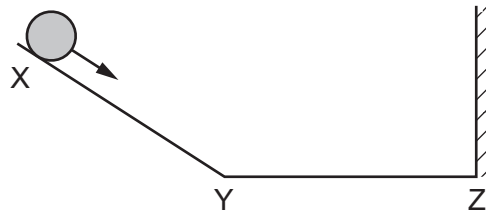
Which time does it take the sprinter to run the race?

- | | | | |
|--------------|---------------|---------------|---------------|
| A 4 s | B 10 s | C 12 s | D 20 s |
|--------------|---------------|---------------|---------------|

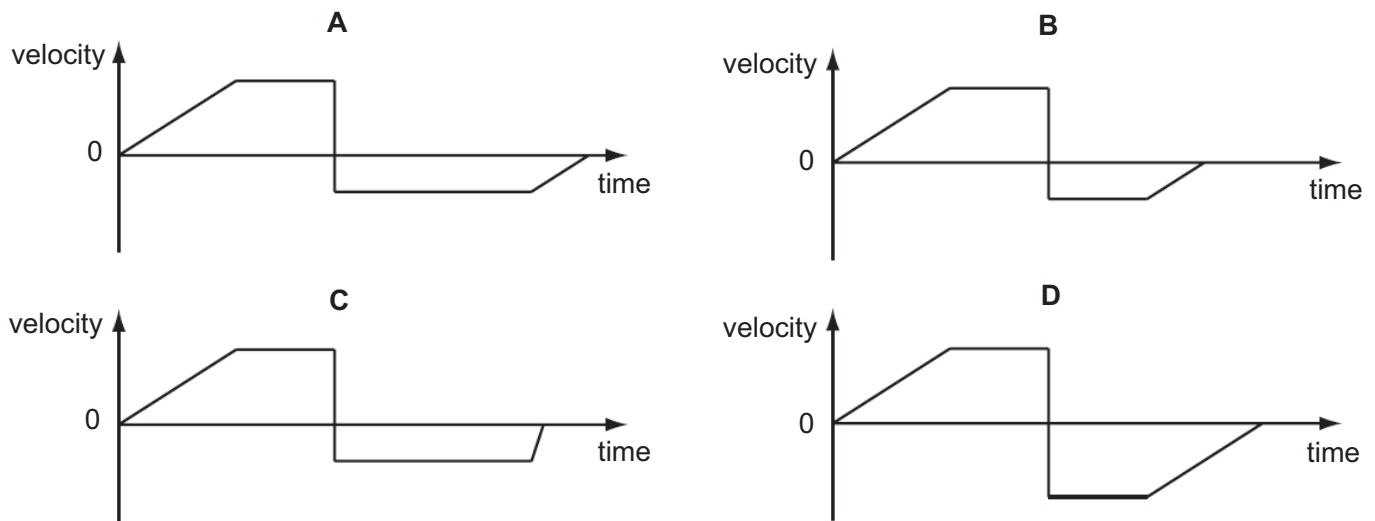
108 9702/13/O/N/14/Q8

A ball is released from rest on a smooth slope XY.

It moves down the slope, along a smooth horizontal surface YZ and rebounds inelastically at Z. Then it moves back to Y and comes to rest momentarily somewhere on XY.



Which velocity-time graph represents the motion of the ball?



109 9702/13/O/N/14/Q9

A man weighs 240 N on Mars where the acceleration of free fall g is 4 ms^{-2} . On the Moon, g is 2 ms^{-2} .

Which statement is correct?

- A The man has a mass on Mars of 60 N.
- B The man has a mass on the Moon of 120 kg.
- C The man weighs 120 N on the Moon.
- D The man weighs 240 N on the Moon.

110 9702/11/M/J/15/Q8

An insect jumps with an initial vertical velocity of 1.0 ms^{-1} , reaching a maximum height of $3.5 \times 10^{-2} \text{ m}$. Assume the deceleration is uniform.

What is the magnitude of the deceleration?

- A 3.6 ms^{-2}
- B 9.8 ms^{-2}
- C 14 ms^{-2}
- D 29 ms^{-2}

111 9702/11/M/J/15/Q9

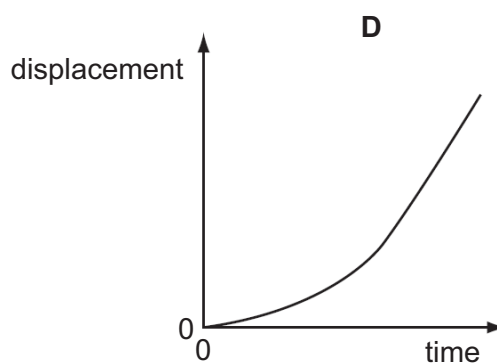
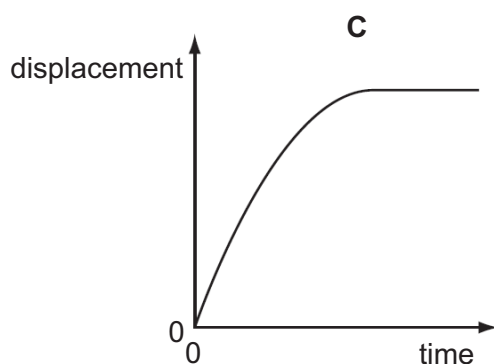
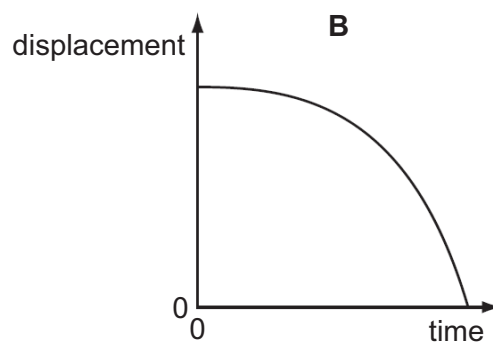
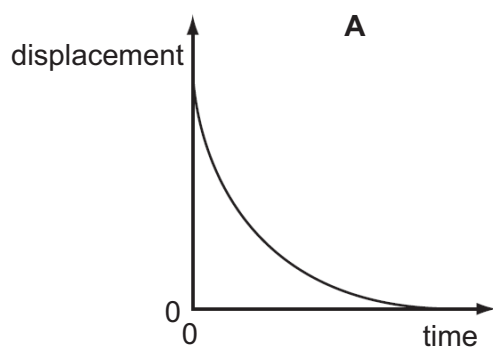
A body having uniform acceleration a increases its velocity from u to v in time t .

Which expression would **not** give a correct value for the body's displacement during time t ?

- A $ut + \frac{1}{2}at^2$
- B $vt - \frac{1}{2}at^2$
- C $\frac{(v+u)(v-u)}{2a}$
- D $\frac{(v-u)t}{2}$

112 9702/11/M/J/15/Q7

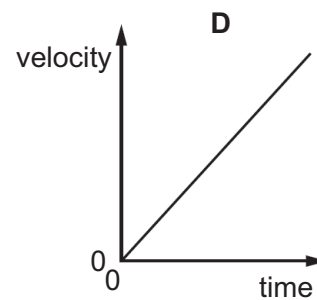
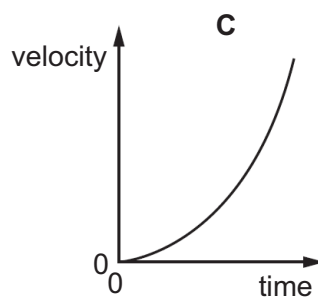
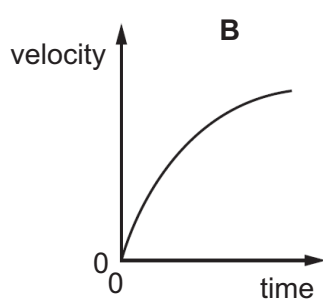
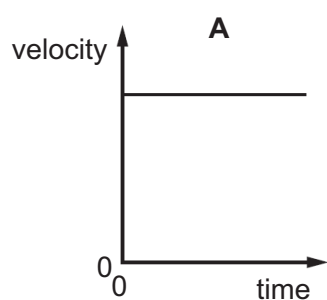
A sphere is released and falls. Its initial acceleration reduces until it eventually begins to travel at constant terminal velocity. Which displacement-time graph best represents the motion of the sphere?



113 9702/12/M/J/15/Q8

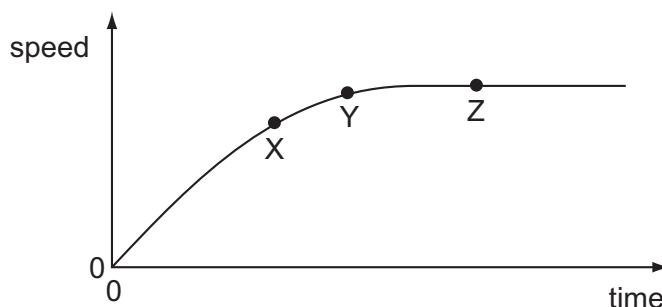
A stone is thrown horizontally from the top of a cliff. Air resistance is negligible.

Which graph shows the variation with time of the vertical component of the stone's velocity?



114 9702/13/M/J/15/Q7

A raindrop falls vertically from rest in air. The variation with time of the speed of the raindrop is shown in the graph.



Which statement about the raindrop is correct?

- A** At point X, the raindrop has an acceleration of 9.81 m s^{-2} .
- B** At point Z, the force on the raindrop due to air resistance has reached its maximum value and so the acceleration of the raindrop has also reached its maximum value.
- C** At point Z, the force due to air resistance is equal and opposite to the weight of the raindrop and so the speed of the raindrop is zero.
- D** The resultant force on the raindrop at point Y is less than the resultant force on the raindrop at point X.

115 9702/11/M/J/17/Q4

The values of displacement, velocity and acceleration of a vehicle can be deduced from graphs representing its motion. Often the areas under these graphs, or the gradients of the graphs, are used.

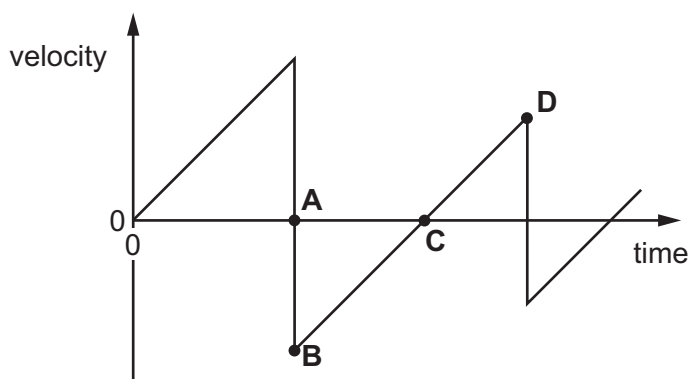
What would **not** give a value for a displacement, a velocity or an acceleration?

- A** area under a velocity-time graph
- B** gradient of a displacement-time graph
- C** gradient of a velocity-time graph
- D** gradient of an acceleration-time graph

116 9702/11/M/J/17/Q5

A ball is released from rest above a hard, horizontal surface. The graph shows how the velocity of the bouncing ball varies with time.

At which point on the graph does the ball reach its maximum height after the first bounce?



117 9702/11/M/J/17/Q6

A ball is kicked upwards at an angle of 45° to horizontal ground. After a short flight, the ball returns to the ground.

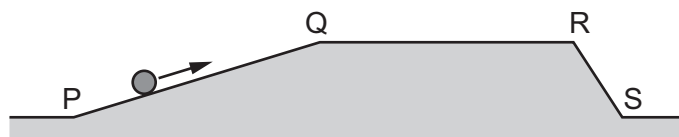
It may be assumed that air resistance is negligible.

What is **never** zero during the flight of the ball?

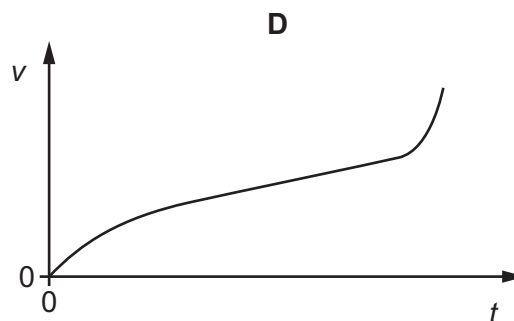
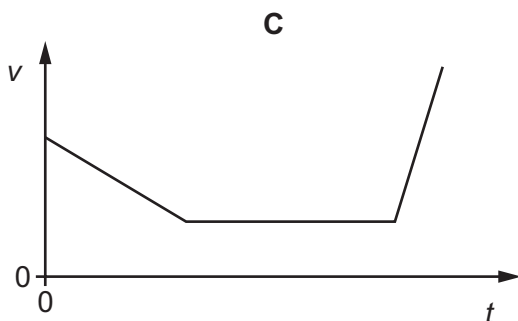
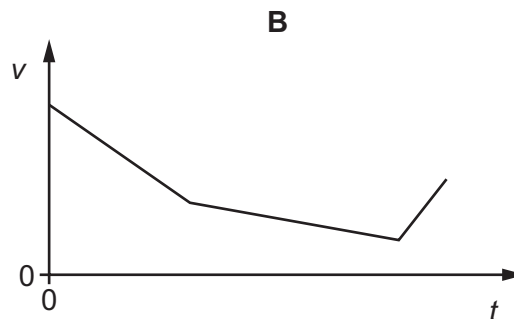
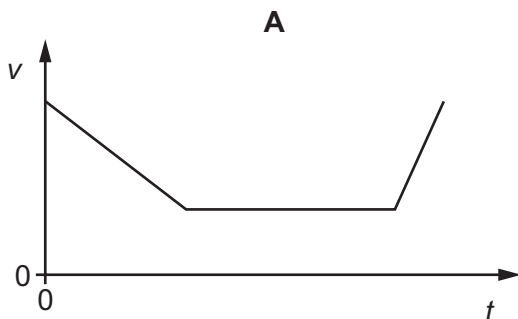
- A the horizontal component of the ball's acceleration
- B the horizontal component of the ball's velocity
- C the vertical component of the ball's momentum
- D the vertical component of the ball's velocity

118 9702/12/M/J/17/Q6

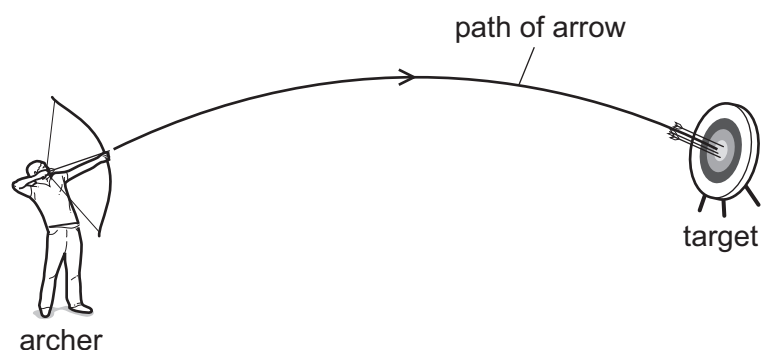
A ball is set in motion at P on a frictionless surface. It moves up slope PQ, along the horizontal surface QR and finally descends slope RS.



Which graph could represent the variation with time t of the ball's speed v as the ball moves from P to S?

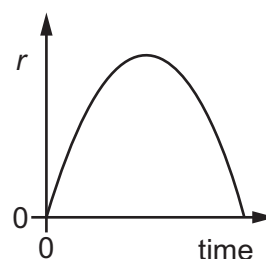
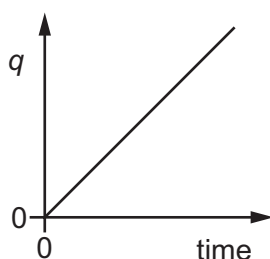
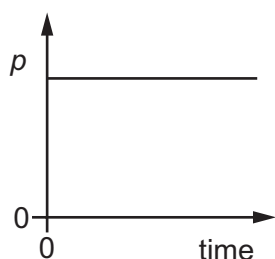


An archer shoots an arrow at a target. The diagram shows the path of the arrow.



Air resistance is assumed to be negligible.

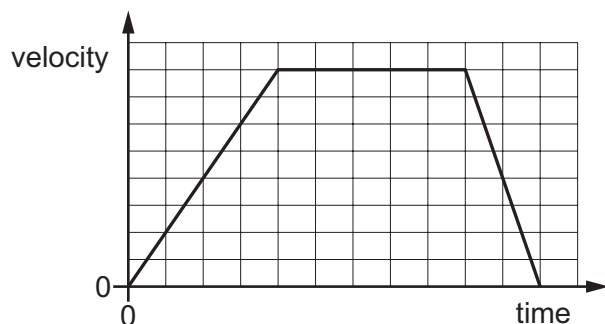
The graphs show how three different quantities p , q and r , relating to the motion of the arrow, vary with time.



Which quantity is the horizontal component of displacement and which quantity is the vertical component of displacement of the arrow?

	horizontal component of displacement	vertical component of displacement
A	p	q
B	q	r
C	r	p
D	r	q

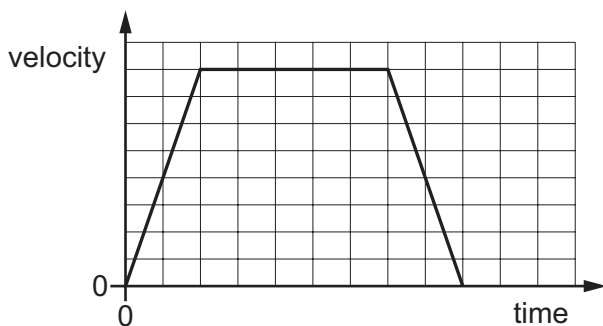
The velocity-time graph for a train starting at one station and stopping at the next is shown.



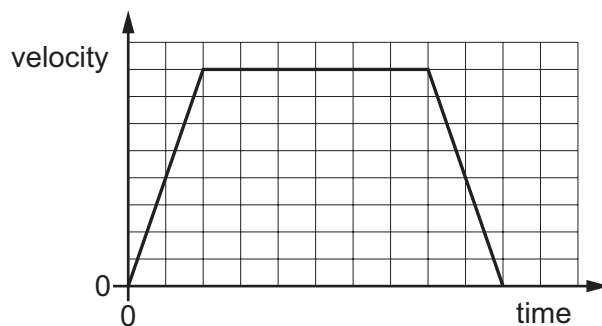
Another train has double the acceleration but the same maximum speed and the same deceleration.

Which velocity-time graph, on the same scale, shows the motion of this train between the same stations?

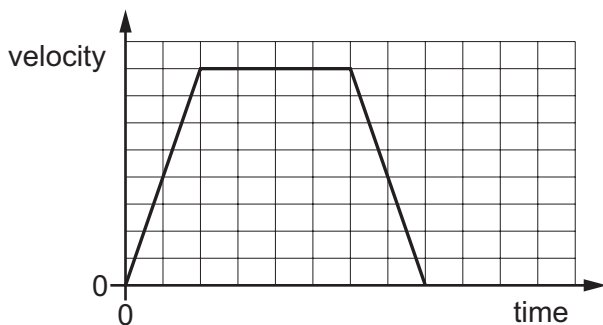
A



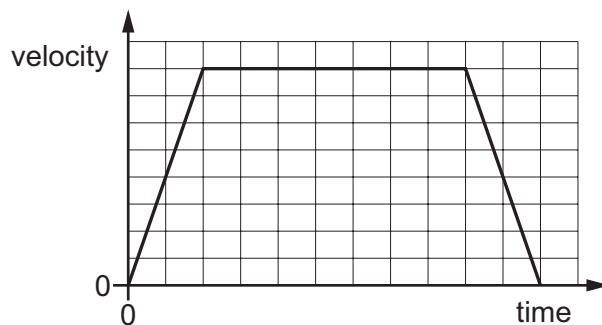
B



C



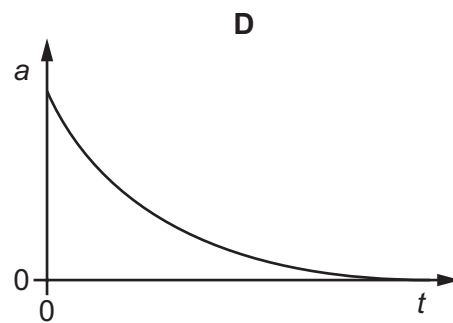
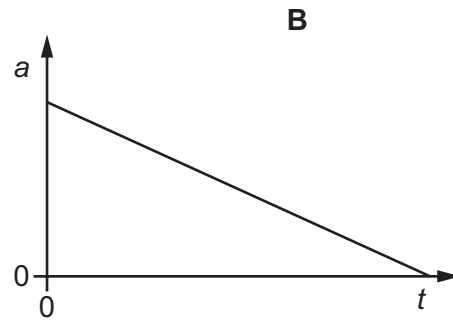
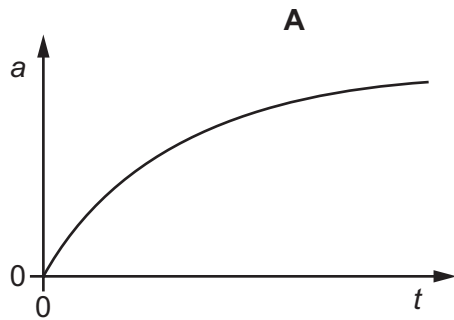
D



121 9702/12/O/N/17/Q8

A stone is released from rest and falls a long distance in air.

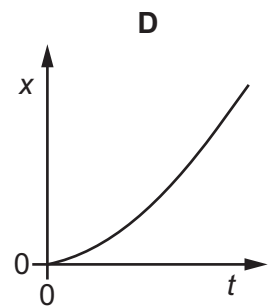
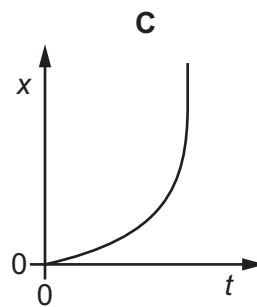
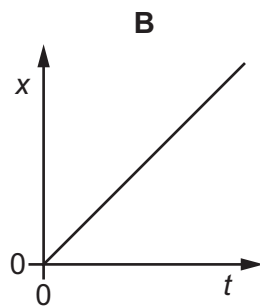
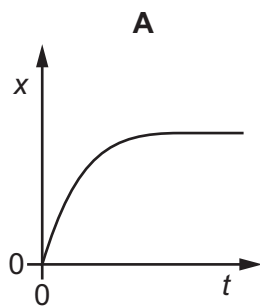
Which graph could show the variation with time t of the acceleration a of the stone?



122 9702/13/O/N/17/Q6

A football falls from the top of a tall building.

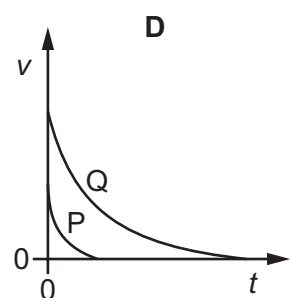
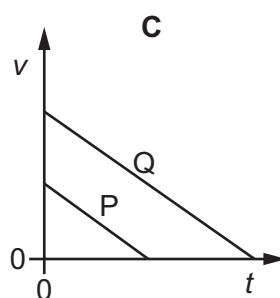
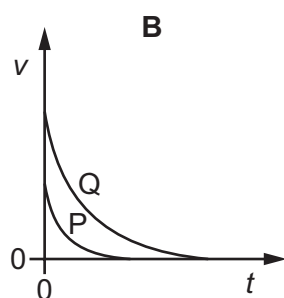
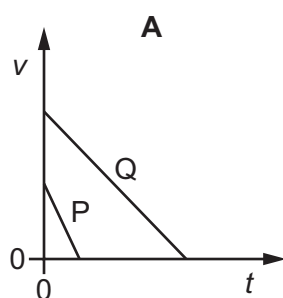
Which graph best represents the way in which the distance x fallen varies with time t ?



123 9702/13/O/N/17/Q7

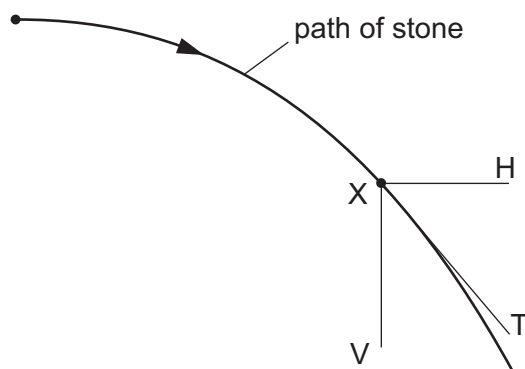
Two identical cars P and Q are travelling along a straight road. Car Q is travelling at twice the speed of car P. The brakes are applied to both cars, producing the same constant deceleration.

Which graph shows how the velocity v of each car varies with time t ?



124 9702/13/O/N/17/Q11

A stone is projected horizontally in a vacuum and moves along the path shown.



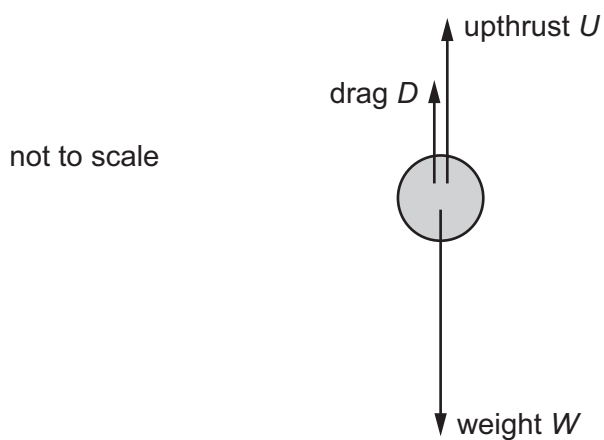
X is a point on this path. XV and XH are vertical and horizontal lines respectively through X. XT is the tangent to the path at X.

Along which directions do forces act on the stone at X?

- A** XV only **B** XH only **C** XV and XH **D** XT only

125 9702/13/O/N/17/Q13

A solid sphere falls at constant (terminal) velocity in a liquid. The three forces acting on the sphere are shown in the diagram.



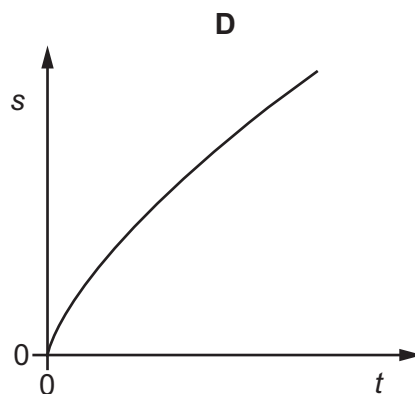
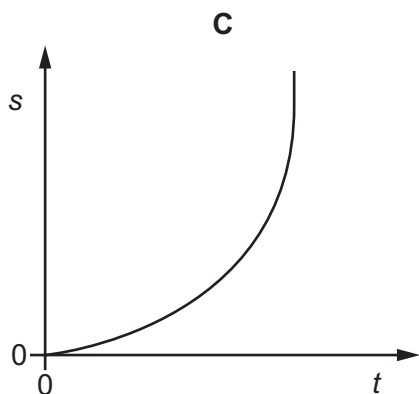
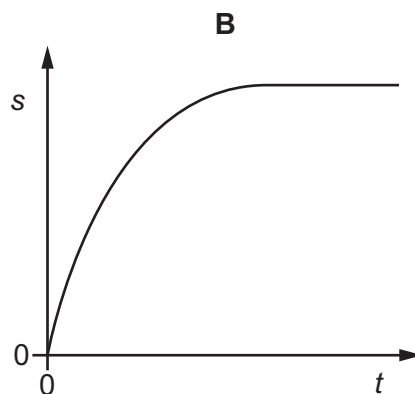
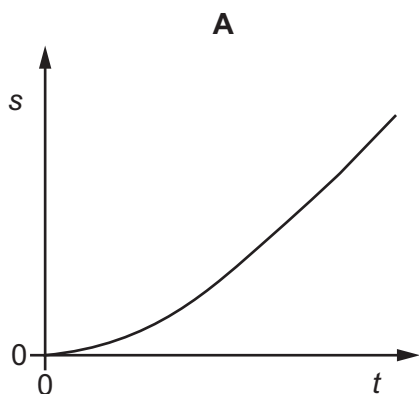
How are the three forces related?

- A** $W + D = U$
B $W > U + D$
C $W - U = D$
D $W < D + U$

126 9702/11/M/J/18/Q6

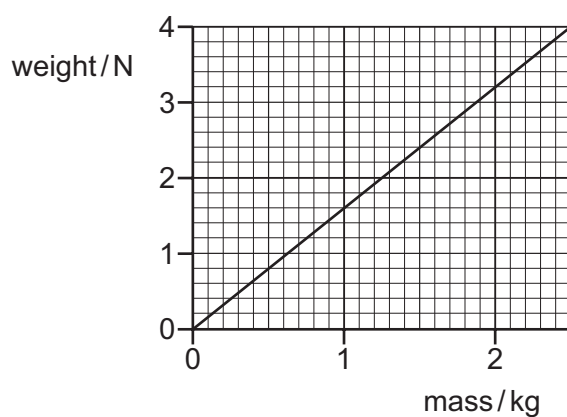
A tennis ball falls freely, in air, from the top of a tall building.

Which graph best represents the variation with time t of the distance s fallen?



127 9702/11/M/J/18/Q7

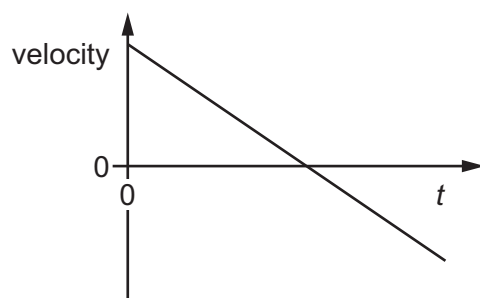
The graph shows the variation with mass of the weight of objects on a particular planet.



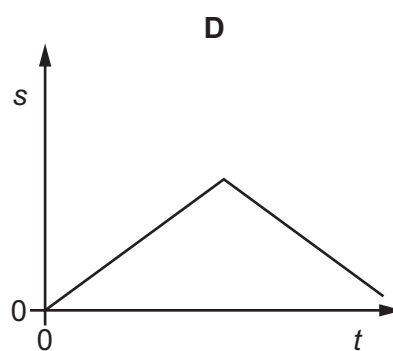
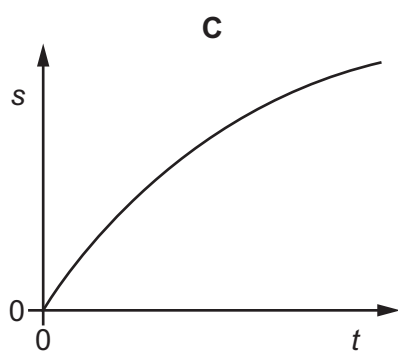
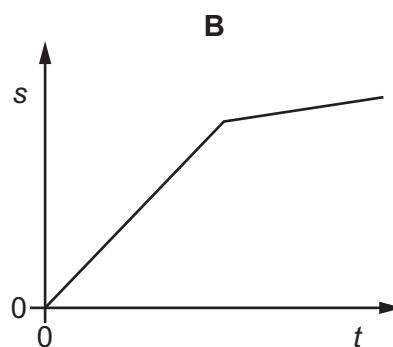
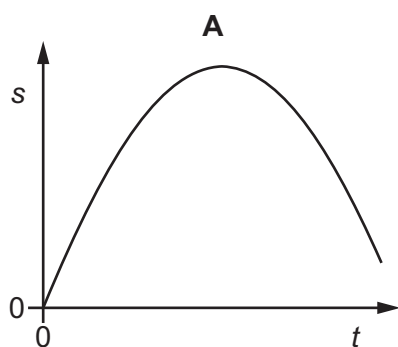
What is the value of the acceleration of free fall on the planet?

- A** 0.63 ms^{-2} **B** 1.6 ms^{-2} **C** 3.2 ms^{-2} **D** 9.8 ms^{-2}

The velocity of an object changes with time t as shown.

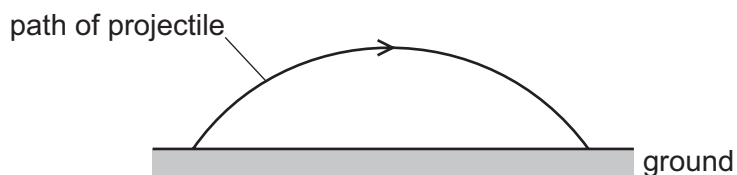


Which graph best shows the variation with time t of the displacement s of the object?



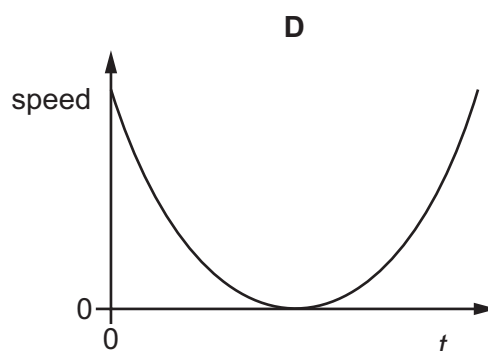
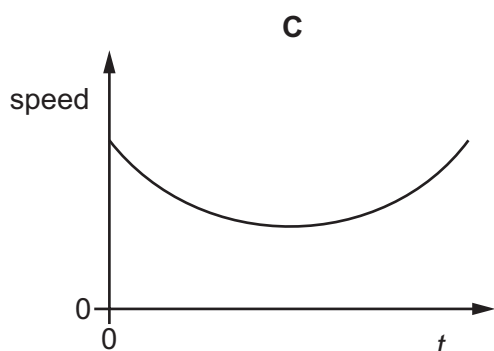
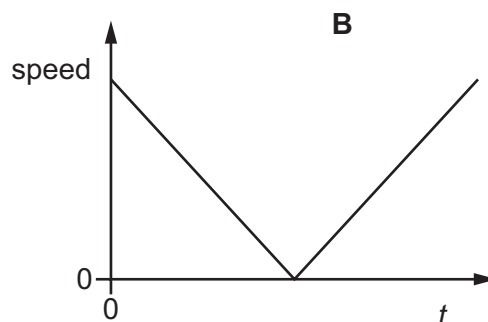
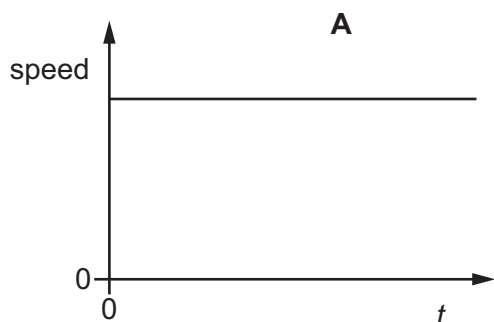
129 9702/12/M/J/18/Q6

A projectile is launched at an angle to the horizontal at time $t = 0$. It travels over horizontal ground, as shown.



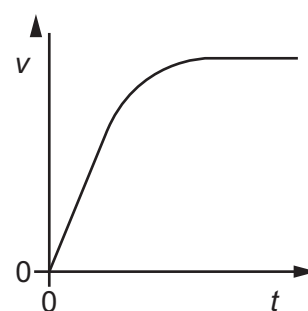
Assume that air resistance is negligible.

Which graph best shows the variation with t of the speed of the projectile from when it is launched to when it lands on the ground?

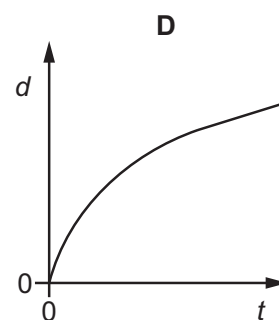
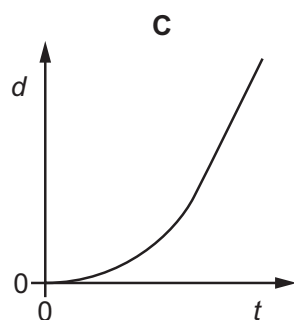
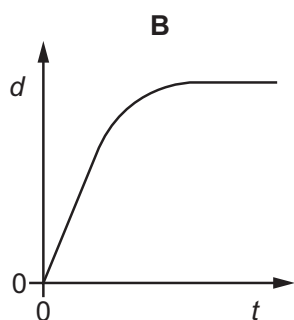
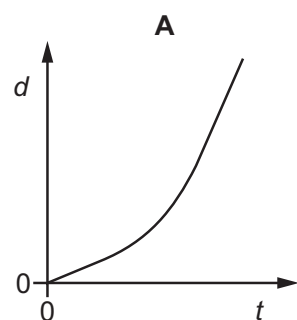


130 9702/12/M/J/18/Q7

A sky-diver falls vertically from a helicopter and reaches constant (terminal) velocity. The graph shows the variation with time t of the speed v of the sky-diver.



Which graph shows the variation with time t of the distance d fallen by the sky-diver?



131 9702/13/M/J/18/Q6

A rock on the surface of Mars is projected vertically upwards with an initial speed of 9.4 ms^{-1} . The rock rises to a height of 12 m above the surface.

Assume there is no atmosphere on Mars.

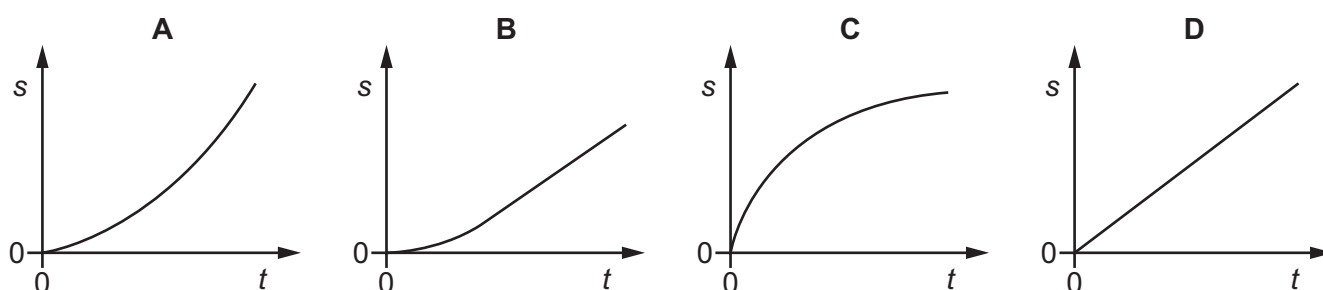
What is the acceleration of free fall near the surface of Mars?

- A** 0.39 ms^{-2} **B** 3.7 ms^{-2} **C** 7.4 ms^{-2} **D** 9.8 ms^{-2}

132 9702/13/M/J/18/Q8

A sky-diver falls from a stationary balloon at time $t = 0$. As the sky-diver falls, her speed and the air resistance increase until the force of the air resistance is equal to her weight.

Which graph best shows the variation with time t of the displacement s for the motion of the sky-diver?



133 9702/12/F/M/18/Q7

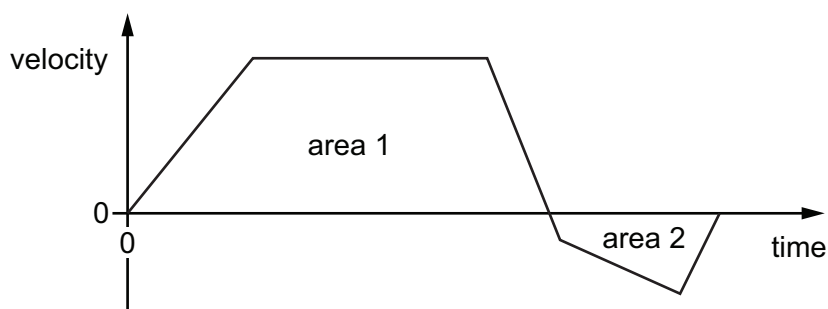
A stone of mass m is dropped from a tall building. There is significant air resistance. The acceleration of free fall is g .

When the stone is falling at a constant (terminal) velocity, which information is correct?

	magnitude of the acceleration of the stone	magnitude of the force of gravity on the stone	magnitude of the force of air resistance on the stone
A	g	zero	mg
B	zero	mg	mg
C	zero	zero	mg
D	zero	mg	zero

134 9702/12/F/M/18/Q8

The velocity-time graph for an object is shown.



How can the total displacement of the object be determined?

- A area 1 – area 2 B $\frac{(\text{area 1} + \text{area 2})}{2}$ C area 1 + area 2 D area 2 – area 1

135 9702/12/F/M/18/Q9

A girl throws a ball vertically upwards. It takes a time of 3.20 s to return to her hand.

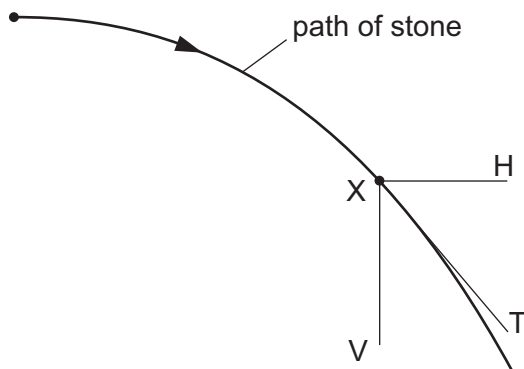
Assume air resistance is negligible.

What is the initial speed with which the ball is thrown?

- A 3.07 ms^{-1} B 7.85 ms^{-1} C 15.7 ms^{-1} D 31.4 ms^{-1}

136 9702/13/O/N/17/Q11

A stone is projected horizontally in a vacuum and moves along the path shown.



X is a point on this path. XV and XH are vertical and horizontal lines respectively through X. XT is the tangent to the path at X.

Along which directions do forces act on the stone at X?

- A XV and XH B XV only C XH only D XT only

137 9702/11/O/N/18/Q6

A tennis ball is thrown horizontally in air from the top of a tall building.

The effect of air resistance is **not** negligible.

What happens to the horizontal and to the vertical components of the ball's velocity?

	horizontal component of velocity	vertical component of velocity
A	constant	constant
B	constant	increases at a constant rate
C	decreases to zero	increases at a constant rate
D	decreases to zero	increases to a maximum value

138 9702/12/O/N/18/Q6

A sprinter runs a 100 m race. The sprinter has a constant acceleration from rest of 2.5 ms^{-2} until reaching a speed of 10 ms^{-1} . The speed then remains constant until the end of the race.

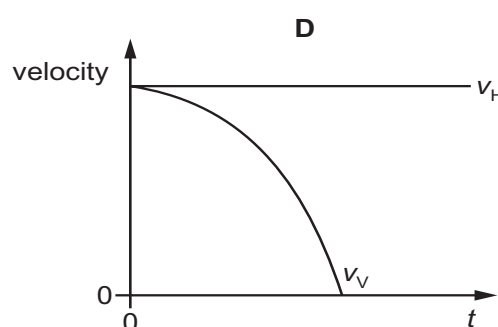
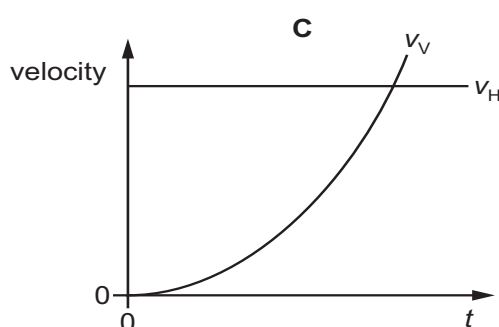
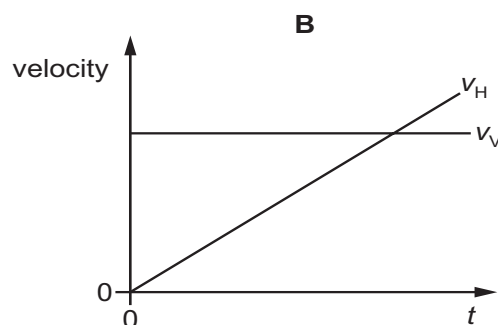
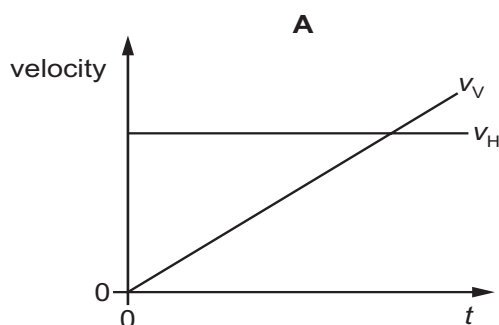
Which time does it take the sprinter to run the race?

- A** 8.9 s **B** 10 s **C** 12 s **D** 14 s

139 9702/13/O/N/18/Q6

A stone is projected horizontally at time $t = 0$ and falls. Air resistance is negligible. The stone has a horizontal component of velocity v_H and a vertical component of velocity v_V .

Which graph shows how v_H and v_V vary with time t ?



140 9702/13/O/N/18/Q8

A mass is placed on a frictionless slope inclined at 30° to the horizontal. The mass is then released.

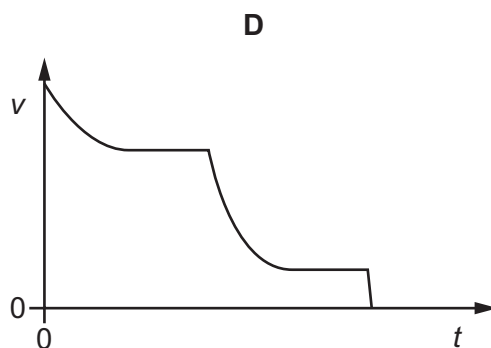
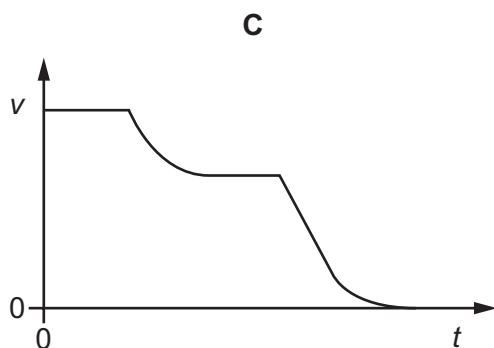
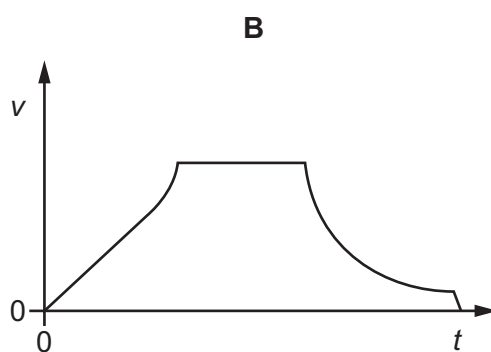
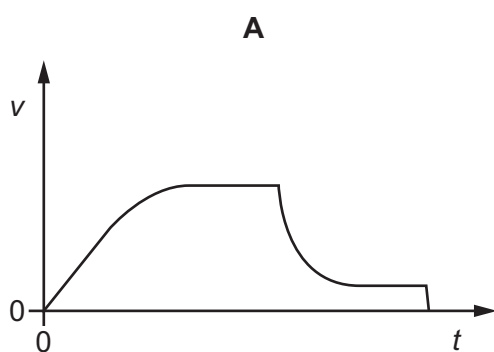
What is its acceleration down the slope?

- A 4.9 ms^{-2} B 5.7 ms^{-2} C 8.5 ms^{-2} D 9.8 ms^{-2}

141 9702/13/O/N/18/Q9

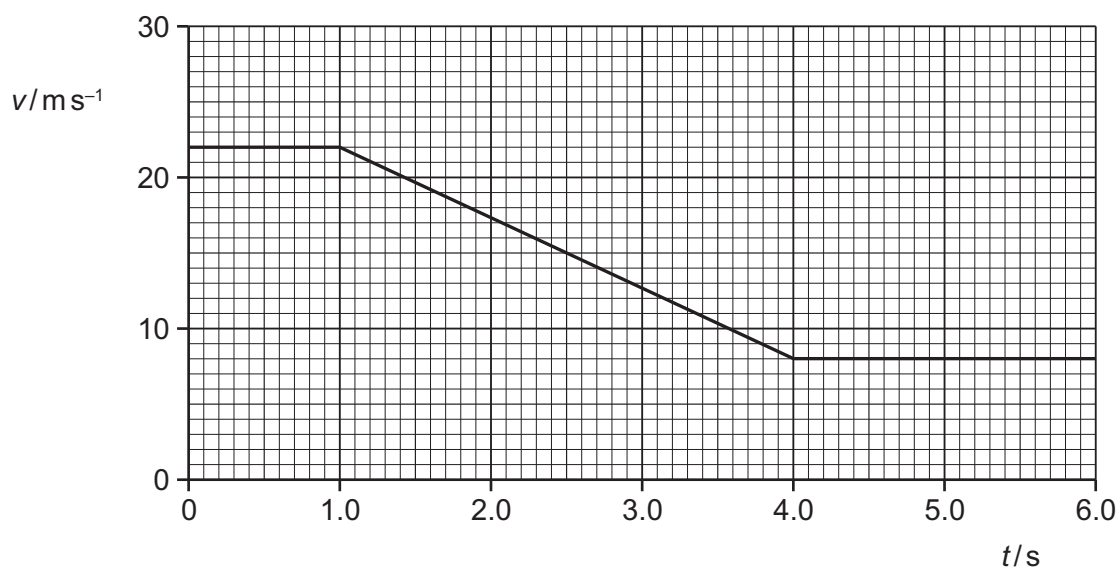
A parachutist falls vertically from rest at time $t = 0$ from a hot-air balloon. She falls for some distance before opening her parachute.

Which graph best shows the variation with time t of the speed v of the parachutist?



142 9702/11/M/J/19/Q6

A car travels along a straight horizontal road. The graph shows the variation of the velocity v of the car with time t for 6.0 s of its journey.



The brakes of the car are applied from $t = 1.0$ s to $t = 4.0$ s.

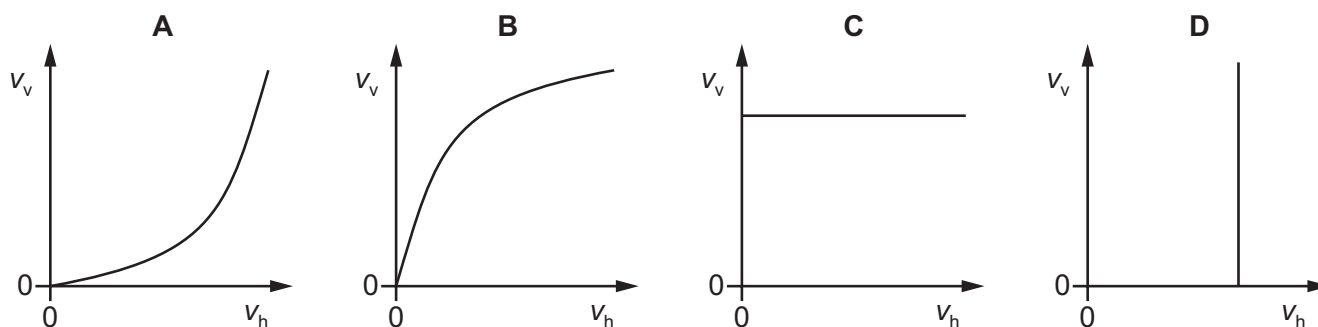
How far does the car travel while the brakes are applied?

- A 21 m B 45 m C 67 m D 83 m

143 9702/11/M/J/19/Q7

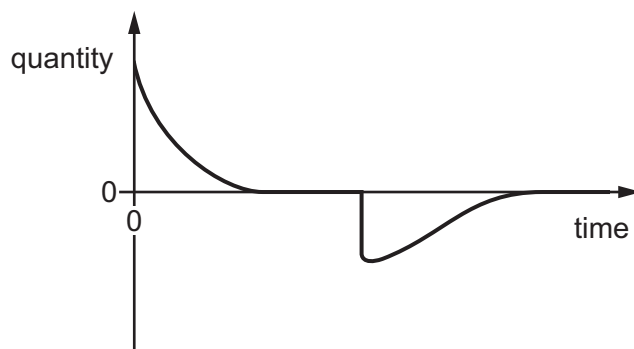
A stone is thrown horizontally from the top of a cliff and falls into the sea some time later. Air resistance is negligible.

Which graph shows how the vertical component v_v of velocity of this stone varies with its horizontal component v_h of velocity as it moves through the air?



144 9702/12/M/J/19/Q8

The graph shows how a physical quantity varies with time.

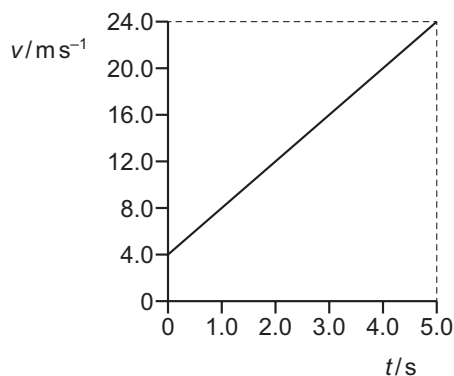


Which event could best be represented by the graph?

- A** the acceleration of a firework rising to a maximum height and falling to the ground
- B** the acceleration of a skydiver leaving an aircraft, falling, opening a parachute and falling to the ground
- C** the speed of a javelin as it leaves an athlete's hand, falls and sinks into the ground
- D** the speed of a high jump athlete leaving the ground, jumping over a bar and descending to the ground

145 9702/13/M/J/19/Q7

The graph shows the variation of velocity v with time t for an object.



The object passes a fixed point at time $t = 0$.

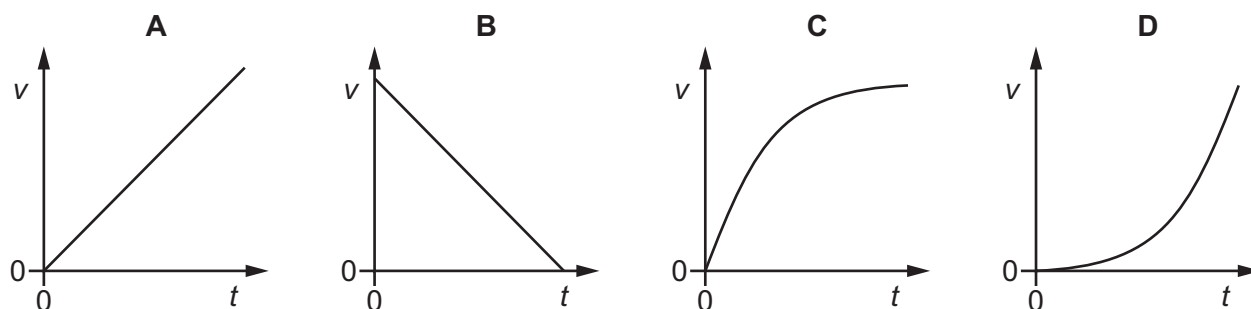
What is the displacement of the object from the fixed point at time $t = 5.0$ s and what is the acceleration of the object?

	displacement /m	acceleration /ms ⁻²
A	60	4.0
B	70	4.0
C	60	4.8
D	70	4.8

146 9702/13/M/J/19/Q8

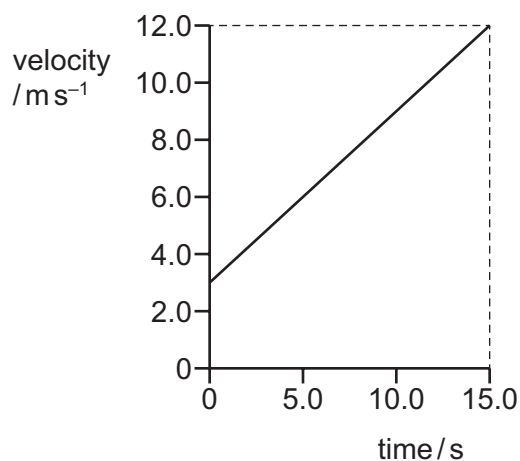
A skydiver jumps from an aeroplane and falls vertically through the air.

Which graph shows the variation with time t of the skydiver's vertical velocity v ?



147 9702/11/O/N/19/Q6

The velocity-time graph for an object of mass 2.5 kg is shown.



What is the resultant force acting on the object?

- A** 0.60 N **B** 0.80 N **C** 1.5 N **D** 2.0 N

148 9702/12/O/N/19/Q6

A ball is thrown vertically upwards from ground level and reaches a maximum height of 12.7 m before falling back to ground level.

Assume air resistance is negligible.

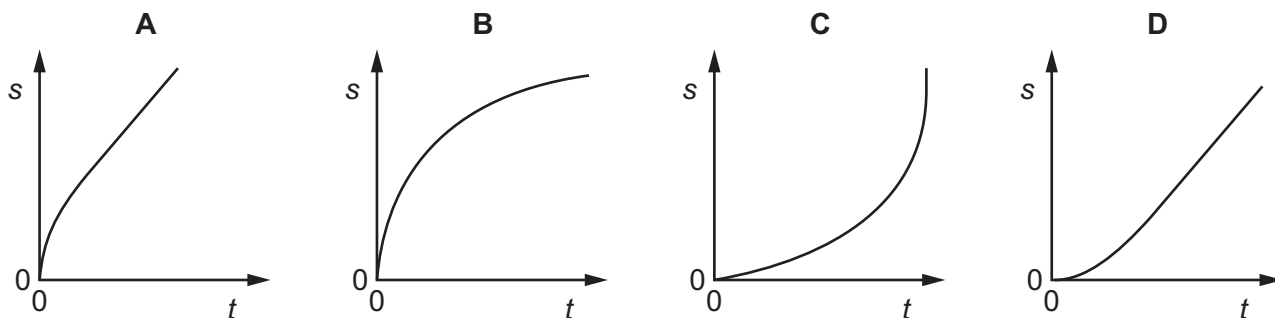
What is the total time for which the ball is in the air?

- A** 1.61 s **B** 3.22 s **C** 3.88 s **D** 5.18 s

149 9702/12/O/N/19/Q8

A tennis ball is released from rest at time $t = 0$ and falls through air for a long time.

Which graph of its displacement s against time t best represents the motion of the ball?



150 9702/13/O/N/19/Q6

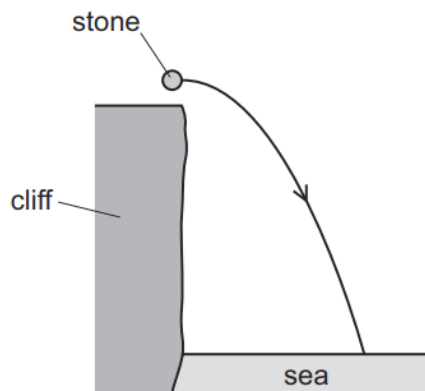
A lead sphere is released from rest at point X, a long way above the surface of a planet. The sphere falls in a vacuum. After a time of 4.0 s, it has fallen through a vertical distance of 3.0 m. Assume the acceleration of free fall is constant.

How far will the sphere have fallen from point X at a time of 20 s after its release?

- A** 15 m **B** 75 m **C** 80 m **D** 2000 m

151. 9702/12/F/M/20/Q6

A stone is thrown horizontally from the top of a cliff and falls into the sea below. Air resistance is negligible. The path of the stone is shown.



In which direction does the resultant force on the stone act during its fall?

- A** horizontally to the right
- B** parallel to its velocity
- C** perpendicular to its velocity
- D** vertically downwards

152. 9702/12/F/M/20/Q7

A car moves with uniform acceleration along a straight road. Oil leaks from the car at the rate of one drop every two seconds. The diagram shows the distances between three successive oil drops on the road.



What is the acceleration of the car?

- A** 0.75 ms^{-2} **B** 1.5 ms^{-2} **C** 3.0 ms^{-2} **D** 6.0 ms^{-2}

153. 9702/12/F/M/20/Q8

A person of mass 60 kg stands on accurate bathroom scales, placed on the floor of an elevator (lift) which operates in a tall building.

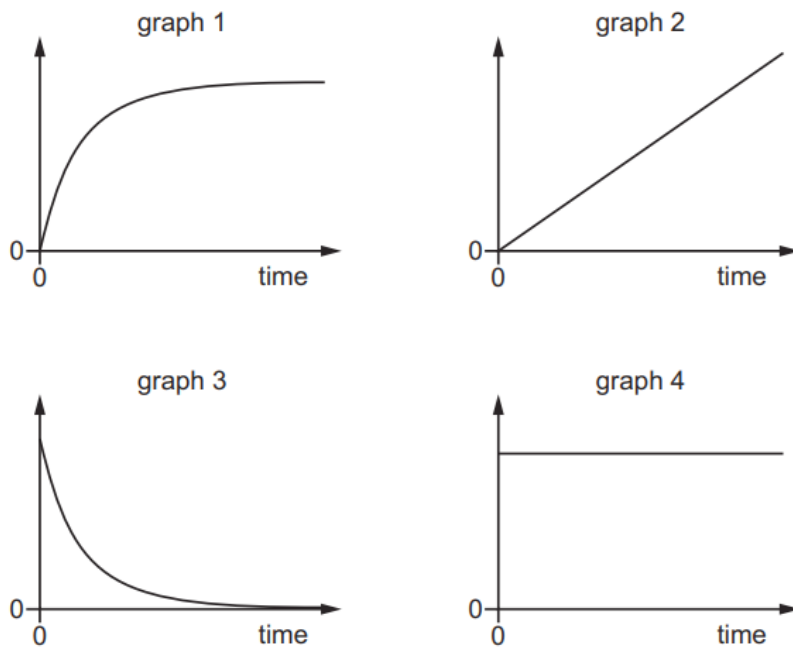
At a certain instant the bathroom scales read 58 kg.

Which row could give the person's direction of movement and type of motion?

	direction	motion
A	downwards	constant speed
B	downwards	slowing down
C	upwards	constant speed
D	upwards	slowing down

154. 9702/12/F/M/20/Q9

The diagram shows graphs of various quantities plotted against time for an object dropped from a stationary balloon high in the atmosphere.

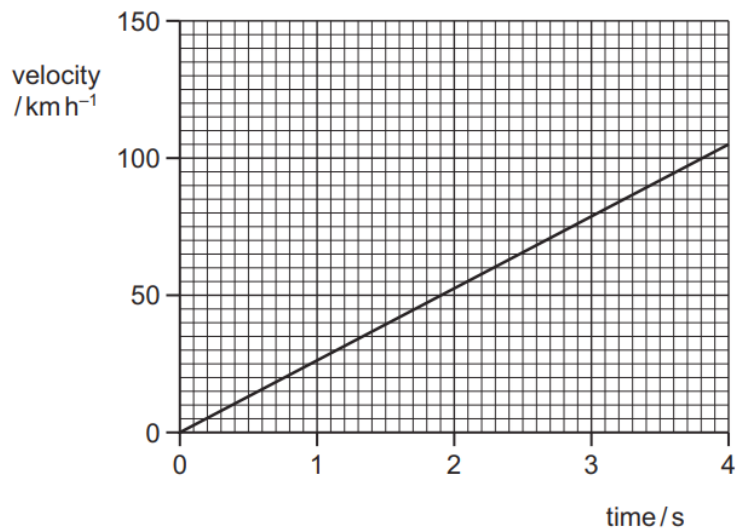


Which statement could be correct?

- A** Graph 1 is acceleration against time and graph 3 is resultant force against time.
- B** Graph 1 is acceleration against time and graph 4 is resultant force against time.
- C** Graph 3 is acceleration against time and graph 1 is velocity against time.
- D** Graph 3 is acceleration against time and graph 2 is velocity against time.

155. 9702/11/M/J/20/Q6

The velocity of an electric car changes as shown.

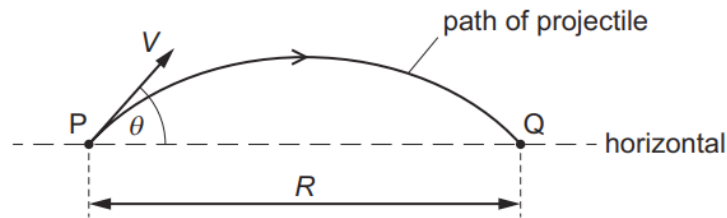


What is the acceleration of the car?

- A** 210 ms^{-2}
- B** 58 ms^{-2}
- C** 26 ms^{-2}
- D** 7.3 ms^{-2}

156. 9702/11/M/J/20/Q7

A projectile is fired from point P with velocity V at an angle θ to the horizontal. It lands at point Q, a horizontal distance R from P, after time T .



The acceleration of free fall is g . Air resistance is negligible.

Which equation is correct?

A $R = VT \cos \theta$

B $R = VT \sin \theta$

C $R = VT \cos \theta - \frac{1}{2}gT^2$

D $R = VT \sin \theta - \frac{1}{2}gT^2$

157. 9702/11/M/J/20/Q9

The resultant force F on a raindrop of mass m falling with velocity v is given by the equation

$$F = mg - kv^2$$

where k is a constant and g is the acceleration of free fall.

What is the velocity of the raindrop when it reaches a constant (terminal) velocity?

A $\sqrt{\frac{k}{mg}}$

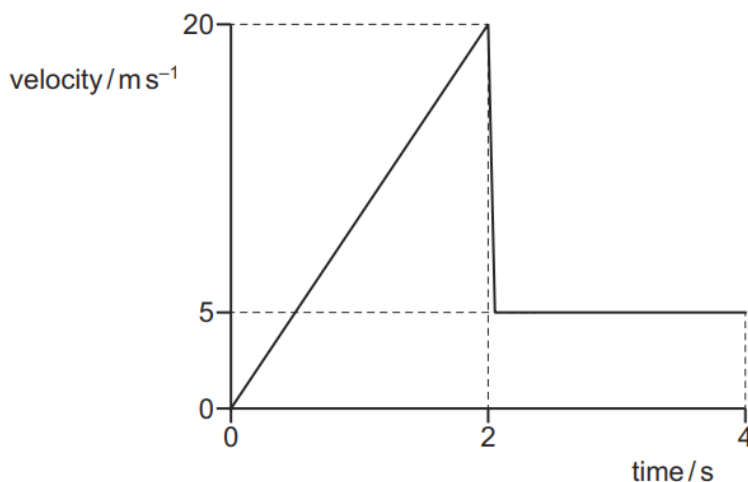
B $\frac{k}{mg}$

C $\sqrt{\frac{mg}{k}}$

D $\frac{mg}{k}$

158. 9702/12/M/J/20/Q5

A stone is dropped from a height of 20 m above water. The graph shows the variation with time of the velocity of the stone.

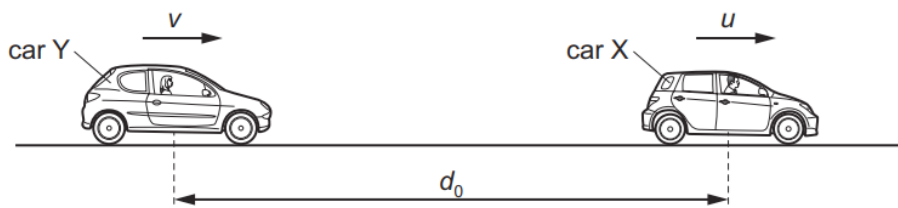


Which statement describes the approximate position of the stone four seconds after it is dropped?

- A** It is at a distance of 10 m above the surface of the water.
- B** It is at a distance of 10 m below the surface of the water.
- C** It is at a distance of 20 m below the surface of the water.
- D** It is at a distance of 30 m below the surface of the water.

159. 9702/12/M/J/20/Q6

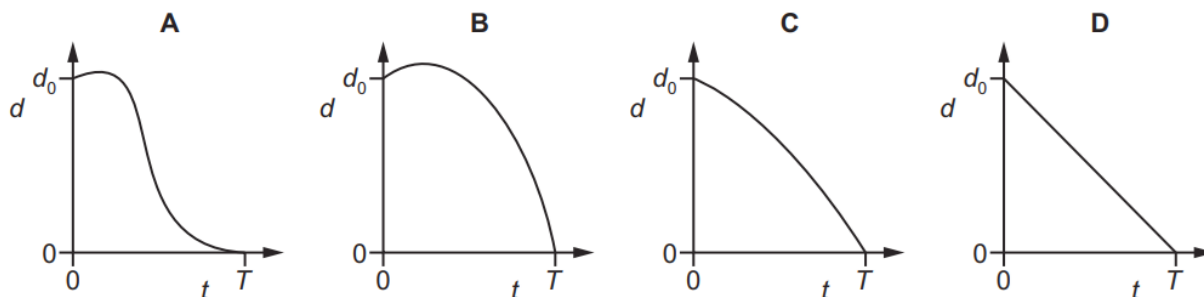
A car X is travelling at a constant speed u along a straight road. At time $t = 0$ a second car Y is a distance d_0 behind car X and travelling at a speed v in the same direction. Speed v is less than speed u .



At time $t = 0$ car Y begins to accelerate with a constant acceleration.

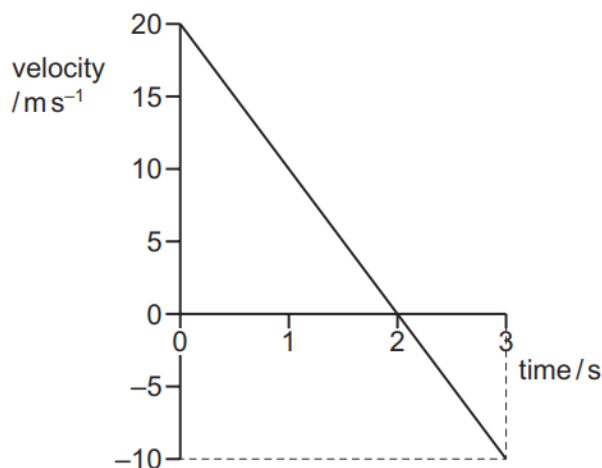
Car Y overtakes car X at time $t = T$.

Which graph could best show the variation with time t of the distance d between the cars?



160. 9702/13/M/J/20/Q6

The graph shows how the velocity of a ball varies with time from the moment it is hit vertically upwards from the ground.



What is the displacement of the ball from the ground after a time of 3.0 s?

- A** 15 m **B** 25 m **C** 30 m **D** 45 m

161. 9702/13/M/J/20/Q7

Two students each throw a ball horizontally from the top of a tower. The two balls are released at the same time.

The first student throws her ball with a speed of 20 m s^{-1} and the second student throws his ball with a speed of 10 m s^{-1} .

Assume air resistance is negligible and that the balls land on horizontal ground.

Which row describes the horizontal distances travelled and the landing times of the two balls on the ground?

	horizontal distances	landing times
A	same	same
B	same	different
C	different	same
D	different	different

162. 9702/13/M/J/20/Q9

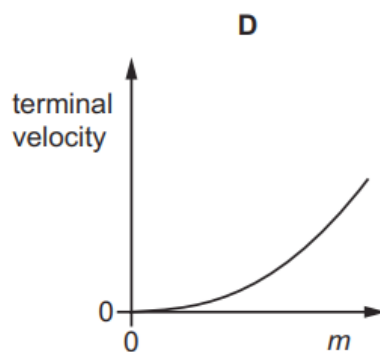
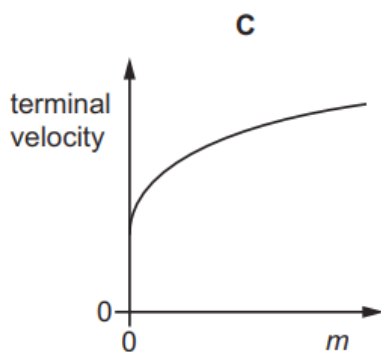
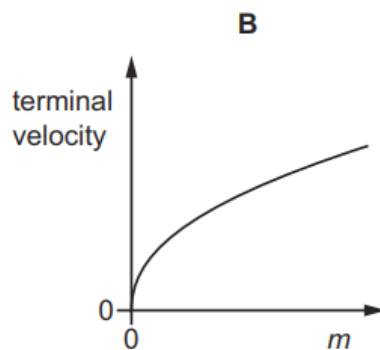
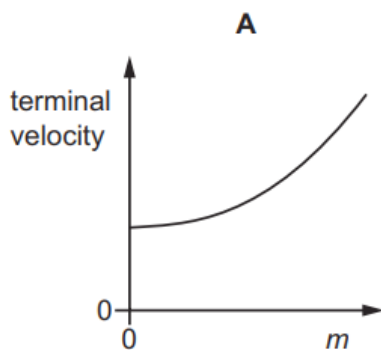
The resultant force F on a raindrop of mass m falling vertically with velocity v is given by the equation

$$F = mg - kv^2$$

where k is a constant and g is the acceleration of free fall.

The falling raindrop eventually reaches a constant (terminal) velocity.

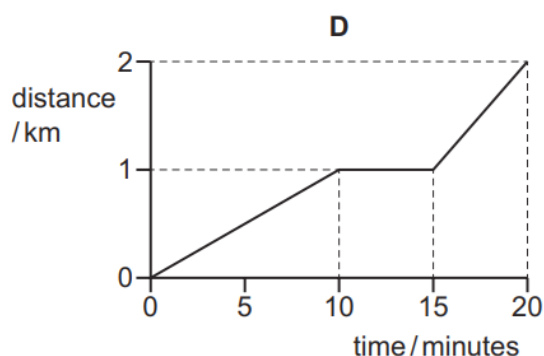
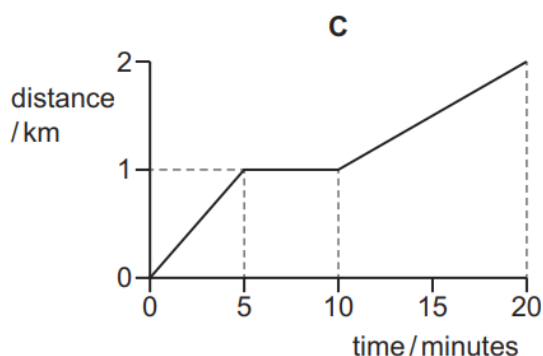
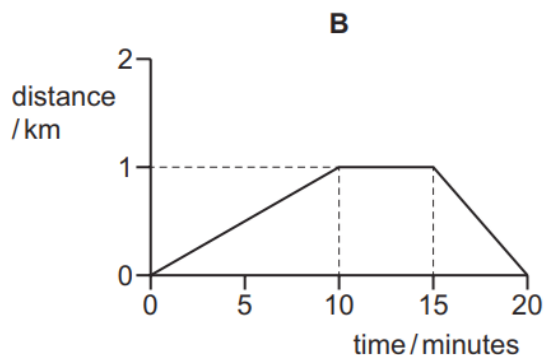
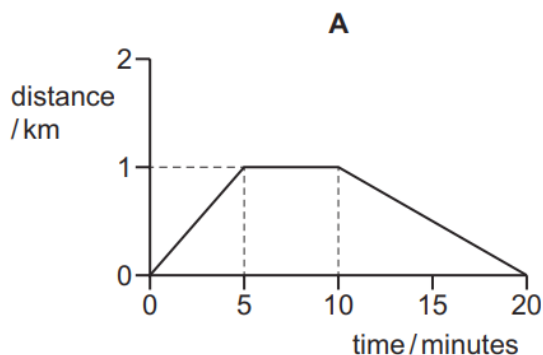
Which graph shows the variation of the terminal velocity of the raindrop with mass m ?



163. 9702/11/O/N/20/Q6

A student cycles uphill from home to a shop, taking 10 minutes. The student then spends 5 minutes in the shop, before cycling home downhill at twice the initial speed.

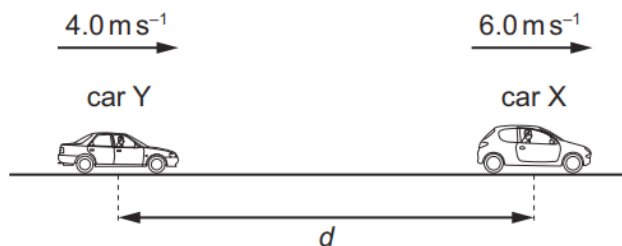
Which graph could show the variation with time of the distance travelled by the cyclist?



164. 9702/11/O/N/20/Q7

Two cars X and Y are travelling along the same straight road. Car X is travelling at a constant speed of 6.0 m s^{-1} . Car Y has a constant acceleration of 0.50 m s^{-2} .

At the instant shown, car X is a distance d ahead of car Y. Car Y is travelling at a speed of 4.0 m s^{-1} .



Car Y is level with car X after a time of 20 seconds.

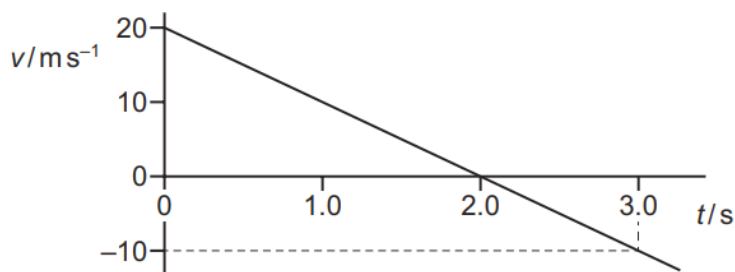
What is the distance d ?

- A** 40 m **B** 60 m **C** 180 m **D** 300 m

165. 9702/12/O/N/20/Q6

A stone is thrown vertically upwards from a point X at time $t = 0$.

The variation with time t of the velocity v of the stone is shown.

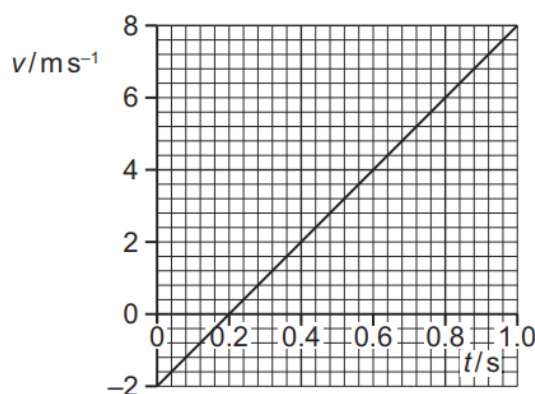


What is the displacement of the stone from point X at time $t = 3.0$ s?

- A** 15 m above X
- B** 15 m below X
- C** 25 m above X
- D** 25 m below X

166. 9702/13/O/N/20/Q6

An object moves in a straight line. The graph shows the variation with time t of the velocity v of the object.



At time $t = 0$ the object is at point X.

What is the displacement of the object from point X at time $t = 0.80$ s?

- A** 1.6 m **B** 1.8 m **C** 2.0 m **D** 3.2 m

167. 9702/13/O/N/20/Q7

An object accelerates uniformly from rest to speed v . It then moves at constant speed v for a time of 8.0 s before decelerating uniformly to rest. The total time taken is 12.0 s, and the total distance travelled is 60 m.

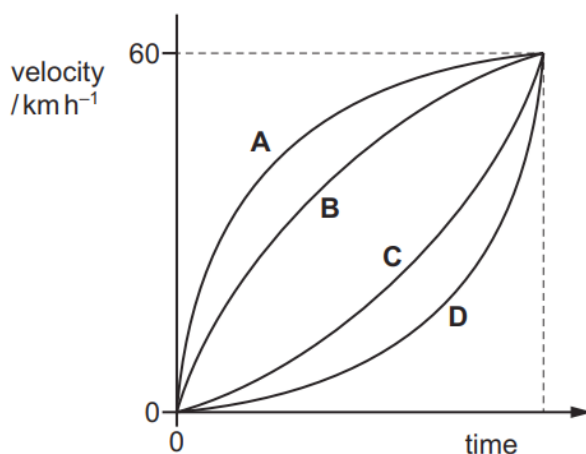
What is the speed v ?

- A** 3.0 m s^{-1} **B** 5.0 m s^{-1} **C** 6.0 m s^{-1} **D** 15 m s^{-1}

168. 9702/12/F/M/21/Q6

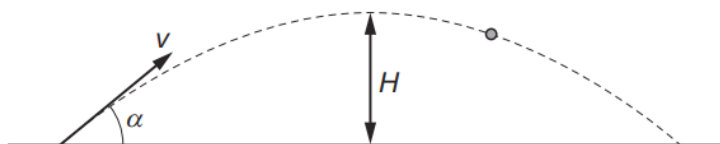
Four cars, **A**, **B**, **C** and **D**, move from rest in a straight line. The cars take the same time to accelerate to a velocity of 60 km h^{-1} . Their velocity–time graphs are shown.

Which car reaches a velocity of 60 km h^{-1} in the shortest distance?



169. 9702/12/F/M/21/Q7

A cannon fires a cannonball with an initial speed v at an angle α to the horizontal.



Which equation is correct for the maximum height H reached?

A $H = \frac{v \sin \alpha}{2a}$ **B** $H = \frac{g \sin \alpha}{2v}$ **C** $H = \frac{(v \sin \alpha)^2}{2a}$ **D** $H = \frac{g^2 \sin \alpha}{2v}$

170. 9702/12/F/M/21/Q9

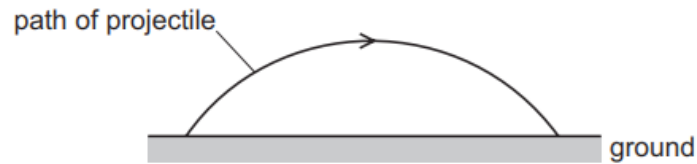
A skydiver, who is falling vertically through the air, opens his parachute.

Which row describes the velocity of the skydiver immediately after he opens his parachute?

	direction of velocity	magnitude of velocity
A	downwards	decreases
B	downwards	increases
C	upwards	decreases
D	upwards	increases

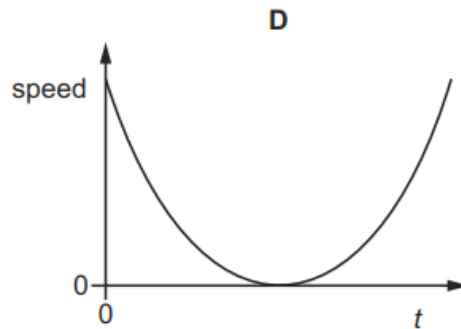
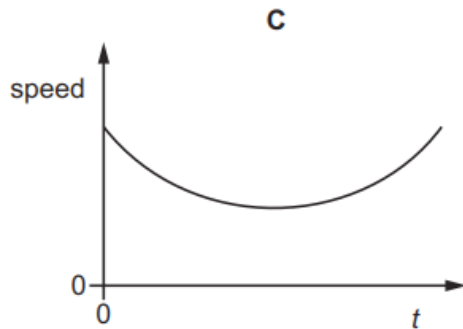
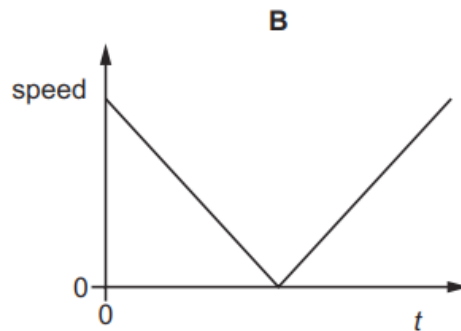
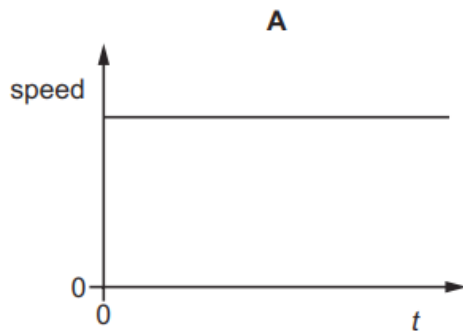
171. 9702/11/M/J/21/Q6

A projectile is launched at an angle to the horizontal at time $t = 0$. It travels over horizontal ground, as shown.



Air resistance is negligible.

Which graph best shows the variation with t of the speed of the projectile from when it is launched to when it lands on the ground?



172. 9702/11/M/J/21/Q7

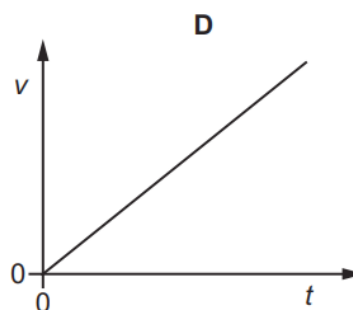
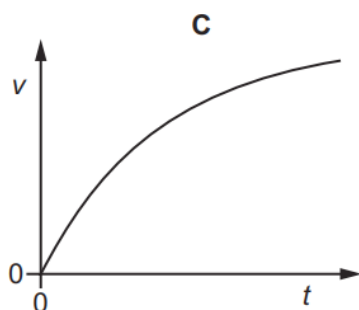
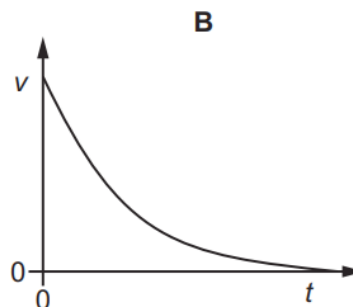
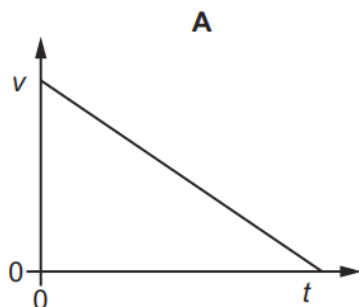
A train, initially at rest at a station, has a uniform acceleration of 0.20 ms^{-2} until it reaches a speed of 20 ms^{-1} . It travels for a time at this constant speed and then has a uniform deceleration of 0.40 ms^{-2} until it comes to rest at the next station. The distance between the two stations is 3000 m.

What is the time taken by the train to travel between the two stations?

- A** 75 s **B** 150 s **C** 230 s **D** 300 s

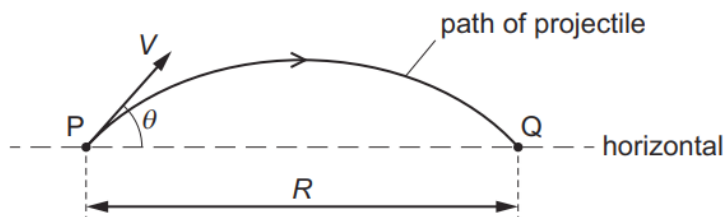
173. 9702/12/M/J/21/Q6

Which graph shows the variation with time t of the velocity v of an object falling vertically downwards in a vacuum?



174. 9702/12/M/J/21/Q7

A projectile is fired from point P with velocity V at an angle θ to the horizontal. It lands at point Q, a horizontal distance R from P. Air resistance is negligible.



The acceleration of free fall is g .

Which equation for R is correct?

A $R = \frac{V^2 \sin \theta \cos \theta}{g}$

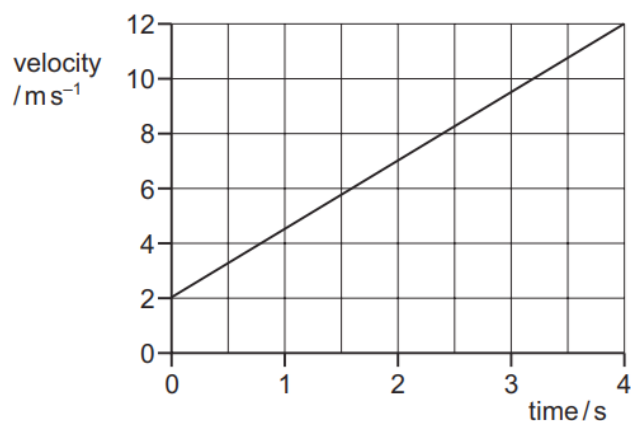
B $R = \frac{2V^2 \sin \theta \cos \theta}{g}$

C $R = \frac{V^2 \sin \theta \cos \theta}{2g}$

D $R = \frac{V^2 g \sin \theta \cos \theta}{2}$

175. 9702/13/M/J/21/Q6

The diagram shows a velocity–time graph for a car.



What is the distance travelled during the first 4.0 s?

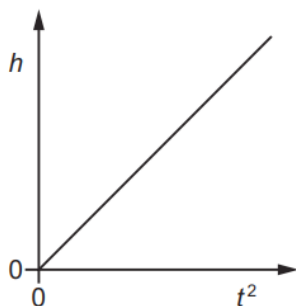
- A** 2.5 m **B** 3.0 m **C** 20 m **D** 28 m

176. 9702/13/M/J/21/Q7

A steel ball is dropped from rest from a height h above the ground. The ball hits the ground after a time t .

This is repeated for a number of different heights.

The graph shows the variation of h with t^2 for the ball.



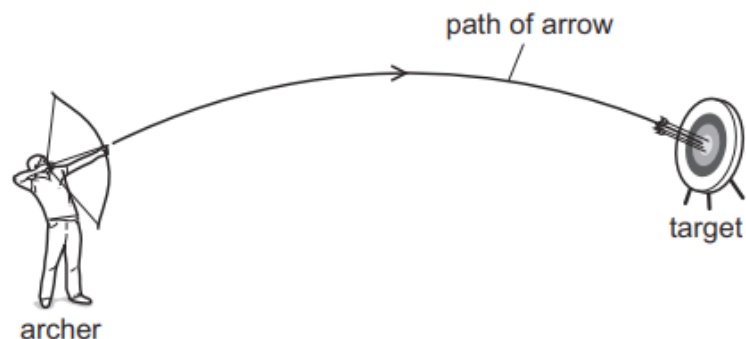
The gradient of the graph is G .

Which expression gives the acceleration of the ball?

- A** $\frac{G}{2}$ **B** G **C** $2G$ **D** G^2

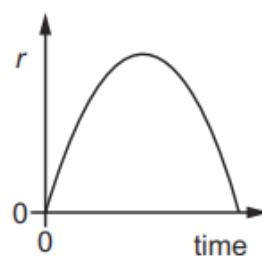
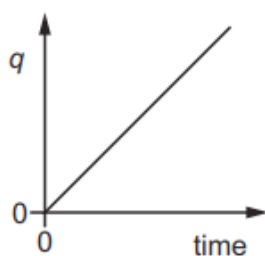
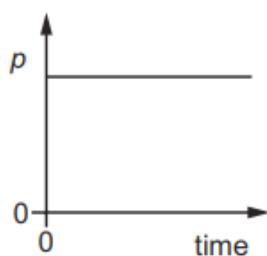
177. 9702/11/O/N/21/Q6

An archer shoots an arrow at a target. The diagram shows the path of the arrow.



Air resistance is negligible.

The graphs show how three different quantities p , q and r vary with time.



Which quantity could be the horizontal component of displacement and which quantity could be the vertical component of displacement of the arrow?

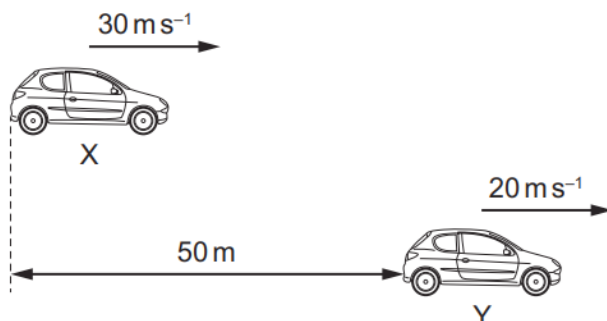
	horizontal component of displacement	vertical component of displacement
A	p	q
B	q	r
C	r	p
D	r	q

178. 9702/11/O/N/21/Q7

Two cars X and Y are positioned as shown at time $t = 0$.

They are travelling in the same direction.

X is 50 m behind Y and has a constant velocity of 30 m s^{-1} . Y has a constant velocity of 20 m s^{-1} .

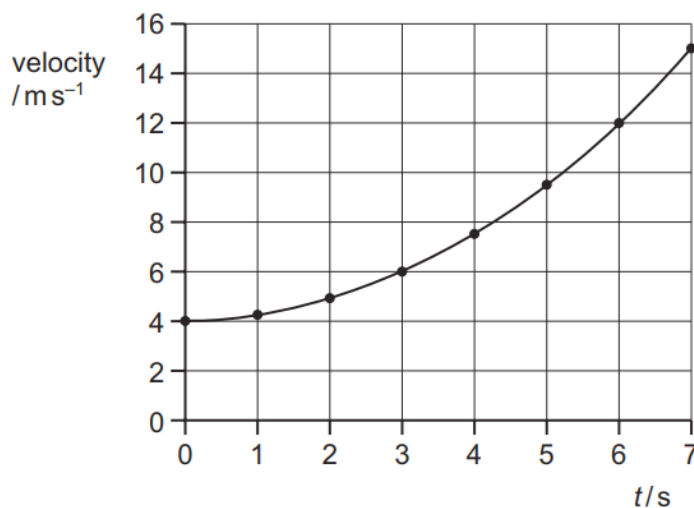


What is the value of t when X is level with Y?

- A** 1.0 s **B** 1.7 s **C** 2.5 s **D** 5.0 s

179. 9702/12/O/N/21/Q6

The graph shows the variation with time t of the velocity of a vehicle moving in a straight line.



The vehicle, moving at 4.0 m s^{-1} , begins to accelerate at time $t = 0$.

What is the vehicle's acceleration at time $t = 3.0 \text{ s}$?

- A** 0.67 m s^{-2} **B** 1.0 m s^{-2} **C** 1.3 m s^{-2} **D** 2.0 m s^{-2}

180. 9702/12/O/N/21/Q7

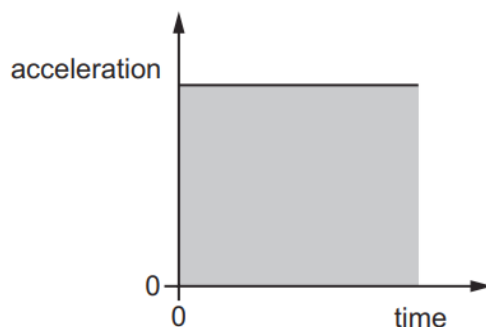
A stone is projected vertically upwards from the ground at an initial speed of 15 ms^{-1} . Air resistance is negligible.

What is the maximum height reached by the stone?

- A** 0.76 m **B** 11 m **C** 23 m **D** 110 m

181. 9702/13/O/N/21/Q6

The graph shows the variation with time of the acceleration of a car.



What **must** the shaded area under the graph represent?

- A** the average velocity of the car
B the change in velocity of the car
C the final velocity of the car
D the initial velocity of the car

182. 9702/13/O/N/21/Q7

A stone is thrown horizontally off a cliff and then lands in the sea. Air resistance is negligible.

Which statement about the stone's motion is **not** correct?

- A** The final displacement of the stone depends upon its initial horizontal velocity.
B The stone travels with a constant horizontal component of velocity until it hits the water.
C The stone travels with an increasing vertical component of velocity.
D The time taken for the stone to hit the surface of the water depends on its initial horizontal velocity.

183. 9702/12/F/M/22/Q5

How can the acceleration of an object be determined?

- A** from the area under a displacement–time graph
B from the area under a velocity–time graph
C from the gradient of a displacement–time graph
D from the gradient of a velocity–time graph

184. 9702/12/F/M/22/Q6

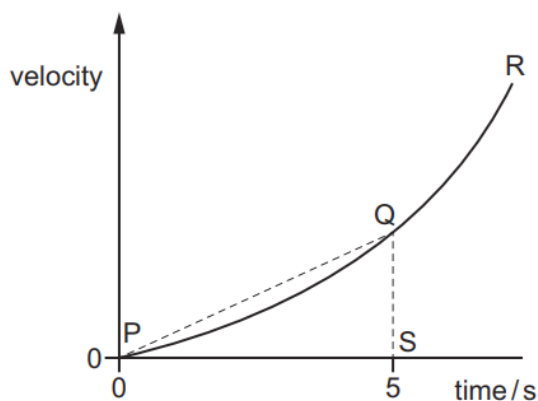
A sprinter takes a time of 11.0 s to run a 100 m race. She first accelerates uniformly from rest, reaching a speed of 10 m s^{-1} . She then runs at a constant speed of 10 m s^{-1} until the finish line.

What is the uniform acceleration of the sprinter for the first part of the race?

- A** 0.5 m s^{-2} **B** 0.91 m s^{-2} **C** 1.7 m s^{-2} **D** 5.0 m s^{-2}

185. 9702/11/M/J/22/Q5

The curved line PQR is the velocity–time graph for a car starting from rest.



What is the average acceleration of the car over the first 5 s?

- A** the area below the curve PQ
B the area of the triangle PQS
C the gradient of the straight line PQ
D the gradient of the tangent at Q

186. 9702/11/M/J/22/Q6

A ball is thrown horizontally with a speed of 10.0 m s^{-1} above horizontal ground. The ball hits the ground after a time of 3.0 s.

Air resistance is negligible.

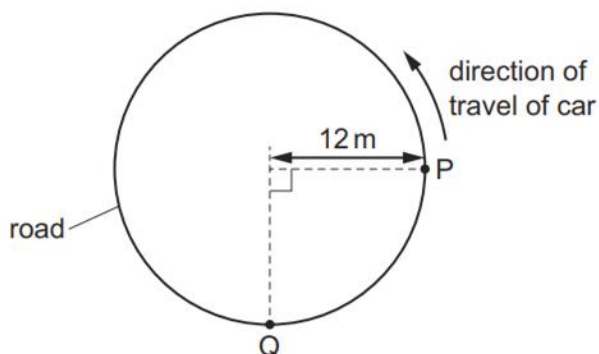
What is the speed of the ball just before it hits the ground?

- A** 10 m s^{-1} **B** 29 m s^{-1} **C** 31 m s^{-1} **D** 39 m s^{-1}

187. 9702/12/M/J/22/Q5

A car travels anticlockwise along a horizontal circular road of radius 12 m, as shown.

The car takes a time of 4.0 s to move from position P to position Q.



What is the magnitude of the average velocity of the car for the journey from P to Q?

- A** 4.2 ms^{-1} **B** 4.7 ms^{-1} **C** 6.0 ms^{-1} **D** 14 ms^{-1}

188. 9702/12/M/J/22/Q6

The water surface in a deep well is 78.0 m below the top of the well. A person at the top of the well drops a heavy stone down the well.

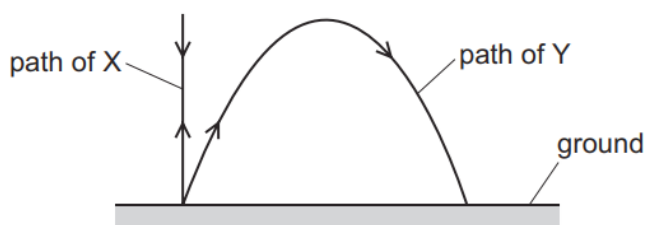
Air resistance is negligible. The speed of sound in the air is 330 ms^{-1} .

What is the time interval between the person dropping the stone and hearing it hitting the water?

- A** 3.75 s **B** 3.99 s **C** 4.19 s **D** 4.22 s

189. 9702/13/M/J/22/Q6

Two projectiles, X and Y, are fired into the air from the same place on level ground and reach the same maximum height, as shown.



Projectile X is fired vertically upwards and projectile Y is fired at an angle to the horizontal.

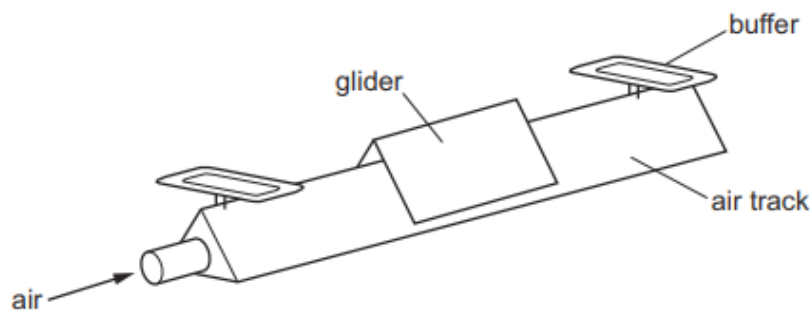
Air resistance is negligible.

Which statement is correct?

- A** X and Y are at rest at their maximum heights.
B X and Y are fired with the same speed.
C X and Y take the same time to return to the ground.
D X and Y travel the same distance.

190. 9702/13/M/J/22/Q7

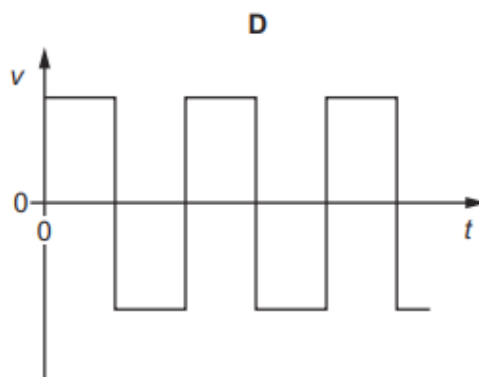
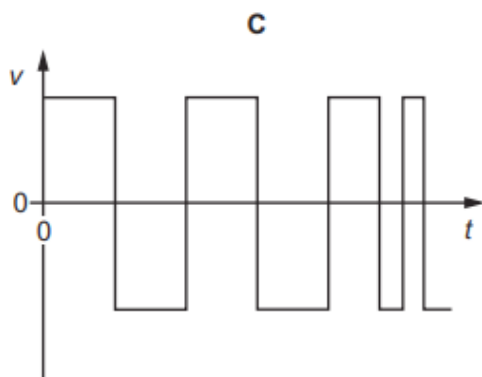
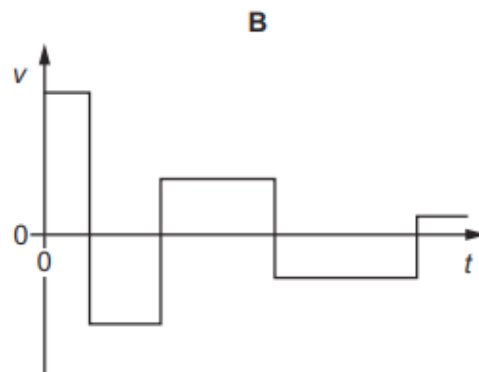
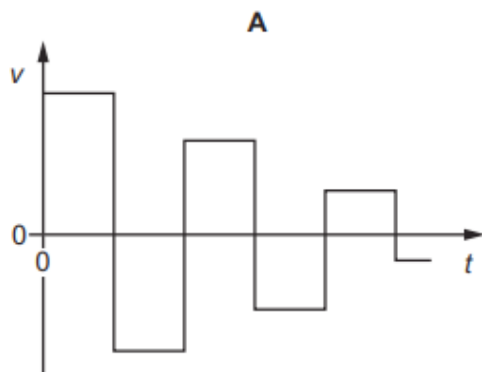
A small glider moves along a horizontal air track as shown.



At each end of the air track, the glider has a perfectly elastic collision with a fixed buffer.

The glider moves at a constant speed between collisions.

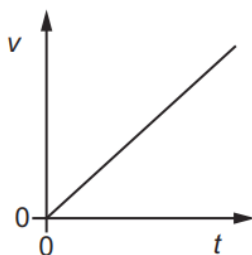
Which graph represents the variation with time t of the velocity v of the glider as it moves between the two buffers?



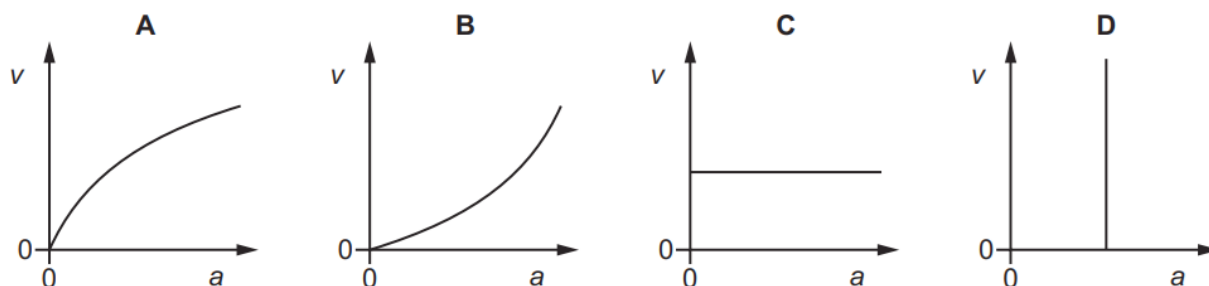
191. 9702/11/O/N/22/Q5

A particle accelerates from rest.

The graph shows the variation of the velocity v of the particle with time t .

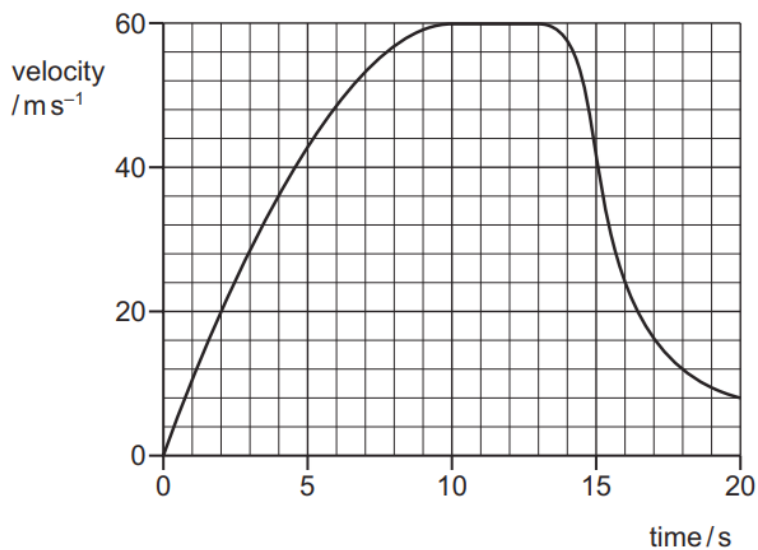


Which graph shows the variation of the velocity v with the acceleration a of the particle?



192. 9702/11/O/N/22/Q6

The graph shows the vertical velocity of a parachutist during the first 20 s of her jump.



Approximately how far does she fall **before** opening the parachute?

- A** 390 m **B** 570 m **C** 710 m **D** 770 m

193. 9702/12/O/N/22/Q5

A toy car travels on a circular track at a constant speed of 0.50 m s^{-1} . It passes a point on the track at time $t = 0$ and takes a time of 40 s to travel once around the track.

The magnitude of the average velocity of the car between $t = 0$ and $t = 20 \text{ s}$ is v_{20} .

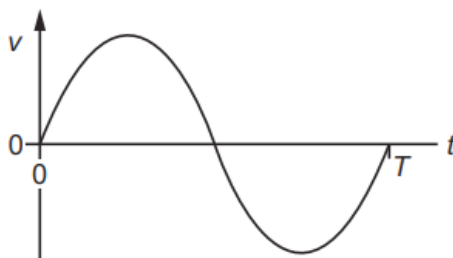
The magnitude of the average velocity of the car between $t = 0$ and $t = 40 \text{ s}$ is v_{40} .

What are v_{20} and v_{40} ?

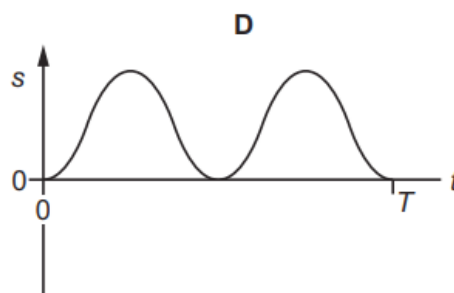
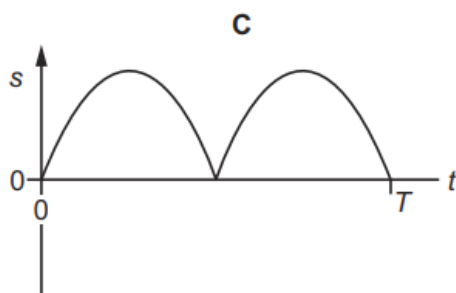
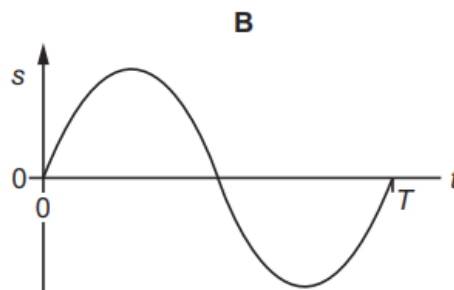
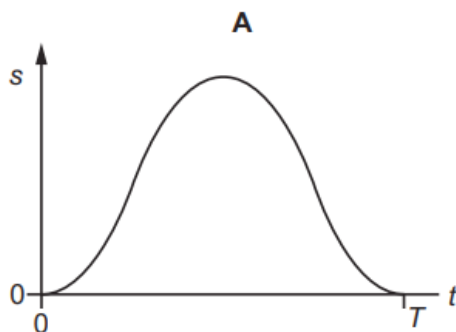
	$v_{20} / \text{m s}^{-1}$	$v_{40} / \text{m s}^{-1}$
A	0.32	0
B	0.32	0.32
C	0.50	0
D	0.50	0.50

194. 9702/12/O/N/22/Q6

The graph shows how the velocity v of an object moving in a straight line varies with time t from $t = 0$ to $t = T$.



Which graph could represent the displacement s of the object from time $t = 0$ to $t = T$?



195. 9702/12/O/N/22/Q7

A goods train passes through a station at a steady speed of 10 ms^{-1} at time $t = 0$. An express train is at rest at the station. The express train leaves the station with a uniform acceleration of 0.5 ms^{-2} just as the goods train goes past. Both trains move in the same direction on straight, parallel tracks.

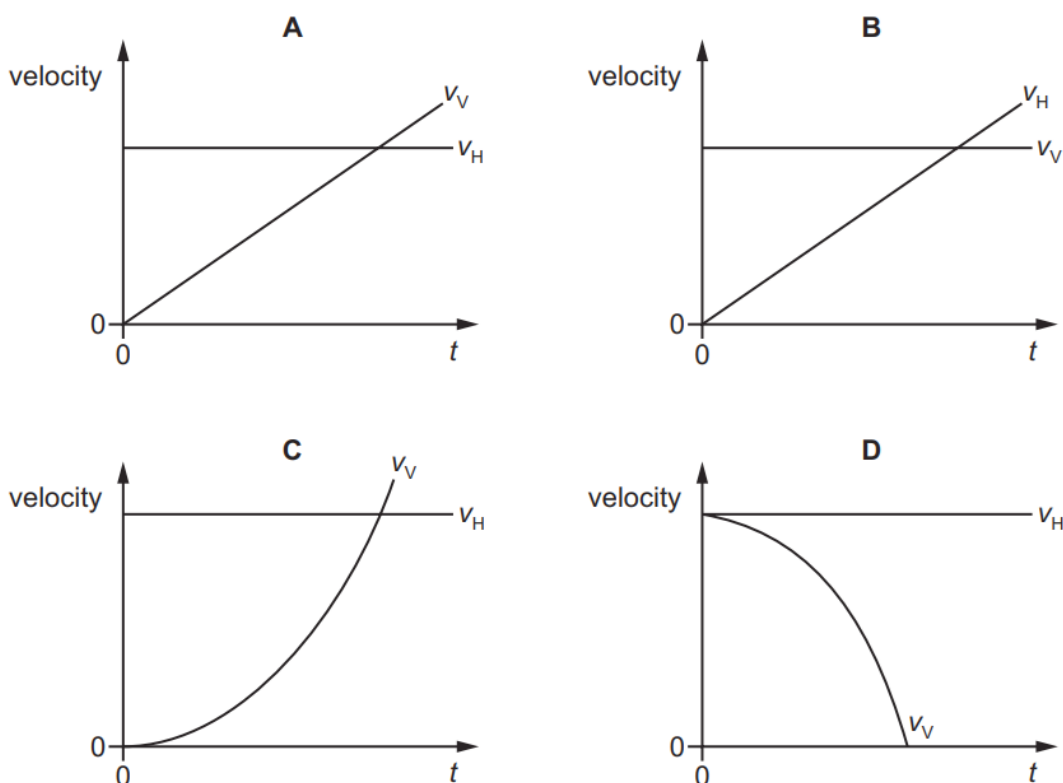
At which time t does the express train overtake the goods train?

- A** 6 s **B** 10 s **C** 20 s **D** 40 s

196. 9702/13/O/N/22/Q5

A stone is projected horizontally at time $t = 0$ and falls. Air resistance is negligible. The stone has a horizontal component of velocity v_H and a vertical component of velocity v_V .

Which graph shows how v_H and v_V vary with time t ?



197. 9702/13/O/N/22/Q6

On the Earth, an object takes time T_E to fall from rest through a vertical distance h .

On the Moon, the same object takes time T_M to fall from rest through the same vertical distance h .

The ratio $\frac{\text{acceleration of free fall on the Earth}}{\text{acceleration of free fall on the Moon}}$ is equal to 6.

Air resistance is negligible for the object on the Earth and on the Moon.

What is the ratio $\frac{T_E}{T_M}$?

- A** $\frac{1}{6}$ **B** $\frac{1}{\sqrt{6}}$ **C** $\sqrt{6}$ **D** 6

198. 9702/13/O/N/22/Q7

A cyclist in still air pedals as hard as she can. She reaches a maximum speed. However, after a certain time her maximum speed increases.

What could be a possible cause for this?

- A** She cycles into a wind.
- B** She cycles over rougher ground.
- C** She sits more upright on the bicycle.
- D** She starts to travel downhill.

199. 9702/12/F/M/23/Q5

A ball is kicked upwards at an angle of 45° to horizontal ground. After a short flight, the ball returns to the ground.

It may be assumed that air resistance is negligible.

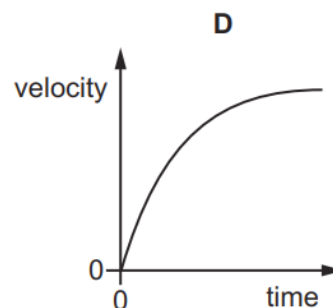
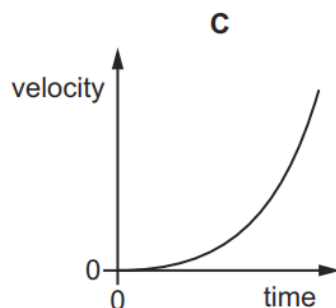
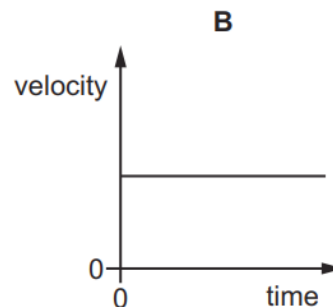
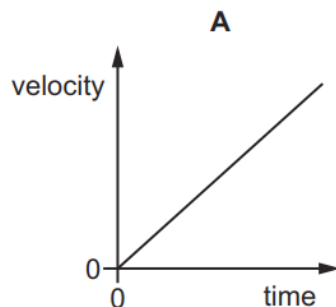
What is **never** zero during the flight of the ball?

- A** the horizontal component of the ball's acceleration
- B** the horizontal component of the ball's velocity
- C** the vertical component of the ball's momentum
- D** the vertical component of the ball's velocity

200. 9702/12/F/M/23/Q9

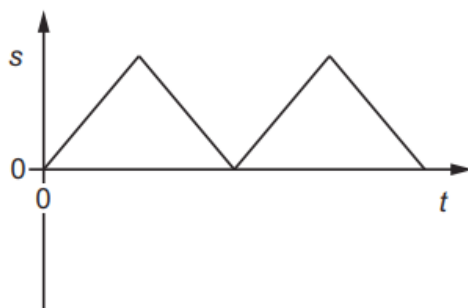
The velocity–time graphs of four different objects are shown.

Which graph represents an object falling from rest through a long distance in the Earth's atmosphere?

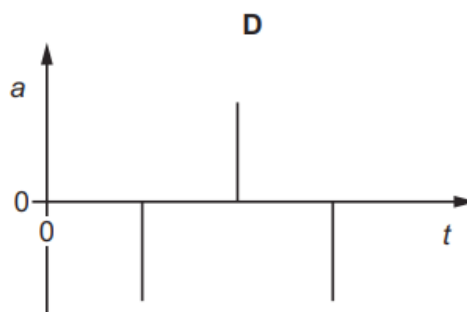
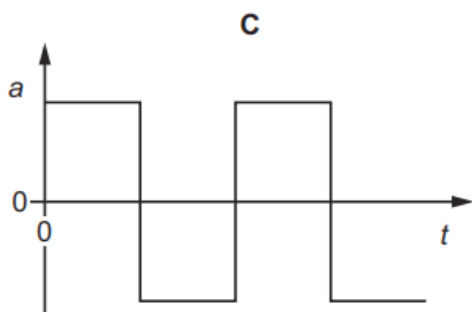
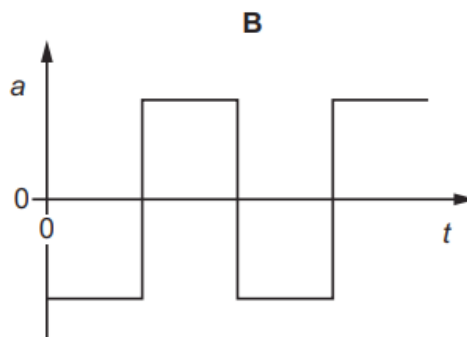
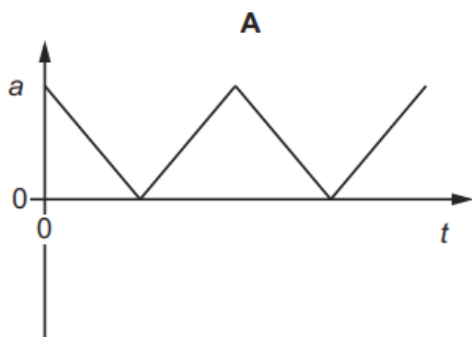


201. 9702/12/F/M/23/Q6

The graph shows the variation with time t of the displacement s of an object.



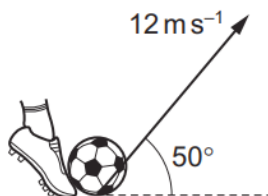
Which graph represents the variation with time t of the acceleration a of the object?



202. 9702/11/M/J/23/Q6

A ball is kicked so that it has an initial velocity of 12 m s^{-1} at an angle of 50° to horizontal ground.

Assume that air resistance is negligible.

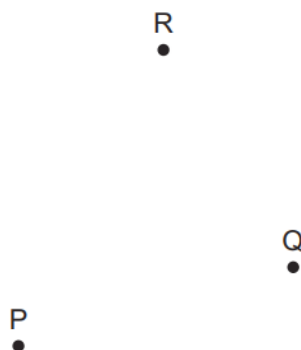


What is the maximum height above the ground that is reached by the ball?

- A** 0.47 m **B** 3.0 m **C** 4.3 m **D** 7.3 m

203. 9702/11/M/J/23/Q5

One object moves directly from P to R.



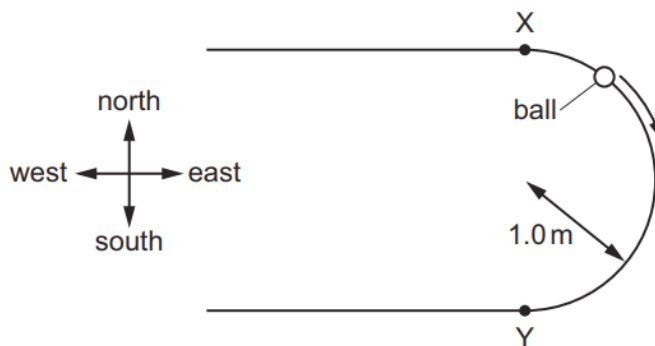
In a shorter time, a second object moves from P to Q to R.

Which statement about the two objects is correct for the journey from P to R?

- A** They have the same average speed.
- B** They have the same average velocity.
- C** They have the same displacement.
- D** They travel the same distance.

204. 9702/12/M/J/23/Q5

A ball travels from point X to point Y around a semicircular track of radius 1.0 m, as shown.



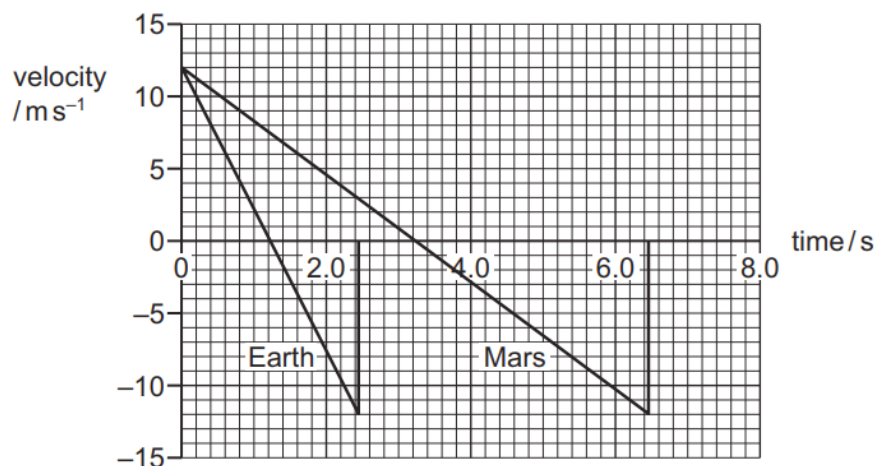
What is the displacement of the ball from X to Y?

- A** 2.0 m
- B** 2.0 m due south
- C** 3.1 m
- D** 3.1 m due south

205. 9702/12/M/J/23/Q6

A rock is launched vertically upwards from the surface of the Earth and an identical rock is launched vertically upwards from the surface of Mars. Each rock is launched with an initial velocity of 12 m s^{-1} .

Each rock then reaches its maximum height above the surface before returning back down to the surface. The velocity–time graph for each rock is shown. In both cases, air resistance is negligible.



What is the difference in the maximum heights of the two rocks?

- A** 12 m **B** 15 m **C** 19 m **D** 24 m

206. 9702/13/M/J/23/Q5

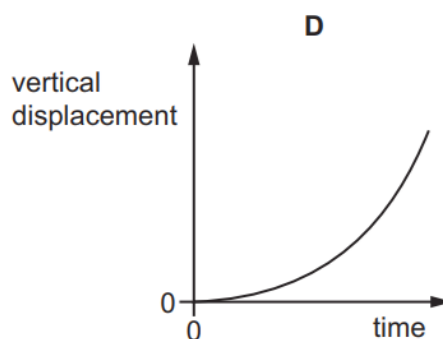
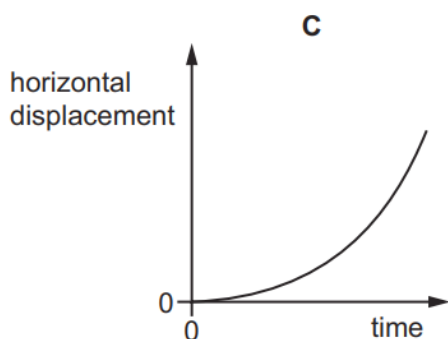
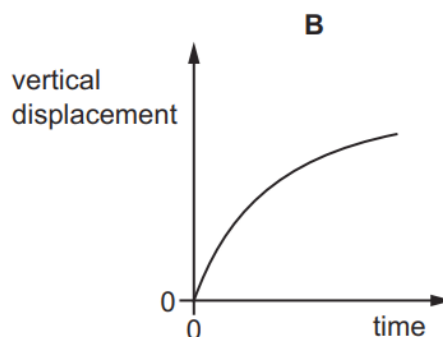
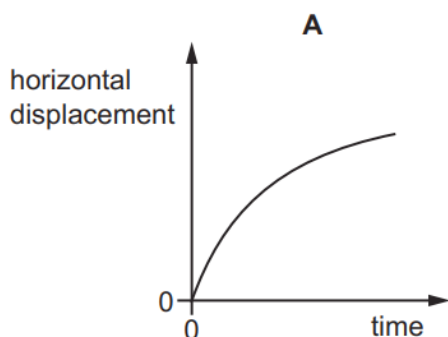
Which equation, representing uniformly accelerated motion in a straight line, can be determined using **only** the definition of acceleration?

- A** $s = ut + \frac{1}{2}at^2$
B $s = \frac{1}{2}(u + v)t$
C $v = u + at$
D $v^2 = u^2 + 2as$

207. 9702/13/M/J/23/Q6

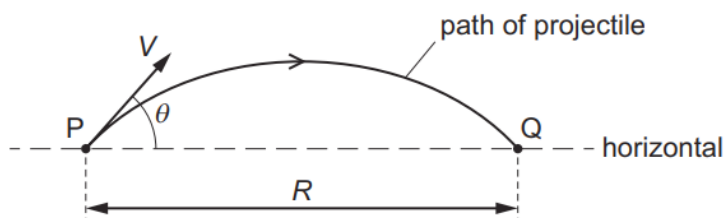
An object moves from rest with uniform velocity horizontally and uniform acceleration vertically.

Which graph showing the variation with time of the displacement of the object from its initial position is correct?



208. 9702/11/O/N/23/Q6

A projectile is fired from point P with velocity V at an angle θ to the horizontal. It lands at point Q, a horizontal distance R from P, after time T .



The acceleration of free fall is g . Air resistance is negligible.

Which equation is correct?

A $R = VT \cos \theta$

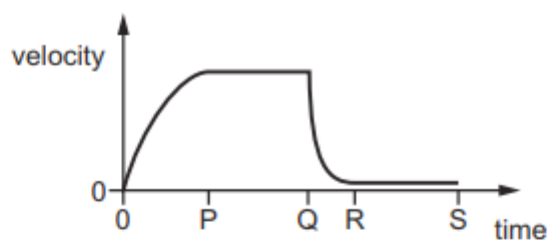
B $R = VT \sin \theta$

C $R = VT \cos \theta - \frac{1}{2} g T^2$

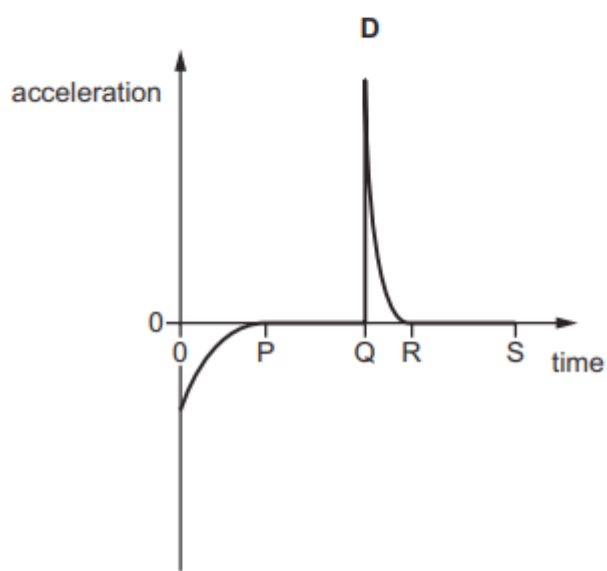
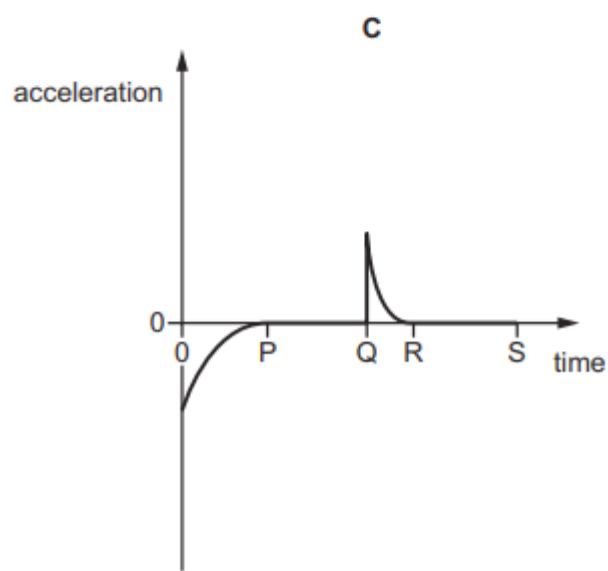
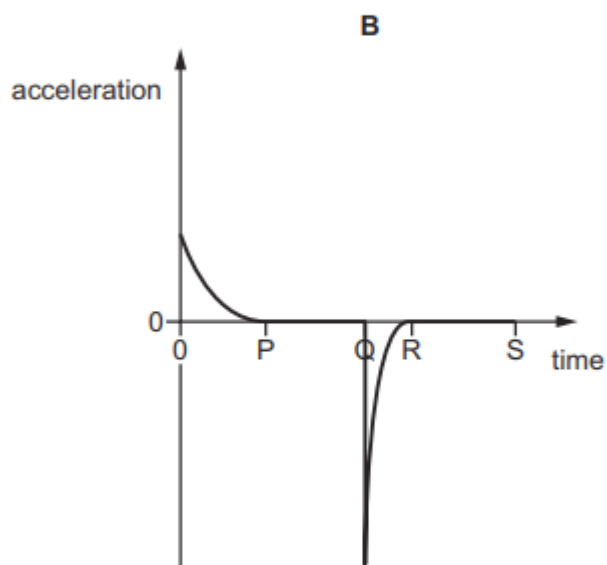
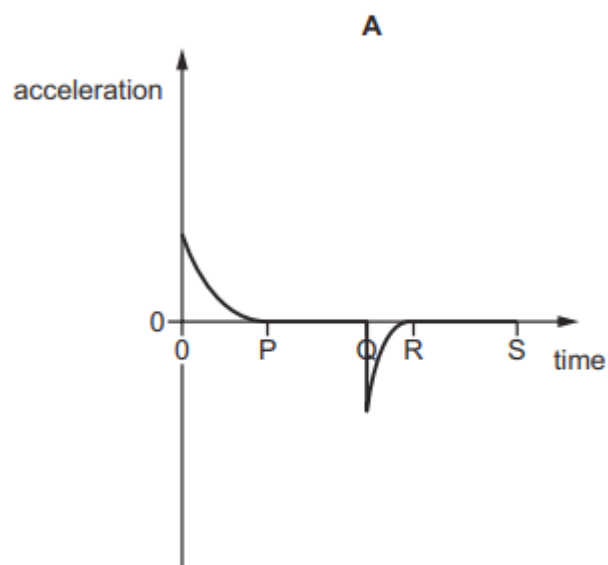
D $R = VT \sin \theta - \frac{1}{2} g T^2$

209. 9702/11/O/N/23/Q5

A parachutist falls from a stationary balloon at time $t = 0$. The velocity–time graph for the parachutist from time $t = 0$ until the time when he is just above the ground is shown.



Which graph best shows the variation with time of the acceleration of the parachutist?



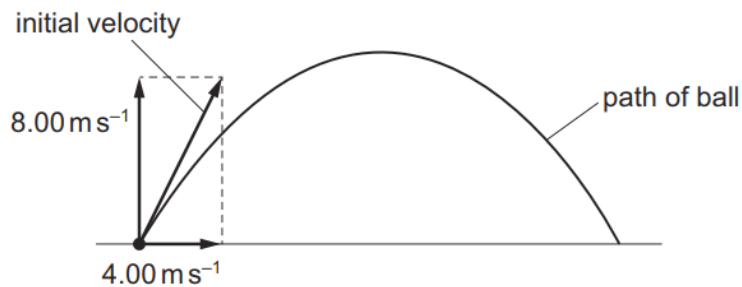
210. 9702/12/O/N/23/Q5

What is the definition of acceleration?

- A** the rate of change of displacement
- B** the rate of change of kinetic energy
- C** the rate of change of momentum
- D** the rate of change of velocity

211. 9702/12/O/N/23/Q6

An astronaut on the Moon, where there is no air resistance, throws a ball. The ball's initial velocity has a vertical component of 8.00 m s^{-1} and a horizontal component of 4.00 m s^{-1} , as shown.



The acceleration of free fall on the Moon is 1.62 m s^{-2} .

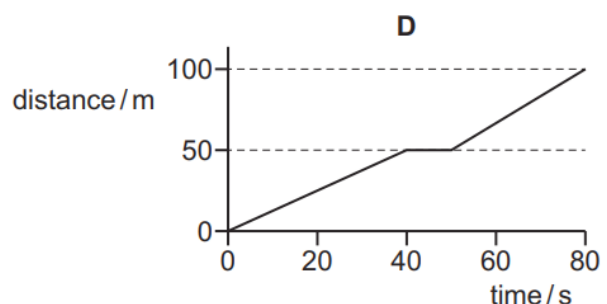
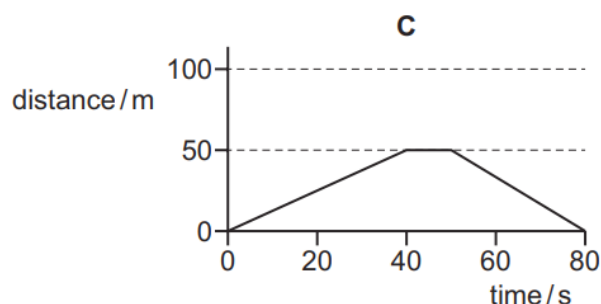
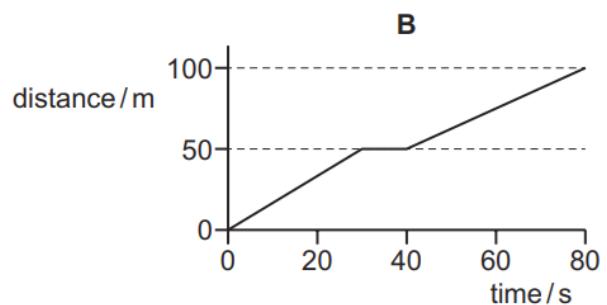
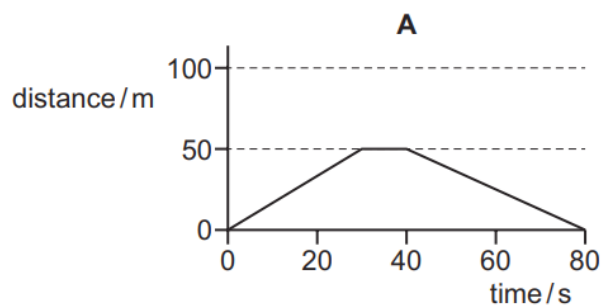
What is the speed of the ball 9.00 s after being thrown?

- A** 6.58 m s^{-1}
- B** 7.70 m s^{-1}
- C** 10.6 m s^{-1}
- D** 14.6 m s^{-1}

212. 9702/13/O/N/23/Q5

A student walks at a constant speed for a distance of 50 m in a time of 40 s . The student rests for a time of 10 s and then walks back to the starting point at a constant speed in a time of 30 s .

What is the distance–time graph for the motion of the student?



213. 9702/13/O/N/23/Q6

The time taken for an object to fall from rest through a certain distance on Mars is T_M . The time taken for the same object to fall from rest through the same distance on Earth is T_E . The acceleration of free fall on Mars is 3.71 m s^{-2} .

Assume that air resistance is negligible on both Earth and Mars.

What is the ratio $\frac{T_M}{T_E}$?

A 0.378

B 0.615

C 1.63

D 2.64

ANSWERS

1. C	50. C	99. D	148. B	195. D
2. A	51. A	100. D	149. D	196. A
3. C	52. C	101. C	150. B	197. B
4. B	53. B	102. A	151. D	198. D
5. B	54. C	103. B	152. A	199. B
6. A	55. B	104. A	153. D	200. D
7. A	56. A	105. D	154. C	201. D
8. D	57. D	106. B	155. D	202. C
9. C	58. D	107. C	156. A	203. C
10. B	59. A	108. A	157. C	204. B
11. C	60. D	109. C	158. B	205. A
12. D	61. A	110. C	159. B	206. C
13. D	62. B	111. D	160. A	207. D
14. C	63. D	112. D	161. C	208. A
15. C	64. A	113. D	162. B	209. B
16. B	65. D	114. D	163. D	210. D
17. D	66. D	115. D	164. B	211. B
18. B	67. D	116. C	165. A	212. D
19. A	68. A	117. B	166. A	213. C
20. D	69. B	118. A	167. C	
21. D	70. B	119. B	168. D	
22. C	71. C	120. B	169. C	
23. C	72. B	121. D	170. A	
24. D	73. D	122. D	171. C	
25. B	74. A	123. C	172. C	
26. D	75. D	124. A	173. D	
27. A	76. A	125. C	174. B	
28. C	77. D	126. A	175. D	
29. D	78. C	127. B	176. C	
30. A	79. D	128. A	177. B	
31. A	80. B	129. C	178. D	
32. D	81. C	130. C	179. C	
33. C	82. A	131. B	180. B	
34. C	83. D	132. B	181. B	
35. D	84. A	133. B	182. D	
36. D	85. D	134. A	183. D	
37. B	86. B	135. C	184. D	
38. C	87. D	136. B	185. C	
39. D	88. A	137. D	186. C	
40. B	89. B	138. C	187. A	
41. C	90. B	139. A	188. D	
42. C	91. B	140. A	189. C	
43. D	92. D	141. A	190. D	
44. A	93. C	142. B	191. D	
45. A	94. A	143. D	192. B	
46. D	95. C	144. B	193. A	
47. C	96. B	145. B	194. A	
48. A	97. B	146. C		
49. B	98. B	147. C		

9702/22/M/J/09/Q2

- 1 An experiment is conducted on the surface of the planet Mars. A sphere of mass 0.78 kg is projected almost vertically upwards from the surface of the planet. The variation with time t of the vertical velocity v in the upward direction is shown in Fig. 2.1.

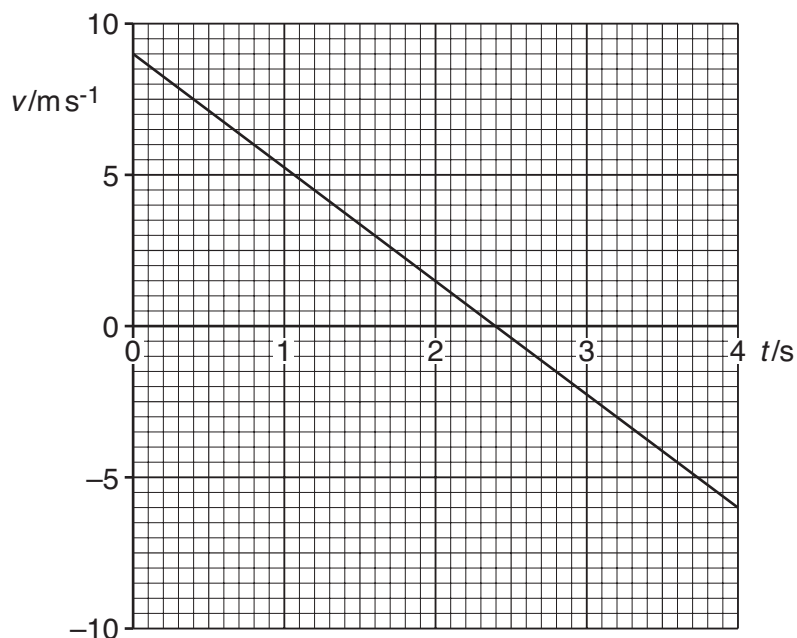


Fig. 2.1

The sphere lands on a small hill at time $t = 4.0$ s.

- (a) State the time t at which the sphere reaches its maximum height above the planet's surface.

$t = \dots\dots\dots$ s [1]

- (b) Determine the vertical height above the point of projection at which the sphere finally comes to rest on the hill.

height = $\dots\dots\dots$ m [3]

2 A sky-diver jumps from a high-altitude balloon.

9702/21/O/N/09/Q2

(a) Explain briefly why the acceleration of the sky-diver

(i) decreases with time,

.....

 [2]

(ii) is 9.8 ms^{-2} at the start of the jump.

.....
 [1]

(b) The variation with time t of the vertical speed v of the sky-diver is shown in Fig. 2.1.

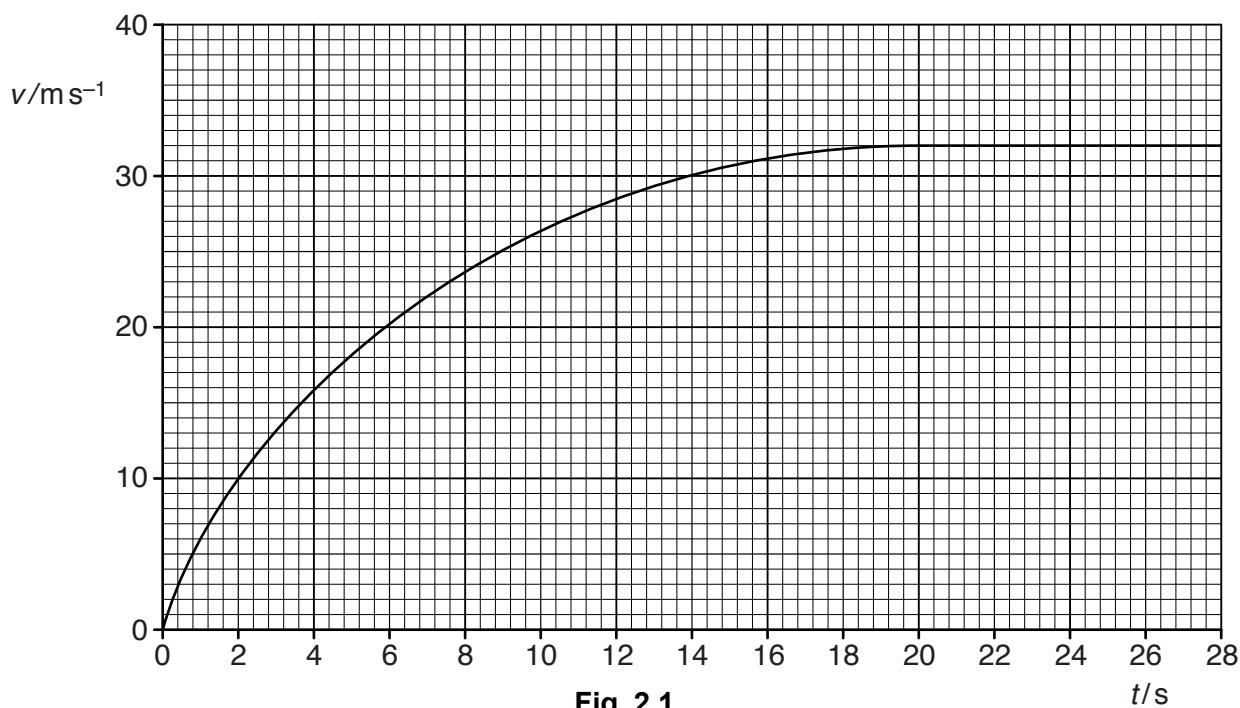


Fig. 2.1

Use Fig. 2.1 to determine the magnitude of the acceleration of the sky-diver at time $t = 6.0 \text{ s}$.

acceleration = m s^{-2} [3]

(c) The sky-diver and his equipment have a total mass of 90 kg.

(i) Calculate, for the sky-diver and his equipment,

1. the total weight,

weight = N [1]

2. the accelerating force at time $t = 6.0$ s.

force = N [1]

(ii) Use your answers in **(i)** to determine the total resistive force acting on the sky-diver at time $t = 6.0$ s.

force = N [1]

9702/22/O/N/09/Q3

- 3 A small ball is thrown horizontally with a speed of 4.0 m s^{-1} . It falls through a vertical height of 1.96 m before bouncing off a horizontal plate, as illustrated in Fig. 3.1.

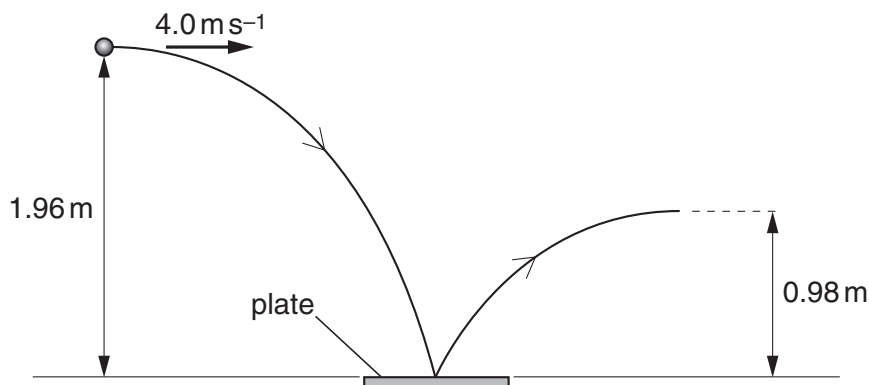


Fig. 3.1

Air resistance is negligible.

- (a) For the ball, as it hits the horizontal plate,
 (i) state the magnitude of the horizontal component of its velocity,

horizontal velocity = m s^{-1} [1]

- (ii) show that the vertical component of the velocity is 6.2 m s^{-1} .

[1]

- (b) The components of the velocity in (a) are both vectors.
 Complete Fig. 3.2 to draw a vector diagram, to scale, to determine the velocity of the ball as it hits the horizontal plate.

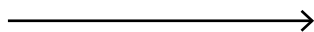


Fig. 3.2

velocity = m s^{-1}

at $^{\circ}$ to the vertical [3]

- (c) After bouncing on the plate, the ball rises to a vertical height of 0.98 m.
(i) Calculate the vertical component of the velocity of the ball as it leaves the plate.

vertical velocity = m s^{-1} [2]

- (ii) The ball of mass 34 g is in contact with the plate for a time of 0.12 s.
Use your answer in (c)(i) and the data in (a)(ii) to calculate, for the ball as it bounces on the plate,

1. the change in momentum,

change = kg m s^{-1} [3]

2. the magnitude of the average force exerted by the plate on the ball due to this momentum change.

force = N [2]

9702/21/M/J/10/Q2

- 4 (a) Complete Fig. 2.1 to show whether each of the quantities listed is a vector or a scalar.

	vector / scalar
distance moved	
speed	
acceleration	

Fig. 2.1

[3]

- (b) A ball falls vertically in air from rest. The variation with time t of the distance d moved by the ball is shown in Fig. 2.2.

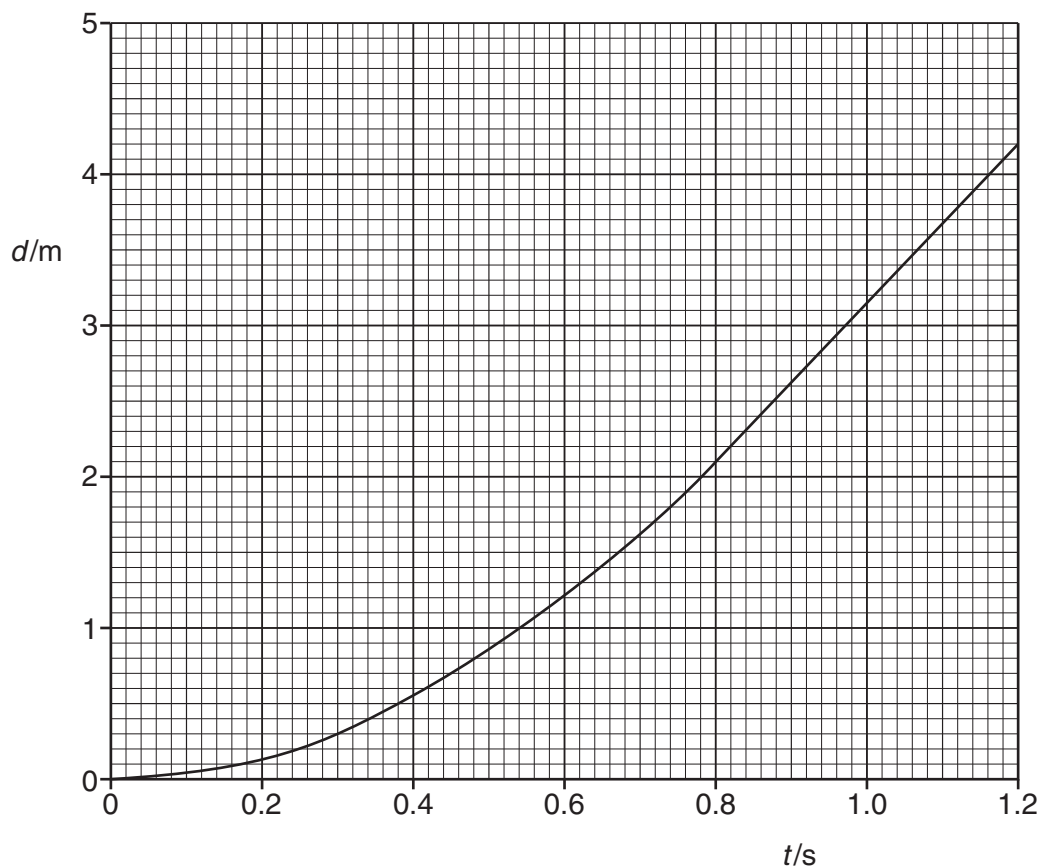


Fig. 2.2

- (i) By reference to Fig. 2.2, explain how it can be deduced that

1. the ball is initially at rest,

.....

 [2]

2. air resistance is not negligible.

.....
.....[1]

- (ii) Use Fig. 2.2 to determine the speed of the ball at a time of 0.40 s after it has been released.

speed = m s^{-1} [2]

- (iii) On Fig. 2.2, sketch a graph to show the variation with time t of the distance d moved by the ball for negligible air resistance. You are not expected to carry out any further calculations.

[3]

9702/22/M/J/10/Q2

- 5 (a)** The distance s moved by an object in time t may be given by the expression

$$s = \frac{1}{2}at^2$$

where a is the acceleration of the object.

State two conditions for this expression to apply to the motion of the object.

1.

.....

2.

.....

[2]

- (b)** A student takes a photograph of a steel ball of radius 5.0 cm as it falls from rest. The image of the ball is blurred, as illustrated in Fig. 2.1. The image is blurred because the ball is moving while the photograph is being taken.

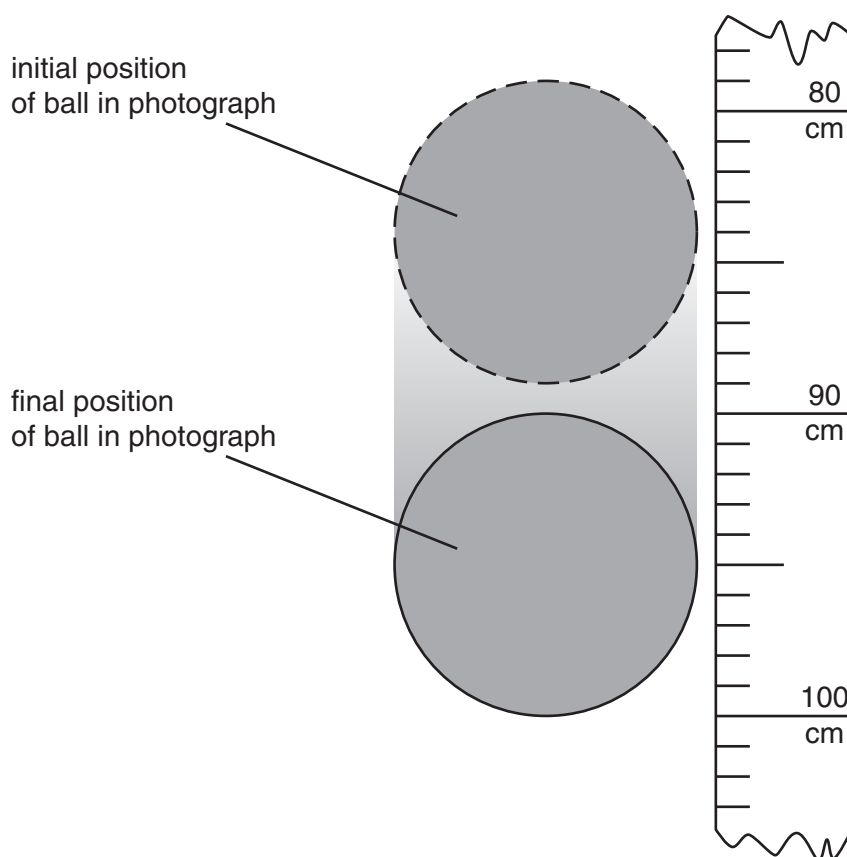


Fig. 2.1

The scale shows the distance fallen from rest by the ball. At time $t = 0$, the top of the ball is level with the zero mark on the scale. Air resistance is negligible.

Calculate, to an appropriate number of significant figures,

- (i) the time the ball falls before the photograph is taken,

time = s [3]

- (ii) the time interval during which the photograph is taken.

time interval = s [3]

- (c) The student in (b) takes a second photograph starting at the same position on the scale. The ball has the same radius but is less dense, so that air resistance is not negligible.

State and explain the changes that will occur in the photograph.

.....
.....
.....
..... [2]

9702/21/O/N/10/Q2

- 6 A ball is thrown horizontally from the top of a building, as shown in Fig. 2.1.

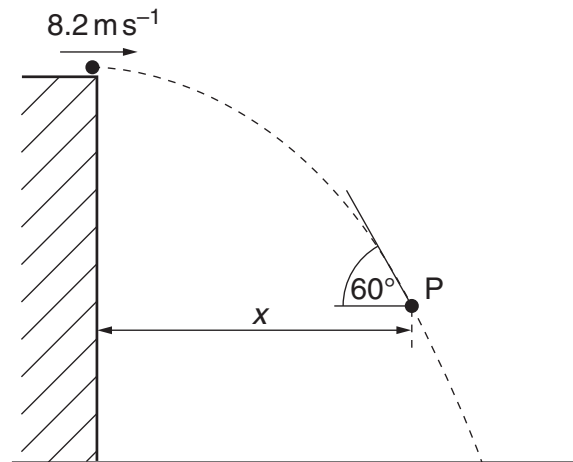


Fig. 2.1

The ball is thrown with a horizontal speed of 8.2 ms^{-1} . The side of the building is vertical. At point P on the path of the ball, the ball is distance x from the building and is moving at an angle of 60° to the horizontal. Air resistance is negligible.

- (a) For the ball at point P,

- (i) show that the vertical component of its velocity is 14.2 ms^{-1} ,

[2]

- (ii) determine the vertical distance through which the ball has fallen,

distance = m [2]

- (iii) determine the horizontal distance x .

$x = \dots\dots\dots$ m [2]

- (b) The path of the ball in (a), with an initial horizontal speed of 8.2 m s^{-1} , is shown again in Fig. 2.2.

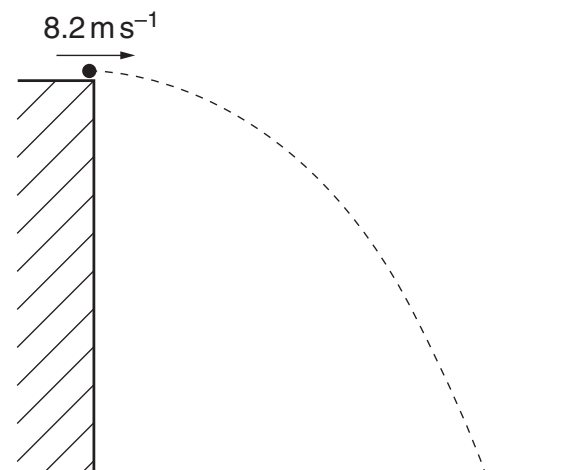


Fig. 2.2

On Fig. 2.2, sketch the new path of the ball for the ball having an initial horizontal speed

- (i) greater than 8.2 m s^{-1} and with negligible air resistance (label this path G), [2]
- (ii) equal to 8.2 m s^{-1} but with air resistance (label this path A). [2]
-

9702/22/O/N/10/Q2

- 7 A ball is thrown from a point P, which is at ground level, as illustrated in Fig. 2.1.

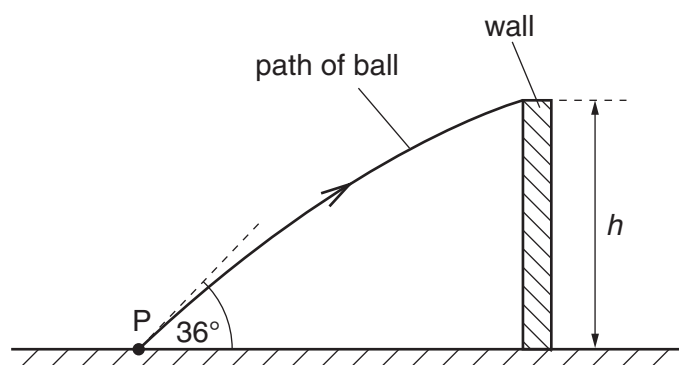


Fig. 2.1

The initial velocity of the ball is 12.4 m s^{-1} at an angle of 36° to the horizontal. The ball just passes over a wall of height h . The ball reaches the wall 0.17 s after it has been thrown.

- (a) Assuming air resistance to be negligible, calculate
- (i) the horizontal distance of point P from the wall,

distance = m [2]

- (ii) the height h of the wall.

$h = \dots\dots\dots \text{ m [3]}$

- (b) A second ball is thrown from point P with the same velocity as the ball in (a). For this ball, air resistance is not negligible. This ball hits the wall and rebounds.

On Fig. 2.1, sketch the path of this ball between point P and the point where it first hits the ground.

[2]

9702/22/M/J/11/Q1(c)

- 8 A stone is thrown with a horizontal velocity of 20 m s^{-1} from the top of a cliff 15 m high. The path of the stone is shown in Fig. 1.1.

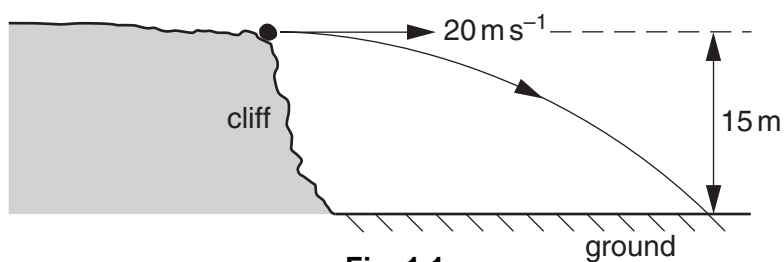


Fig. 1.1

Air resistance is negligible.

For this stone,

- (i) calculate the time to fall 15 m ,

time = s [2]

- (ii) calculate the magnitude of the resultant velocity after falling 15 m ,

resultant velocity = m s^{-1} [3]

9702/22/M/J/11/Q2

- 9 (a) A sphere of radius R is moving through a fluid with constant speed v . There is a frictional force F acting on the sphere, which is given by the expression

$$F = 6\pi DRv$$

where D depends on the fluid.

- (i) Show that the SI base units of the quantity D are $\text{kg m}^{-1} \text{s}^{-1}$.

[3]

- (ii) A raindrop of radius 1.5 mm falls vertically in air at a velocity of 3.7 m s^{-1} . The value of D for air is $6.6 \times 10^{-4} \text{ kg m}^{-1} \text{s}^{-1}$. The density of water is 1000 kg m^{-3} .

Calculate

1. the magnitude of the frictional force F ,

$$F = \dots\dots\dots \text{ N [1]}$$

2. the acceleration of the raindrop.

$$\text{acceleration} = \dots\dots\dots \text{ ms}^{-2} [3]$$

(b) The variation with time t of the speed v of the raindrop in (a) is shown in Fig. 2.1.

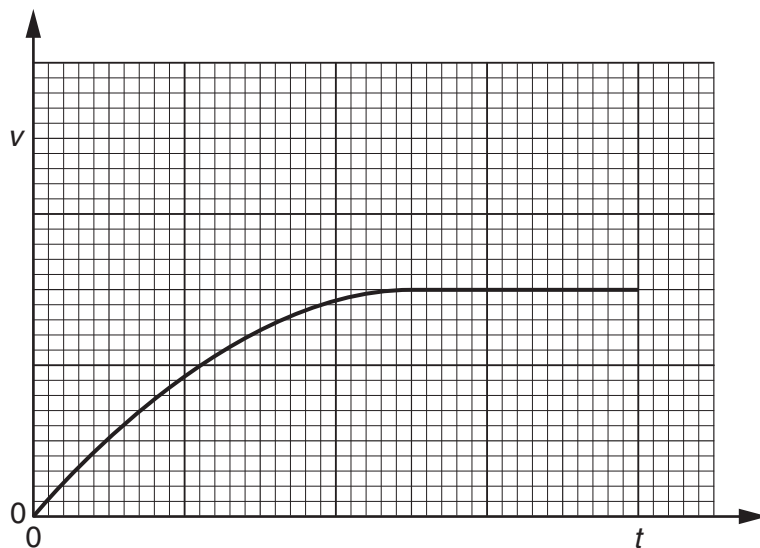


Fig. 2.1

(i) State the variation with time of the **acceleration** of the raindrop.

.....

.....

.....

.....

..... [3]

(ii) A second raindrop has a radius that is smaller than that given in (a). On Fig. 2.1, sketch the variation of speed with time for this second raindrop.

[2]

9702/21/O/N/11/Q3

- 10** A ball is thrown against a vertical wall. The path of the ball is shown in Fig. 3.1.

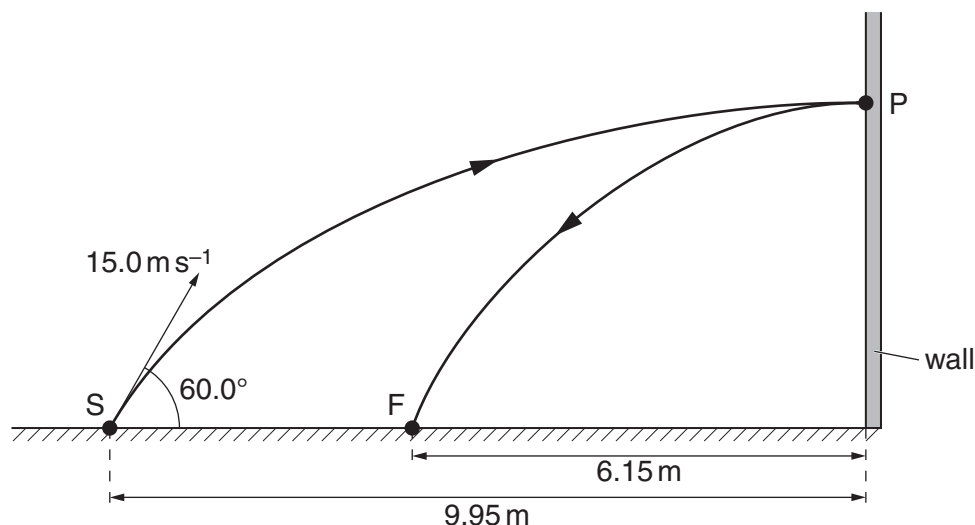


Fig. 3.1 (not to scale)

The ball is thrown from S with an initial velocity of 15.0 ms^{-1} at 60.0° to the horizontal. Assume that air resistance is negligible.

- (a)** For the ball at S, calculate

- (i)** its horizontal component of velocity,

horizontal component of velocity = ms^{-1} [1]

- (ii)** its vertical component of velocity.

vertical component of velocity = ms^{-1} [1]

- (b)** The horizontal distance from S to the wall is 9.95 m . The ball hits the wall at P with a velocity that is at right angles to the wall. The ball rebounds to a point F that is 6.15 m from the wall.

Using your answers in **(a)**,

- (i)** calculate the vertical height gained by the ball when it travels from S to P,

height = m [1]

- (ii) show that the time taken for the ball to travel from S to P is 1.33s,

[1]

- (iii) show that the velocity of the ball immediately after rebounding from the wall is about 4.6 m s^{-1} .

[1]

- (c) The mass of the ball is $60 \times 10^{-3} \text{ kg}$.

- (i) Calculate the change in momentum of the ball as it rebounds from the wall.

change in momentum = N s [2]

- (ii) State and explain whether the collision is elastic or inelastic.

.....

.....

..... [1]

9702/22/O/N/11/Q1

- 11** The variation with time t of the displacement s for a car is shown in Fig. 1.1.

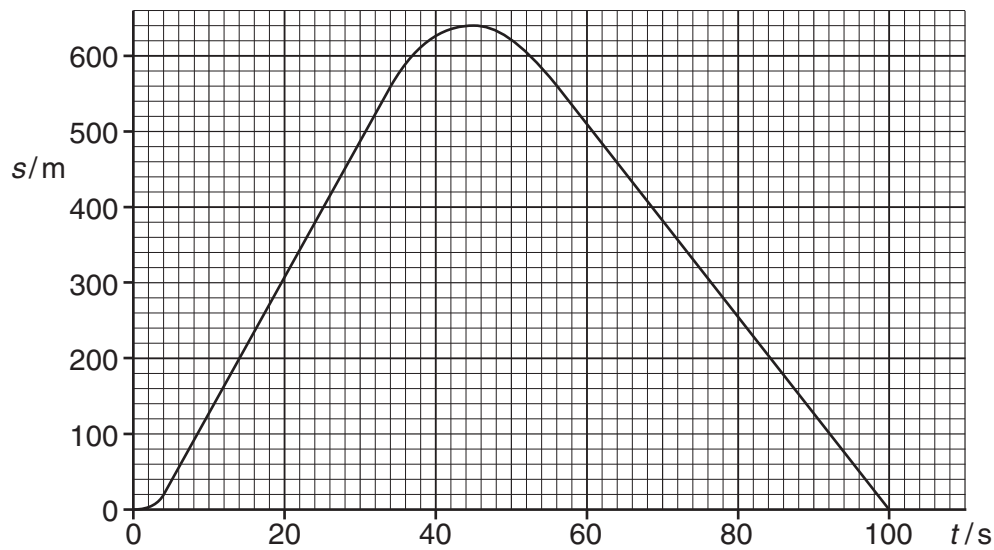


Fig. 1.1

- (a) Determine the magnitude of the average velocity between the times 5.0 s and 35.0 s.

average velocity = ms^{-1} [2]

- (b) On Fig. 1.2, sketch the variation with time t of the velocity v for the car.

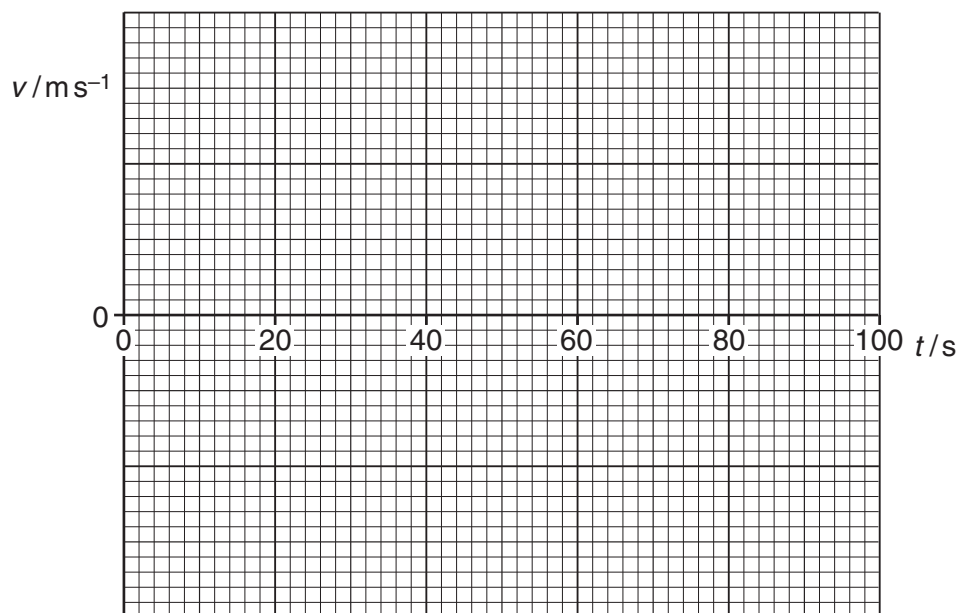


Fig. 1.2

[4]

9702/21/M/J/12/Q2

- 12** A ball is thrown vertically down towards the ground with an initial velocity of 4.23 m s^{-1} . The ball falls for a time of 1.51 s before hitting the ground. Air resistance is negligible.

(a) (i) Show that the downwards velocity of the ball when it hits the ground is 19.0 m s^{-1} .

[2]

(ii) Calculate, to three significant figures, the distance the ball falls to the ground.

distance = m [2]

- (b) The ball makes contact with the ground for 12.5 ms and rebounds with an upwards velocity of 18.6 m s^{-1} . The mass of the ball is 46.5 g.

(i) Calculate the average force acting on the ball on impact with the ground.

magnitude of force = N

direction of force

[4]

- (ii) Use conservation of energy to determine the maximum height the ball reaches after it hits the ground.

height = m [2]

- (c) State and explain whether the collision the ball makes with the ground is elastic or inelastic.

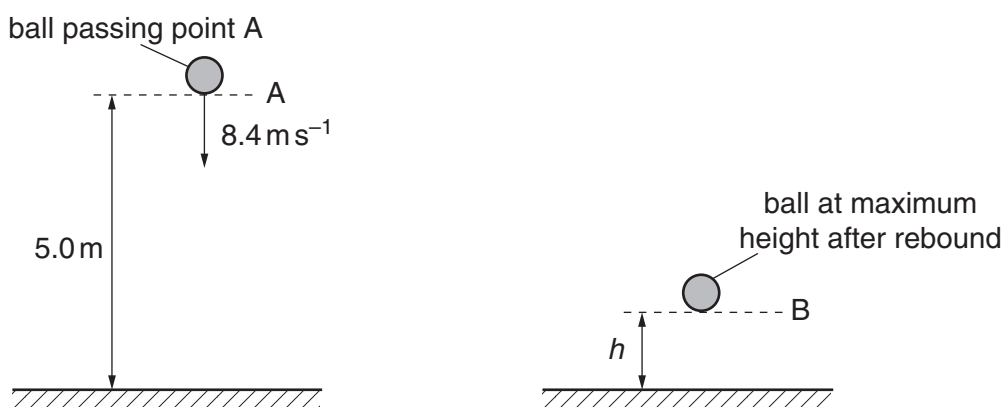
.....

.....

..... [1]

9702/22/M/J/12/Q2

- 13 (a)** A ball is thrown vertically down towards the ground and rebounds as illustrated in Fig. 2.1.

**Fig. 2.1**

As the ball passes A, it has a speed of 8.4 ms^{-1} . The height of A is 5.0 m above the ground. The ball hits the ground and rebounds to B. Assume that air resistance is negligible.

- (i) Calculate the speed of the ball as it hits the ground.

speed = ms^{-1} [2]

- (ii) Show that the time taken for the ball to reach the ground is 0.47 s .

[1]

- (b) The ball rebounds vertically with a speed of 4.2 ms^{-1} as it leaves the ground. The time the ball is in contact with the ground is 20 ms . The ball rebounds to a maximum height h .

The ball passes A at time $t = 0$. On Fig. 2.2, plot a graph to show the variation with time t of the velocity v of the ball. Continue the graph until the ball has rebounded from the ground and reaches B.

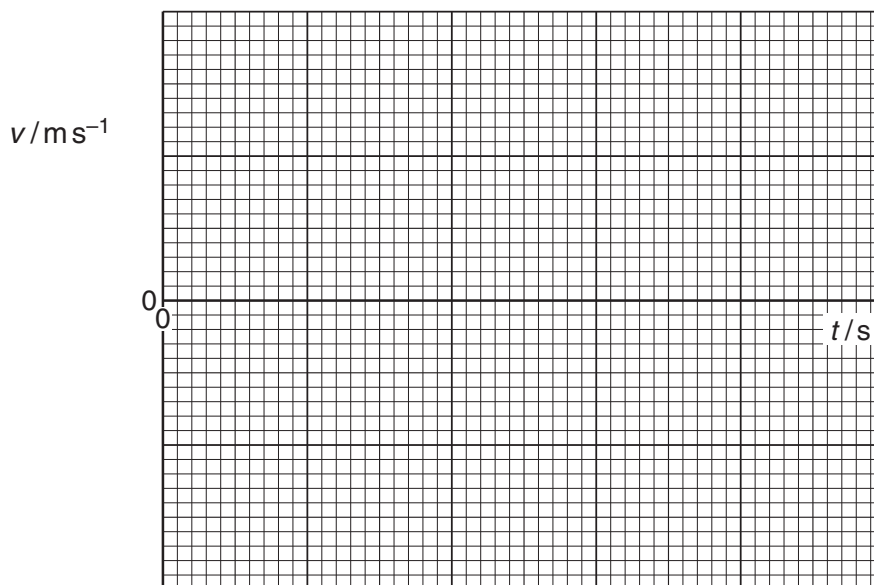


Fig. 2.2

[3]

- (c) The ball has a mass of 0.050 kg . It moves from A and reaches B after rebounding.

- (i) For this motion, calculate the change in

1. kinetic energy,

change in kinetic energy = J [2]

2. gravitational potential energy.

change in potential energy = J [3]

- (ii) State and explain the total change in energy of the ball for this motion.

.....

 [2]

9702/23/M/J/12/Q1

- 14 (a) Explain the differences between the quantities *distance* and *displacement*.

.....

[2]

- (b) State Newton's first law.

.....

[1]

- (c) Two tugs pull a tanker at constant velocity in the direction XY, as represented in Fig. 1.1.

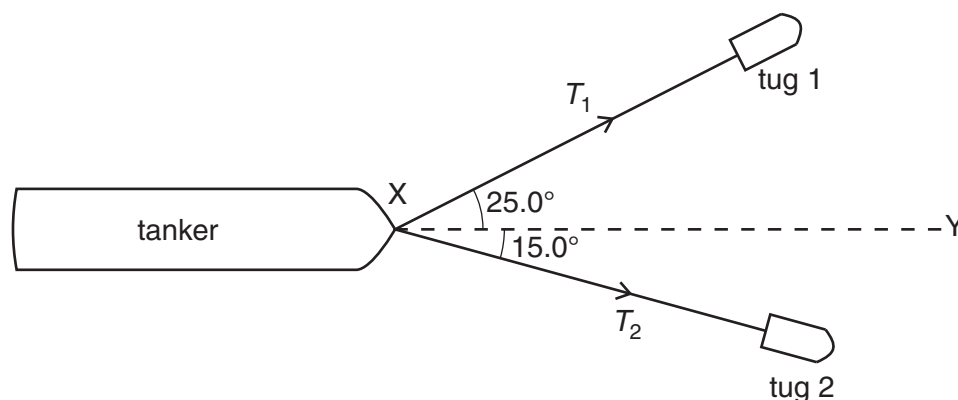


Fig. 1.1

Tug 1 pulls the tanker with a force T_1 at 25.0° to XY. Tug 2 pulls the tanker with a force of T_2 at 15.0° to XY. The resultant force R due to the two tugs is $25.0 \times 10^3 \text{ N}$ in the direction XY.

- (i) By reference to the forces acting on the tanker, explain how the tanker may be described as being in equilibrium.

.....

[2]

- (ii) 1. Complete Fig. 1.2 to draw a vector triangle for the forces R , T_1 and T_2 . [2]

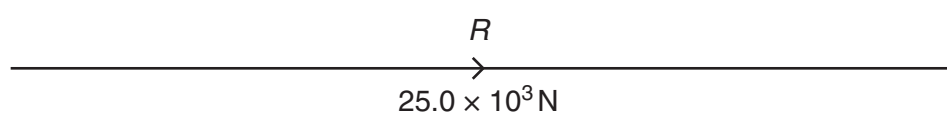


Fig. 1.2

2. Use your vector triangle in Fig. 1.2 to determine the magnitude of T_1 and of T_2 .

$T_1 = \dots\dots\dots$ N

$T_2 = \dots\dots\dots$ N

[2]

15 (a) (i) Define *acceleration*.

9702/21/O/N/12/Q1

.....
..... [1]

(ii) State Newton's first law of motion.

.....
..... [1]

(b) The variation with time t of vertical speed v of a parachutist falling from an aircraft is shown in Fig. 1.1.

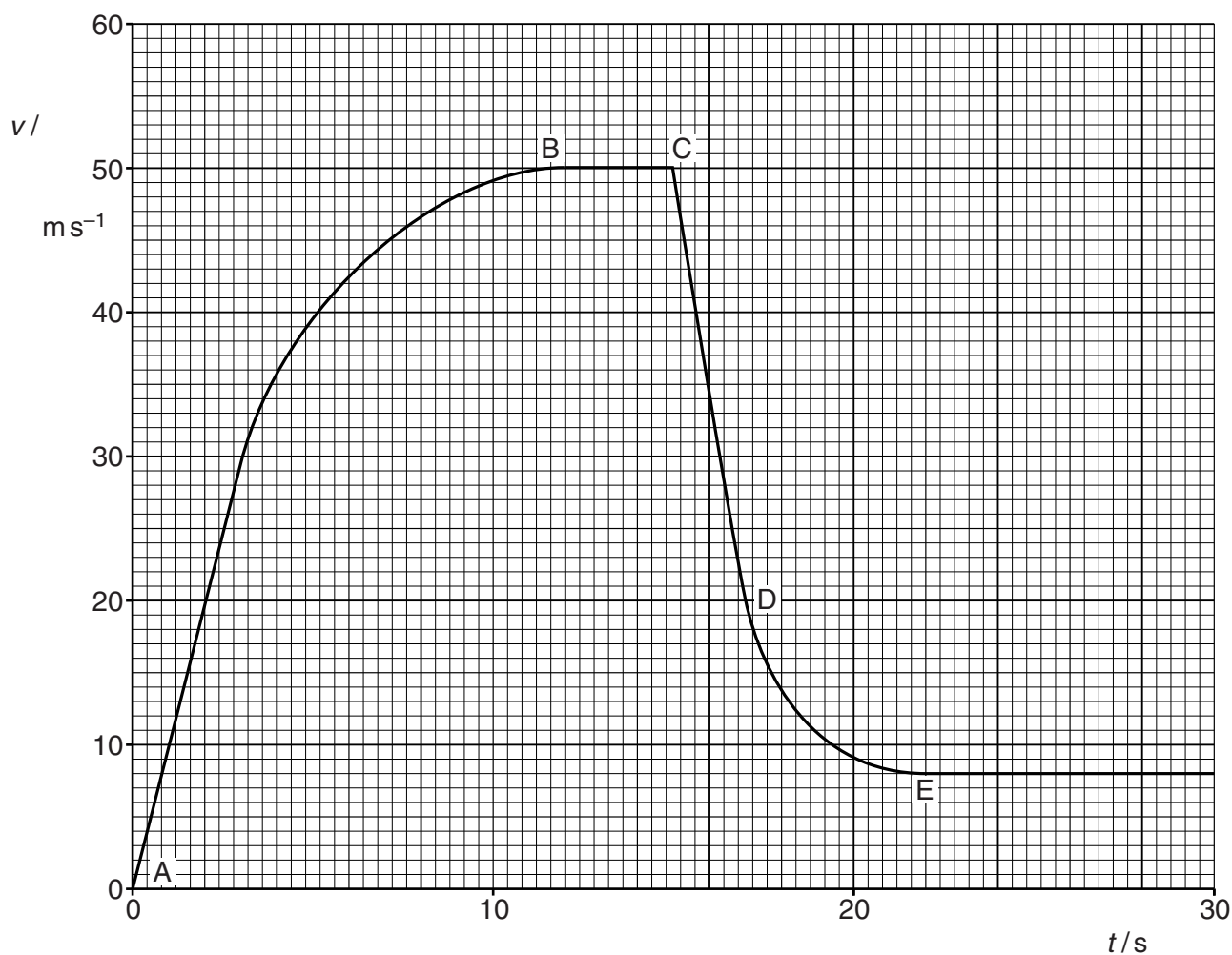


Fig. 1.1

- (i) Calculate the distance travelled by the parachutist in the first 3.0s of the motion.

distance = m [2]

- (ii) Explain the variation of the resultant force acting on the parachutist from $t = 0$ (point A) to $t = 15$ s (point C).

.....

 [3]

- (iii) Describe the changes to the frictional force on the parachutist

1. at $t = 15$ s (point C),

.....
 [1]

2. between $t = 15$ s (point C) and $t = 22$ s (point E).

.....
 [1]

- (iv) The mass of the parachutist is 95 kg.

Calculate, for the parachutist between $t = 15$ s (point C) and $t = 17$ s (point D),

1. the average acceleration,

acceleration = ms^{-2} [2]

2. the average frictional force.

frictional force = N [3]

9702/22/O/N/12/Q1

- 16 (a)** The drag force D on an object of cross-sectional area A , moving with a speed v through a fluid of density ρ , is given by

$$D = \frac{1}{2} C \rho A v^2$$

where C is a constant.

Show that C has no unit.

[2]

- (b)** A raindrop falls vertically from rest. Assume that air resistance is negligible.

- (i)** On Fig. 1.1, sketch a graph to show the variation with time t of the velocity v of the raindrop for the first 1.0 s of the motion.

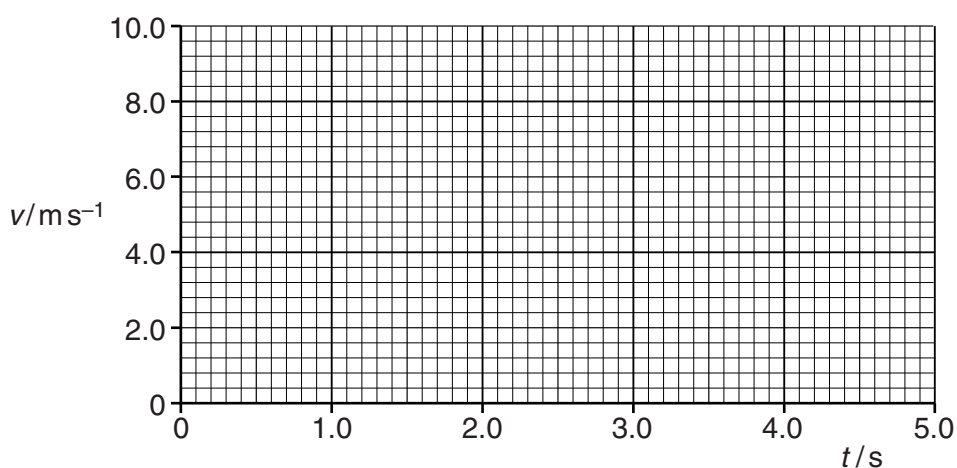


Fig. 1.1

[1]

- (ii)** Calculate the velocity of the raindrop after falling 1000 m.

velocity = ms^{-1} [2]

(c) In practice, air resistance on raindrops is not negligible because there is a drag force. This drag force is given by the expression in (a).

(i) State an equation relating the forces acting on the raindrop when it is falling at terminal velocity.

[1]

(ii) The raindrop has mass $1.4 \times 10^{-5} \text{ kg}$ and cross-sectional area $7.1 \times 10^{-6} \text{ m}^2$. The density of the air is 1.2 kg m^{-3} and the initial velocity of the raindrop is zero. The value of C is 0.60.

1. Show that the terminal velocity of the raindrop is about 7 m s^{-1} .

[2]

2. The raindrop reaches terminal velocity after falling approximately 10 m. On Fig. 1.1, sketch the variation with time t of velocity v for the raindrop. The sketch should include the first 5 s of the motion.

[2]

9702/23/O/N/12/Q2

- 17 Two planks of wood AB and BC are inclined at an angle of 15° to the horizontal. The two wooden planks are joined at point B, as shown in Fig. 2.1.

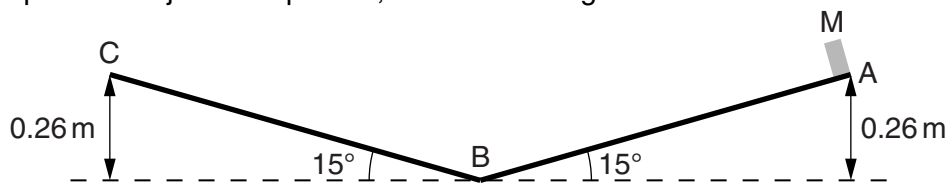


Fig. 2.1

A small block of metal M is released from rest at point A. It slides down the slope to B and up the opposite side to C. Points A and C are 0.26 m above B. Assume frictional forces are negligible.

- (a) (i) Describe and explain the acceleration of M as it travels from A to B and from B to C.

.....

 [3]

- (ii) Calculate the time taken for M to travel from A to B.

(iii) Calculate the speed of M at B. time = s [3]

speed = ms^{-1} [2]

- (b) The plank BC is adjusted so that the angle it makes with the horizontal is 30° . M is released from rest at point A and slides down the slope to B. It then slides a distance along the plank from B towards C.

Use the law of conservation of energy to calculate this distance. Explain your working.

distance = m [2]

- 18 (a)** A student walks from A to B along the path shown in Fig. 2.1.

9702/23/M/J/13/Q2

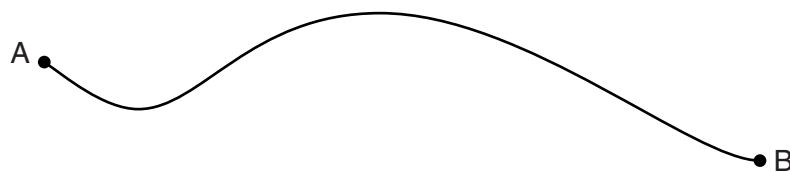


Fig. 2.1

The student takes time t to walk from A to B.

- (i)** State the quantity, apart from t , that must be measured in order to determine the average value of

- 1.** speed,

.....
..... [1]

- 2.** velocity.

.....
..... [1]

- (ii)** Define *acceleration*.

..... [1]

- (b) A girl falls vertically onto a trampoline, as shown in Fig. 2.2.

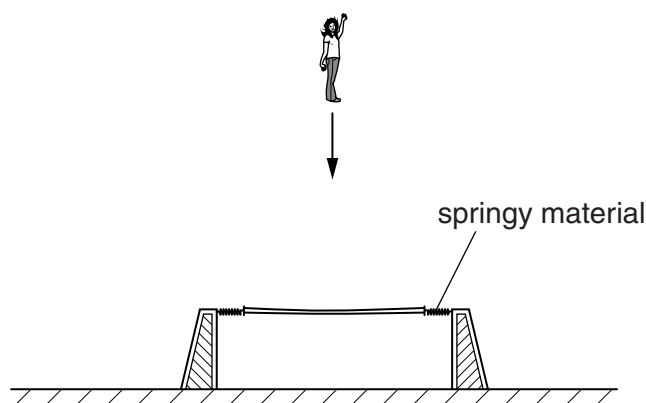


Fig. 2.2

The trampoline consists of a central section supported by springy material. At time $t = 0$ the girl starts to fall. The girl hits the trampoline and rebounds vertically. The variation with time t of velocity v of the girl is illustrated in Fig. 2.3.

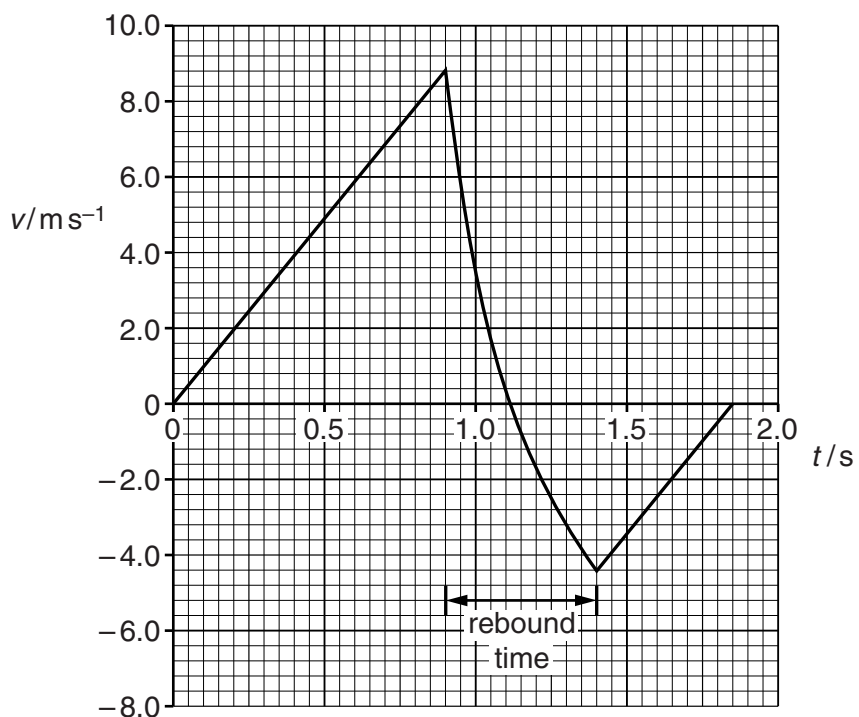


Fig. 2.3

For the motion of the girl, calculate

- (i) the distance fallen between time $t = 0$ and when she hits the trampoline,

distance = m [2]

- (ii) the average acceleration during the rebound.

acceleration = ms^{-2} [2]

- (c) (i) Use Fig. 2.3 to compare, without calculation, the accelerations of the girl before and after the rebound. Explain your answer.

.....
.....
.....[2]

- (ii) Use Fig. 2.3 to compare, without calculation, the potential energy of the girl at $t = 0$ and $t = 1.85\text{ s}$. Explain your answer.

.....
.....
.....[2]

9702/23/O/N/13/Q3

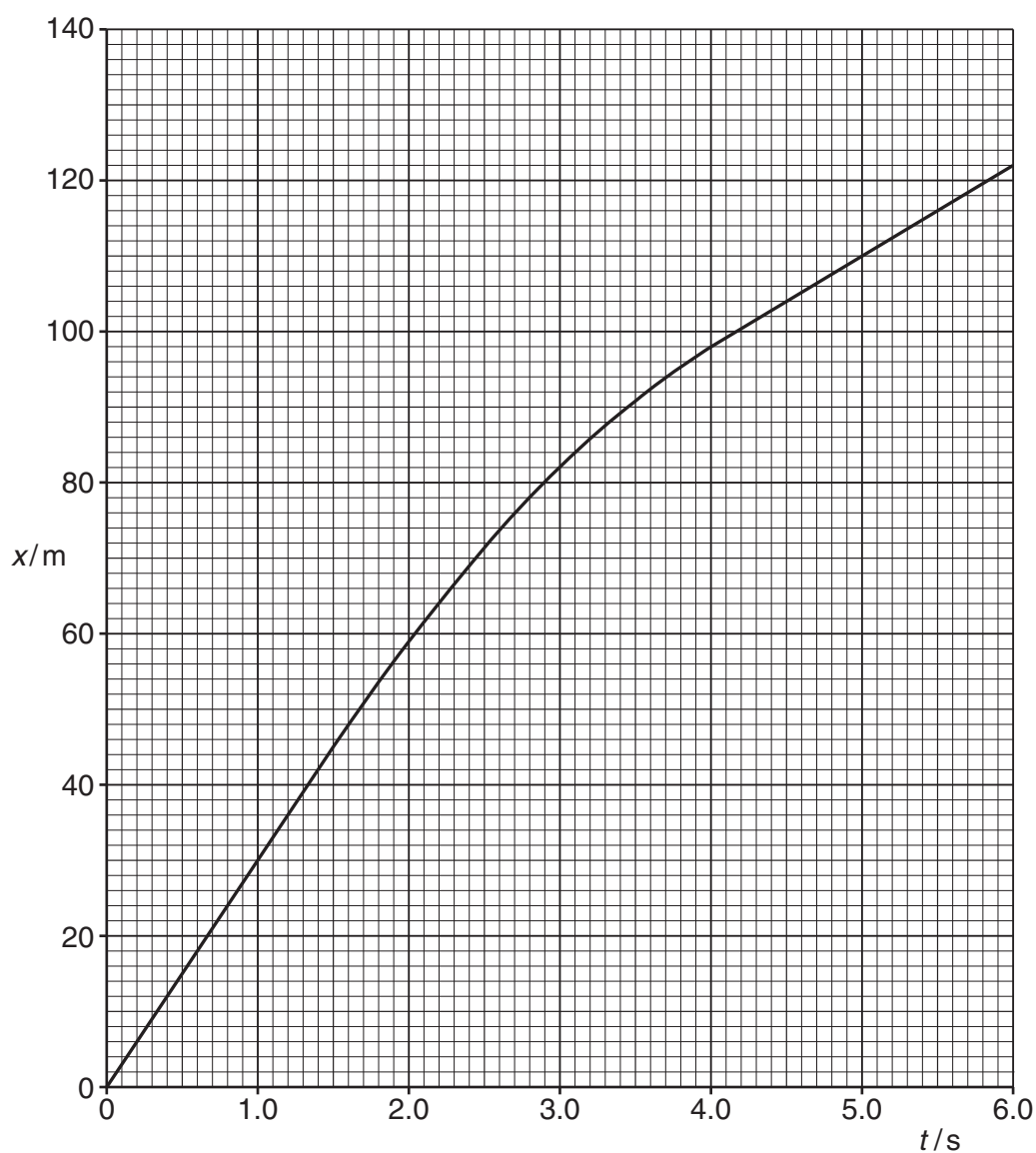
19 (a) Define(i) *velocity*,

.....
..... [1]

(ii) *acceleration*.

.....
..... [1]

(b) A car of mass 1500 kg travels along a straight horizontal road.
The variation with time t of the displacement x of the car is shown in Fig. 3.1.

**Fig. 3.1**

- (i) Use Fig. 3.1 to describe qualitatively the velocity of the car during the first six seconds of the motion shown.
Give reasons for your answers.

.....
.....
.....
.....
.....[3]

- (ii) Calculate the average velocity during the time interval $t = 0$ to $t = 1.5$ s.

average velocity = ms^{-1} [1]

- (iii) Show that the average acceleration between $t = 1.5$ s and $t = 4.0$ s is -7.2 ms^{-2} .

[2]

- (iv) Calculate the average force acting on the car between $t = 1.5$ s and $t = 4.0$ s.

force = N [2]

9702/21/M/J/14/Q1

20 (a) (i) Define *velocity*.

.....
[1]

(ii) Distinguish between *speed* and *velocity*.

.....
[2]

(b) A car of mass 1500 kg moves along a straight, horizontal road. The variation with time t of the velocity v for the car is shown in Fig. 1.1.

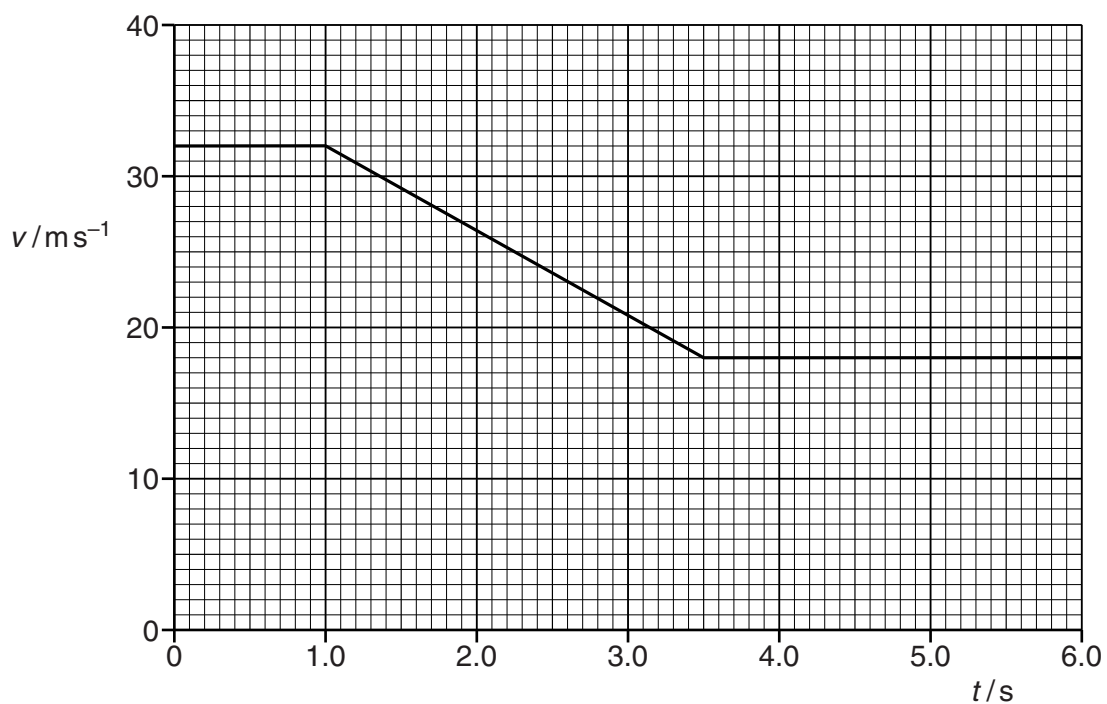


Fig. 1.1

The brakes of the car are applied from $t = 1.0$ s to $t = 3.5$ s.
 For the time when the brakes are applied,

(i) calculate the distance moved by the car,

distance = m [3]

- (ii) calculate the magnitude of the resultant force on the car.

resultant force = N [3]

- (c) The direction of motion of the car in (b) at time $t = 2.0$ s is shown in Fig. 1.2.

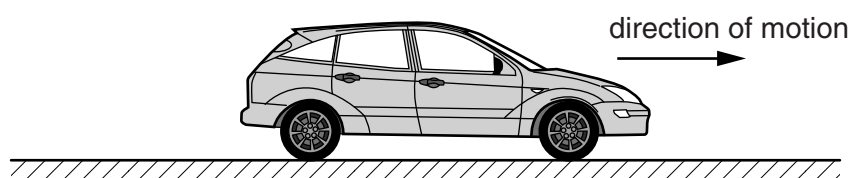


Fig. 1.2

On Fig. 1.2, show with arrows the directions of the acceleration (label this arrow A) and the resultant force (label this arrow F).

[1]

9702/23/M/J/14/Q2

- 21 (a)** Explain what is meant by a *scalar* quantity and by a *vector* quantity.

scalar:

.....

vector:

.....

[2]

- (b)** A ball leaves point P at the top of a cliff with a horizontal velocity of 15 ms^{-1} , as shown in Fig. 2.1.

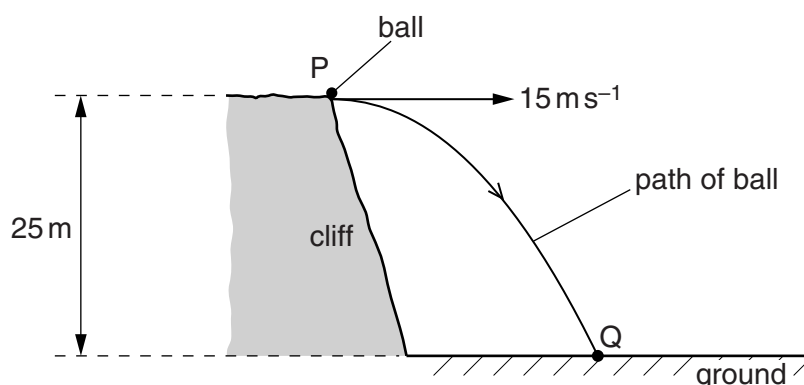


Fig. 2.1

The height of the cliff is 25 m. The ball hits the ground at point Q.
Air resistance is negligible.

- (i)** Calculate the vertical velocity of the ball just before it makes impact with the ground at Q.

vertical velocity = ms^{-1} [2]

- (ii)** Show that the time taken for the ball to fall to the ground is 2.3 s.

[1]

- (iii)** Calculate the magnitude of the displacement of the ball at point Q from point P.

displacement = m [4]

- (iv)** Explain why the distance travelled by the ball is different from the magnitude of the displacement of the ball.

.....

.....

.....[2]

9702/21/O/N/14/Q4

- 22** A trolley moves down a slope, as shown in Fig. 4.1.

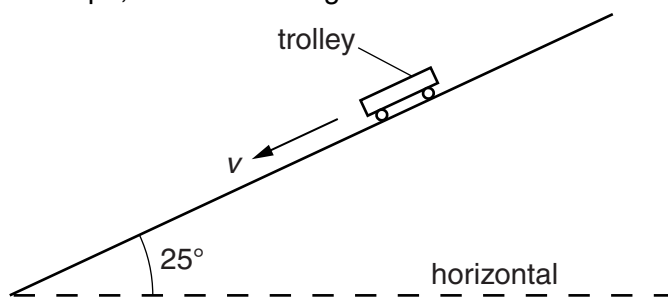


Fig. 4.1

The slope makes an angle of 25° with the horizontal. A constant resistive force F_R acts up the slope on the trolley.

At time $t = 0$, the trolley has velocity $v = 0.50 \text{ m s}^{-1}$ down the slope.

At time $t = 4.0 \text{ s}$, $v = 12 \text{ m s}^{-1}$ down the slope.

- (a) (i)** Show that the acceleration of the trolley down the slope is approximately 3 m s^{-2} .

[2]

- (ii)** Calculate the distance x moved by the trolley down the slope from time $t = 0$ to $t = 4.0 \text{ s}$

$x = \dots\dots\dots \text{ m}$ [2]

- (iii)** On Fig. 4.2, sketch the variation with time t of distance x moved by the trolley.

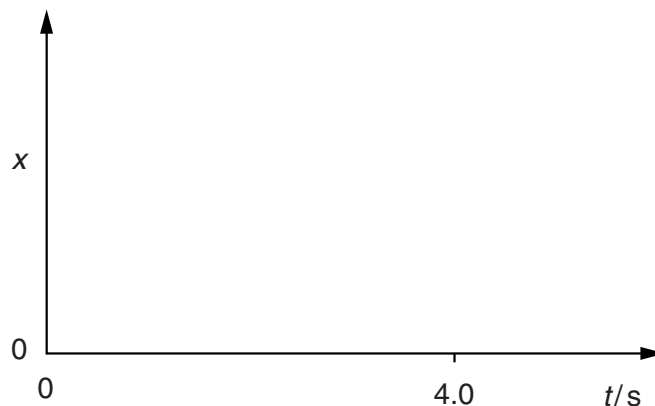


Fig. 4.2

[2]

- (b)** The mass of the trolley is 2.0 kg .

- (i)** Show that the component of the weight of the trolley down the slope is 8.3 N .

- (ii)** Calculate the resistive force F_R .

[1]

$F_R = \dots\dots\dots \text{ N}$ [2]

9702/22/O/N/14/Q2

- 23 A ball is thrown from A to B as shown in Fig. 2.1.

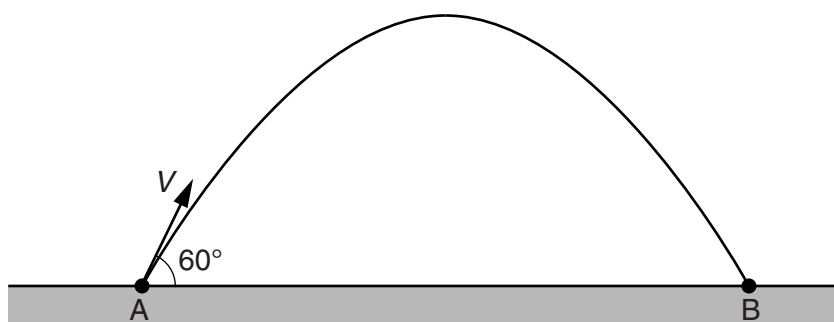


Fig. 2.1

The ball is thrown with an initial velocity V at 60° to the horizontal.

The variation with time t of the vertical component V_v of the velocity of the ball from $t = 0$ to $t = 0.60$ s is shown in Fig. 2.2.

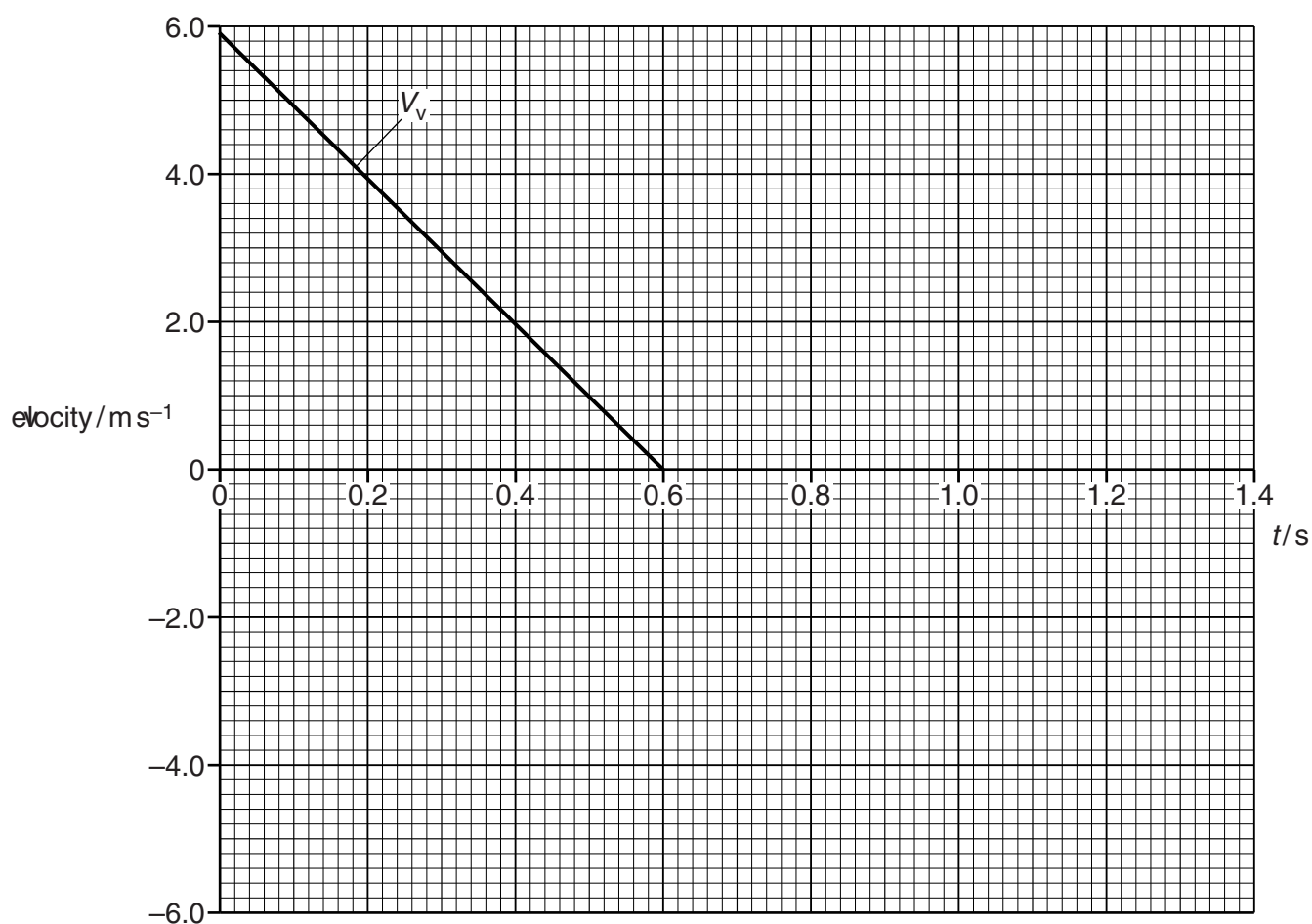


Fig. 2.2

Assume air resistance is negligible.

- (a) (i) Complete Fig. 2.2 for the time until the ball reaches B.

[2]

- (ii) Calculate the maximum height reached by the ball.

height =m [2]

- (iii) Calculate the horizontal component V_h of the velocity of the ball at time $t = 0$.

$V_h = \dots\dots\dots \text{ms}^{-1}$ [2]

- (iv) On Fig. 2.2, sketch the variation with t of V_h . **Label** this sketch V_h . [1]

- (b) The ball has mass 0.65 kg.

Calculate, for the ball,

- (i) the maximum kinetic energy,

maximum kinetic energy =J [3]

- (ii) the maximum potential energy above the ground.

maximum potential energy =J [2]

9702/21/M/J/15/Q2

- 24 (a)** Define *speed* and *velocity* and use these definitions to explain why one of these quantities is a scalar and the other is a vector.

speed:

velocity:

.....

.....[2]

- (b)** A ball is released from rest and falls vertically. The ball hits the ground and rebounds vertically, as shown in Fig. 2.1.

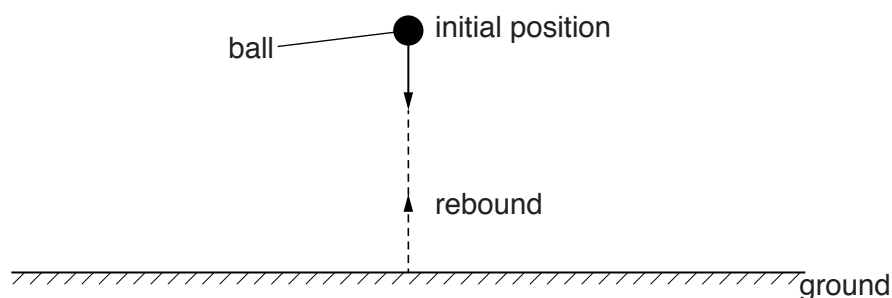


Fig. 2.1

The variation with time t of the velocity v of the ball is shown in Fig. 2.2.

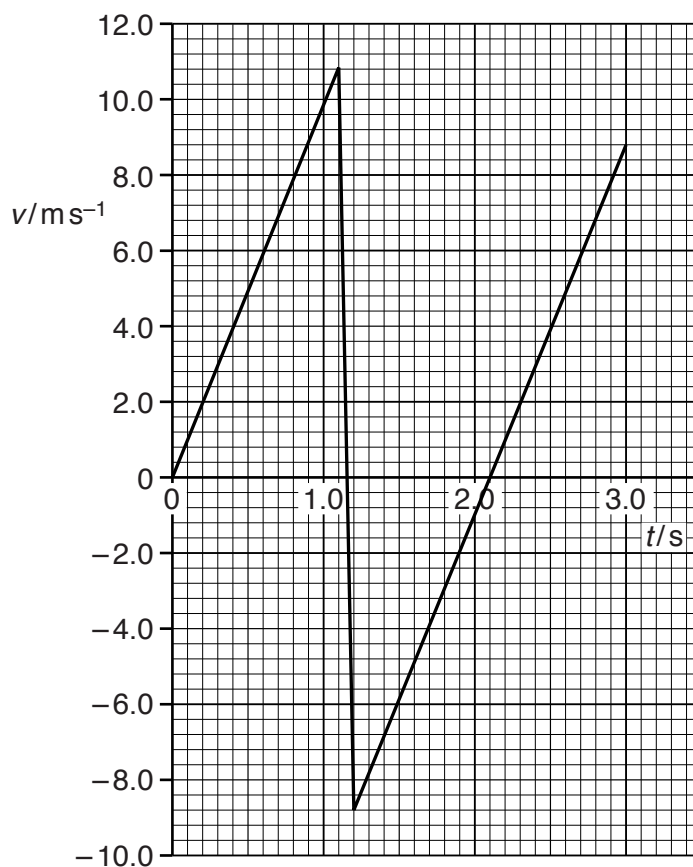


Fig. 2.2

Air resistance is negligible.

- (i) Without calculation, use Fig. 2.2 to describe the variation with time t of the velocity of the ball from $t = 0$ to $t = 2.1$ s.

.....

.....

.....

.....

.....[3]

- (ii) Calculate the acceleration of the ball after it rebounds from the ground. Show your working.

acceleration = ms^{-2} [3]

- (iii) Calculate, for the ball, from $t = 0$ to $t = 2.1$ s,

1. the distance moved,

distance = m [3]

2. the displacement from the initial position.

displacement = m [2]

(iv) On Fig. 2.3, sketch the variation with t of the speed of the ball.

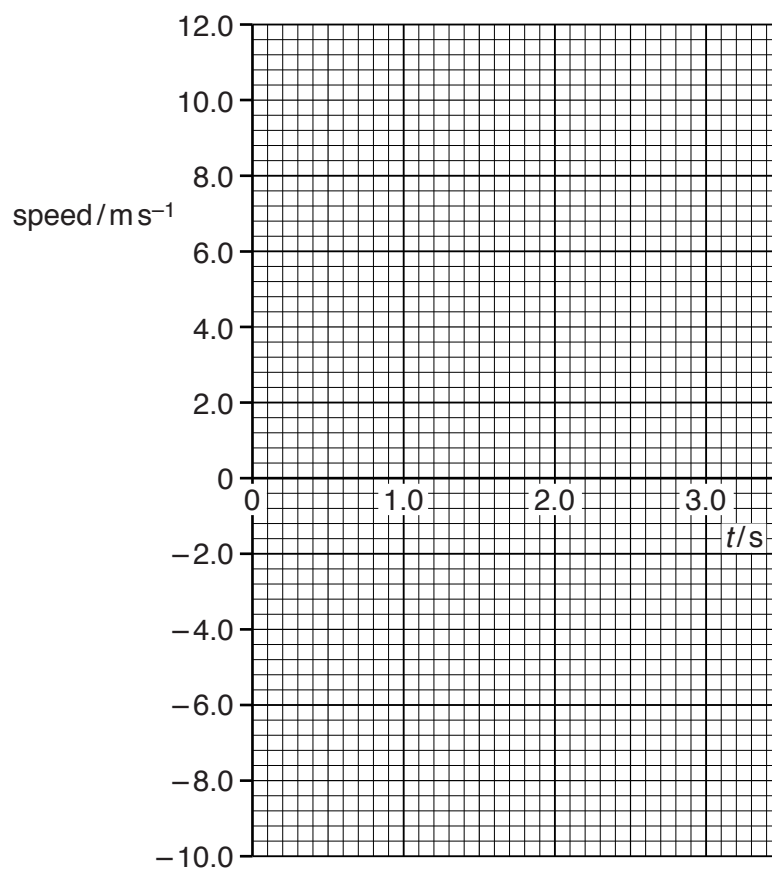


Fig. 2.3

[2]

9702/22/M/J/15/Q2

- 25** A stone is thrown vertically upwards. The variation with time t of the displacement s of the stone is shown in Fig. 2.1.

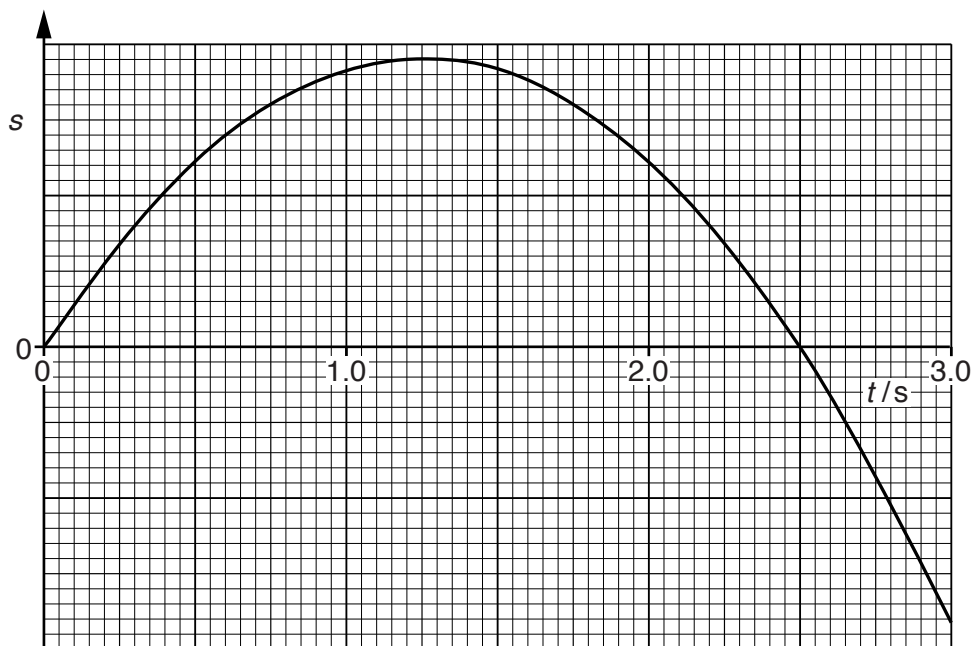


Fig. 2.1

- (a)** Use Fig. 2.1 to describe, without calculation, the speed of the stone from $t = 0$ to $t = 3.0$ s.

.....

.....

..... [2]

- (b)** Assume air resistance is negligible and therefore the stone has constant acceleration.

Calculate, for the stone,

- (i)** the speed at 3.0 s,

speed = ms^{-1}
[3]

(ii) the distance travelled from $t = 0$ to $t = 3.0$ s,

distance = m [3]

(iii) the displacement from $t = 0$ to $t = 3.0$ s.

displacement = m

direction [2]

(c) On Fig. 2.2, draw the variation with time t of the velocity v of the stone from $t = 0$ to $t = 3.0$ s.

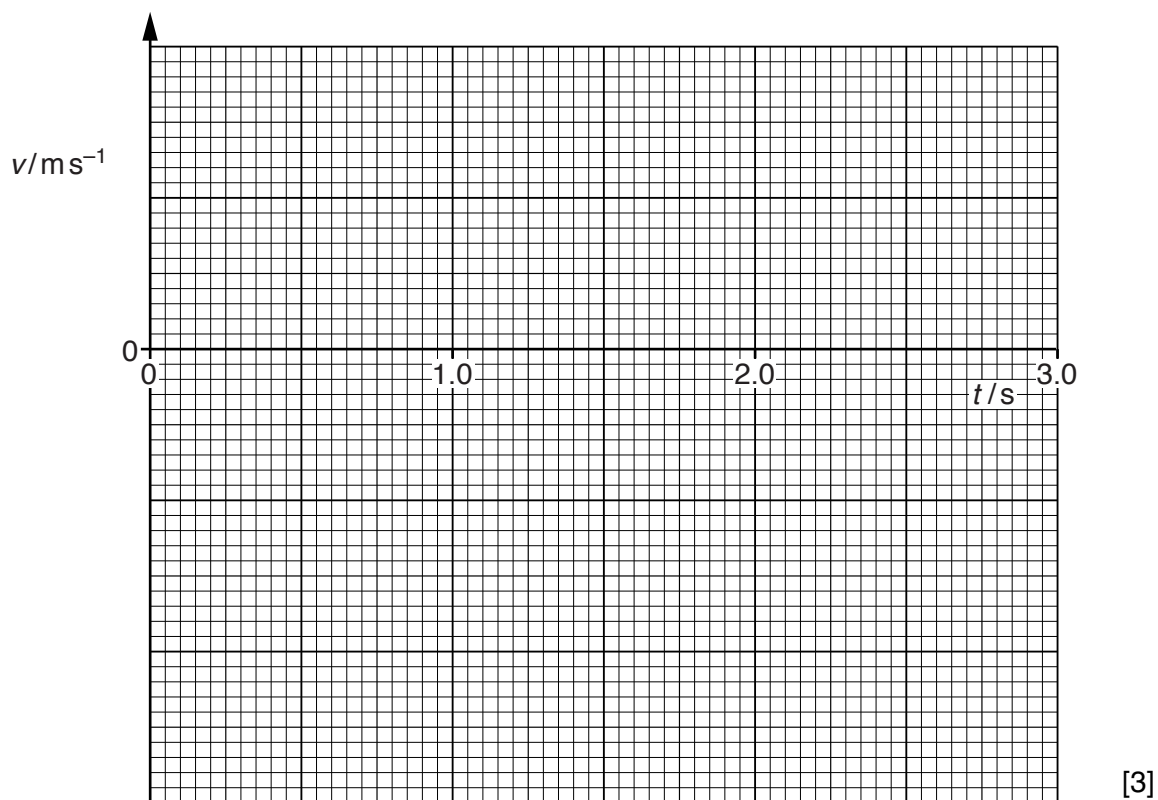


Fig. 2.2

9702/23/M/J/15/Q2

- 26** The variation with time t of the velocity v of a ball is shown in Fig. 2.1.

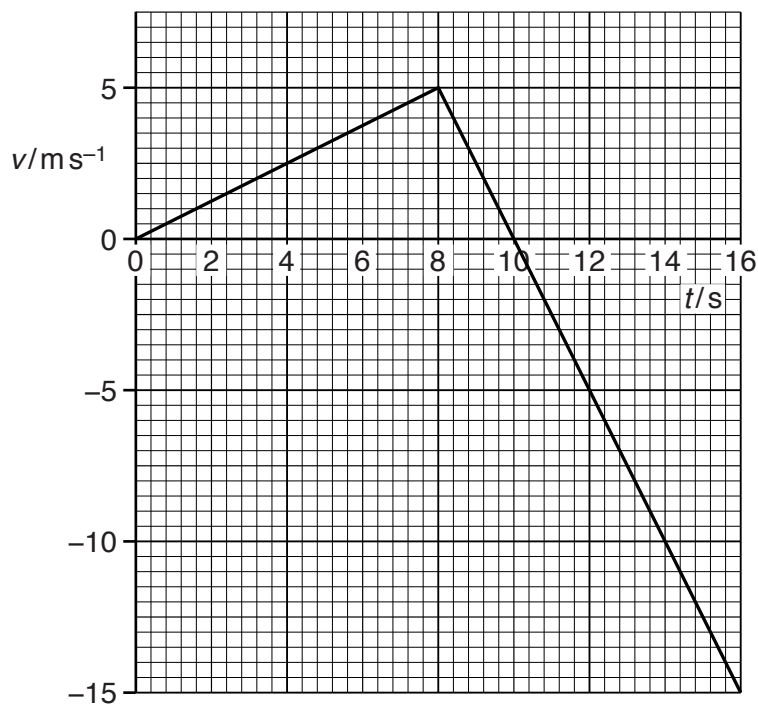


Fig. 2.1

The ball moves in a straight line from a point P at $t = 0$. The mass of the ball is 400 g.

- (a)** Use Fig. 2.1 to describe, without calculation, the velocity of the ball from $t = 0$ to $t = 16$ s.

.....

[2]

- (b)** Use Fig. 2.1 to calculate, for the ball,

- (i)** the displacement from P at $t = 10$ s,

displacement = m [2]

(ii) the acceleration at $t = 10\text{ s}$,

acceleration = ms^{-2} [2]

(iii) the maximum kinetic energy.

kinetic energy = J [2]

(c) Use your answers in (b)(i) and (b)(ii) to determine the time from $t = 0$ for the ball to return to P.

time = s [2]

9702/23/M/J/12/Q2

- 27** A motor drags a log of mass 452 kg up a slope by means of a cable, as shown in Fig. 2.1.

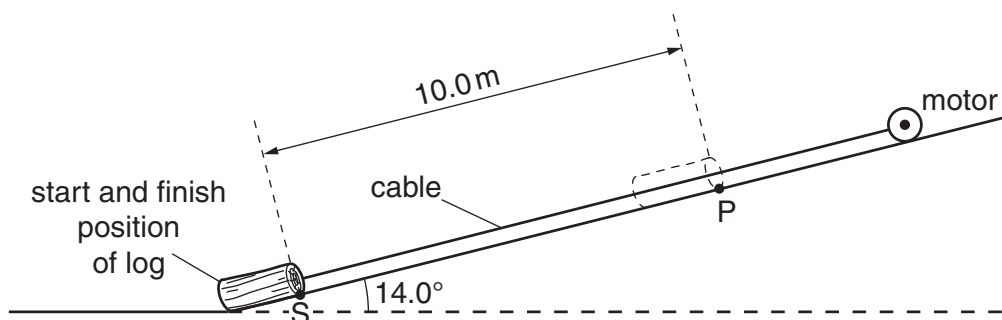


Fig. 2.1

The slope is inclined at 14.0° to the horizontal.

- (a) Show that the component of the weight of the log acting down the slope is 1070 N.

[1]

- (b) The log starts from rest. A constant frictional force of 525 N acts on the log. The log accelerates up the slope at 0.130 m s^{-2} .

- (i) Calculate the tension in the cable.

tension = N [3]

- (ii) The log is initially at rest at point S. It is pulled through a distance of 10.0m to point P.

Calculate, for the log,

1. the time taken to move from S to P,

time = s [2]

2. the magnitude of the velocity at P.

velocity = ms^{-1} [1]

- (c) The cable breaks when the log reaches point P. On Fig. 2.2, sketch the variation with time t of the velocity v of the log. The graph should show v from the start at S until the log returns to S. [4]



Fig. 2.2

28 (a) Define *acceleration*.

9702/22/F/M/16/Q2

.....
 [1]

(b) A ball is kicked from horizontal ground towards the top of a vertical wall, as shown in Fig. 2.1.

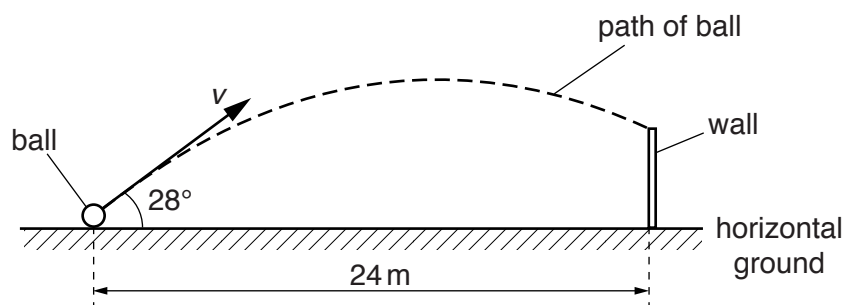


Fig. 2.1 (not to scale)

The horizontal distance between the initial position of the ball and the base of the wall is 24 m. The ball is kicked with an initial velocity v at an angle of 28° to the horizontal. The ball hits the top of the wall after a time of 1.5 s. Air resistance may be assumed to be negligible.

(i) Calculate the initial horizontal component v_x of the velocity of the ball.

$$v_x = \dots\dots\dots \text{ms}^{-1} \quad [1]$$

(ii) Show that the initial vertical component v_y of the velocity of the ball is 8.5 ms^{-1} .

[2]

(iii) Calculate the time taken for the ball to reach its maximum height above the ground.

$$\text{time} = \dots\dots\dots \text{s} \quad [2]$$

- (iv) The ball is kicked at time $t = 0$. On Fig. 2.2, sketch the variation with time t of the vertical component v_y of the velocity of the ball until it hits the wall. It may be assumed that velocity is positive when in the upwards direction.

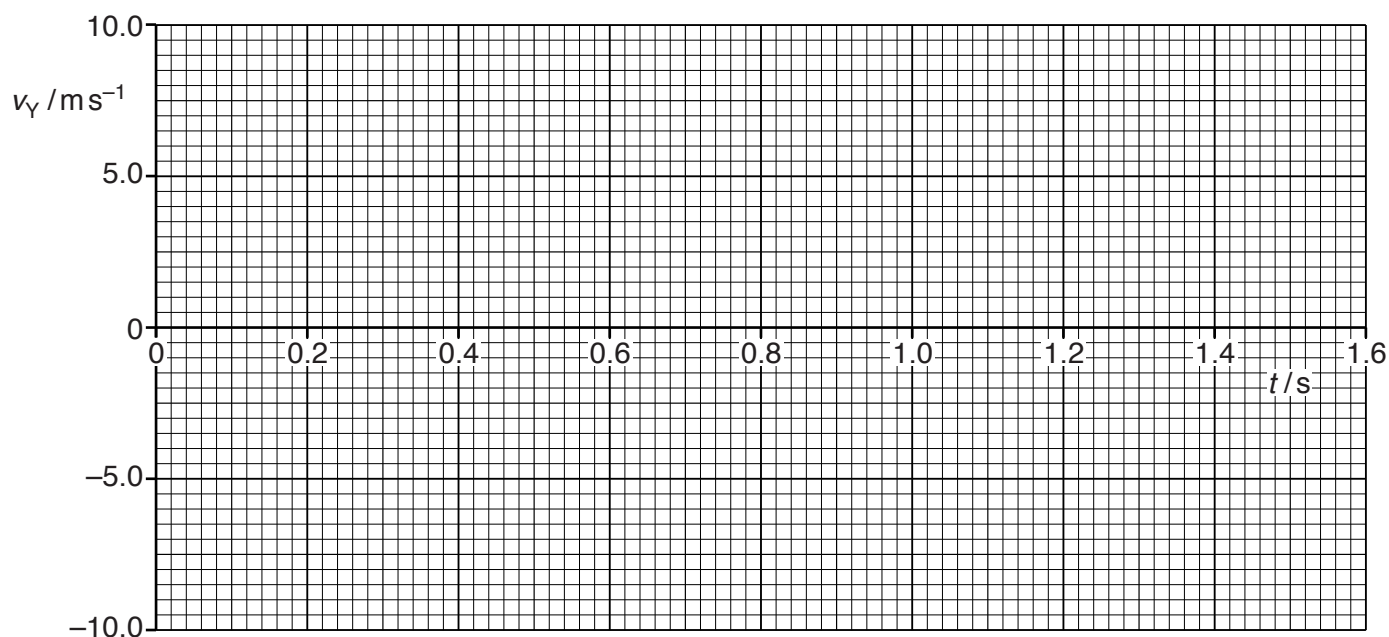


Fig. 2.2

[2]

- (c) (i) Use the information in (b) to determine the maximum height of the ball above the ground.

maximum height = m [2]

- (ii) The maximum gravitational potential energy of the ball above the ground is 22 J. Calculate the mass of the ball.

mass = kg [2]

- (d) A ball of greater mass is kicked with the same velocity as the ball in (b). State and explain the effect, if any, of the increased mass on the maximum height reached by the ball. Air resistance is still assumed to be negligible.

.....
 [1]

9702/21/M/J/16/Q2

- 29 A ball is thrown from a point P with an initial velocity u of 12 m s^{-1} at 50° to the horizontal, as illustrated in Fig. 2.1.

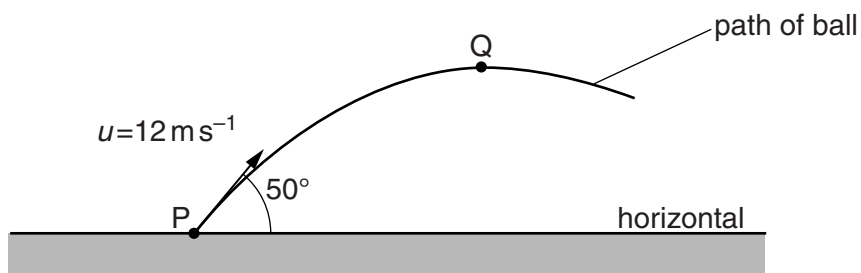


Fig. 2.1

The ball reaches maximum height at Q.

Air resistance is negligible.

(a) Calculate

- (i) the horizontal component of u ,

horizontal component = m s^{-1} [1]

- (ii) the vertical component of u .

vertical component = m s^{-1} [1]

- (b) Show that the maximum height reached by the ball is 4.3 m.

[2]

- (c) Determine the magnitude of the displacement PQ.

displacement = m [4]

30 (a) Define *velocity*.

9702/22/M/J/17/Q2

.....
 [1]

(b) A ball of mass 0.45 kg leaves the edge of a table with a horizontal velocity v , as shown in Fig. 2.1.

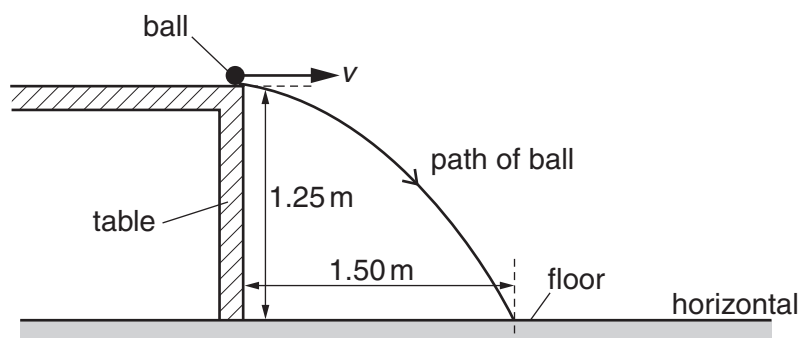


Fig. 2.1

The height of the table is 1.25 m. The ball travels a distance of 1.50 m horizontally before hitting the floor.

Air resistance is negligible.

Calculate, for the ball,

(i) the horizontal velocity v as it leaves the table,

$v = \dots\dots\dots \text{ms}^{-1}$ [3]

(ii) the velocity just as it hits the floor,

magnitude of velocity = $\dots\dots\dots \text{ms}^{-1}$

angle to the horizontal = $\dots\dots\dots^\circ$
 [4]

9702/21/O/N/17/Q2

- 31 The variation with time t of the velocity v of two cars P and Q is shown in Fig. 2.1.

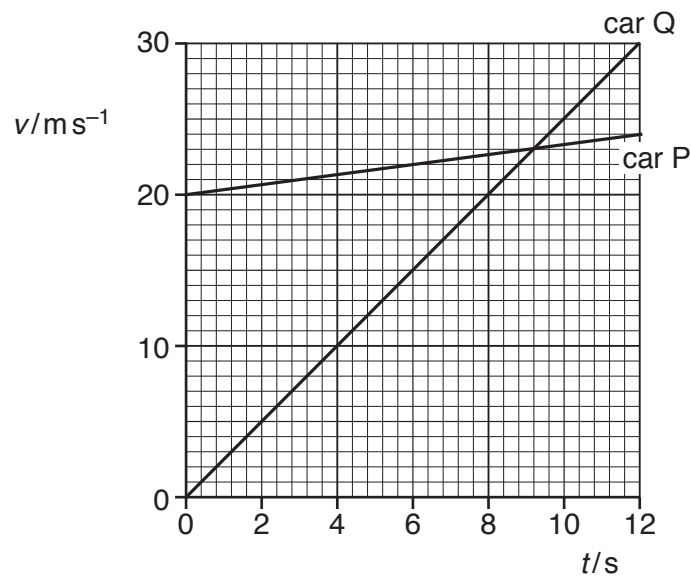


Fig. 2.1

The cars travel in the same direction along a straight road.
Car P passes car Q at time $t = 0$.

- (a) The speed limit for cars on the road is 100 km h^{-1} . State and explain whether car Q exceeds the speed limit.

.....[1]

- (b) Calculate the acceleration of car P.

acceleration = m s^{-2} [2]

- (c) Determine the distance between the two cars at time $t = 12\text{ s}$.

distance = m [3]

- (d) From time $t = 12\text{ s}$, the velocity of each car remains constant at its value at $t = 12\text{ s}$.

Determine the time t at which car Q passes car P.

$t =$ s [2]

9702/22/M/J/18/Q3

- 32** A child on a sledge slides down a steep hill and then travels in a straight line up an ice-covered slope, as illustrated in Fig. 3.1.

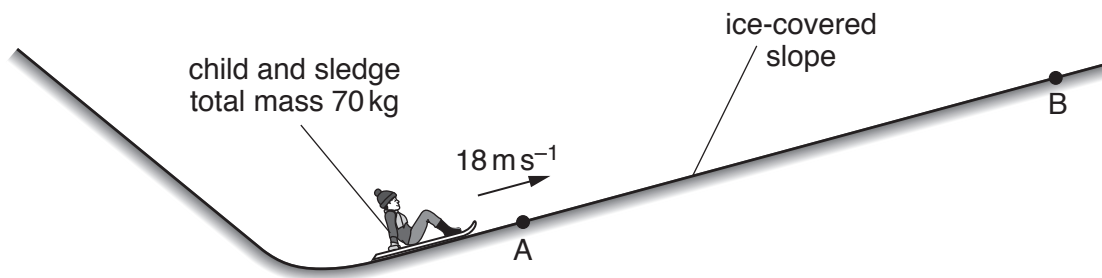


Fig. 3.1 (not to scale)

The sledge passes point A with speed 18 m s^{-1} at time $t = 0$ and then comes to rest at point B. The child applies a brake to the sledge at point B. The brake does not keep the sledge stationary and it immediately slides back down the slope towards A.

The variation with time t of the velocity v of the sledge from $t = 0$ to $t = 24 \text{ s}$ is shown in Fig. 3.2.

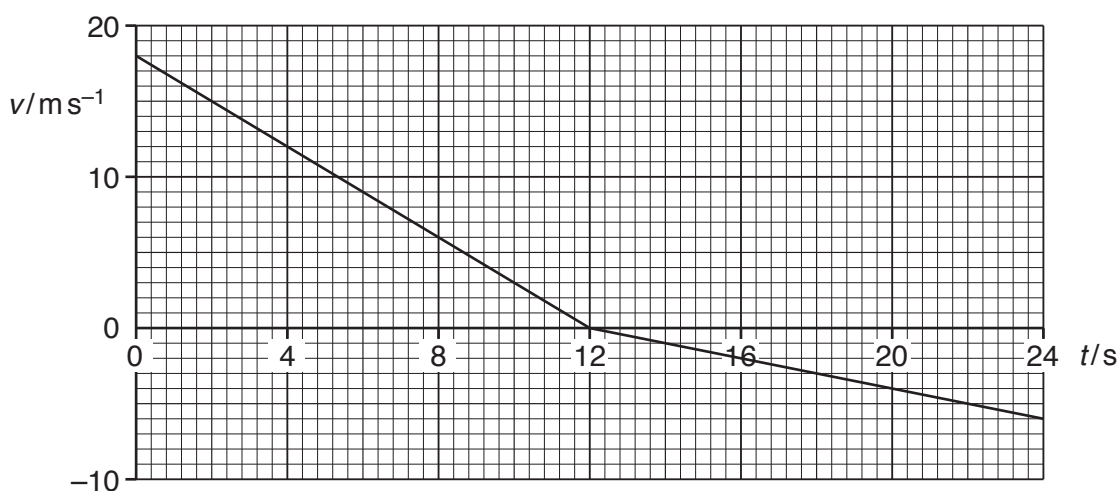


Fig. 3.2

- (a)** State the time taken for the sledge to travel from A to B.

time = s [1]

- (b)** Determine the displacement of the sledge up the slope from point A at time $t = 24$ s.

displacement =m [3]

- (c)** Show that the acceleration of the sledge as it moves from B back towards A is 0.50 m s^{-2} .

[2]

- (d)** The child and sledge have a total mass of 70 kg. The component of the total weight of the child and sledge that acts down the slope is 80 N.

Determine

- (i)** the frictional force on the sledge as it moves from B towards A,

frictional force = N [2]

- (ii)** the angle θ of the slope to the horizontal.

$\theta =$ $^{\circ}$ [2]

- (e) The child on the sledge blows a whistle between $t = 4.0\text{ s}$ and $t = 8.0\text{ s}$. The whistle emits sound of frequency 900 Hz . The speed of the sound in the air is 340 m s^{-1} . A man standing at point A hears the sound.

Use Fig. 3.2 to

- (i) determine the initial frequency of the sound heard by the man,

initial frequency = Hz [2]

- (ii) describe and explain qualitatively the variation, if any, in the frequency of the sound heard by the man.

.....
[1]

- 33 Two vertical metal plates in a vacuum are separated by a distance of 0.12 m . Fig. 4.1 shows a side view of this arrangement.

9702/23/M/J/19/Q4

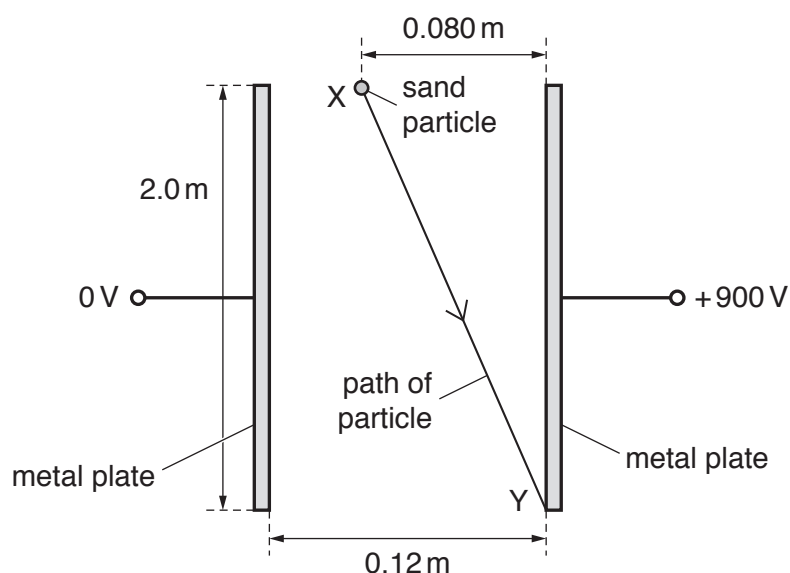


Fig. 4.1 (not to scale)

Each plate has a length of 2.0 m . The potential difference between the plates is 900 V . The electric field between the plates is uniform.

A negatively charged sand particle is released from rest at point X, which is a horizontal distance of 0.080 m from the top of the positively charged plate. The particle then travels in a straight line and collides with the positively charged plate at its lowest point Y, as illustrated in Fig. 4.1.

- (a) Describe the pattern of the field lines (lines of force) between the plates.

.....

[2]

- (b) State the names of the **two** forces acting on the particle as it moves from X to Y.

.....[1]

- (c) By considering the vertical motion of the sand particle, show that the time taken for the particle to move from X to Y is 0.64 s.

[2]

- (d) Calculate the horizontal component of the acceleration of the particle.

horizontal component of acceleration = ms^{-2} [2]

- (e) (i)** Calculate the magnitude of the electric field strength.

electric field strength = NC^{-1} [2]

- (ii)** The sand particle has mass m and charge q . Use your answers in **(d)** and **(e)(i)** to determine the ratio $\frac{q}{m}$.

ratio = C kg^{-1} [2]

- (f)** Another particle has a smaller magnitude of the ratio $\frac{q}{m}$ than the sand particle. This particle is also released from point X.

For the movement of this particle, state the effect, if any, of the decreased magnitude of the ratio on:

- (i)** the vertical component of the acceleration

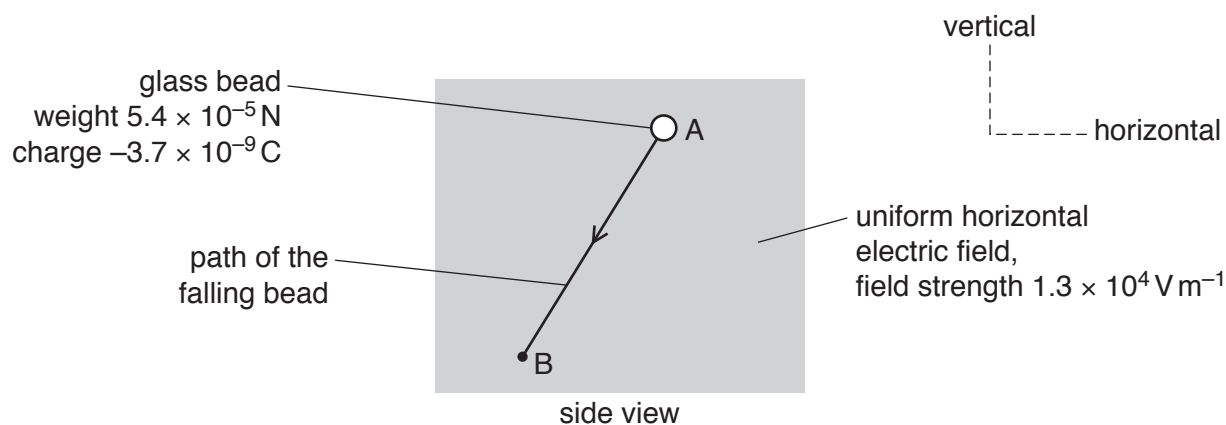
.....[1]

- (ii)** the horizontal component of the acceleration.

.....[1]

9702/21/O/N/19/Q2

- 34** A small charged glass bead of weight $5.4 \times 10^{-5} \text{ N}$ is initially at rest at point A in a vacuum. The bead then falls through a uniform horizontal electric field as it moves in a straight line to point B, as illustrated in Fig. 2.1.

**Fig. 2.1** (not to scale)

The electric field strength is $1.3 \times 10^4 \text{ V m}^{-1}$. The charge on the bead is $-3.7 \times 10^{-9} \text{ C}$.

- (a)** Describe how two metal plates could be used to produce the electric field. Numerical values are not required.

.....

.....

..... [2]

- (b)** Determine the magnitude of the electric force acting on the bead.

electric force = N [2]

- (c) Use your answer in (b) and the weight of the bead to show that the resultant force acting on it is $7.2 \times 10^{-5} \text{ N}$.

[1]

- (d) Explain why the resultant force on the bead of $7.2 \times 10^{-5} \text{ N}$ is constant as the bead moves along path AB.

.....

.....

.....

..... [2]

- (e) (i) Calculate the magnitude of the acceleration of the bead along the path AB.

acceleration = ms^{-2} [2]

- (ii) The path AB has length 0.58 m.

Use your answer in (i) to determine the speed of the bead at point B.

speed = ms^{-1} [2]

9702/22/O/N/19/Q2

35 (a) Define *acceleration*.

..... [1]

(b) A steel ball of diameter 0.080 m is released from rest and falls vertically in air, as illustrated in Fig. 2.1.

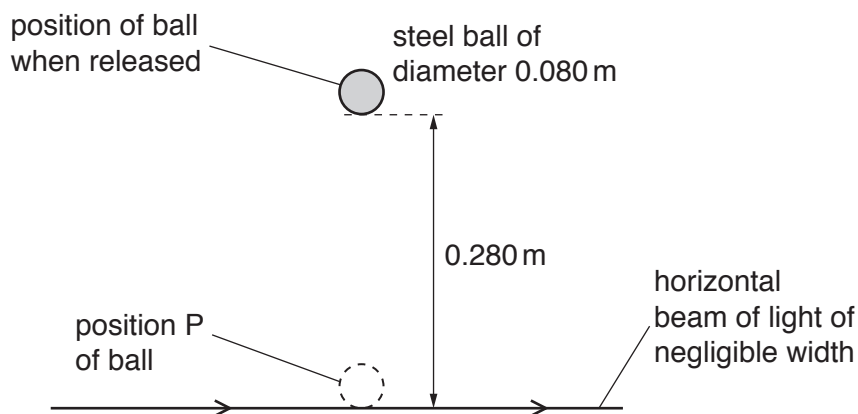


Fig. 2.1 (not to scale)

A horizontal beam of light of negligible width is a vertical distance of 0.280 m below the bottom of the ball when it is released. The ball falls through and breaks the beam of light.

(i) Explain why the force due to air resistance acting on the ball may be neglected when calculating the time taken for the ball to reach the beam of light.

.....
 [1]

(ii) Calculate the time taken for the ball to fall from rest to position P where the bottom of the ball touches the beam of light.

time taken = s [2]

- (iii) Determine the time interval during which the beam of light is broken by the ball.

time interval = s [2]

- (c) A different ball is released from the same position as the steel ball in (b). This ball has the same diameter but a much lower density. For this ball, the force due to air resistance cannot be neglected as the ball falls.

State and explain the change, if any, to the time interval during which the beam of light is broken by the ball.

.....
.....
.....
..... [2]

9702/23/O/N/19/Q2

- 36 (a) State what is meant by *work done*.

.....
 [1]

- (b) A lift (elevator) of weight 13.0 kN is connected by a cable to a motor, as shown in Fig. 2.1.

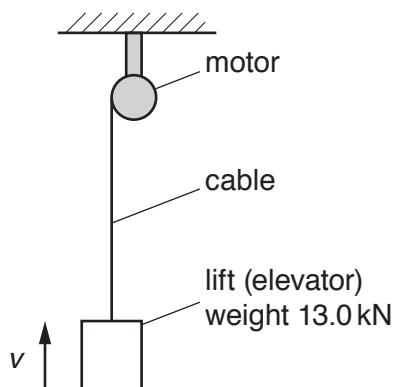


Fig. 2.1

The lift is pulled up a vertical shaft by the cable. A constant frictional force of 2.0 kN acts on the lift when it is moving. The variation with time t of the speed v of the lift is shown in Fig. 2.2.

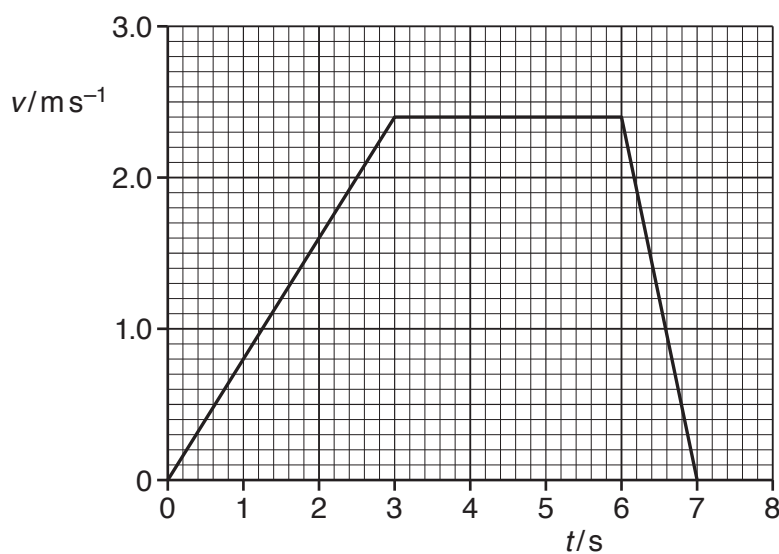


Fig. 2.2

(i) Use Fig. 2.2 to determine:

1. the acceleration of the lift between time $t = 0$ and $t = 3.0$ s

acceleration = m s^{-2} [2]

2. the work done by the motor to raise the lift between time $t = 3.0$ s and $t = 6.0$ s.

work done = J [2]

(ii) The motor has an efficiency of 67%. The tension in the cable is 1.6×10^4 N at time $t = 2.5$ s.

Determine the input power to the motor at this time.

input power = W [3]

(iii) State and explain whether the increase in gravitational potential energy of the lift from time $t = 0$ to $t = 7.0$ s is less than, the same as, or greater than the work done by the motor. A calculation is not required.

.....
 [1]

9702/22/M/J/20/Q1

37 (a) Define *velocity*.

.....
 [1]

(b) The drag force F_D acting on a car moving with speed v along a straight horizontal road is given by

$$F_D = v^2 Ak$$

where k is a constant and A is the cross-sectional area of the car.

Determine the SI base units of k .

SI base units [2]

(c) The value of k , in SI base units, for the car in (b) is 0.24. The cross-sectional area A of the car is 5.1 m^2 .

The car is travelling with a constant speed along a straight road and the output power of the engine is $4.8 \times 10^4 \text{ W}$. Assume that the output power of the engine is equal to the rate at which the drag force F_D is doing work against the car.

Determine the speed of the car.

speed = ms^{-1} [3]

9702/23/M/J/20/Q1

- 38 (a) State **one** similarity and **one** difference between *distance* and *displacement*.

similarity:

.....

difference:

.....

[2]

- (b) A student takes several measurements of the same quantity. This set of measurements has high precision, but low accuracy.

Describe what is meant by:

- (i) *high precision*

.....

..... [1]

- (ii) *low accuracy*.

.....

..... [1]

39. 9702/21/O/N/20/Q2

A small block is lifted vertically upwards by a toy aircraft, as illustrated in Fig. 2.1.

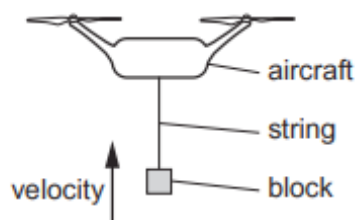


Fig. 2.1

As the block is moving upwards, the string breaks at time $t = 0$. The block initially continues moving upwards and then falls and hits the ground at time $t = 0.90\text{ s}$. The variation with time t of the velocity v of the block is shown in Fig. 2.2.

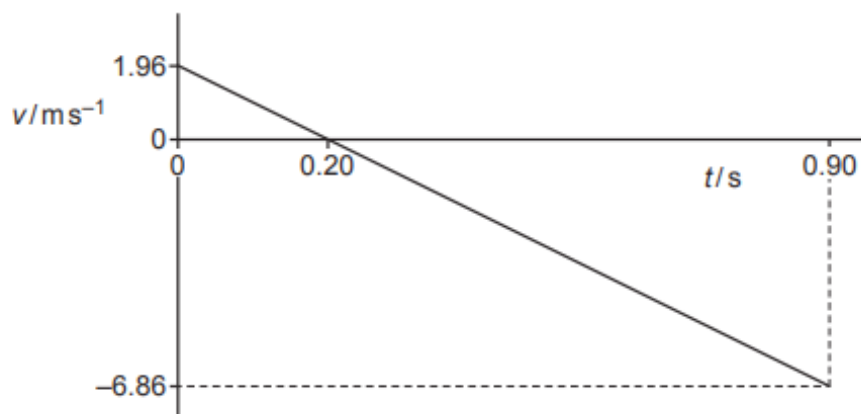


Fig. 2.2

Air resistance is negligible.

- (a) State the feature of the graph in Fig. 2.2 that shows the block has a constant acceleration.

..... [1]

- (b) Use Fig. 2.2 to determine the height of the block above the ground when the string breaks at time $t = 0$.

height = m [3]

- (c) The block has a weight of 0.86 N.

Calculate the difference in gravitational potential energy of the block between time $t = 0$ and time $t = 0.90$ s.

difference in gravitational potential energy = J [2]

- (d) On Fig. 2.3, sketch a line to show the variation of the distance moved by the block with time t from $t = 0$ to $t = 0.20$ s. Numerical values of distance are not required.

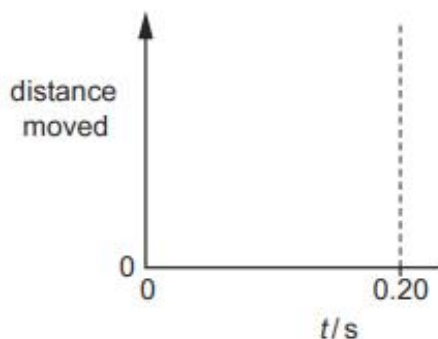


Fig. 2.3

[2]

- (e) A block of greater mass is now released from the same height with the same upward velocity. Air resistance is still negligible.

State and explain the effect, if any, of the increased mass on the speed with which the block hits the ground.

.....

..... [1]

[Total: 9]

40. 9702/22/O/N/20/Q1

- (a) Complete Table 1.1 by putting a tick (✓) in the appropriate column to indicate whether the listed quantities are scalars or vectors.

Table 1.1

quantity	scalar	vector
acceleration		
density		
temperature		
momentum		

[2]

- (b) A toy train moves along a straight section of track. Fig. 1.1 shows the variation with time t of the distance d moved by the train.

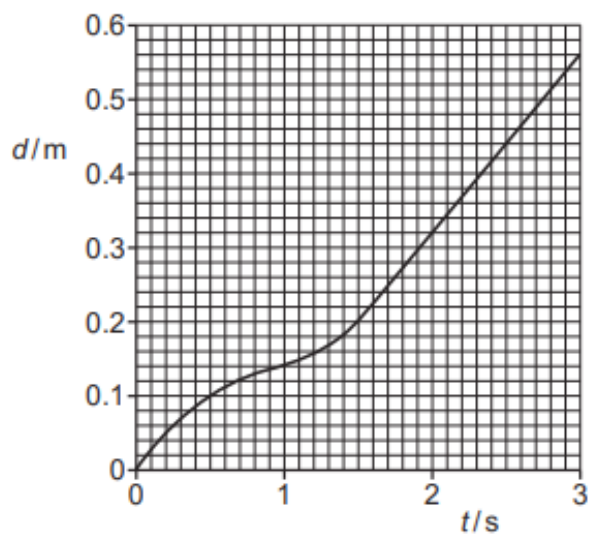


Fig. 1.1

- (i) Describe qualitatively the motion of the train between time $t = 0$ and time $t = 1.0$ s.

.....
 [1]

(ii) Determine the speed of the train at time $t = 2.0$ s.

speed = ms^{-1} [2]

(c) The straight section of track in (b) is part of the loop of track shown in Fig. 1.2.



Fig. 1.2

The train completes exactly one lap of the loop.

State and explain the average velocity of the train over the one complete lap.

.....
.....
..... [1]

[Total: 6]

41. 9702/23/O/N/20/Q3

A ball is fired horizontally with a speed of 41.0 m s^{-1} from a stationary cannon at the top of a hill. The ball lands on horizontal ground that is a vertical distance of 57 m below the cannon, as shown in Fig. 3.1.

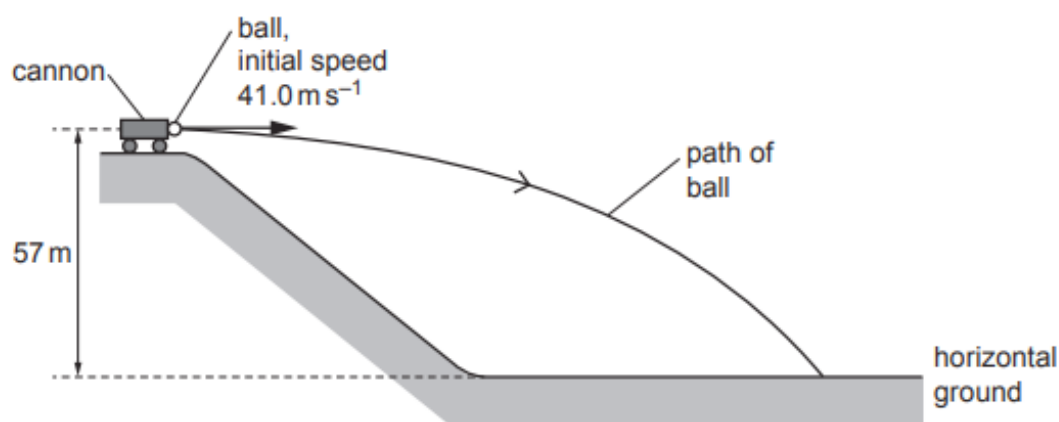


Fig. 3.1 (not to scale)

Assume air resistance is negligible.

- (a) Show that the time taken for the ball to reach the ground, after being fired, is 3.4 s.

[2]

- (b) Calculate the horizontal distance of the ball from the cannon at the point where the ball lands on the ground.

horizontal distance = m [1]

- (c) Determine the magnitude of the displacement of the ball from the cannon at the point where the ball lands on the ground.

displacement = m [2]

- (d) The ball leaves the cannon at time $t = 0$.

On Fig. 3.2, sketch a graph to show the variation of the magnitude v of the vertical component of the velocity of the ball with time t from $t = 0$ to $t = 3.4$ s. Numerical values are not required.

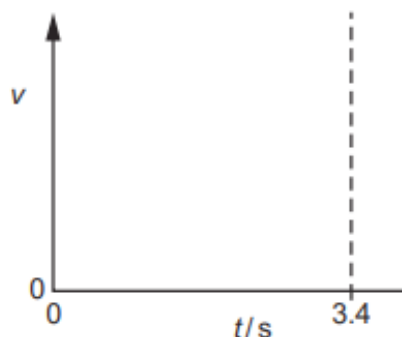


Fig. 3.2

[1]

- (e) The cannon recoils horizontally with a speed of 0.340 ms^{-1} when it fires the ball. The total mass of the ball and the cannon is 1480 kg . Assume that no external horizontal forces act on the ball-cannon system.

Determine, to three significant figures, the mass of the ball.

mass = kg [2]

- (f) The cannon now fires a ball of smaller mass. Assume that air resistance is still negligible.

State and explain the change, if any, to the graph in Fig. 3.2 due to the decreased mass of the ball.

.....

 [2]

[Total: 10]

42. 9702/22/F/M/21/Q1

(a) Complete Table 1.1 by stating whether each of the quantities is a vector or a scalar.

Table 1.1

quantity	vector or scalar
acceleration	
power	
work	

[2]

(b) The variation with time t of the velocity v of an object is shown in Fig. 1.1.

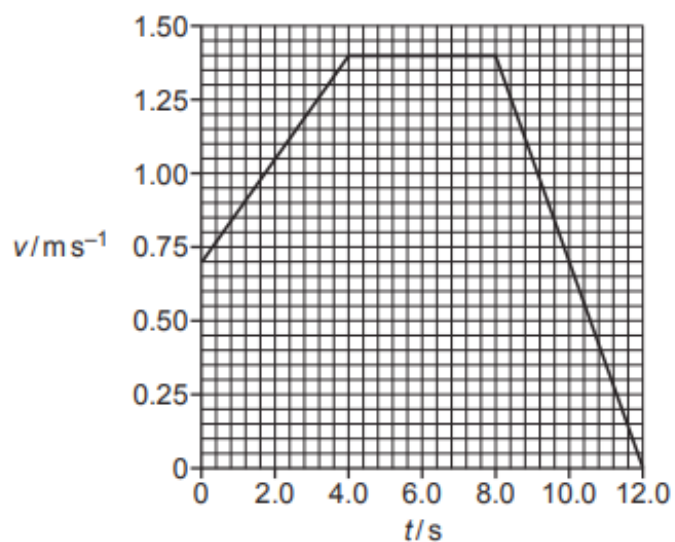


Fig. 1.1

(i) Determine the acceleration of the object from time $t = 0$ to time $t = 4.0$ s.

acceleration = ms^{-2} [2]

- (ii) Determine the distance moved by the object from time $t = 0$ to time $t = 4.0$ s.

distance = m [2]

- (c) (i) Define *force*.

.....
 [1]

- (ii) The motion represented in Fig. 1.1 is caused by a resultant force F acting on the object.

On Fig. 1.2, sketch the variation of F with time t from $t = 0$ to $t = 12.0$ s.
 Numerical values of F are not required.

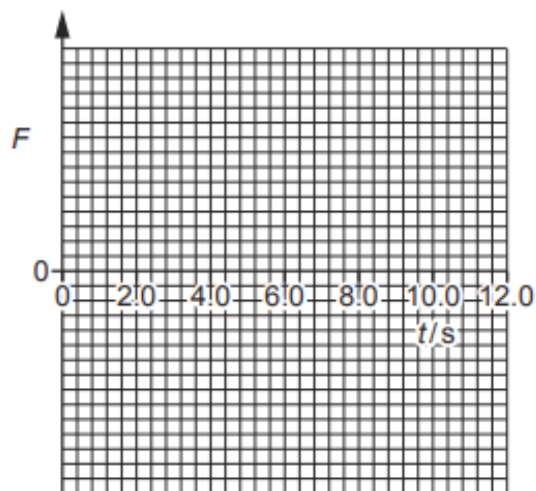


Fig. 1.2

[3]

[Total: 10]

43. 9702/23/M/J/21/Q2

(a) Define *acceleration*.

.....
..... [1]

(b) A stone falls vertically from the top of a cliff. Fig. 2.1 shows the variation with time t of the velocity v of the stone.

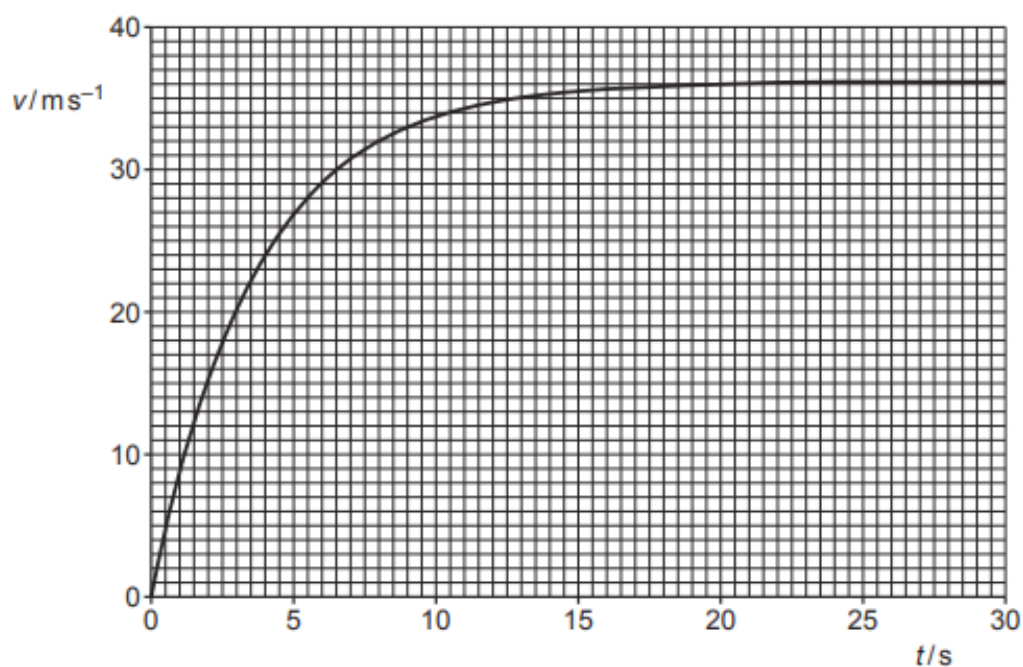


Fig. 2.1

(i) Explain, with reference to forces acting on the stone, the shape of the curve in Fig. 2.1.

.....
.....
.....
.....
..... [3]

(ii) Use Fig. 2.1 to determine the speed of the stone when the resultant force on it is zero.

speed = ms^{-1} [1]

- (iii) Use Fig. 2.1 to calculate the approximate height through which the stone falls between $t = 0$ and $t = 30$ s.

height = m [3]

- (iv) On Fig. 2.2, sketch the variation with t of the acceleration a of the stone between $t = 0$ and $t = 30$ s.

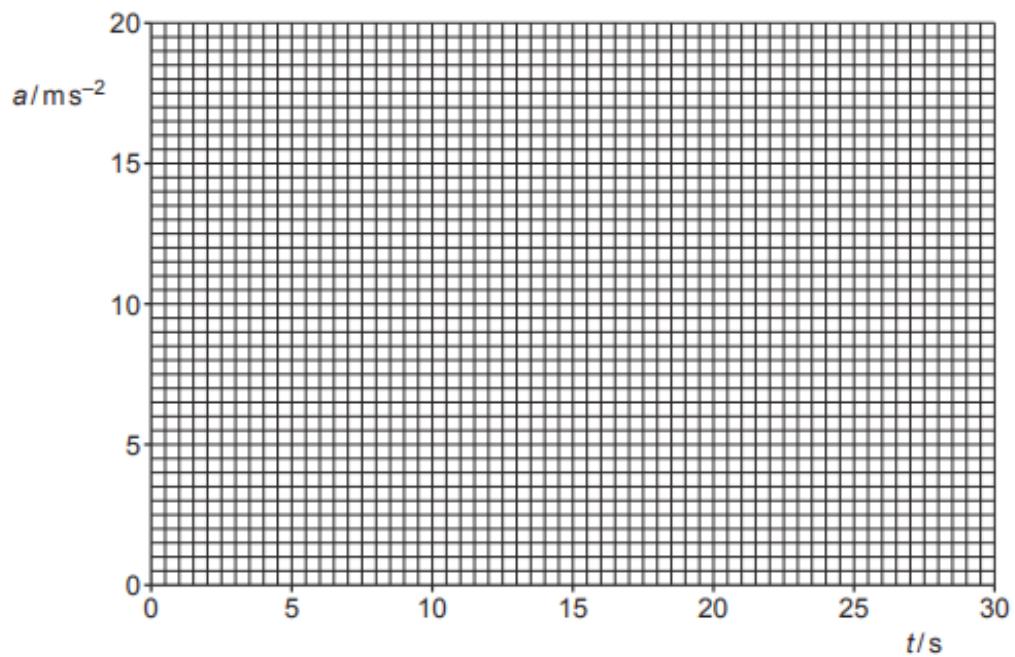


Fig. 2.2

[3]

[Total: 11]

44. 9702/22/O/N/21/Q4

A child sits on the ground next to a remote-controlled toy car. At time $t = 0$, the car begins to move in a straight line directly away from the child. The variation with time t of the velocity of the car along this line is shown in Fig. 4.1.

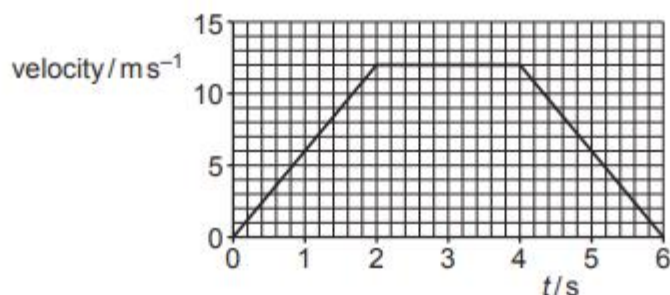


Fig. 4.1

The car's horn continually emits sound of frequency 925 Hz between time $t = 0$ and time $t = 6.0$ s. The speed of the sound in the air is 338 m s^{-1} .

- (a) Describe qualitatively the variation, if any, in the frequency of the sound heard, by the child, that was emitted from the car horn:

- (i) from time $t = 0$ to time $t = 2.0$ s

..... [1]

- (ii) from time $t = 4.0$ s to time $t = 6.0$ s.

..... [1]

- (b) Determine the frequency, to three significant figures, of the sound heard, by the child, that was emitted from the car horn at time $t = 3.0$ s.

frequency = Hz [2]

(c) Determine the time taken for the sound emitted at time $t = 4.0$ s to travel to the child.

time taken = s [2]

[Total: 6]

45. 9702/23/O/N/21/Q3

(a) Define *velocity*.

.....
..... [1]

(b) A remote-controlled toy aircraft is flying horizontally in a wind. Fig. 3.1 shows the velocity vectors, to scale, of the wind and of the aircraft in still air.

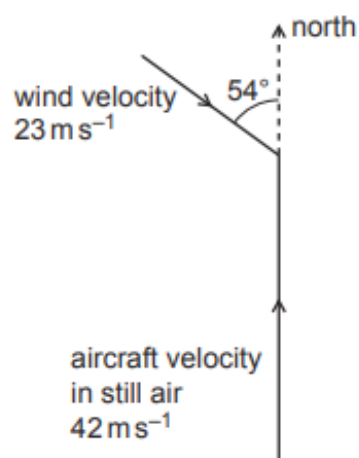


Fig. 3.1

The velocity of the aircraft in still air is 42 m s^{-1} to the north. The velocity of the wind is 23 m s^{-1} in a direction of 54° east of south.

Determine the magnitude of the resultant velocity of the aircraft.

magnitude of velocity = m s^{-1} [2]

46. 9702/22/F/M/22/Q2

Water leaves the end of a hose pipe at point P with a horizontal velocity of 6.6 m s^{-1} , as shown in Fig. 2.1.

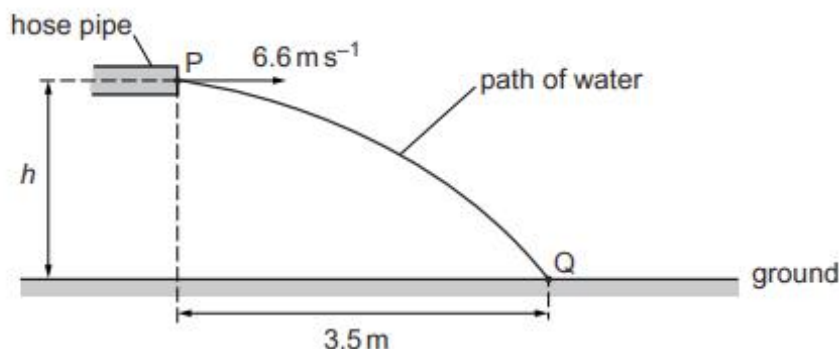


Fig. 2.1 (not to scale)

Point P is at height h above the ground. The water hits the ground at point Q. The horizontal distance from P to Q is 3.5 m .

Air resistance is negligible. Assume that the water between P and Q consists of non-interacting droplets of water and that the only force acting on each droplet is its weight.

- (a) Explain, briefly, why the horizontal component of the velocity of a droplet of water remains constant as it moves from P to Q.

.....
 [1]

- (b) Show that the time taken for a droplet of water to move from P to Q is 0.53 s .

[1]

- (c) Calculate height h .

$h = \dots\dots\dots \text{ m}$ [2]

- (d) For the movement of a droplet of water from P to Q, state and explain whether the displacement of the droplet is less than, more than or the same as the distance along its path.

.....
.....
..... [1]

- (e) Calculate the magnitude of the displacement of a droplet of water that moves from P to Q.

displacement = m [2]

[Total: 7]

47. 9702/22/M/J/22/Q3

A man standing on a wall throws a small ball vertically upwards with a velocity of 5.6 ms^{-1} . The ball leaves his hand when it is at a height of 3.1 m above the ground, as shown in Fig. 3.1.

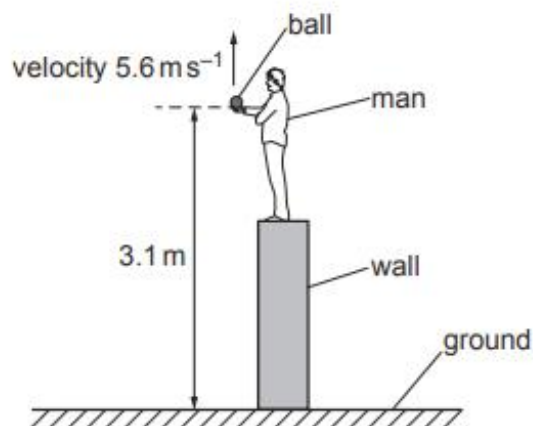


Fig. 3.1 (not to scale)

Assume that air resistance is negligible.

- (a)** Show that the ball reaches a maximum height above the ground of 4.7 m .

[2]

- (b)** The man does not catch the ball as it falls.

Calculate the time taken for the ball to fall from its maximum height to the ground.

time taken = s [2]

- (c) The ball leaves the man's hand at time $t = 0$ and hits the ground at time $t = T$.

On Fig. 3.2, sketch a graph to show the variation of the velocity v of the ball with time t from $t = 0$ to $t = T$. Numerical values of v and t are not required. Assume that v is positive in the upward direction.

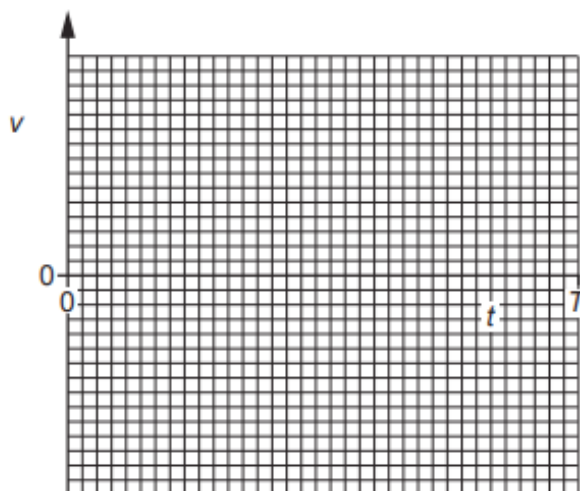


Fig. 3.2

[3]

- (d) State what is represented by the gradient of the graph in (c).

..... [1]

- (e) The man now throws a second ball with the same velocity and from the same height as the first ball. The mass of the second ball is greater than that of the first ball. Assume that air resistance is still negligible.

For the first and second balls, compare:

- (i) the magnitudes of their accelerations

..... [1]

- (ii) the speeds with which they hit the ground.

..... [1]

[Total: 10]

48.9702/23/M/J/22/Q2

An archer releases an arrow towards a target at a velocity of 65.0 m s^{-1} at an angle of 4.30° above the horizontal, as shown in Fig. 2.1.

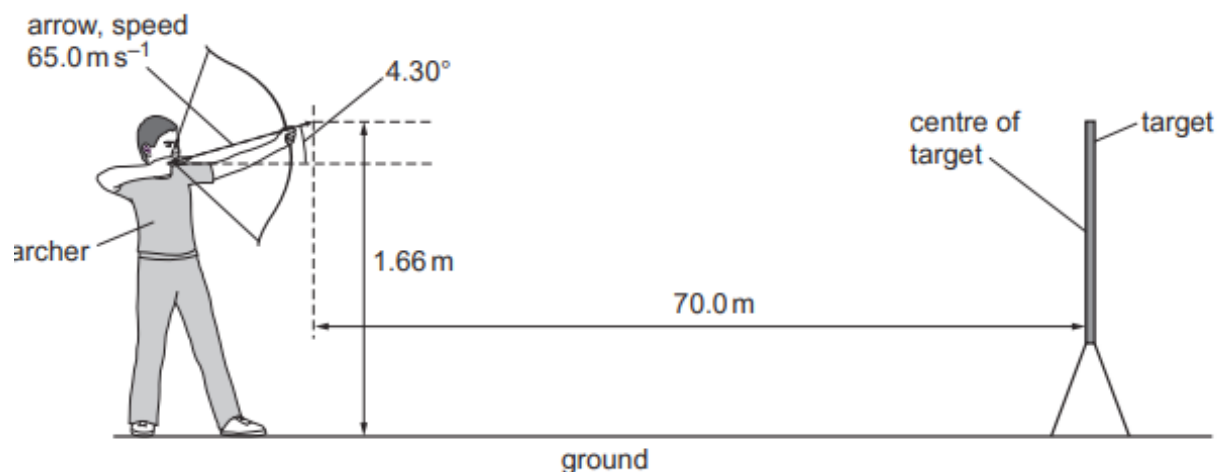


Fig. 2.1 (not to scale)

When released, the tip of the arrow is a horizontal distance of 70.0 m from the target and 1.66 m above the horizontal ground.

The arrow hits the centre of the target.

Assume that air resistance is negligible and that all the mass of the arrow is at its tip.

(a) Show that the time taken for the arrow to reach the target is 1.08 s .

[2]

(b) Calculate the height of the centre of the target above the ground.

height above ground = m [3]

- (c) By considering energy changes, state and explain how the final kinetic energy of the arrow as it hits the target compares with its initial kinetic energy immediately after release. A numerical calculation is not required.

.....

.....

.....

..... [2]

[Total: 7]

49. 9702/23/O/N/22/Q2

The engine of a toy rocket pushes gases vertically downwards and this results in the rocket accelerating vertically upwards from the ground.

The rocket starts to move from rest at time $t = 0$. The variation with time t of the vertical velocity v of the rocket for the first 0.30 s of the flight is shown in Fig. 2.1.

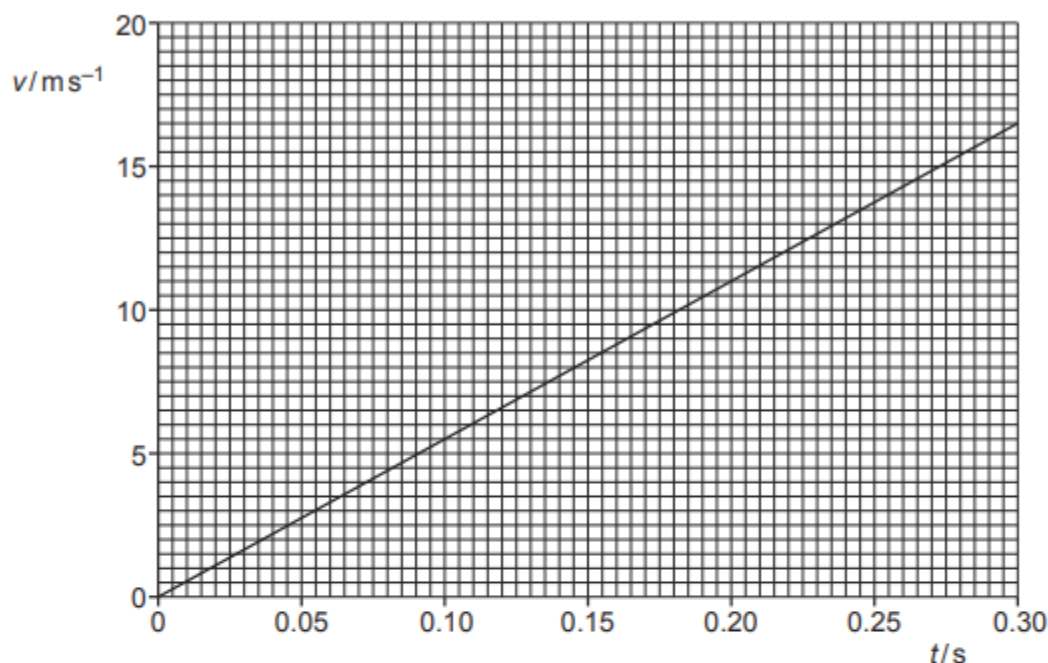


Fig. 2.1

As the rocket moves, the thrust force T provided by the rocket engine is 16 N. Assume that the mass of the rocket is constant for this part of its flight.

Assume that air resistance is negligible.

(a) For this part of the rocket's flight:

- (i) show that the acceleration of the rocket is 55 ms^{-2}

[1]

- (ii) state an expression for the resultant force F experienced by the rocket in terms of the thrust force T and the weight W of the rocket

[1]

(iii) calculate the mass of the rocket.

mass = kg [2]

(b) At time $t = 0.30$ s, a small piece of metal separates from the rocket.

Calculate:

(i) the height of the rocket above the ground at $t = 0.30$ s

height = m [2]

(ii) the speed at which the piece of metal strikes the ground.

speed = ms^{-1} [3]

[Total: 9]

50. 9702/21/O/N/23/Q3

A trolley A moves along a horizontal surface at a constant velocity towards another trolley B which is moving at a lower constant speed in the same direction. Fig. 3.1 shows the trolleys at time $t = 0$.



Fig. 3.1

Table 3.1 shows data for the trolleys.

Table 3.1

trolley	mass/kg	initial speed/ m s^{-1}
A	0.25	0.48
B	0.75	0.12

The two trolleys collide elastically and then separate. Resistive forces are negligible.

Fig. 3.2 shows the variation with time t of the velocity v for trolley B.

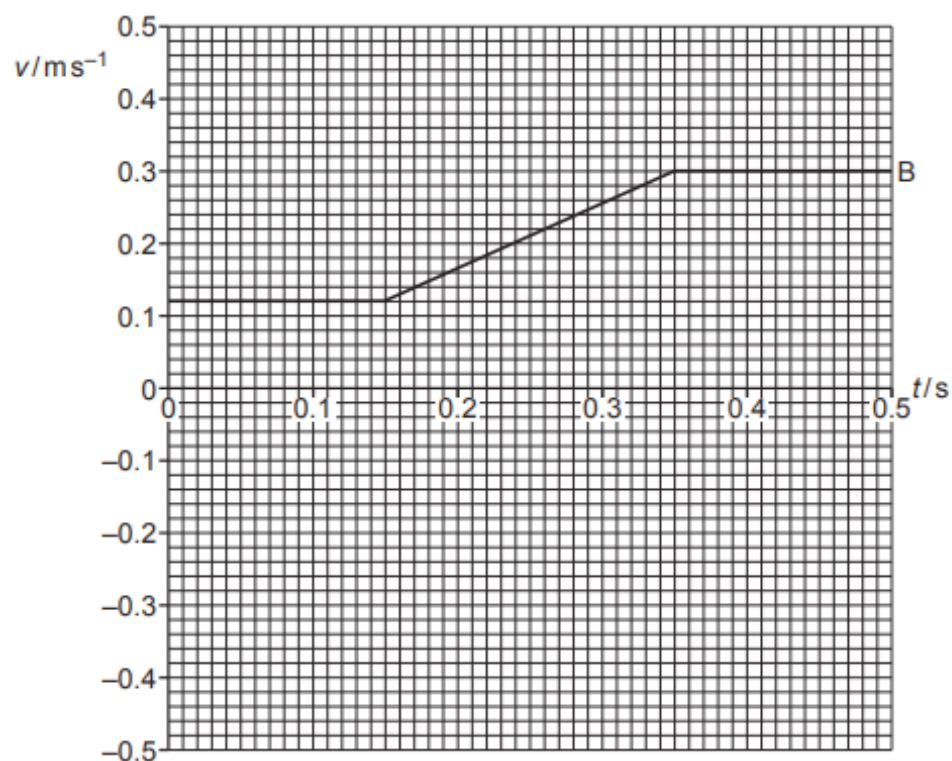


Fig. 3.2

(a) State what is represented by the area under a velocity–time graph.

..... [1]

(b) Use Table 3.1 and Fig. 3.2 to determine:

(i) the acceleration of trolley B during the collision

acceleration of B = ms^{-2} [2]

(ii) the magnitude and direction of the final velocity of trolley A.

magnitude = ms^{-1}

direction [3]

(c) On Fig. 3.2, sketch the variation of the velocity of trolley A with time t from $t = 0$ to $t = 0.50$ s. [3]

[Total: 9]

51. 9702/22/O/N/23/Q1

(a) In the following list, underline **all** quantities that are SI base quantities.

charge

electric current

force

time

[1]

(b) Under certain conditions, the distance s moved in a straight line by an object in time t is given by

$$s = \frac{1}{2}at^2$$

where a is the acceleration of the object.

State **two** conditions under which the above expression applies to the motion of the object.

1

2

[2]

(c) The variation with time t of the velocity v of a car that is moving in a straight line is shown in Fig. 1.1.

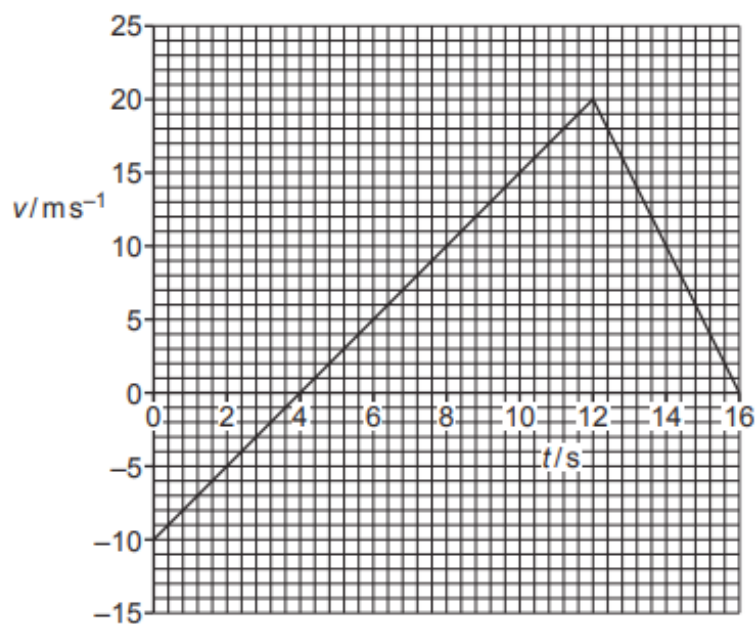


Fig. 1.1

- (i) Compare, qualitatively, the acceleration of the car at time $t = 8.0$ s and at time $t = 14.0$ s in terms of:

- magnitude

.....

.....

- direction.

.....

.....

[2]

- (ii) Determine the magnitude of the acceleration of the car at time $t = 4.0$ s.

acceleration = ms^{-2} [2]

- (iii) The car is at point X at time $t = 0$.

Determine the magnitude of the displacement of the car from X at time $t = 12.0$ s.

displacement = m [2]

[Total: 9]

52. 9702/23/O/N/23/Q2

A ball on horizontal ground is kicked towards a vertical wall. Fig. 2.1 shows the path of the ball.

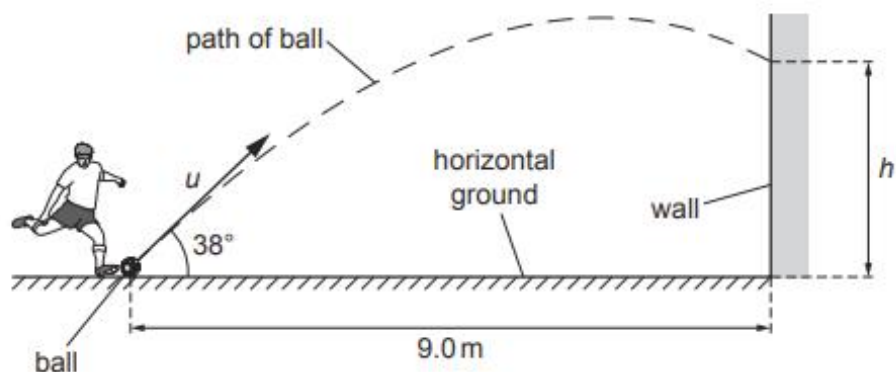


Fig. 2.1 (not to scale)

The ball has an initial velocity u at an angle of 38° to the ground. The ball travels a horizontal distance of 9.0 m before striking the wall at a height h above the ground. The horizontal component u_H of the initial velocity of the ball is 9.5 m s^{-1} .

Air resistance is negligible.

- (a) (i) Show that the time t for the ball to reach the wall is 0.95 s.

[1]

- (ii) Calculate the vertical component u_V of the initial velocity of the ball.

$$u_V = \dots \text{ ms}^{-1} \quad [2]$$

- (iii) Determine h .

$$h = \dots \text{ m} \quad [2]$$

- (b) The speed of the ball just after striking the wall is less than its speed just before striking the wall.

State what this indicates about the nature of the collision of the ball with the wall.

.....

..... [1]

[Total: 6]

ANSWERS

- 1 (a)** 2.4s A1 [1]
- (b)** in **(b)** and **(c)**, allow answers as (+) or (–)
- recognises distance travelled as area under graph line C1
- height = $(\frac{1}{2} \times 2.4 \times 9.0) - (\frac{1}{2} \times 1.6 \times 6.0)$ C1
- = 6.0m (allow 6m) A1 [3]
- (answer 15.6 scores 2 marks)
- answer 10.8 or 4.8 scores 1 mark)
- alternative solution: $s = ut - \frac{1}{2}at^2$
- = $(9 \times 4) - \frac{1}{2} \times (9 / 2.4) \times 4^2$
- = 6.0m
-
- 2 (a) (i)** (air) resistance increases with speed M1
- resultant / accelerating force decreases A1 [2]
- (ii)** either (air) resistance is zero
- or weight / gravitational force is only force B1 [1]
- (b)** use of gradient of a tangent M1
- acceleration = $1.9 \pm 0.2 \text{ m s}^{-2}$ A2 [3]
- (for values $> \pm 0.2$ but ≤ 0.4 , allow 1 mark)
- (answer 3.3 m s^{-2} scores no marks)
- (c) (i)** 1 weight = $90 \times 9.8 = 880 \text{ N}$ A1 [1]
- (use of $g = 10 \text{ m s}^{-2}$ then deduct mark but once only in the Paper)
- 2 accelerating force = $90 \times 1.9 = 170 \text{ N}$... (allow ecf) A1 [1]
- (ii)** resistive force = $880 - 170 = 710 \text{ N}$ A1 [1]
- (allow ecf but only if resistive force remains positive)
-
- 3 (a) (i)** speed = 4.0 m s^{-1} ... (allow 1 s.f.) A1 [1]
- (ii)** $v^2 = 2gh$
- = $2 \times 9.8 \times 1.96$ M1
- $v = 6.2 \text{ m s}^{-1}$ A0 [1]
- (use of $g = 10 \text{ m s}^{-2}$ loses the mark)
- (b)** correct basic shape with correct directions for vectors M1
- speed = $(7.4 \pm 0.2) \text{ m s}^{-1}$ A1
- at $(33 \pm 2)^\circ$ to the vertical A1 [3]
- (for credit to be awarded, speed and angle must be correct on the diagram – not calculated)
- (c) (i)** either $v^2 = 2 \times 9.8 \times 0.98$ or $v = 6.2 / \sqrt{2}$ C1
- speed = 4.4 m s^{-1} A1 [2]
- (allow calculation of $t = 0.447 \text{ s}$, then $v = 4.4 \text{ m s}^{-1}$)
- (ii)** 1 momentum = mv C1
- change in momentum = $0.034 (6.2 + 4.4)$ C1
- = 0.36 kg m s^{-1} A1 [3]
- (use of $0.034 (6.2 - 4.4)$ loses last two marks)
- 2 force = $\Delta p / \Delta t$ (however expressed) C1
- = $\frac{0.36}{0.12}$
- = 3.0 N (allow 1 s.f.) A1 [2]

4	(a)	scalar	B1	
		scalar	B1	
		vector	B1 [3]	
	(b) (i) 1	gradient (of graph) is the speed/velocity (<i>can be scored here or in 2</i>).....	B1	
		<u>initial gradient</u> is zero	B1 [2]	
		2 gradient (of line/graph) becomes constant	B1 [1]	
		(ii) speed = $(2.8 \pm 0.1) \text{ m s}^{-1}$	A2 [2]	
		(if answer > ± 0.1 but $\leq \pm 0.2$, then award 1 mark)		
	(iii)	curved line never below given line and starts from zero	B1	
		continuous curve with increasing gradient	B1	
line never vertical or straight		B1 [3]		
5	(a)	e.g. initial speed is zero		
		constant acceleration		
		straight line motion		
		(any two, one mark each)	B2 [2]	
	(b) (i)	$s = \frac{1}{2}at^2$		
		$0.79 = \frac{1}{2} \times 9.8 \times t^2$	C1	
		$t = 0.40 \text{ s}$ allow 1 SF or greater	A1	
		2 or 3 SF answer	A1 [3]	
		(ii)	distance travelled by end of time interval = 90 cm	C1
			$0.90 = \frac{1}{2} \times 9.8 \times t^2$	
			$t = 0.43 \text{ s}$ allow 2 SF or greater	C1
		time interval = 0.03 s	A1 [3]	
		(c)	(air resistance) means ball's speed/acceleration is less	M1
			length of image is shorter	A1 [2]
	6	(a) (i)	horizontal speed constant at 8.2 m s^{-1}	C1
			vertical component of speed = $8.2 \tan 60^\circ$	M1
			$= 14.2 \text{ m s}^{-1}$	A0 [2]
(ii)		$14.2^2 = 2 \times 9.8 \times h$ (using $g = 10$ then -1)	C1	
		vertical distance = 10.3 m	A1 [2]	
(iii)		time of descent = $14.2 / 9.8 = 1.45 \text{ s}$	C1	
		$x = 1.45 \times 8.2$		
		$= 11.9 \text{ m}$	A1 [2]	
(b) (i)		smooth path curved and above given path	M1	
		hits ground at more acute angle	A1 [2]	
		(ii)	smooth path curved and below given path	M1
			hits ground at steeper angle	A1 [2]

7	(a) (i) $V_H = 12.4 \cos 36^\circ (= 10.0 \text{ m s}^{-1})$ distance = 10.0×0.17 = 1.7 m	C1	A1 [2]
	(ii) $V_V = 12.4 \sin 36^\circ (= 7.29 \text{ m s}^{-1})$ $h = 7.29 \times 0.17 - \frac{1}{2} \times 9.81 \times 0.17^2$ = 1.1 m	C1	C1 A1 [3]
	(b) smooth curve with ball hitting wall below original smooth curve showing rebound to ground with correct reflection at wall	B1	B1 [2]
8	(i) $s = ut + \frac{1}{2} at^2$ $15 = 0.5 \times 9.81 \times t^2$ $T = 1.7 \text{ s}$ if $g = 10$ is used then -1 but only once on paper (ii) vertical component v_v : $v_v^2 = u^2 + 2as = 0 + 2 \times 9.81 \times 15$ or $v_v = u + at = 9.81 \times 1.7(5)$ $v_v = 17.16$ resultant velocity: $v^2 = (17.16)^2 + (20)^2$ $v = 26 \text{ m s}^{-1}$ If $u = 20$ is used instead of $u = 0$ then 0/3 Allow the solution using: initial (potential energy + kinetic energy) = final kinetic energy	C1	A1 [2]
9	(a) (i) base units of D : force: kg m s^{-2} radius: m velocity: m s^{-1} base units of D : $[F / (R \times v)] \text{ kg m s}^{-2} / (\text{m} \times \text{m s}^{-1})$ = $\text{kg m}^{-1} \text{s}^{-1}$ (ii) 1. $F = 6\pi \times D \times R \times v = [6\pi \times 6.6 \times 10^{-4} \times 1.5 \times 10^{-3} \times 3.7]$ = $6.9 \times 10^{-5} \text{ N}$ 2. $mg - F = ma$ hence $a = g - [F / m]$ $m = \rho \times V = \rho \times \frac{4}{3} \pi R^3 = (1.4 \times 10^{-5})$ $a = 9.81 - [6.9 \times 10^{-5} / \rho \times \frac{4}{3} \pi \times (1.5 \times 10^{-3})^3]$ (9.81 – 4.88) $a = 4.9(3) \text{ m s}^{-2}$ (b) (i) $a = g$ at time $t = 0$ a decreases (as time increases) a goes to zero (ii) Correct shape below original line sketch goes to terminal velocity earlier	B1	B1
		M1	A0 [3]
		A1	[1]
		C1	M1
		A1	[3]
		B1	B1
		B1	[3]
		M1	A1
		A1	[2]

- 10 (a) (i)** horizontal velocity = $15 \cos 60^\circ = 7.5 \text{ ms}^{-1}$ A1 [1]
(ii) vertical velocity = $15 \sin 60^\circ = 13 \text{ ms}^{-1}$ A1 [1]
(b) (i) $v^2 = u^2 + 2as$
 $s = (13)^2 / (2 \times 9.81) = 8.6(1) \text{ m}$ A1 [1]
 using $g = 10$ then max. 1
(ii) $t = 13 / 9.81 = 1.326 \text{ s}$ or $t = 9.95 / 7.5 = 1.327 \text{ s}$ A1 [1]
(iii) velocity = $6.15 / 1.33$ M1
 $= 4.6 \text{ ms}^{-1}$ A0 [1]
(c) (i) change in momentum = $60 \times 10^{-3} [-4.6 - 13]$ C1
 $= (-)1.06 \text{ N s}$ A1 [2]
(ii) final velocity / kinetic energy is less after the collision or
 relative speed of separation < relative speed of approach M1
 hence inelastic A0 [1]

- 11 (a)** average velocity = $540 / 30$ C1
 $= 18 \text{ ms}^{-1}$ A1 [2]
(b) velocity zero at time $t = 0$ B1
 positive value and horizontal line for time $t = 5 \text{ s}$ to 35 s B1
 line / curve through $v = 0$ at $t = 45 \text{ s}$ to negative velocity B1
 negative horizontal line from 53 s with magnitude less than positive value and
 horizontal line to time = 100 s B1 [4]

- 12 (a) (i)** $v = u + at$ C1
 $= 4.23 + 9.81 \times 1.51$ M1
 $= 19.0(4) \text{ ms}^{-1}$ (Allow 2 s.f.) A0 [2]
 (Use of $-g$ max 1/2. Use of $g = 10$ max 1/2. Allow use of 9.8. Allow 19 ms^{-1})
(ii) either $s = ut + \frac{1}{2}at^2$ (or $v^2 = u^2 + 2as$ etc.) C1
 $= 4.23 \times 1.51 + 0.5 \times 9.81 \times (1.51)^2$ A1 [2]
 $= 17.6 \text{ m}$ (or 17.5 m)
 (Use of $-g$ here wrong physics (0/2))
(b) (i) $F = \Delta P / \Delta t$ need idea of change in momentum C1
 $= [0.0465 \times (18.6 + 19)] / 12.5 \times 10^{-3}$ C1
 $= 140 \text{ N}$ A1
 (Use of $-$ sign max 2/4. Ignore $-ve$ sign in answer)
 Direction: upwards B1 [4]
(ii) $h = \frac{1}{2} \times (18.6)^2 / 9.81$ C1
 $= 17.6 \text{ m}$ (2 s.f. -1) A1 [2]
 (Use of 19 ms^{-1} , 0/2 wrong physics)
(c) either kinetic energy of the ball is not conserved on impact
 or speed before impact is not equal to speed after hence inelastic B1 [1]

- 13 (a) (i)** $v^2 = u^2 + 2as$
 $= (8.4)^2 + 2 \times 9.81 \times 5$ C1
 $= 12.99 \text{ m s}^{-1}$ (allow 13 to 2 s.f. but not 12.9) A1 [2]
- (ii)** $t = (v - u) / a$ or $s = ut + \frac{1}{2}at^2$
 $= (12.99 - 8.4) / 9.81$ or $5 = 8.4t + \frac{1}{2} \times 9.81t^2$ M1
 $t = 0.468 \text{ s}$ A0 [1]
- (b)** reasonable shape M1
suitable scale A1
correctly plotted 1st and last points at (0,8.4) and (0.88 – 0.96,0)
with non-vertical line at 0.47 s A1 [3]
- (c) (i)** 1. kinetic energy at end is zero so $\Delta KE = \frac{1}{2}mv^2$ or $\Delta KE = \frac{1}{2}mu^2 - \frac{1}{2}mv^2$ C1
 $= \frac{1}{2} \times 0.05 \times (8.4)^2$
 $= (-) 1.8 \text{ J}$ A1 [2]
2. final maximum height $= (4.2)^2 / (2 \times 9.8) = (0.9 \text{ (m)})$
change in PE $= mgh_2 - mgh_1$ C1
 $= 0.05 \times 9.8 \times (0.9 - 5)$ C1
 $= (-) 2.0 \text{ J}$ A1 [3]
- (ii)** change is – 3.8 (J) B1
energy lost to ground (on impact) / energy of deformation of the ball /
thermal energy in ball B1 [2]
-
- 14 (a)** displacement is a vector, distance is a scalar B1
displacement is straight line between two points / distance is sum of lengths
moved / example showing difference B1 [2]
(either one of the definitions for the second mark)
- (b)** a body continues at rest or at constant velocity unless acted on by a resultant
(external) force B1 [1]
- (c) (i)** sum of T_1 and T_2 equals frictional force B1
these two forces are in opposite directions B1 [2]
(allow for 1/2 for travelling in straight line hence no rotation / no resultant
torque)
- (ii)** 1. scale vector triangle with correct orientation / vector triangle with correct
orientation both with arrows B1
scale given or mathematical analysis for tensions B1 [2]
2. $T_1 = 10.1 \times 10^3 (\pm 0.5 \times 10^3) \text{ N}$ A1
 $T_2 = 16.4 \times 10^3 (\pm 0.5 \times 10^3) \text{ N}$ A1 [2]
-

- 15 (a) (i)** acceleration = change in velocity / time (taken)
or acceleration = rate of change of velocity B1 [1]
- (ii)** a body continues at constant velocity unless acted on by a resultant force B1 [1]
- (b) (i)** distance is represented by the area under graph C1
- distance = $\frac{1}{2} \times 29.5 \times 3 = 44.3 \text{ m}$ (accept 43.5 m for 29 to 45 m for 30) A1 [2]
- (ii)** resultant force = weight – frictional force B1
- frictional force increases with speed B1
- at start frictional force = 0 / at end weight = frictional force B1 [3]
- (iii) 1.** frictional force increases B1 [1]
- 2.** frictional force (constant) and then decreases B1 [1]
- (iv) 1.** acceleration = $(v_2 - v_1) / t = (20 - 50) / (17 - 15)$ C1
- = $(-) 15 \text{ m s}^{-2}$ A1 [2]
- 2.** $W - F = ma$ C1
- $W = 95 \times 9.81 (= 932)$ C1
- $F = (95 \times 15) + 932 = 2400 (2360) (2357) \text{ N}$ A1 [3]
-
- 16 (a)** units for D identified as kg m s M1
- all other units shown: units for A : m^2 units for v^2 : $\text{m}^2 \text{ s}^{-2}$ units for ρ : kg m^{-3}
- $C = \frac{\text{kg m s}^{-2}}{\text{kg m}^{-3} \text{ m}^2 \text{ m}^2 \text{ s}^{-2}}$ with cancelling / simplification to give C no units A1 [2]
- (b) (i)** straight line from (0,0) to (1,9.8) $\pm$ half a square B1 [1]
- (ii)** $\frac{1}{2} mv^2 = mgh$ or using $v^2 = 2as$ C1
- $v = (2 \times 9.81 \times 1000)^{1/2} = 140 \text{ m s}^{-1}$ A1 [2]
- (c) (i)** weight = drag (D) (+ upthrust) B1 [1]
- Allow mg or W for weight and D or expression for D for drag
- (ii) 1.** $mg = 1.4 \times 10^{-5} \times 9.81$ C1
- $1.4 \times 10^{-5} \times 9.81 = 0.5 \times 0.6 \times 1.2 \times 7.1 \times 10^{-6} \times v^2$ M1
- $v = 7.33 \text{ m s}^{-1}$ A0 [2]
- 2.** line from (0,0) correct curvature to a horizontal line at velocity of 7 m s^{-1} M1
- line reaches 7 m s^{-1} between 1.5 s and 3.5 s A1 [2]
-
- 17 (a) (i)** accelerations (A to B and B to C) are same magnitude B1
- accelerations (A to B and B to C) are opposite directions B1
- or both accelerations are toward B B1 [3]
- (A to B and B to C) the component of the weight down the slope provides the acceleration C1
- (ii)** acceleration = $g \sin 15^\circ$ C1
- $s = 0 + \frac{1}{2} at^2$ $s = 0.26 / \sin 15^\circ = 1.0$ C1
- $t^2 = \frac{1.0 \times 2}{9.8 \times \sin 15^\circ}$ $t = 0.89 \text{ s}$ A1 [3]
- (iii)** $v = 0 + g \sin 15^\circ t$ or $v^2 = 0 + 2g \sin 15^\circ \times 1.0$ C1
- $v = 2.26 \text{ m s}^{-1}$ A1 [2]
- (using loss of GPE = gain KE can score full marks)

- (b) loss of GPE at A = gain in GPE at C or loss of KE at B = gain in GPE at C B1
 $h_1 = h_2 = 0.26 \text{ m}$ or $\frac{1}{2}mv^2 = mgh$ $h_2 = 0.5 \times (2.26)^2 / 9.81 = 0.26 \text{ m}$
 $x = 0.26 / \sin 30^\circ = 0.52 \text{ m}$ A1 [2]

- 18 (a) (i) 1. distance of path / along line AB B1 [1]
 2. shortest distance between AB / distance in straight line between AB or displacement from A to B B1 [1]
 (ii) acceleration = rate of change of velocity A1 [1]
 (b) (i) distance = area under line or $(v/2)t$ or $s = (8.8)^2 / (2 \times 9.81)$ C1
 $= 8.8 / 2 \times 0.90 = 3.96 \text{ m}$ or $s = 3.95 \text{ m} = 4(.0) \text{ m}$ A1 [2]
 (ii) acceleration = $(-4.4 - 8.8) / 0.50$ C1
 $= (-) 26(.4) \text{ m s}^{-2}$ A1 [2]
 (c) (i) the accelerations are constant as straight lines B1
 the accelerations are the same as same gradient or
 no air resistance as acceleration is constant or
 change of speed in opposite directions (one speeds up one slows down) B1 [2]
 (ii) area under the lines represents height
 or KE at trampoline equals PE at maximum height B1
 second area is smaller / velocity after rebound smaller hence KE less B1
 hence less height means loss in potential energy A0 [2]

- 19 (a) (i) velocity = rate of change of displacement
 OR displacement change / time (taken) A1 [1]
 (ii) acceleration = rate of change of velocity
 OR change in velocity / time (taken) A1 [1]
 (b) (i) initial constant velocity as straight line / gradient constant B1
 middle section deceleration/ speed / velocity decreases / slowing down as
 gradient decreases B1
 last section lower velocity (than at start) as gradient (constant and) smaller B1 [3]
 [special case: all three stages correct descriptions but no reasons 1/3]
 (ii) velocity = $45 / 1.5 = 30 \text{ m s}^{-1}$ A1 [1]
 (iii) velocity at 4.0 s is $(122 - 98) / 2.0 = 12 \text{ (m s}^{-1})$ (allow 12 to 13) B1
 acceleration = $(12 - 30) / 2.5 = -7.2 \text{ m s}^{-2}$ (if answer not this value then
 comment needed to explain why, e.g. difficulty in drawing tangent) A1 [2]
 (iv) $F = ma$ C1
 $= (-)1500 \times 7.2 = (-)11000 \text{ (10800) N}$ A1 [2]

- 20 (a) (i)** *either* rate of change of displacement
or (change in) displacement/time (taken) B1 [1]
- (ii)** speed has magnitude only B1
velocity has magnitude and direction B1 [2]
- (b) (i)** idea of area under graph/use of $s = \frac{(u+v)}{2} \times t$ C1
- $$s = \frac{(18+32)}{2} \times 2.5$$
- $$= 62.5 \text{ m}$$
- C1 A1 [3]
- (ii)** $a = (18 - 32)/2.5 (= -5.6)$ C1
 $F = ma$ C1
 $F = 1500 \times (-) 5.6 = (-) 8400 \text{ N}$ A1 [3]
- (c)** arrow labelled A and arrow labelled F both to the left B1 [1]

- 21 (a)** scalar has magnitude only B1
vector has magnitude and direction B1 [2]
- (b) (i)** $v^2 = 0 + 2 \times 9.81 \times 25$ (or using $\frac{1}{2}mv^2 = mgh$) C1
 $v = 22(.1) \text{ ms}^{-1}$ A1 [2]
- (ii)** $22.1 = 0 + 9.81 \times t$ (or $25 = \frac{1}{2} \times 9.81 \times t^2$) M1
 $t (=22.1/9.81) = 2.26 \text{ s}$ or $t [= (5.097)^{1/2}] = 2.26 \text{ s}$ A0 [1]
- (iii)** horizontal distance $= 15 \times t$
 $= 15 \times 2.257 = 33.86$ (allow $15 \times 2.3 = 34.5$) C1
(displacement) $^2 =$ (horizontal distance) $^2 +$ (vertical distance) 2 C1
 $= (25)^2 + (33.86)^2$ C1
displacement $= 42$ (42.08) m (allow 43 (42.6) m, allow 2 or more s.f.) A1 [4]
- (iv)** distance is the actual (curved) path followed by ball B1
displacement is the straight line/minimum distance P to Q B1 [2]

- 22 (a) (i)** acceleration $= (v - u) / t$ or $(12 - 0.5) / 4$ C1
 $= (12 - 0.5) / 4 = 2.9$ (2.875) (= approximately 3 ms^{-2}) M1 [2]
- (ii)** $x = (u + v)t / 2$
 $= [(12 + 0.5) \times 4] / 2$ C1
 $= 25 \text{ m}$ A1 [2]
- (iii)** line with increasing gradient M1
non-zero gradient at origin A1 [2]
- (b) (i)** weight down slope $= 2 \times 9.81 \times \sin 25^\circ = 8.29 / 8.3$ M1 [1]
- (ii)** ($F = ma$) $8.3 - F_R = 2 \times 2.9$ C1
 $F_R = 2.5$ (2.3 if 3 used for a) N A1 [2]

23 (a) (i) straight line from $t = 0.60\text{ s}$ to $t = 1.2\text{ s}$ and $ V_v = 5.9$ at $t = 1.2\text{ s}$ $V_v = -5.9$ at $t = 1.2\text{ s}$ i.e. line is for negative values of V_v	M1 A1 [2]
(ii) $s = 0 + \frac{1}{2} \times 9.81 \times (0.6)^2$ or area of graph $= (5.9 \times 0.6) / 2$ $= 1.8 (1.77)\text{ m}$ $= 1.8 (1.77)\text{ m}$	C1 A1 [2]
(iii) $V_h = V \cos 60^\circ$ <u>and</u> $V_v = V \sin 60^\circ$ or $V_h = 5.9 / \tan 60^\circ$ or $V_h = 5.9 \tan 30^\circ$ $V_h = 3.4\text{ ms}^{-1}$	C1 A1 [2]
(iv) horizontal line at 3.4 from $t = 0$ to $t = 1.2\text{ s}$ [to half a small square]	B1 [1]
(b) (i) $KE = \frac{1}{2}mv^2$ $= \frac{1}{2} \times 0.65 \times (6.81)^2$ [allow if valid method to find v] $= 15 (15.1)\text{ J}$	C1 C1 A1 [3]
(ii) $PE = 0.65 \times 9.81 \times 1.77$ $= 11(11.3)\text{ J}$	C1 A1 [2]
24 (a) speed = distance/time and velocity = displacement/time speed is a scalar as distance has no direction and velocity is a vector as displacement has direction	B1 B1 [2]
(b) (i) <u>constant</u> acceleration or linear/uniform increase in velocity until 1.1 s rebounds or bounces or changes direction decelerates to zero velocity at the same acceleration as initial value	B1 B1 B1 [3]
(ii) $a = (v - u)/t$ or use of gradient implied $= (8.8 + 8.8)/1.8$ or appropriate values from line or $= (8.6 + 8.6)/1.8$ $= 9.8 (9.78)\text{ ms}^{-2}$ $\text{or} = 9.6\text{ ms}^{-2}$	C1 B1 A1 [3]
(iii) 1. distance = first area above graph + second area below graph $= (1.1 \times 10.8)/2 + (0.9 \times 8.8)/2 (= 5.94 + 3.96)$ $= 9.9\text{ m}$ 2. displacement = first area above graph – second area below graph $= (1.1 \times 10.8)/2 - (0.9 \times 8.8)/2$ $= 2.0 (1.98)\text{ m}$	C1 C1 A1 [3] C1 A1 [2]
(iv) correct shape with straight lines and all lines above the time axis or all below correct times for zero speeds (0.0, 1.15 s, 2.1 s) and peak speeds (10.8 ms^{-1} at 1.1 s and 8.8 ms^{-1} at 1.2 s and 3.0 s)	M1 A1 [2]

25 (a)	speed decreases/stone decelerates <u>to rest/zero</u> at 1.25 s	B1
	speed then increases/stone accelerates (in opposite direction)	B1 [2]
(b) (i)	$v = u + at$ (or $s = ut + \frac{1}{2}at^2$ and $v^2 = u^2 + 2as$)	C1
	$= 0 + (3.00 - 1.25) \times 9.81$	C1
	$= 17.2$ (17.17) m s^{-1}	A1 [3]
(ii)	$s = ut + \frac{1}{2}at^2$	
	$s = \frac{1}{2} \times 9.81 \times (1.25)^2 [= 7.66]$	C1
	$s = \frac{1}{2} \times 9.81 \times (1.75)^2 [= 15.02]$	C1
	(distance = 7.66 + 15.02)	
	$[v = u + at = 0 + 9.81 \times (2.50 - 1.25) = 12.26 \text{ m s}^{-1}]$	
	or	
	$s = \frac{1}{2} \times 9.81 \times (1.25)^2 [= 7.66]$	(C1)
	$s = 12.26 \times 0.50 + \frac{1}{2} \times 9.81 \times (3.00 - 2.50)^2 [= 7.36]$	(C1)
	(distance = 2 × 7.66 + 7.36)	
	<i>Example alternative method:</i>	
	$s = (v^2 - u^2) / 2a = (12.26^2 - 0) / 2 \times 9.81 [= 7.66]$	(C1)
	$s = (v^2 - u^2) / 2a = (17.17^2 - 12.26^2) / 2 \times 9.81 [= 7.36]$	(C1)
	(distance = 2 × 7.66 + 7.36)	
	22.7 (22.69 or 23) m	A1 [3]
(iii)	$(s = 15.02 - 7.66 =) 7.4$ (7.36) m (<i>ignore sign in answer</i>)	A1
	down	A1 [2]
(c)	straight line from positive value of v to t axis	M1
	same straight line <u>crosses</u> t axis at $t = 1.25$ s	A1
	same straight line continues with same gradient to $t = 3.0$ s	A1 [3]
26	D <u>constant</u> rate of increase in velocity/acceleration from $t = 0$ to $t = 8$ s	B1
	<u>constant</u> deceleration from $t = 8$ s to $t = 16$ s or constant rate of increase in	
	velocity in the opposite direction from $t = 10$ s to $t = 16$ s	B1 [2]
(b) (i)	area under lines to 10 s	C1
	(displacement =) $(5.0 \times 8.0) / 2 + (5.0 \times 2.0) / 2 = 25$ m	
	or $\frac{1}{2} (10.0 \times 5.0) = 25$ m	A1 [2]
(ii)	$a = (v - u) / t$ or gradient of line	C1
	$= (-15.0 - 5.0) / 8.0$	
	$= (-) 2.5 \text{ m s}^{-2}$	A1 [2]
(iii)	$\text{KE} = \frac{1}{2}mv^2$	C1
	$= 0.5 \times 0.4 \times (15.0)^2 = 45$ J	A1 [2]
(c)	(distance =) 25 (m) ($= ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 2.5 \times t^2$)	C1
	$(t = 4.5$ (4.47) s therefore) time to return = 14.5 s	A1 [2]

27 (a)	weight = 452×9.81 component down the slope = $452 \times 9.81 \times \sin 14^\circ$ = $1072.7 = 1070 \text{ N}$	M1 A0 [1]
(b) (i)	$F = ma$ $T - (1070 + 525) = 452 \times 0.13$ $T = 1650 \text{ (1653.76) N}$ any forces missing 1/3	C1 C1 A1 [3]
(ii) 1.	$s = ut + \frac{1}{2}at^2$ hence $10 = 0 + \frac{1}{2} \times 0.13t^2$ $t = [(2 \times 10) / 0.13]^{1/2} = 12.4 \text{ or } 12 \text{ s}$	C1 A1 [2]
2.	$v = (0 + 2 \times 0.13 \times 10)^{1/2} = 1.61 \text{ or } 1.6 \text{ ms}^{-1}$	A1 [1]
(c)	straight line from the origin line down to zero velocity in short time compared to stage 1 line less steep negative gradient final velocity larger than final velocity in the first part – at least $2\times$	B1 B1 B1 B1 [4]
28 (a)	change in velocity / time (taken) or rate of change of velocity	B1
(b) (i)	$v_x = (24 / 1.5) = 16 \text{ (ms}^{-1}\text{)}$	A1
(ii)	$\tan 28^\circ = v_y / v_x$ or $v_x = v \cos 28^\circ$ and $v_y = v \sin 28^\circ$ $v_y = 16 \tan 28^\circ$ or $v_y = 16 \times (\sin 28^\circ / \cos 28^\circ)$ so $v_y = 8.5 \text{ (ms}^{-1}\text{)}$	C1 A1
(iii)	$v = u + at$ $t = (0 - 8.5) / (-9.81)$ $= 0.87 \text{ (s)}$	C1 A1
(iv)	straight line from positive v_y at $t = 0$ to negative v_y at $t = 1.5 \text{ s}$ line starts at $(0, 8.5)$ and crosses t -axis at $(0.87, 0)$ and does not go beyond $t = 1.5 \text{ s}$.	M1 A1
(c) (i)	$(v^2 = u^2 + 2as)$ $0 = 8.5^2 + 2(-9.81)s$ or $(s = ut + \frac{1}{2}at^2)$ $s = 8.5 \times 0.87 + \frac{1}{2} \times (-9.81) \times 0.87^2$ or $(s = vt - \frac{1}{2}at^2)$ $s = 0 - \frac{1}{2} \times (-9.81) \times 0.87^2$ or $(s = \frac{1}{2}(u + v)t$ or area under graph) $s = 0.5 \times 8.5 \times 0.87$ $s = 3.7 \text{ (m)}$	C1 A1
(ii)	$\Delta E_p = mg\Delta h$ (allow $E = mgh$) $m = 22 / (9.81 \times 3.7)$ $= 0.61 \text{ (kg)}$	C1 A1
(d)	acceleration (of freefall) is unchanged / not dependent on mass, and so no effect (on maximum height) or explanation in terms of energy: (initial) $\text{KE} \propto \text{mass}$, $(\Delta)\text{KE} = (\Delta)\text{PE}$, (max) $\text{PE} \propto \text{mass}$, and so no effect (on maximum height)	B1

- 29 (a) (i)** horizontal component ($= 12 \cos 50^\circ$) $= 7.7 \text{ m s}^{-1}$ A1 [1]
 (ii) vertical component ($= 12 \sin 50^\circ$ or $7.7 \tan 50^\circ$) $= 9.2 \text{ m s}^{-1}$ A1 [1]
(b) $v^2 = u^2 + 2as$ and $v = 0$ or $mgh = \frac{1}{2}mv^2$ or $s = v^2 \sin^2 \theta / 2g$ C1
 $9.2^2 = 2 \times 9.81 \times h$ hence $h = 4.3$ (4.31) m A1 [2]
 alternative methods using time to maximum height of 0.94 s:
 $s = ut + \frac{1}{2}at^2$ and $t = 0.94$ (s) (C1)
 $s = 9.2 \times 0.94 - \frac{1}{2} \times 9.81 \times 0.94^2$ hence $s = 4.3$ m (A1)
 or
 $s = vt - \frac{1}{2}at^2$ and $t = 0.94$ (s) (C1)
 $s = \frac{1}{2} \times 9.81 \times 0.94^2$ hence $s = 4.3$ m (A1)
 or
 $s = \frac{1}{2}(u + v)t$ and $t = 0.94$ (s) (C1)
 $s = \frac{1}{2} \times 9.2 \times 0.94$ hence $s = 4.3$ m (A1)
(c) $t (= 9.2/9.81) = 0.94$ (0.938) s C1
 horizontal distance $= 0.938 \times 7.7$ ($= 7.23$ m) C1
 displacement $= [4.3^2 + 7.23^2]^{1/2}$ C1
 $= 8.4$ m A1 [4]
-

- 30 (a)** rate of change of displacement **or** change in displacement/time taken B1
 E L $s = ut + \frac{1}{2}at^2$ C1
 $t = [(2 \times 1.25) / 9.81]^{1/2}$ ($= 0.5048$ s) C1
or $v^2 = u^2 + 2as$ (C1)
 $v_{\text{vert}} = (2 \times 9.81 \times 1.25)^{1/2}$ ($= 4.95$)
 $t = [2s / (u + v)] = 2 \times 1.25 / 4.95$ ($= 0.5048$ s) (C1)
 $v = d / t = 1.5 / 0.50(48)$ A1
 $= 3.0$ (2.97) m s^{-1}
(ii) vertical velocity $= at$ C1
 $= 9.81 \times 0.5048$ ($= 4.95$) [using $t = 0.50$ gives 4.9]
 velocity $= [(v_h)^2 + (v_v)^2]^{1/2}$ C1
 $= [(2.97)^2 + (4.95)^2]^{1/2}$ A1
 $= 5.8$ (5.79) [using $t = 0.50$ leads to 5.7]
 direction $(= \tan^{-1} 4.95/2.97) = 59^\circ$ A1
-

31	D	$30 \text{ ms}^{-1} = 108 \text{ kmh}^{-1}$ or $100 \text{ kmh}^{-1} = 28 \text{ ms}^{-1}$ and so exceeds speed limit	B1
	E	acceleration = gradient or $\Delta v / (\Delta)t$ or $(v - u) / t$ e.g. acceleration = $(24 - 20) / 12$ [other points on graph line may be used] $= 0.33 \text{ ms}^{-2}$	C1 A1
	F	distance travelled by Q = $\frac{1}{2} \times 12 \times 30$ (= 180 m) distance travelled by P = $\frac{1}{2} \times (20 + 24) \times 12$ (= 264 m) distance between cars = $264 - 180$	C1 C1 A1
	G	$30 - 24 = 6 \text{ ms}^{-1}$ = 84 m 'extra' time $T = 84 / 6$ (= 14 s) or $180 + 30T = 264 + 24T$ 'extra' time $T = 84 / 6$ (= 14 s) $t = 12 + 14 = 26 \text{ s}$	C1 A1
	D	time = 12 s	A1
	E	distance (up slope) = $\frac{1}{2} \times 12 \times 18$ (= 108) distance (down slope) = $\frac{1}{2} \times 12 \times 6$ (= 36) displacement from A = $108 - 36$ = 72 m	C1 C1 A1
	F	$v = u + at$ or $a = \text{gradient}$ or $a = \Delta v / (\Delta)t$ $a = 6 / 12 = 0.50 \text{ (ms}^{-2}\text{)}$ (other points from the line may be used) or $v^2 = u^2 + 2as$ and $u = 0$ (C1) or $s = ut + \frac{1}{2}at^2$ and $u = 0$ (C1) or $s = vt - \frac{1}{2}at^2$ $a = 2 \times (6 \times 12 - 36) / 12^2 = 0.50 \text{ (ms}^{-2}\text{)}$	C1 A1 or $v^2 = 2as$ $a = 6.0^2 / (2 \times 36) = 0.50 \text{ (ms}^{-2}\text{)}$ (A1) or $s = \frac{1}{2}at^2$ $a = 2 \times 36 / 12^2 = 0.50 \text{ (ms}^{-2}\text{)}$ (A1) (C1) (A1)
	(d) (i)	$F = 70 \times 0.50$ (= 35) frictional force = $80 - 35$ = 45 N	C1 A1
	(ii)	$\sin \theta = 80 / (70 \times 9.81)$ $\theta = 6.7^\circ$	C1 A1
	(e) (i)	$f_0 = (900 \times 340) / (340 + 12)$ = 870 Hz	C1 A1
	(ii)	speed/velocity (of sledge) decreases and (so) frequency increases	B1

33 (a)	straight (horizontal) lines and from the +0.90 kV plate/to the 0 V plate (lines are) equally spaced	B1 B1
(b)	weight/gravitational force and electric force	B1
(c)	$s = \frac{1}{2}at^2$ or $s = ut + \frac{1}{2}at^2$ and $u = 0$ $2.0 = \frac{1}{2} \times 9.81 \times t^2$ so $t = 0.64$ s	C1 A1
(d)	$0.080 = \frac{1}{2} \times a \times 0.64^2$ $a = 0.39 \text{ m s}^{-2}$	C1 A1
(e) (i)	$E = (\Delta)V / (\Delta)d$ $E = 0.90 \times 10^3 / 0.12$ $= 7.5 \times 10^3 \text{ N C}^{-1}$	C1 A1
(ii)	$ma = Eq$ or $F = ma$ and $F = Eq$ $q/m = 0.39 / 7.5 \times 10^3$ $= 5.2 \times 10^{-5} \text{ C kg}^{-1}$	C1 A1
(f) (i)	no effect	B1
(ii)	decreases/smaller	B1

34 (a)	the (two) plates are <u>vertical</u> (and separated) left plate positively charged and right plate negatively charged/earthed or right plate negatively charged and left plate positively charged/earthed	B1 B1
(b)	$F = Eq$ $= 1.3 \times 10^4 \times 3.7 \times 10^{-9}$ $= 4.8 \times 10^{-5} \text{ N}$	C1 A1
(c)	$F^2 = (4.8 \times 10^{-5})^2 + (5.4 \times 10^{-5})^2$ so $F = 7.2 \times 10^{-5} \text{ N}$ or $F = [(4.8 \times 10^{-5})^2 + (5.4 \times 10^{-5})^2]^{0.5}$ so $F = 7.2 \times 10^{-5} \text{ N}$	A1
(d)	electric force is constant (because field strength/ E is constant) weight is constant (and so resultant force constant)	B1 B1
(e) (i)	$m = 5.4 \times 10^{-5} / 9.81 (= 5.5 \times 10^{-6})$ $a = 7.2 \times 10^{-5} / (5.5 \times 10^{-6})$ $= 13 \text{ m s}^{-2}$	C1 A1
(ii)	$v^2 = u^2 + 2as$ $v^2 = 2 \times 13 \times 0.58$ $v = 3.9 \text{ m s}^{-1}$	C1 A1

35 (a)	change in velocity / time (taken)	A1
(b) (i)	weight $\gg$ (force due to) air resistance or (force due to) air resistance is negligible <u>compared to weight</u>	B1
(ii)	$s = ut + \frac{1}{2}at^2$ $0.280 = \frac{1}{2} \times 9.81 \times t^2$ $t = 0.24 \text{ s}$	C1 A1
(iii)	total distance fallen = $0.280 + 0.080 = 0.360$ $0.360 = \frac{1}{2} \times 9.81 \times t^2$ $t = 0.27 \text{ s}$ time taken = $0.27 - 0.24$ or = 0.03 s $v = 9.81 \times 0.239$ or $(2 \times 9.81 \times 0.280)^{0.5}$ or $(2 \times 0.280) / 0.239$ = $2.34 \text{ (ms}^{-1}\text{)}$ $0.080 = 2.34t + \frac{1}{2} \times 9.81 \times t^2$ solving quadratic equation gives $t = 0.03 \text{ s}$ <i>allow any correct method using equations of uniform accelerated motion</i>	C1 A1 (C1) (A1)
(c)	(average) resultant force/acceleration/speed/velocity (of low-density ball) is less (so) time interval is longer	B1 B1
36 (a)	(work done =) force $\times$ distance <u>moved</u> in direction of force	B1
(b) (i)	1. acceleration = gradient or $a = (v - u) / t$ or $a = \Delta v / t$ e.g. $a = 2.4 / 3.0$ = 0.80 m s^{-2} 2. tension in cable = $(13.0 + 2.0) \times 10^3$ work done = $15 \times 10^3 \times (3.0 \times 2.4)$ = $1.1 \times 10^5 \text{ J}$	C1 A1 C1 A1
(ii)	power = Fv $v = 2.0 \text{ (ms}^{-1}\text{)}$ input power = $(1.6 \times 10^4 \times 2.0) / 0.67$ = $4.8 \times 10^4 \text{ W}$	C1 C1 A1
(iii)	work is done against friction so (increase in) GPE is less (than work done by motor) or energy is lost or transferred or converted to heat/thermal energy due to friction or resistance force or work is done lifting the cable so GPE is less	A1
37 (a)	(velocity =) change in displacement / time (taken)	B1
(b)	units of F : kg m s^{-2} units of k : $\text{kg m s}^{-2} / [\text{m}^2 \times (\text{m s}^{-1})^2]$ = kg m^{-3}	C1 A1
(c)	$P = Fv$ $4.8 \times 10^4 = 0.24 \times 5.1 \times v^3$ $v = 34 \text{ m s}^{-1}$	C1 C1 A1

- 38 (a)** similarity: both have magnitude **B1**
 difference: distance is a scalar/does not have direction **B1**
or
 displacement is a vector/has direction
- (b)(i)** the measurements have a small range **B1**
(ii) the (average of the) measurements is not close to the true value **B1**

39

- | | |
|-----|---|
| (a) | constant gradient |
| (b) | (displacement until 0.20 s =) $\frac{1}{2} \times 1.96 \times 0.20$ (= 0.196 m)
or
(displacement after 0.20 s =) $\frac{1}{2} \times 6.86 \times 0.70$ (= 2.401 m)
height = 2.401 – 0.196
= 2.2 m
<i>(alternative methods are possible using equations of uniformly accelerated motion)</i> |
| (c) | $(\Delta)E = mg(\Delta)h$ or $W(\Delta)h$
$(\Delta)E = 0.86 \times 2.2$
= 1.9 J |
| (d) | curved line from the origin
gradient of curved line decreases and is zero at $t = 0.20$ s only |
| (e) | acceleration (of free fall) is unchanged/is not dependent on mass <u>and</u> (so) no effect |

40

- | | |
|---------|--|
| (a) | density and temperature indicated as scalars
acceleration and momentum indicated as vectors |
| (b)(i) | decelerates
or
speed/velocity decreases |
| (b)(ii) | speed = $(\Delta)d / (\Delta)t$ or gradient
= e.g. $(0.56 - 0.20) / 1.5$
= 0.24 ms^{-1} |
| (c) | displacement is zero (so) average velocity is zero |

41.

(a)	$s = \frac{1}{2}at^2$ $57 = \frac{1}{2} \times 9.81 \times t^2$ and $t = 3.4$ (s)
(b)	horizontal distance = 41×3.4 $= 140$ m
(c)	$(\text{displacement})^2 = 57^2 + 140^2$ $\text{displacement} = (57^2 + 140^2)^{0.5}$ $= 150$ m
(d)	straight line from the origin with positive gradient
(e)	$(1480 - m) \times 0.340 = m \times 41.0$ $m = 12.2$ kg or $m_c 0.34 = m_b 41$ and $m_c + m_b = 1480$ $m_c = (41 / 0.34)m_b$ $(41 / 0.34)m_b + m_b = 1480$ $m_b = 12.2$ kg
(f)	acceleration (of free fall) is unchanged/is not dependent on mass (so) no change (to the graph)

42.

(a)	acceleration: vector work: scalar power: scalar <i>Three correct scores 2 marks. Two correct scores 1 mark.</i>
(b)(i)	$a = (v - u) / t$ or $a = \text{gradient}$ or $a = \Delta v / (\Delta)t$ e.g. $a = (1.40 - 0.70) / 4.0$ $= 0.18 \text{ m s}^{-2}$
(b)(ii)	$\text{distance} = 0.5 \times (0.70 + 1.40) \times 4.0$ or $(0.70 \times 4.0) + (0.5 \times 0.70 \times 4.0)$ $= 4.2$ m
(c)(i)	(force equal to) rate of change of momentum
(c)(ii)	horizontal line starting from $t = 0$ and ending at $t = 4.0$ s at a positive value of F horizontal line starting from $t = 4.0$ s and ending at $t = 8.0$ s at $F = 0$ horizontal line starting from $t = 8.0$ s and ending at $t = 12.0$ s at a negative value of F <u>and</u> the magnitude of F is larger than from $t = 0$ to 4.0 s

43.

(a)	change in velocity / time (taken)
(b)(i)	air resistance increases (with speed/with time)
	resultant force decreases (as speed increases/with time) so acceleration decreases (as speed increases/with time)
	when air resistance equals the weight the speed/velocity/ v becomes constant
(b)(ii)	speed = 36 m s^{-1}
(b)(iii)	height given by area under the curve
	height = 950 m
	<i>Round to two significant figures and award 2 marks for a value in the range 920–980 m and 1 mark for a value in the range 900–910 m or 990–1000 m.</i>
(b)(iv)	line starting at (0, 9.8)
	curve with negative gradient between $t = 0$ and $t = 20 \text{ s}$
	line showing zero acceleration between $t = 20 \text{ s}$ and $t = 30 \text{ s}$

44.

(a)(i)	decrease(s)
(a)(ii)	increase(s)
(b)	$f_o = f_s v / (v + v_s)$
	$= 925 \times 338 / (338 + 12)$
	$= 893 \text{ Hz}$
(c)	distance = $(\frac{1}{2} \times 2 \times 12) + (2 \times 12)$
	$(= 36 \text{ m})$
	time taken = $36 / 338$
	$= 0.11 \text{ s}$

45.

(a)	change in displacement / time (taken)
(b)	by calculation:
	$v^2 = 42^2 + 23^2 - (2 \times 42 \times 23 \times \cos 54^\circ)$
	or
	$v^2 = (42 - 23 \cos 54^\circ)^2 + (23 \sin 54^\circ)^2$
	or
	$v^2 = (42 - 23 \sin 36^\circ)^2 + (23 \cos 36^\circ)^2$
	$v = 34 \text{ m s}^{-1}$
	or
	by scale diagram: triangle of vector velocities drawn
	$v = 34 \text{ m s}^{-1}$ (allow $\pm 1 \text{ m s}^{-1}$ if scale diagram used)

46.

(a)	force (on droplet of water) in horizontal direction is zero.
(b)	(time taken =) $3.5 / 6.6 = 0.53$ (s)
(c)	$s = ut + \frac{1}{2}at^2$ $s = \frac{1}{2} \times 9.81 \times 0.53^2$ $h = 1.4$ m
(d)	displacement is straight-line distance (from P to Q) so less (than distance along path) or displacement is the shortest distance (from P to Q).
(e)	(displacement) ² = $3.5^2 + 1.4^2$ displacement = 3.8 m

47.

(a)	$v^2 = u^2 + 2as$ $s = 5.6^2 / (2 \times 9.81)$ (max height =) $3.1 + 5.6^2 / (2 \times 9.81) = 4.7$ (m)
(b)	$s = ut + \frac{1}{2}at^2$ $4.7 = \frac{1}{2} \times 9.81 \times t^2$ $t = 0.98$ s
(c)	line drawn from a non-zero speed at $t = 0$ to a greater speed at $t = T$ a single sloping straight line drawn from $t = 0$ to $t = T$ line starts with a positive non-zero value of v and ends with a negative non-zero value of v
(d)	acceleration (of the ball)
(e)(i)	(magnitudes of accelerations are) equal / same
(e)(ii)	(speeds are) equal / same

48.

(a)	(time =) displacement / velocity
	(time =) $70.0 / 65.0 \cos 4.30^\circ = 1.08 \text{ (s)}$
(b)	$s = ut + \frac{1}{2}at^2$
	$= (65 \times \sin 4.30^\circ \times 1.08) - (0.5 \times 9.81 \times 1.08^2)$
	$s = -0.46 \text{ (m)}$
	height above ground = $1.66 - 0.46$ $= 1.2 \text{ m}$
(c)	GPE has decreased
	(total energy conserved so) KE has increased

49.

(a)(i)	(e.g. $a =$) $[16.5 - 0] / [0.30 - 0] = 55 \text{ (m s}^{-2}\text{)}$
(a)(ii)	$(F =) T - W$
(a)(iii)	$ma = T - mg$
	$m = 16 / (55 + 9.81)$
	$m = 0.25 \text{ kg}$
(b)(i)	$s = ut + \frac{1}{2}at^2$ or $v^2 = u^2 + 2as$ or $s = vt - \frac{1}{2}at^2$ or $s = \frac{1}{2}(u + v)t$ or $s = \text{area under graph}$
	$s = \frac{1}{2} \times 55 \times 0.30^2$
	or
	$s = 16.5^2 / (2 \times 55)$
	or
	$s = 16.5 \times 0.30 - \frac{1}{2} \times 55 \times 0.30^2$
(b)(ii)	or
	$s = \frac{1}{2} \times 16.5 \times 0.30$
	$s = 2.5 \text{ m}$
	$u = 16.5 \text{ (m s}^{-1}\text{)}$
(b)(ii)	$v^2 = u^2 + 2as$
	$v^2 = 16.5^2 + 2 \times 9.81 \times 2.5$
	$v = 18 \text{ m s}^{-1}$

50.

(a)	displacement
(b)(i)	$a = \text{gradient}$ or $a = \Delta v / (\Delta)t$ or $a = (v - u) / t$ e.g. $a = (0.30 - 0.12) / (0.35 - 0.15)$ $a = 0.90 \text{ m s}^{-2}$
(b)(ii)	$(0.25 \times 0.48) + (0.75 \times 0.12) = (0.25 v) + (0.75 \times 0.30)$ or $(0.48 - 0.12) = (0.30 - v)$ or $(\frac{1}{2} \times 0.25 \times 0.48^2) + (\frac{1}{2} \times 0.75 \times 0.12^2) = (\frac{1}{2} \times 0.25 \times v^2) + (\frac{1}{2} \times 0.75 \times 0.30^2)$ $v = (-)0.060 \text{ m s}^{-1}$ direction: to the left / from the right / opposite to (its) initial velocity / opposite to (initial / final) velocity of B
(c)	sketch: horizontal line from (0, 0.48) to (0.15, 0.48) horizontal line from (0.35, -0.06) to (0.5, -0.06) straight line between (0.15, 0.48) and (0.35, -0.06)

51.

(a)	only electric current and time underlined
(b)	initial speed / velocity is zero (non-zero magnitude of) acceleration is constant / uniform (and in a straight line)
(c)(i)	- magnitude of acceleration at $t = 8.0 \text{ s}$ is less than that at $t = 14.0 \text{ s}$ - direction of acceleration at $t = 8.0 \text{ s}$ is opposite to that at $t = 14.0 \text{ s}$
(c)(ii)	$a = \text{gradient}$ or $a = (v - u) / t$ or $a = \Delta v / (\Delta)t$ $a = \text{e.g. } (20 + 10) / 12$ or $(0 + 10) / 4$ or $(20 - 0) / (12 - 4)$ $a = 2.5 \text{ m s}^{-2}$
(c)(iii)	displacement = $[\frac{1}{2} \times (12 - 4) \times 20] - [\frac{1}{2} \times 4 \times 10]$ or displacement = $(-10 \times 12) + (\frac{1}{2} \times 2.5 \times 12^2)$ or displacement = $(20 \times 12) - (\frac{1}{2} \times 2.5 \times 12^2)$ or displacement = $\frac{1}{2} \times (20 - 10) \times 12$ displacement = 60 m

52.

(a)(i)	(time / t) $9(.0) / 9.5 = 0.95$ (s)
(a)(ii)	$(u_V) = 9.5 \tan 38^\circ$ or $9.5 / \tan 52^\circ$ or $9.5 = u \cos 38^\circ$ and $u_V = u \sin 38^\circ$ or $9.5 = u \cos 38^\circ$ and $u_V = (u^2 - 9.5^2)^{1/2}$
	$u_V = 7.4 \text{ m s}^{-1}$
(a)(iii)	$s = ut + \frac{1}{2}at^2$
	$(h =) 7.4 \times 0.95 - \frac{1}{2} \times 9.81 \times 0.95^2$
	$h = 2.6 \text{ m}$
(b)	(collision is) inelastic